

Project Data Page:





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DESIGN CALCULATION

In Accordance with ASME Section VIII Division 1

ASME Code Version : 2025

Analysis Performed by : SL User

Job File : ...\\175-C.pvdb

Date of Analysis : Sep 9, 2026 9:45am

PV Elite 28, January 2026



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FileName : 175-C

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Vessel Design Summary:

Step: 32

9:45am

Sep 9, 2026

Vessel Design Summary:

ASME Code, Section VIII Division 1, 2025

Diameter Spec : 1100.000 mm ID
Vessel Design Length, Tangent to Tangent 7259.00 mm
Specified Datum Line Distance 0.00 mm
Shell Side Design Temperature 275 °C
Channel Side Design Temperature 430 °C
Shell Side Design Pressure 1080.000 KPa.
Channel Side Design Pressure 5900.000 KPa.
Shop Shell Side Test Pressure 2476.714 KPa.
Field Shell Side Test Pressure 1404.000 KPa.
Shop Channel Side Test Pressure 9179.087 KPa.
Field Channel Side Test Pressure 7721.800 KPa.
Wind Design Code ASCE/SEI 7-16
Earthquake Design Code ASCE/SEI 7-16

Materials of Construction:

Component Type	Material	Class	Thickness	UNS #	Material Treatment Options
Shell	SA-387 11	2	...	K11789	
Shell	SA-516 70	...	...	K02700	
Head	SA-516 70	...	...	K02700	
Flange	SA-336 F11	3	...	K11572	
Nozzle	SA-335 P11	...	...	K11597	
Nozzle	SA-106 B	...	...	K03006	
Nozzle	SA-105	...	...	K03504	
Re-Pad	SA-387 11	2	...	K11789	
Re-Pad	SA-516 70	...	...	K02700	
Nozzle Flg	SA-336 F11	3	...	K11572	
Nozzle Flg	SA-182 F11	2	...	K11572	
Nozzle Flg	SA-105	...	...	K03504	
Tubes	SA-213 T11	...	...	K11597	
Tubesheet	SA-336 F11	3	...	K11572	
Flg Bolting	SA-193 B16	...	<= 64	K14072	
Hrz Bolting	SA-307 A	...	6 <= t <= 100		
Saddles	SA-516 70	...	...	K02700	

+FG - Fine Grained, +R - rolled, +T - Tempered, +N - Normalized

+Q - quenched, +CBV - Cb & V added, +IT - Impact tested

Element Pressures and MAWP (KPa. & mm):

Element Description or Type	Design Pressure + Stat. head	Ext. Press.	Element M.A.W.P	Total Corrosion Allowance	Str. Flg. Gov.	In Creep Range
CH COVER	5900.000	101.30	6332.000	3.0000	N/A	No
CH Flange	5900.000	101.30	7473.000	3.0000	N/A	No
Channel shell	5910.812	101.30	6342.000	3.0000	N/A	No
Shell	1090.812	0.00	3120.000	3.0000	N/A	No
Shell head	1090.812	0.00	1260.000	3.0000	No	No

Liquid Level: 1100.00 mm Dens.: 0.001 kg./cm³ Sp. Gr.: 1.000



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Vessel Design Summary:

Step: 32

9:45am

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Element Types and Properties:

Element Type	"To" Elev mm	Element Length mm	Nominal Thickness mm	Finished Thickness mm	Reqd Thk Internal mm	Reqd Thk External mm	Long Eff	Circ Eff
Body Flg	174.5	170.0	178.0	170.0	165.3	...	1.00	1.00
Body Flg	439.0	260.0	213.0	205.0	200.1	...	1.00	1.00
Cylinder	1639.0	1200.0	32.0	32.0	29.9	...	0.85	0.85
Cylinder	7209.0	5400.0	18.0	18.0	8.2	...	0.85	0.85
Ellipse	7259.0	50.0	14.0	9.0	8.1	...	0.85	0.85

External Pressure Calculations:

From	To	External Actual T. mm	External Required T. mm	External Design Pressure KPa.	External M.A.W.P. KPa.
10	20	170	141.58	101.3	No Calc
20	30	205	200.101	101.3	No Calc
30	Tsht	32	6.80561	101.3	3569
Tsht	40	32	No Calc	101.3	No Calc
40	50	18	No Calc	...	655
50	60	9	No Calc	...	444

External Pressure Calculations:

From	To	Actual Length Bet. Stiffeners mm	Allowable Length Bet. Stiffeners mm	Ring Inertia Required cm^4	Ring Inertia Available cm^4
10	20	No Calc	No Calc	No Calc	No Calc
20	30	No Calc	No Calc	No Calc	No Calc
30	Tsht	1199.97	34620.8	No Calc	No Calc
Tsht	40	No Calc	No Calc	No Calc	No Calc
40	50	5541.7	No Calc	No Calc	No Calc
50	60	No Calc	No Calc	No Calc	No Calc

Local Stress Analysis Results:

Description	Analysis Type	Max Stress Ratio	High Stress Location	Pass Fail
N3- Tube inlet	WRC-297	0.823	Shell	Passed
Tube outlet	WRC-297	0.823	Shell	Passed
Shell inlet-N1	WRC-297	0.371	Nozzle	Passed
Shell Inlet	WRC-297	0.990	Shell	Passed

Foundation/Support Summary

Saddle Parameters:

Saddle Width	203.000	mm
Saddle Bearing Angle	120.000	deg.
Centerline Dimension	1000.000	mm
Wear Pad Width	360.000	mm
Wear Pad Thickness	15.000	mm
Wear Pad Bearing Angle	132.000	deg.
Distance from Saddle to Tangent	1050.000	mm
Baseplate Length	1150.000	mm
Baseplate Thickness	20.000	mm
Baseplate Width	392.000	mm



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Vessel Design Summary:

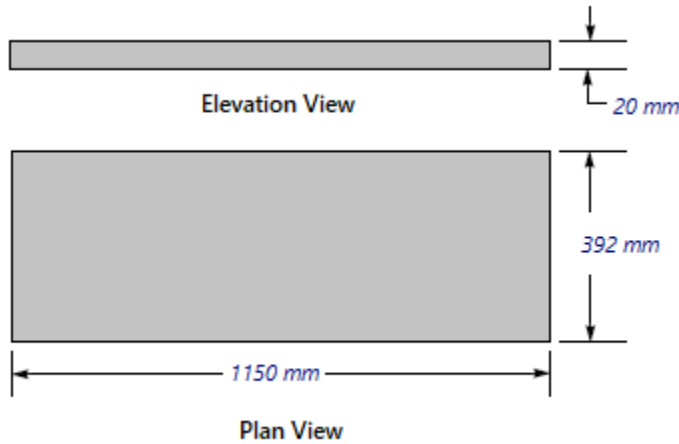
Step: 32

9:45am

Sep 9, 2026

Number of Ribs (including outside ribs)	3
Rib Thickness	10.000 mm
Web Thickness	10.000 mm
Height of Center Web	306.000 mm
Number of Bolts in Baseplate	4

Baseplate Sketch



Maximum Tensile Bolt Load 15. Kg

Summary of Maximum Saddle Loads, Operating Case, Un-Factored:

Maximum Vertical Saddle Load	22227.83	Kgf
Maximum Transverse Saddle Shear Load	906.98	Kgf
Maximum Longitudinal Saddle Shear Load	1243.70	Kgf

Summary of Maximum Saddle Loads, Pressure Test Case:

Maximum Vertical Saddle Load	19838.33	Kgf
Maximum Transverse Saddle Shear Load	127.13	Kgf
Maximum Longitudinal Saddle Shear Load	100.20	Kgf

Weights:

Fabricated - Bare w/o removable internals	13608.4	kg.
Shop Test - Fabricated + water (full)	20073.8	kg.
Shipping - Fab. + rem. intls.+ shipping app.	13608.4	kg.
Erected - Fab. + rem. intls.+ insul. (etc)	14108.4	kg.
Empty - Fab. + intls. + details + wghts.	15389.5	kg.
Operating - Empty + operating liquid (no ca)	21598.2	kg.
Field Test - Empty weight + test medium (full)	21355.0	kg.

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Nozzle Summary:

Step: 30 9:45am Sep 9,2026

Nozzle Calculation Summary (KPa. & mm):

Description	MAWP	Ext	MAPNC	UG-45	[tr]	Weld Path	Areas or Stresses
N3- Tube inlet	6342	OK	...	OK	8.26	OK	Passed
Tube outlet	6342	OK	...	OK	7.80	OK	Passed
Shell inlet-N1	3120	...	...	OK	7.53	OK	Passed
Shell Inlet	3120	...	...	OK	7.49	OK	Passed
Vent	3120	...	...	OK	7.49	OK	No Calc[*]
Drain	3120	...	...	OK	7.53	OK	No Calc[*]

Nozzle MAWP Summary:

Minimum MAWP Nozzles : 3120 Nozzle : Shell inlet-N1 [Shellside]
 Minimum MAWP Nozzles : 6342 Nozzle : N3- Tube inlet [Tubeside]

[*] - This was a small opening and the areas were not computed.

Multiple output lines for the same nozzle indicates required Code calculations in both the longitudinal and circumferential planes of reinforcement where applicable.

Check the spatial relationship between the nozzles:

From Node	Nozzle Description	X Coordinate mm	Layout Angle deg	Dia. Limit mm
30	N3- Tube inlet	839.000	0.000	180.681
30	Tube outlet	939.000	180.000	146.900
40	Shell inlet-N1	6609.000	180.000	923.250
40	Shell Inlet	2809.000	0.000	923.250
40	Vent	3808.999	0.000	113.600
40	Drain	3309.000	180.000	113.600

The nozzle spacing is computed by the following:

$$= \sqrt{ll^2 + lc^2} \text{ where}$$

ll - Arc length along the inside vessel surface in the long. direction.

lc - Arc length along the inside vessel surface in the circ. direction

If any interferences/violations are found, they will be noted below.

No interference violations have been detected!

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Nozzle Schedule:

Step: 29 9:45am Sep 9, 2026

Nozzle Schedule:

Description	Nominal or Actual Size	Schd or FVC Type	Flg Type	Nozzle O/Dia in	Wall Thk mm	Reinforcing Diameter	Pad Thk mm	Cut Length mm	Flg Class
Vent	2.000 in	Actual	LWN	3.060	13.462	...	...	228.92	150
Drain	2.000 in	Actual	LWN	3.060	13.462	...	...	228.92	150
Tube outlet	3.000 in	XXS	WNF	3.500	15.240	214.30	28.00	283.80	...
N3- Tube inlet	4.000 in	XXS	WNF	4.500	17.120	214.30	28.00	284.98	...
Shell inlet-N1	20.000 in	80	WNF	20.000	26.187	808.00	20.00	342.16	...
Shell Inlet	20.000 in	80	WNF	20.000	26.187	808.00	16.00	330.16	150

General Notes for the above table:

The cut length is the outside projection + inside projection + drop + in-plane shell thickness. This value does not include weld gaps, nor does it account for shrinkage.

In the case of oblique nozzles, the outside diameter must be increased. The re-pad width around the nozzle is calculated as follows:

Width of Pad = (Pad Outside Dia. (per above) - Nozzle Outside Dia.)/2

For hub nozzles, the thickness and diameter shown are those of the smaller and thinner section.

Nozzle Material and Weld Fillet Leg Size Details (mm):

Description	Material	Shl Grve Weld	Noz Shl/Pad Weld	Pad OD Weld	Pad Grve Weld	Inside Weld
Vent	SA-105	14.000	20.000	...	...	...
Drain	SA-105	14.000	20.000	...	...	...
Tube outlet	SA-335 P11	26.000	15.000	15.000	28.000	...
N3- Tube in	SA-335 P11	26.000	15.000	15.000	28.000	...
Shell inlet	SA-106 B	14.000	20.000	10.000	20.000	...
Shell Inlet	SA-106 B	14.000	16.000	15.000	6.000	...

Note: The Outside projections below do not include the flange thickness.

Nozzle Miscellaneous Data:

Description	Elev/Distance From Datum mm	Layout Angle deg	Proj Outside mm	Proj Inside mm	Installed in Component
Vent	3808.999	0.0	209.55	0.00	Shell
Drain	3308.999	180.0	209.55	0.00	Shell
Tube outlet	939.000	180.0	250.00	0.00	Channel shell
N3- Tube inlet	839.000	0.0	250.00	0.00	Channel shell
Shell inlet-N1	6609.000	180.0	250.00	0.00	Shell
Shell Inlet	2808.999	0.0	250.00	0.00	Shell

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Bill of Material:

Step: 28 9:45am Sep 9,2026

Bill of Materials:

QTY	DESCRIPTION	MATERIAL
1	BLIND FLANGE: 178.0mm THK X 1460.0mm OD	SA-336 F11
1	WELD NECK FLANGE: 213.0mm THK X 1460.0mm OD	SA-336 F11
1	ELLIPTICAL HEAD: 2.0 X 1, 14.0mm THK X 1100.0mm ID X 50.0mm	SA-516 70
1	INSULATION: 151mm X 150mm THK	Hot
1	INSULATION: 181mm X 150mm THK	Hot
1	INSULATION: 1200mm X 150mm THK	Hot
2	BAFFLES: 1040.0mm DIA X 13.0mm THK	SA-387 11
8	TIE RODS: 14mm DIA X 5200mm	SA-193 B7
1	TUBESHEET: 1156.0mm X 170.0mm THK	SA-336 F11
446	TUBES: 5200.0mm X 19.0mm DIA X 2.1mm THK	SA-213 T11
2	SADDLE: 203.0mm X 120 DEG	SA-516 70
1	INSULATION: 5400mm X 100mm THK	Hot
1	INSULATION: 312mm X 100mm THK	Hot
1	GASKET: 1176.0mm OD X 1132.0mm ID	...
32	BODY FLANGE BOLTS: 57.1mm DIA	SA-193 B16
64	NUTS FOR BODY FLANGE BOLTS: 57.1mm DIA	...
2	CYLINDRICAL SEGMENT 32.0mm THK X 1372.7mm X 1200.0mm	SA-387 11
1	CYLINDRICAL SEGMENT 32.0mm THK X 710.3mm X 1200.0mm	SA-387 11
1	CYLINDRICAL SEGMENT 18.0mm THK X 1334.3mm X 5400.0mm	SA-516 70
1	CYLINDRICAL SEGMENT 18.0mm THK X 1324.7mm X 5400.0mm	SA-516 70
1	CYLINDRICAL SEGMENT 18.0mm THK X 796.7mm X 5400.0mm	SA-516 70
1	NAMEPLATE	...



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MDMT Summary:

Step: 31 9:45am Sep 9,2026

Minimum Design Metal Temperature Results Summary (°C):

Description	Notes	Curve	Basic MDMT	Reduced MDMT	UG-20(f) MDMT	Thickness ratio	Gov Thk mm	E*	PWHT reqd
Shell	[8]	B	-11	-104	-29	0.300	18.000	0.85	No
Shell head	[10]	B	-29	-43	-29	0.740	9.000	0.85	No
Shell head	[7]	B	-18	-48	-29	0.409	14.000	0.85	No
Left Saddle	[24]	B	-16	-104	-29	0.300	15.000	0.85	No
Right saddle	[24]	B	-16	-104	-29	0.300	15.000	0.85	No
Shell inlet-N1	[1]	B	16	-104		0.121	22.914	1.00	No
Shell Inlet	[1]	B	16	-104		0.110	22.914	1.00	No
Nozzle Flg	[5]	A	-18	-43					
Vent	[1]	A	1	-104		0.022	13.462	1.00	No
Nozzle Flg	[5]	A	-18	-43					
Drain	[1]	A	1	-104		0.022	13.462	1.00	No
Nozzle Flg	[5]	A	-18	-42					
Tubesheet: SS	[13]	B	13	-104		0.184	42.500	1.00	No
Warmest MDMT:			16	-42					
CH COVER	[11]	B	13	5		0.853	42.500	1.00	No
CH Flange	[11]	B	6	-8		0.747	32.000	1.00	No
Channel shell	[8]	A	25	16		0.826	32.000	0.85	Yes
N3- Tube inlet	[1]	A	22	12		0.822	28.000	1.00	Yes
Tube outlet	[1]	A	22	12		0.823	28.000	1.00	Yes
Tubesheet: CS	[14]	A	33	21		0.796	42.500	1.00	No
Bolting	[21]		-29						
Warmest MDMT:			33	21					

Exchanger Side	Computed MDMT	Required MDMT	Pass/Fail
Shell	-42.0	-12.0	Pass
Channel/Tube	21.0	-12.0	Fail

Notes:

- [!] - This was an impact tested material.
- [1] - Governing Nozzle Weld.
- [4] - ASME Flange MDMT Calcs; Thickness ratio per UCS-66(b)(1)(-c).
- [5] - ASME Flange MDMT Calcs; Thickness ratio per UCS-66(b)(1)(-b).
- [6] - MDMT Calculations at the Shell/Head Joint.
- [7] - MDMT Calculations for the Straight Flange.
- [8] - Cylinder/Cone/Flange Junction MDMT.
- [9] - Calculations in the Spherical Portion of the Head.
- [10] - Calculations in the Knuckle Portion of the Head.
- [11] - Calculated (Body Flange) Flange MDMT.
- [12] - Calculated Flat Head MDMT per UCS-66.3
- [13] - Tubesheet MDMT, shell side, if applicable
- [14] - Tubesheet MDMT, tube side, if applicable
- [15] - Nozzle Material
- [16] - Shell or Head Material
- [17] - Impact Testing required
- [18] - Impact Testing not required, see UCS-66(b)(3)
- [20] - Cylinder/Cone Junction MDMT based on Longitudinal Stress considerations
- [21] - Body Flange Bolting Material
- [22] - Nozzle Flange Bolting Material
- [23] - Stiffening Ring to Shell Weld
- [24] - Saddle to Shell Weld

UG-84(b)(2) was not considered.

UCS-66(g) was not considered.



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MDMT Summary:

Step: 31 9:45am Sep 9, 2026

UCS-66(i) was not considered.

Notes:

Impact test temps were not entered in and not considered in the analysis.

UCS-66(i) applies to impact tested materials not by specification and

UCS-66(g) applies to materials impact tested per UG-84.1 General Note (c).

The Basic MDMT includes the (30F) PWHT credit if applicable.

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Warnings and Errors:

Step: 0 9:45am Sep 9, 2026

Class From To : Basic Element Checks.

Class From To: Check of Additional Element Data

There were no geometry errors or warnings.

PV Elite performs all calculations internally in Imperial Units to remain compliant with the ASME Code and any built in assumptions in the ASME Code formulas. The finalized results are reflected to show the set of selected units for this analysis.

Information Regarding Nozzle Loads

Please note that nozzle loadings, if included, are assumed to be local in nature and will not contribute to or create a net section bending moment. Therefore, the addition of nozzle loads will not affect the support load calculation. If you wish to create a load on the support(s) from nozzle loads, you can enable that feature in tools->configuration->Job Specific Settings.



One or more nozzles in this model was specified using the actual thickness basis and a standard flange is also specified. In many cases a matching nominal diameter cannot be determined and the weight of the flange and optional blind flange cannot be determined. When actual thickness basis is specified, you must calculate the weight of the assembly (by hand) and enter it into the 'Overriding Weight' field in the nozzle dialog when there is a standard flange specified. However, for FVC nozzles that have their nominal diameter specified in the FVC dialog, entering the overriding weight for that nozzle is not required.



The ASME Code Section VIII Division 1 does not explicitly require stresses on elements to be computed under test conditions. Please see interpretation VIII-1-10-12 for more information. PV Elite also computes weight due to test fluid and uses the weight in various calculations such as saddle and leg loads. PV Elite also computes static pressure due to operating and test liquids. The static pressure is used in conjunction with the design and test pressures to determine a total pressure in the determination of required thickness, MAWP etc. This ensures compliance with UG-22 requirements.

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Input Echo:

Step: 1 9:45am Sep 9, 2026

Units used in this Analysis (MM):

Name	System Unit	Constant	User Unit
Length	Feet	304.8000	mm
Force	Pounds	0.4536	Kgf
Mass	Pounds	0.4536	kg.
Area	sq. inches	6.4516	cm ²
Moment	ft. lbs.	0.1383	Kg-m.
Stress	lbs./sq.in.	0.0069	N./mm ²
Temperature	Degrees F	0.5556	°C
Pressure	psig	6.8946	KPa.
Elast. Modulus	lbs./sq.in.	6.8946	KPa.
Pipe Density	lbs./cu.in.	0.0277	kg./cm ³
Ins. Density	lbs./cu.ft.	0.1600E-04	kg./cm ³
Fluid Density	lbs./cu.ft.	0.1600E-04	kg./cm ³
Wind Speed	miles/hr	1.6093	Km/hr
Tray Weight	lbs./sq.ft.	0.0005	Kgs/cm ²
Inertia	in.4	41.6231	cm4
G Load	G's	1.0000	g's
Wind Load	lbs./sq.ft.	4.8825	Kgs/m ²
Elevation	Feet	304.8000	mm
Volume	in.3	16.3900	cm ³
Diameter	inches	25.4000	mm
Thickness	inches	25.4000	mm

PV Elite Vessel Analysis Program: Input Data

Exchanger Design Pressures and Temperatures

Shell Side Design Pressure	1080	KPa.
Channel Side Design Pressure	5900	KPa.
Shell Side Design Temperature	275.0	°C
Channel Side Design Temperature	430.0	°C
Radiography, Shell Side	RT-1	
Radiography, Channel Side	RT-1	
Service Type, Shell Side	Air/Water/Steam	
Service Type, Channel Side	Air/Water/Steam	
MDMT (CET), Shell Side	-12.0	°C
MDMT (CET), Tube Side	-12.0	°C
User defined MAWP, Shell Side	0	KPa.
User defined MAWP, Channel Side	0	KPa.
User defined MAPnc, Shell Side	0	KPa.
User defined MAPnc, Channel Side	0	KPa.
User defined Test Pres., Shell Side	0	KPa.
User defined Test Pres., Channel Side	0	KPa.

Projection of Nozzle from Vessel Top	0	mm
Projection of Nozzle from Vessel Bottom	0	mm
Type of Construction	Welded	
Use Higher Longitudinal Stresses (Flag)	Y	
Select t for Internal Pressure (Flag)	N	
Select t for External Pressure (Flag)	N	
Select t for Axial Stress (Flag)	N	
Select Location for Stiff. Rings (Flag)	N	
Consider Vortex Shedding	N	

Shop Pressure Test:

Type of Pressure Test	UG-99(c)
Pressure Test Position	Horizontal
Test Performed in Corroded Condition	No



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Input Echo:

Step: 1 9:45am Sep 9,2026

Field Pressure Test:

Type of Pressure Test UG-99(b) Note [35]
 Pressure Test Position Horizontal
 Test Performed in Corroded Condition Yes

Load Case 1	NP+EW+WI+FW+BW
Load Case 2	NP+EW+EE+FS+BS
Load Case 3	NP+OW+WI+FW+BW
Load Case 4	NP+OW+EQ+FS+BS
Load Case 5	NP+HW+HI
Load Case 6	NP+HW+HE
Load Case 7	IP+OW+WI+FW+BW
Load Case 8	IP+OW+EQ+FS+BS
Load Case 9	EP+OW+WI+FW+BW
Load Case 10	EP+OW+EQ+FS+BS
Load Case 11	HP+HW+HI
Load Case 12	HP+HW+HE
Load Case 13	IP+WE+EW
Load Case 14	IP+WF+CW
Load Case 15	IP+VO+OW
Load Case 16	IP+VE+EW
Load Case 17	NP+VO+OW
Load Case 18	FS+BS+IP+OW
Load Case 19	FS+BS+EP+OW
Load Case 20	BL+IP+OW

Wind Design Code	ASCE/SEI 7-16
Wind Load Reduction Scale Factor	1.100
Basic Wind Speed [V]	97.2 Km/hr
Surface Roughness Category	C: Open Terrain
Type of Surface	Moderately Smooth
Base Elevation	300 mm
Percent Wind for Hydrotest	50.0
Using User defined Wind Press. Vs Elev.	N
Height of Hill or Escarpment H or Hh	0 mm
Distance Upwind of Crest Lh	0 mm
Distance from Crest to the Vessel x	0 mm
Type of Terrain (Hill, Escarpment)	Flat
Damping Factor (Beta) for Wind (Ope)	0.0100
Damping Factor (Beta) for Wind (Empty)	0.0000
Damping Factor (Beta) for Wind (Filled)	0.0000
Site Elevation above Sea Level	0 mm

Seismic Design Code	ASCE/SEI 7-16
Seismic Load Reduction Scale Factor	1.000
Importance Factor	1.250
Table Value Fa	1.600
Table Value Fv	2.400
Max. Mapped Res. Acceleration [Ss]	0.147
Max. Eff. Ground Acceleration [S]	0.065
Force Modification Factor R	3.000
Site Class	D
Component Elevation Ratio z/h	0.000
Amplification Factor Ap	1.000
Force Factor	0.000
Consider Vertical Acceleration	Yes
Minimum Acceleration Multiplier	0.000
User Value of Sds (used if > 0)	0.000
User Value of Sd1 (used if > 0)	0.000
Moment Reduction Factor Tau	1.000

M.A.W.P + Static Head to Nozzle	Y
Consider MAP New and Cold in Noz. Design	N
Consider External Loads for Nozzle Des.	Y



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Input Echo:

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Use ASME VIII-1 Appendix 1-9	N
Perform Blast Load Analysis	No
Material Database Year	Current w/Addenda or Code Year

Configuration Directives:

g0 is taken as min(g0, attached shell thk)	Yes
Do not use Nozzle MDMT Interpretation VIII-1 01-37	No
Use Table G instead of exact equation for "A"	Yes
Shell Head Joints are Tapered	Yes
Compute "K" in corroded condition	Yes
Use Code Case 2286	No
Use the MAWP to compute the MDMT	Yes
For thickness ratios <= 0.35, MDMT will be -155F (-104C)	Yes
For PWHT & P1 Materials the MDMT can be < -55F (-48C)	No
Using Metric Material Databases, ASME II D	Yes
Calculate B31.3 type stress for Nozzles with Loads	Yes
Reduce the MDMT due to lower membrane stress	Yes
Consider Longitudinal Stress in MDMT Calculations	Yes

Complete Listing of Vessel Elements and Details:

Element#	1/5
Element From Node	10
Element To Node	20
Element Type	Flange
Description	CH COVER
Distance "FROM" to "TO"	170 mm
Flange Outside Diameter	1460 mm
Element Thickness	170 mm
Internal Corrosion Allowance	3 mm
Nominal Thickness	178 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	5900 KPa.
Design Temperature Internal Pressure	430 °C
Design External Pressure	101.3 KPa.
Design Temperature External Pressure	200 °C
Effective Diameter Multiplier	1.2
Material Name	SA-336 F11
Allowable Stress, Ambient	148 N./mm ²
Allowable Stress, Operating	147 N./mm ²
Allowable Stress, Hydrotest	278.99 N./mm ²
Material Density	0.00775 kg./cm ³
P Number Thickness	16.002 mm
Yield Stress, Operating	223 N./mm ²
UCS-66 Chart Curve Designation	B
External Pressure Chart Name	CS-3
UNS Number	K11572
Class / Thickness / Grade	3 ::
Product Form	Forgings
Perform Flange Stress Calculation (Y/N)	Y
Weld is pre-Heated	No
Element From Node	10
Detail Type	Insulation
Detail ID	Ins: 10
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Insulation	152 mm
Thickness of Insulation	150 mm
Density	0.000352 kg./cm ³



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Input Echo:

Step: 1 9:45am Sep 9,2026

Element#	2/5
Element From Node	20
Element To Node	30
Element Type	Flange
Description	CH Flange
Distance "FROM" to "TO"	260 mm
Flange Inside Diameter	1100 mm
Element Thickness	205 mm
Internal Corrosion Allowance	3 mm
Nominal Thickness	213 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	5900 KPa.
Design Temperature Internal Pressure	430 °C
Design External Pressure	101.3 KPa.
Design Temperature External Pressure	200 °C
Effective Diameter Multiplier	1.2
Material Name	SA-336 F11
Perform Flange Stress Calculation (Y/N)	Y
Weld is pre-Heated	No
Element From Node	20
Detail Type	Insulation
Detail ID	Ins: 20
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Insulation	182 mm
Thickness of Insulation	150 mm
Density	0.000352 kg./cm ³

Element#	3/5
Element From Node	30
Element To Node	40
Element Type	Cylinder
Description	Channel shell
Distance "FROM" to "TO"	1200 mm
Inside Diameter	1100 mm
Element Thickness	32 mm
Internal Corrosion Allowance	3 mm
Nominal Thickness	32 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	5900 KPa.
Design Temperature Internal Pressure	430 °C
Design External Pressure	101.3 KPa.
Design Temperature External Pressure	200 °C
Effective Diameter Multiplier	1.2
Material Name	SA-387 11
Allowable Stress, Ambient	148 N./mm ²
Allowable Stress, Operating	147 N./mm ²
Allowable Stress, Hydrotest	278.99 N./mm ²
Material Density	0.00775 kg./cm ³
P Number Thickness	16.002 mm
Yield Stress, Operating	223 N./mm ²
UCS-66 Chart Curve Designation	A
External Pressure Chart Name	CS-3
UNS Number	K11789
Class / Thickness / Grade	2 ::
Product Form	Plate
Efficiency, Longitudinal Seam	0.85
Efficiency, Circumferential Seam	0.85
Weld is pre-Heated	No
Element From Node	30
Detail Type	Liquid
Detail ID	Liquid: 20



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Input Echo:

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Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Liquid	1100	mm
Liquid Density	0.0009984	kg./cm ³

Element From Node	30
Detail Type	Insulation
Detail ID	Ins: 40
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Insulation	1200 mm
Thickness of Insulation	150 mm
Density	0.000352 kg./cm ³

Element From Node	30
Detail Type	Nozzle
Detail ID	N3- Tube inlet
Dist. from "FROM" Node / Offset dist	400 mm
Nozzle Diameter	4 in.
Nozzle Schedule	XXS
Nozzle Class	600
Layout Angle	0.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	27.429 Kg
Grade of Attached Flange	None
Nozzle Matl	SA-335 P11

Element From Node	30
Detail Type	Nozzle
Detail ID	Tube outlet
Dist. from "FROM" Node / Offset dist	500 mm
Nozzle Diameter	3 in.
Nozzle Schedule	XXS
Nozzle Class	600
Layout Angle	180.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	20.358 Kg
Grade of Attached Flange	None
Nozzle Matl	SA-335 P11

Element From Node	30
Detail Type	Weight
Detail ID	Pass Partition
Dist. from "FROM" Node / Offset dist	500 mm
Miscellaneous Weight	260 Kg
Offset from Element Centerline	0 mm

Element#	4/5
Element From Node	40
Element To Node	50
Element Type	Cylinder
Description	Shell
Distance "FROM" to "TO"	5400 mm
Inside Diameter	1100 mm
Element Thickness	18 mm
Internal Corrosion Allowance	3 mm
Nominal Thickness	18 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	1080 KPa.
Design Temperature Internal Pressure	275 °C
Design External Pressure	0 KPa.
Design Temperature External Pressure	25 °C
Effective Diameter Multiplier	1.2
Material Name	SA-516 70
Allowable Stress, Ambient	138 N./mm ²



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Input Echo:

Step: 1 9:45am Sep 9, 2026

Allowable Stress, Operating	138	N./mm ²
Allowable Stress, Hydrotest	234	N./mm ²
Material Density	0.00775	kg./cm ³
P Number Thickness	32.004	mm
Yield Stress, Operating	262	N./mm ²
UCS-66 Chart Curve Designation	B	
External Pressure Chart Name	CS-2	
UNS Number	K02700	
Product Form	Plate	
Efficiency, Longitudinal Seam	0.85	
Efficiency, Circumferential Seam	0.85	
Weld is pre-Heated	No	

Element From Node	40	
Detail Type	Saddle	
Detail ID	Left Saddle	
Dist. from "FROM" Node / Offset dist	1050	mm
Width of Saddle	203	mm
Height of Saddle at Bottom	1000	mm
Saddle Contact Angle	120.0	
Height of Composite Ring Stiffener	0	mm
Width of Wear Plate	360	mm
Thickness of Wear Plate	15	mm
Contact Angle, Wear Plate (degrees)	132.0	
Friction coefficient	0.0	
Moment Factor	3.0	
Dimension E at base (optional)	0	mm
Circumferential Eff. over Saddle	1.0	
Circumferential Eff. at Midspan	1.0	
Tangent to Tangent dist. (optional)	0	mm

Element From Node	40	
Detail Type	Saddle	
Detail ID	Right saddle	
Dist. from "FROM" Node / Offset dist	4150	mm
Width of Saddle	203	mm
Height of Saddle at Bottom	1000	mm
Saddle Contact Angle	120.0	
Height of Composite Ring Stiffener	0	mm
Width of Wear Plate	360	mm
Thickness of Wear Plate	15	mm
Contact Angle, Wear Plate (degrees)	132.0	
Friction coefficient	0.40000001	
Moment Factor	3.0	
Dimension E at base (optional)	0	mm
Circumferential Eff. over Saddle	1.0	
Circumferential Eff. at Midspan	1.0	
Tangent to Tangent dist. (optional)	0	mm

Element From Node	40	
Detail Type	Liquid	
Detail ID	Liquid: 40	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Liquid	1100	mm
Liquid Density	0.0009984	kg./cm ³

Element From Node	40	
Detail Type	Insulation	
Detail ID	Ins: 40	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Insulation	5400	mm
Thickness of Insulation	100	mm
Density	0.000352	kg./cm ³

Element From Node	40	
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Detail Type	Nozzle
Detail ID	Shell inlet-N1
Dist. from "FROM" Node / Offset dist	4800 mm
Nozzle Diameter	20 in.
Nozzle Schedule	80
Nozzle Class	150
Layout Angle	180.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	217.27 Kg
Grade of Attached Flange	None
Nozzle Matl	SA-106 B

Element From Node	40
Detail Type	Nozzle
Detail ID	Shell Inlet
Dist. from "FROM" Node / Offset dist	1000 mm
Nozzle Diameter	20 in.
Nozzle Schedule	80
Nozzle Class	150
Layout Angle	0.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	217.27 Kg
Grade of Attached Flange	GR 1.1
Nozzle Matl	SA-106 B

Element From Node	40
Detail Type	Nozzle
Detail ID	Vent
Dist. from "FROM" Node / Offset dist	2000 mm
Nozzle Diameter	2 in.
Nozzle Schedule	None
Nozzle Class	150
Layout Angle	0.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	7.8799 Kg
Grade of Attached Flange	GR 1.1
Nozzle Matl	SA-105

Element From Node	40
Detail Type	Nozzle
Detail ID	Drain
Dist. from "FROM" Node / Offset dist	1500 mm
Nozzle Diameter	2 in.
Nozzle Schedule	None
Nozzle Class	150
Layout Angle	180.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	7.8799 Kg
Grade of Attached Flange	GR 1.1
Nozzle Matl	SA-105

Element From Node	40
Detail Type	Weight
Detail ID	Fire proofing
Dist. from "FROM" Node / Offset dist	1050 mm
Miscellaneous Weight	250 Kg
Offset from Element Centerline	1000 mm

Element From Node	40
Detail Type	Weight
Detail ID	Fire proofing
Dist. from "FROM" Node / Offset dist	4150 mm
Miscellaneous Weight	250 Kg
Offset from Element Centerline	1000 mm



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Input Echo:

Step: 1 9:45am Sep 9,2026

Element#	5/5
Element From Node	50
Element To Node	60
Element Type	Elliptical
Description	Shell head
Distance "FROM" to "TO"	50 mm
Inside Diameter	1100 mm
Element Thickness	9 mm
Internal Corrosion Allowance	3 mm
Nominal Thickness	14 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	1080 KPa.
Design Temperature Internal Pressure	275 °C
Design External Pressure	0 KPa.
Design Temperature External Pressure	25 °C
Effective Diameter Multiplier	1.2
Material Name	SA-516 70
Efficiency, Longitudinal Seam	0.85
Efficiency, Circumferential Seam	0.85
Elliptical Head Factor	2.0
Weld is pre-Heated	No
Element From Node	50
Detail Type	Liquid
Detail ID	Liquid: 50
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Liquid	1100 mm
Liquid Density	0.0009984 kg./cm ³
Element From Node	50
Detail Type	Insulation
Detail ID	Ins: 40
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Insulation	312.5 mm
Thickness of Insulation	100 mm
Density	0.000352 kg./cm ³

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XY Coordinate Calculations: Step: 2 9:45am Sep 9,2026

XY Coordinate Calculations:

From	To	X (Horiz.) mm	Y (Vert.) mm	DX (Horiz.) mm	DY (Vert.) mm
CH COVER		174.5	...	170	...
CH Flange		439	...	260	...
Channel shell		1639	...	1200	...
Shell		7209	...	5400	...
Shell head		7259	...	50	...

The calculated dimensions are based on the given element lengths. Due to variability in manufacturing (weld gaps etc.), the Tangent to Tangent length may not be exact. The length includes gasket thicknesses and raised face heights.

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Flg Calc [Int P]: New Flange

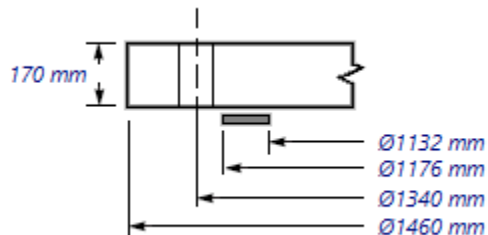
Flng: 5 9:45am Sep 9,2026

Flange Input Data Values

Description: New Flange :

CH COVER

Flange Type	TEMA Channel Cover		
Design Pressure	P	5900.00	KPa.
Design Temperature		430	°C
Internal Corrosion Allowance	ci	3.0000	mm
External Corrosion Allowance	ce	0.0000	mm
Use Corrosion Allowance in Thickness Calcs.		Yes	
Flange Outside Diameter	A	1460.000	mm
Flange Thickness	t	170.0000	mm
Flange Material	SA-336 F11		
Flange Material UNS number	K11572		
Flange Allowable Stress At Temperature	Sfo	147.00	N./mm ²
Flange Allowable Stress At Ambient	Sfa	148.00	N./mm ²
Bolt Material	SA-193 B16		
Bolt Allowable Stress At Temperature	Sb	170.41	N./mm ²
Bolt Allowable Stress At Ambient	Sa	172.00	N./mm ²
Perimeter along the Center of the Bolts	L	4209.734	mm
Youngs Modulus for Blind Flange Matl.	Sy	176184.59	N./mm ²
Allowed Channel Cover Deflection	dfAllow	1.3750	mm
Diameter of Bolt Circle	C	1340.000	mm
Nominal Bolt Diameter	a	57.1500	mm
Type of Threads	TEMA Thread Series		
Number of Bolts	32		
Flange Face Outside Diameter	Fod	1187.000	mm
Flange Face Inside Diameter	Fid	1100.000	mm
Flange Facing Sketch	Sketch 1a		
Gasket Outside Diameter	Go	1176.000	mm
Gasket Inside Diameter	Gi	1132.000	mm
Gasket Factor	m	3.0000	
Gasket Design Seating Stress	y	68.95	N./mm ²
Gasket Thickness	tg	4.5000	mm
Raised face depth	Fd	0.0000	mm



ASME Code, Section VIII Division 1, 2025

Gasket Contact Width,	$N = (Go - Gi)/2$	22.000 mm
Basic Gasket Width,	$bo = N/2$	11.000 mm
Effective Gasket Width,	$b = Cb \cdot \sqrt{bo}$	8.358 mm
Gasket Reaction Diameter,	$G = Go - 2 \cdot b$	1159.285 mm



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Flg Calc [Int P]: New Flange

Flng: 5 9:45am

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Basic Flange and Bolt Loads:

Hydrostatic End Load due to Pressure [H or Wo,1]:

$$= \pi/4 \cdot G^2 \cdot P$$

$$= \pi/4 \cdot 1159^2 \cdot 5900$$

$$= 635065.750 \text{ Kgf}$$

Contact Load on Gasket Surfaces [Hp or Wo,2]:

$$= 2 \cdot b \cdot \pi \cdot G \cdot m \cdot P$$

$$= 2 \cdot 8.36 \cdot \pi \cdot 1159 \cdot 3 \cdot 5900$$

$$= 109881.148 \text{ Kgf}$$

Design Bolt Load for the Operating Condition [Wo]:

$$= \max(W_{o,1} + W_{o,2} + H'p, 0)$$

$$= \max(635066 + 109881 + 0, 0)$$

$$= 744946.875 \text{ Kgf}$$

Design Bolt Load for the Gasket Seating Condition [Wg]:

$$= y \cdot b \cdot \pi \cdot G + y_{Part} \cdot b_{Part} \cdot l_p$$

$$= 68.9 \cdot 8.36 \cdot \pi \cdot 1159 + 0 \cdot 0 \cdot 0$$

$$= 214007.469 \text{ Kgf}$$

Required Bolt Area [Am]:

$$= \max((W_o + F_a + 4Me/G) / S_b, W_g / S_a)$$

$$= \max((744947 + 0 + 4 \cdot 0 / 1159) / 170, 214007 / 172)$$

$$= 428.709 \text{ cm}^2$$

Bolting Information for TEMA Imperial Thread Series (Non Mandatory):

	Minimum	Actual	Maximum
Bolt Area:	428.709	706.682	
Radial Distance between Bolts and Edge:	57.150	60.000	
Circ. Spacing between the Bolts:	120.650	131.343	...

Min. Gasket Contact Width (Brownell Young) [Not an ASME Calc] [Nmin]:

$$= A_b \cdot S_a / (y \cdot \pi (G_o + G_i))$$

$$= 707 \cdot 172 / (68.9 \cdot \pi (1176 + 1132))$$

$$= 24.312 \text{ mm [Note: Exceeds actual gasket width, 22.000]}$$

Flange Design Bolt Load, Gasket Seating [W or Wgs]:

$$= S_a \cdot A_b$$

$$= 172 \cdot 707$$

$$= 1239413.50 \text{ Kgf}$$

Gasket Load for the Operating Condition [HG]:

$$= W_o$$

$$= 744946.88 \text{ Kgf}$$

Moment Arm Calculations:

Distance to Gasket Load Reaction [hg]:

$$= (C - G) / 2$$

$$= (1340 - 1159) / 2$$

$$= 90.3576 \text{ mm}$$

Flange Stresses

Tangential Flange Stress, Flat Head (4.6.2.3), Operating [STo]:

$$= 1.9 \cdot W_{m1} \cdot h_G \cdot B_{sc} / (t^2 \cdot G) + C \cdot Z \cdot P_{eq} \cdot G^2 / t^2$$

$$= 1.9 \cdot 744947 \cdot 90.4 \cdot 1 / (167^2 \cdot 1159) + 0.3 \cdot 1 \cdot 5900 \cdot 1159^2 / 167^2$$

$$= 124.09 \text{ N./mm}^2$$

Tangential Flange Stress, Flat Head (4.6.2.4), Seating [STa]:



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Flg Calc [Int P]: New Flange

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$$= 1.9 \cdot W \cdot hG \cdot Bsc / (t^2 \cdot G)$$

$$= 1.9 \cdot 1239413 \cdot 90.4 \cdot 1 / (167^2 \cdot 1159)$$

$$= 64.54 \text{ N./mm}^2$$

Bolt Stress, Operating [BS_o]:

$$= Wm1 / Ab$$

$$= 744947 / 707$$

$$= 103.38 \text{ N./mm}^2$$

Bolt Stress, Seating [BS_a]:

$$= (Wm2 / Ab)$$

$$= (214007 / 707)$$

$$= 29.70 \text{ N./mm}^2$$

Flange Stress Analysis Results: N./mm²

	Actual	Operating Allowed	Actual	Gasket Seating Allowed
Tangential Flange	124.09	147.00	64.54	148.00
Bolting	103.38	170.41	29.70	172.00

TEMA Channel Cover Deflection Calculations:

Actual deflection at center per TEMA formula 1.3336 mm

Allowed deflection at center of Cover 1.3750 mm

Reqd. Blind Flange Thickness at Center 156.435 mm

Reqd. Blind Flange Thickness at Gasket 113.283 mm

Min. Required Flange Thickness per TEMA 165.306 mm

Estimated M.A.W.P. (Operating) 6332 KPa.

Estimated Finished Weight of Flange at given Thk. 2205.8 kg.

Estimated Unfinished Weight of Forging at given Thk 2205.8 kg.

Minimum Design Metal Temperature Results:

Thickness Ratio = 0.85, Temperature Reduction per Fig. UCS 66.1 = 8 °C

Min Metal Temp. w/o impact per UCS-66, Curve B 13 °C

Min Metal Temp. at Required thickness (UCS 66.1) 5 °C

Note: UCS-66(b)(-c) was considered in the flange MDMT calculation.

Per ASME VIII-2, 4.16.2.3, Perform Calculations in the New Condition:

Flange Stress Analysis Results: N./mm²

	Actual	Operating Allowed	Actual	Gasket Seating Allowed
Tangential Flange	119.75	148.00	62.28	148.00
Bolting	103.38	172.00	29.70	172.00

Minimum Req'd Thickness, ambient, no Corrosion: 157.683 mm

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Flg Calc [Int P]: New Flange

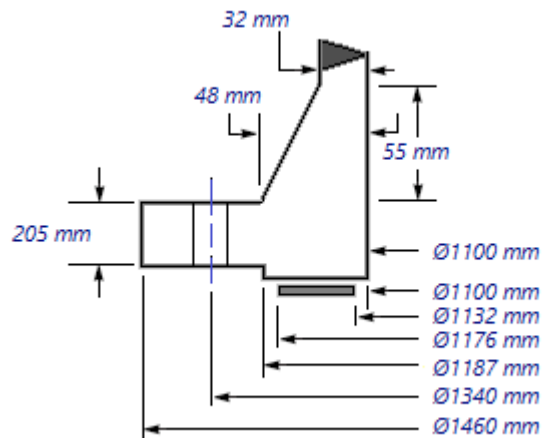
Flng: 6 9:45am Sep 9,2026

Flange Input Data Values

Description: New Flange :

CH Flange

Flange Type	Integral Weld Neck		
Design Pressure	P	5900.00	KPa.
Design Temperature		430	°C
Internal Corrosion Allowance	ci	3.0000	mm
External Corrosion Allowance	ce	0.0000	mm
Use Corrosion Allowance in Thickness Calcs.		Yes	
Flange Inside Diameter	B	1100.000	mm
Flange Outside Diameter	A	1460.000	mm
Flange Thickness	t	205.0000	mm
Thickness of Hub at Small End	go	32.0000	mm
Thickness of Hub at Large End	gl	48.0000	mm
Length of Hub	h	55.0000	mm
Flange Material	SA-336 F11		
Flange Material UNS number	K11572		
Flange Allowable Stress At Temperature	Sfo	147.00	N./mm ²
Flange Allowable Stress At Ambient	Sfa	148.00	N./mm ²
Bolt Material	SA-193 B16		
Bolt Allowable Stress At Temperature	Sb	170.41	N./mm ²
Bolt Allowable Stress At Ambient	Sa	172.00	N./mm ²
Diameter of Bolt Circle	C	1340.000	mm
Nominal Bolt Diameter	a	57.1500	mm
Type of Threads	TEMA Thread Series		
Number of Bolts	32		
Flange Face Outside Diameter	Fod	1187.000	mm
Flange Face Inside Diameter	Fid	1100.000	mm
Flange Facing Sketch	Sketch 1a		
Gasket Outside Diameter	Go	1176.000	mm
Gasket Inside Diameter	Gi	1132.000	mm
Gasket Factor	m	3.0000	
Gasket Design Seating Stress	y	68.95	N./mm ²
Gasket Thickness	tg	4.5000	mm
Raised face depth	Fd	0.0000	mm





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Flg Calc [Int P]: New Flange

Flng: 6 9:45am

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ASME Code, Section VIII Division 1, 2025

Hub Small End Required Thickness due to Internal Pressure:

$$= (P \cdot (D/2 + Ca)) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (5900 \cdot (1100/2 + 3)) / (147 \cdot 1 - 0.6 \cdot 5900) + Ca$$

$$= 25.7436 \text{ mm}$$

Small End Hub MAWP:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (147 \cdot 1 \cdot 29) / (553 + 0.6 \cdot 29)$$

$$= 7473.497 \text{ KPa.}$$

Hub Small End Hub MAPnc:

$$= (Sa \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (148 \cdot 1 \cdot 32) / (550 + 0.6 \cdot 32)$$

$$= 8319.817 \text{ KPa.}$$

Corroded Flange Thickness, $t_c = T - c_i$	202.000 mm
Corroded Flange ID, $B_{cor} = B + 2 \cdot c_i$	1106.000 mm
Corroded Large Hub, $g1_{cor} = g1 - c_i$	45.000 mm
Corroded Small Hub, $g0_{cor} = g0 - c_i$	29.000 mm
Code R Dimension, $R = ((C - B_{cor})/2) - g1_{cor}$	72.000 mm

Gasket Contact Width, $N = (G_o - G_i)/2$	22.000 mm
Basic Gasket Width, $bo = N/2$	11.000 mm
Effective Gasket Width, $b = C_b \cdot \sqrt{bo}$	8.358 mm
Gasket Reaction Diameter, $G = G_o - 2 \cdot b$	1159.285 mm

Basic Flange and Bolt Loads:

Hydrostatic End Load due to Pressure [H or Wo,1]:

$$= \pi/4 \cdot G^2 \cdot P$$

$$= \pi/4 \cdot 1159^2 \cdot 5900$$

$$= 635065.750 \text{ Kg}$$

Contact Load on Gasket Surfaces [Hp or Wo,2]:

$$= 2 \cdot b \cdot \pi \cdot G \cdot m \cdot P$$

$$= 2 \cdot 8.36 \cdot \pi \cdot 1159 \cdot 3 \cdot 5900$$

$$= 109881.148 \text{ Kg}$$

Hydrostatic End Load at Flange ID [Hd]:

$$= \pi/4 \cdot B_{cor}^2 \cdot P$$

$$= \pi/4 \cdot 1106^2 \cdot 5900$$

$$= 578027.750 \text{ Kg}$$

Pressure Force on Flange Face [Ht]:

$$= H - Hd$$

$$= 635066 - 578028$$

$$= 57037.988 \text{ Kg}$$

Design Bolt Load for the Operating Condition [Wo]:

$$= \max(Wo,1 + Wo,2 + H'p, 0)$$

$$= \max(635066 + 109881 + 0, 0)$$

$$= 744946.875 \text{ Kg}$$

Design Bolt Load for the Gasket Seating Condition [Wg]:

$$= y \cdot b \cdot \pi \cdot G + y_{Part} \cdot b_{Part} \cdot l_p$$

$$= 68.9 \cdot 8.36 \cdot \pi \cdot 1159 + 0 \cdot 0$$

$$= 214007.469 \text{ Kg}$$

Required Bolt Area [Am]:

$$= \max((Wo + Fa + 4Me/G)/S_b, Wg/S_a)$$

$$= \max((744947 + 0 + 4 \cdot 0/1159)/170, 214007/172)$$

$$= 428.709 \text{ cm}^2$$



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Flg Calc [Int P]: New Flange

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ASME Maximum Circumferential Spacing between Bolts, Table 4.16.11 [Bsmax]:

$$= 2a + 6t/(m + 0.5)$$

$$= 2 \cdot 57.1 + 6 \cdot 202/(3 + 0.5)$$

$$= 460.586 \text{ mm}$$

Actual Circumferential Bolt Spacing [Bs]:

$$= C \cdot \sin(\pi / n)$$

$$= 1340 \cdot \sin(\pi/32)$$

$$= 131.343 \text{ mm}$$

ASME Moment Multiplier for Bolt Spacing per Table 4.16.11 [Bsc]:

$$= \max(\sqrt{Bs/(2a + t)}, 1)$$

$$= \max(\sqrt{131/(2 \cdot 57.1 + 202)}, 1)$$

$$= 1.0000$$

Bolting Information for TEMA Imperial Thread Series (Non Mandatory):

	Minimum	Actual	Maximum
Bolt Area:	428.709	706.682	
Radial Distance between Hub and Bolts:	69.850	72.000	
Radial Distance between Bolts and Edge:	57.150	60.000	
Circ. Spacing between the Bolts:	120.650	131.343	460.586

Min. Gasket Contact Width (Brownell Young) [Not an ASME Calc] [Nmin]:

$$= Ab \cdot Sa / (y \cdot \pi(Go + Gi))$$

$$= 707 \cdot 172 / (68.9 \cdot \pi(1176 + 1132))$$

$$= 24.312 \text{ mm [Note: Exceeds actual gasket width, 22.000]}$$

Recommended Minimum Gasket Contact Width per Table 4.16.2

Sheet Gaskets including laminated sheet gaskets with or without a metal core: 19 mm
Preformed comp. gaskets incl. spiral wound, jacketed, & solid flat metal gaskets: 16 mm

Flange Design Bolt Load, Gasket Seating [W or Wgs]:

$$= Sa \cdot Ab$$

$$= 172 \cdot 707$$

$$= 1239413.50 \text{ Kgf}$$

Gasket Load for the Operating Condition [HG]:

$$= Wo - H$$

$$= 744947 - 635066$$

$$= 109881.14 \text{ Kgf}$$

Moment Arm Calculations:

Distance to Gasket Load Reaction [hg]:

$$= (C - G) / 2$$

$$= (1340 - 1159) / 2$$

$$= 90.3576 \text{ mm}$$

Distance to Face Pressure Reaction [ht]:

$$= 0.5 ((C - B) / 2 + hg)$$

$$= 0.5 ((1340 - 1106) / 2 + 90.4)$$

$$= 103.6788 \text{ mm}$$

Distance to End Pressure Reaction [hd]:

$$= (C - B - g1) / 2$$

$$= (1340 - 1106 - 45) / 2$$

$$= 94.5000 \text{ mm}$$

Summary of Moments for Internal Pressure: (Kg-m.)

Loading	Force	Distance	Bolt Corr	Moment
---------	-------	----------	-----------	--------



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Flg Calc [Int P]: New Flange Flng: 6 9:45am Sep 9, 2026

End Pressure, Md	578028.	94.5000	1.0000	54625.
Face Pressure, Mt	57038.	103.6788	1.0000	5914.
Gasket Load, Mg	109881.	90.3576	1.0000	9929.
Gasket Seating, Matm	1239414.	90.3576	1.0000	111993.

Total Moment for Operation, Mo 70467. Kg-m.
Total Moment for Gasket seating, Mg 111993. Kg-m.

Effective Hub Length, ho = sqrt(Bcor*goCor) 179.092 mm
Hub Ratio, h/h0 = HL / H0 0.307
Thickness Ratio, g1/g0 = (g1Cor/goCor) 1.552

Flange Factors for Integral Flange:

Factor F per 4.16.5 0.878
Factor V per 4.16.5 0.358
Factor f per 4.16.5 1.231

Stress factor from Table 4.16.4 [K]:

= A / B
= 1460/1106
= 1.320

Stress factor from Table 4.16.4 [Y]:

= (0.66845 + (5.7169(K² • log10(K))) / (K² - 1))/(K - 1)
= (0.66845 + (5.7169(1.32² • log10(1.32)))/(1.32² - 1))/(1.32 - 1)
= 7.143

Stress factor from Table 4.16.4 [T]:

= (K²(1 + 8.55246 • log10(K)) - 1)/((1.0472 + 1.9448 • K²) • (K - 1))
= (1.32²(1 + 8.55246 • log10(1.32)) - 1)/((1.0472 + 1.9448 • 1.32²) • (1.32 - 1))
= 1.789

Stress factor from Table 4.16.4 [U]:

= (K²(1 + 8.55246 • log10(K)) - 1)/(1.36136(K² - 1) • (K - 1))
= (1.32²(1 + 8.55246 • log10(1.32)) - 1)/(1.36136(1.32² - 1) • (1.32 - 1))
= 7.850

Stress factor from Table 4.16.4 [Z]:

= (K² + 1)/(K² - 1)
= (1.32² + 1)/(1.32² - 1)
= 3.693

Stress factor from Table 4.16.4 for integral flanges [d]:

= U • g0² • h0 / V
= 7.85 • 29² • 179/0.36
= 3300758.500 mm³

Stress factor from Table 4.16.4 for integral flanges [e]:

= F / h0
= 0.88/179
= 0.005 1/mm

Stress factor from Table 4.16.4 [L]:

= (t • e + 1) / T + t³/d
= (202 • 0.0049 + 1)/1.79 + 202³/3300758
= 3.610

Flange Stresses

Longitudinal Hub Stress, Operating [SHo]:

= (f • Mop / Bcor) / (L • g1²)
= (1.23•70467/1106)/(3.61•45²)
= 105.23 N./mm²

Longitudinal Hub Stress, Seating [SHa]:



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Flg Calc [Int P]: New Flange

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$$= (f \cdot \text{Matm} / \text{Bcor}) / (L \cdot g^{1^2})$$

$$= (1.23 \cdot 111993 / 1106) / (3.61 \cdot 45^2)$$

$$= 167.24 \text{ N./mm}^2$$

Radial Flange Stress, Operating [SRo]:

$$= (\beta \cdot \text{Mop} / \text{Bcor}) / (L \cdot t^2)$$

$$= (2.32 \cdot 70467 / 1106) / (3.61 \cdot 202^2)$$

$$= 9.85 \text{ N./mm}^2$$

Radial Flange Stress, Seating [SRa]:

$$= (\beta \cdot \text{Matm} / \text{Bcor}) / (L \cdot t^2)$$

$$= (2.32 \cdot 111993 / 1106) / (3.61 \cdot 202^2)$$

$$= 15.65 \text{ N./mm}^2$$

Tangential Flange Stress, Operating [STo]:

$$= (Y \cdot \text{Mo} / (t^2 \cdot \text{Bcor})) - Z \cdot \text{SRo}$$

$$= (7.14 \cdot 70467 / (202^2 \cdot 1106)) - 3.69 \cdot 9.85$$

$$= 73.02 \text{ N./mm}^2$$

Tangential Flange Stress, Seating [STa]:

$$= (y \cdot \text{Matm} / (t^2 \cdot \text{Bcor})) - Z \cdot \text{SRa}$$

$$= (7.14 \cdot 111993 / (202^2 \cdot 1106)) - 3.69 \cdot 15.6$$

$$= 116.05 \text{ N./mm}^2$$

Average Flange Stress, Operating [SAo]:

$$= (\text{SHo} + \max(\text{SRo}, \text{STo})) / 2$$

$$= (105 + \max(9.85, 73)) / 2$$

$$= 89.12 \text{ N./mm}^2$$

Average Flange Stress, Seating [SAa]:

$$= (\text{SHa} + \max(\text{SRa}, \text{STa})) / 2$$

$$= (167 + \max(15.6, 116)) / 2$$

$$= 141.65 \text{ N./mm}^2$$

Bolt Stress, Operating [BSo]:

$$= \text{Wm1} / \text{Ab}$$

$$= 744947 / 707$$

$$= 103.38 \text{ N./mm}^2$$

Bolt Stress, Seating [BSa]:

$$= (\text{Wm2} / \text{Ab})$$

$$= (214007 / 707)$$

$$= 29.70 \text{ N./mm}^2$$

Flange Stress Analysis Results: N./mm²

	Actual	Operating Allowed	Gasket Actual	Seating Allowed
Longitudinal Hub	105.23	220.51	167.24	222.00
Radial Flange	9.85	147.00	15.65	148.00
Tangential Flange	73.02	147.00	116.05	148.00
Maximum Average	89.12	147.00	141.65	148.00
Bolting	103.38	170.41	29.70	172.00

Minimum Required Flange Thickness 200.101 mm

Estimated M.A.W.P. (Operating) 7473 KPa.

Estimated Finished Weight of Flange at given Thk. 1211.1 kg.

Estimated Unfinished Weight of Forging at given Thk 1458.6 kg.

Flange Stress results per WRC-538 at computed Sfmax (N./mm²):



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Flg Calc [Int P]: New Flange

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		Actual	Operating Allowed	stress ratio	Gasket Actual	Seating Allowed	stress ratio
Longitudinal Hub	Sh	505.41	509.72	0.992	505.41	619.99	0.815
Tangential Flange	St	352.26	382.29	0.921	352.26	464.99	0.758
Tangential Flange	Sto	145.07	254.86	0.569	145.07	309.99	0.468
Combined	Sh/f + St	764.58	764.58	1.000	764.58	929.98	0.822
Combined	Sr + St	399.75	764.58	0.523	399.75	929.98	0.430
Combined	Sh + Sto	650.49	764.58	0.851	650.49	929.98	0.699
Radial Flange	Sr	47.50	509.72	0.093	47.50	619.99	0.077
Maximum Stress Ratio				1.000			0.822

Coefficients for f and f_2

	f	f_2	f_3
a	-0.71375912	-0.01447638	-0.06456312
b	-0.12846279	-0.27814745	-0.08232239
c	1.08037907	0.01035395	0.13691677
d	0.99766848	1.37984158	-0.77879888
e	1.21823466	-0.34346764	0.14909019
f	0.01024612	0.02536192	0.00660062
g	0.4262313	-0.0014508	0.0616252
h	1.4204976	2.25543162	0.74639834
i	-0.70278181	-0.60889597	-0.079595
j	-0.02483937	-0.15735462	0.09851673
k	-1.59460436	1.22516062	-0.1121436

$$f_n = \frac{\left[a + c \left(\frac{g_1}{g_0} \right) + e \left(\frac{h}{h_0} \right) + g \left(\frac{g_1}{g_0} \right)^2 + i \left(\frac{h}{h_0} \right)^2 + k \left(\frac{g_1}{g_0} \right) \left(\frac{h}{h_0} \right) \right]}{\left[1 + b \left(\frac{g_1}{g_0} \right) + d \left(\frac{h}{h_0} \right) + f \left(\frac{g_1}{g_0} \right)^2 + h \left(\frac{h}{h_0} \right)^2 + j \left(\frac{g_1}{g_0} \right) \left(\frac{h}{h_0} \right) \right]}$$

Hub Stress Correction Factor [f]:

= max(1; F1num / F1den)

= max(1; 1.54/1.25)

= 1.2258

Hub Stress Correction Factor [f2]:

= max(0.25; fn2) if $h/h_0 < 0.35$, else max (0.25; fn3)

= max(0.25; 0.42/1.19) if $h/h_0 < 0.35$, else max (0.25; 0.28/0.77)

= 0.3518

In this case, the hub ratio h/h_0 , 0.3071 was < 0.35 .

PV Elite iterates on the assembly bolt load stress, Sfmax, until it exceeds the limits provided by WRC 538.



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Flg Calc [Int P]: New Flange

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Bolting and Gasket Calculations per ASME PCC-1 Appendix O:

Input Data:

Maximum permissible bolt stress	Sbmax	507.5 N./mm ²
Minimum permissible bolt stress	Sbmin	290.0 N./mm ²
Max. permissible bolt stress prior to flange damage	Sfmax	522.1 N./mm ²
Target assembly gasket stress	SgT	207.0 N./mm ²
Maximum permissible gasket stress	Sgmax	276.0 N./mm ²
Minimum gasket operating stress	Sgmin-O	97.0 N./mm ²
Minimum gasket seating stress	Sgmin-S	140.0 N./mm ²
Nut Factor	K	0.200
Fraction of Gasket Load remaining after relaxation	Phig	0.700
Root Area of a Single Bolt	Abs	22.0838 cm ²
Max. permissible single flange rotation for gasket	Thetagmax	1.000 deg

Calculated Results:

Flange Material Ambient Yield Stress	Sya	310.0 N./mm ²
Flange Material Operating Yield Stress	Syo	223.0 N./mm ²
Single flange rotation at Sfmax	Thetafmax	0.440 deg

Gasket Contact Area [Ag]:

$$= \pi/4 (GO.D.^2 - GI.D.^2)$$

$$= \pi/4 (1176^2 - 1132^2)$$

$$= 797.6 \text{ cm}^2$$

Target Bolt Stress [Sbsel]: (Eq. O-1)

$$= SgT \cdot Ag / (nb \cdot Abs)$$

$$= 207 \cdot 798 / (32 \cdot 22.1)$$

$$= 233.6 \text{ N./mm}^2$$

Upper Bolt Stress Limit [Sbsel]: (Eq. O-4)

$$= \min(Sbsel; Sbmax)$$

$$= \min(234; 508)$$

$$= 233.6 \text{ N./mm}^2$$

Lower Bolt Stress Limit [Sbsel]: (Eq. O-5)

$$= \max(Sbsel; Sbmin)$$

$$= \max(234; 290)$$

$$= 290.0 \text{ N./mm}^2$$

Upper Bolt Stress Limit including the Flange [Sbsel]: (Eq. O-6)

$$= \min(Sbsel; Sfmax)$$

$$= \min(290; 522)$$

$$= 290.0 \text{ N./mm}^2$$

Check if Gasket Assembly Seating Stress is achieved. (Eq. O-7)

$$:: Sbsel \geq Sgmin-S (Ag / (nb \cdot Abs))$$

$$:: 290 \geq 140 (798 / (32 \cdot 22.1))$$

$$:: 290.0 \geq 158.0 \text{ N./mm}^2 \text{ Check Passes}$$

Check if Gasket Operating Stress is maintained. (Eq. O-8)

$$:: Sbsel \geq (Sgmin-O \cdot Ag + \pi/4 \cdot Pmax \cdot GI.D.^2) / (\phi G \cdot Abs \cdot nb)$$

$$:: 290 \geq (97 \cdot 798 + \pi/4 \cdot 5900 \cdot 1132^2) / (0.7 \cdot 22.1 \cdot 32)$$

$$:: 290.0 \geq 276.4 \text{ N./mm}^2 \text{ Check Passes}$$

Check if Gasket Maximum Stress is achieved. (Eq. O-9)

$$:: Sbsel \leq Sgmax (Ag / (nb \cdot Abs))$$

$$:: 290 \leq 276 (798 / (32 \cdot 22.1))$$

$$:: 290.0 \leq 311.5 \text{ N./mm}^2 \text{ Check Passes}$$

Check if Flange Rotation Limit is exceeded. (Eq. O-10)

$$:: Sbsel \leq Sfmax (thetagmax / thetafmax)$$

$$:: 290 \leq 522 (1/0.44)$$



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Flg Calc [Int P]: New Flange

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:: 290.0 ≤ 1187.0 N./mm² Check Passes

Assembly Bolt Torque [Tb]: (Eq. O-2)

= Sbsel • K • Abs • ϕb

= 290 • 0.2 • 22.1 • 57.1

= 746.4 Kg-m.

Bolt Allowable Stress for: SA-193 B8 Cl. 2 S_bB8 77 N./mm²

Bolt Allowable Stress for: SA-193 B16 S_b 170 N./mm²

Note: Since S_b ≥ S_bB8, bolt allowable stress comparison check Passes (see Div 1 UG-44(b)).

Flange rigidity based on required thickness [ASME]:

Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:

= 52.14 • M_a / B_{sc} • C_{nv}fac • V / (λ • E_{amb} • g_o² • h_o • K_i)

= 52.14 • 111993 / 1 • 9807 • 0.36 / (3.42 • 204092 • 29² • 179 • 0.3)

= 0.650 (should be ≤ 1)

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:

= 52.14 • M_o / B_{sc} • C_{nv}fac • V / (λ • E_{op} • g_o² • h_o • K_i)

= 52.14 • 70467 / 1 • 9807 • 0.36 / (3.42 • 176185 • 29² • 179 • 0.3)

= 0.474 (should be ≤ 1)

Flange rigidity based on given thickness [ASME]:

Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:

= 52.14 • M_a / B_{sc} • C_{nv}fac • V / (λ • E_{amb} • g_o² • h_o • K_i)

= 52.14 • 111993 / 1 • 9807 • 0.36 / (3.61 • 204092 • 29² • 179 • 0.3)

= 0.616 (should be ≤ 1)

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:

= 52.14 • M_o / B_{sc} • C_{nv}fac • V / (λ • E_{op} • g_o² • h_o • K_i)

= 52.14 • 70467 / 1 • 9807 • 0.36 / (3.61 • 176185 • 29² • 179 • 0.3)

= 0.449 (should be ≤ 1)

Minimum Design Metal Temperature Results:

Thickness Ratio = 0.75, Temperature Reduction per Fig. UCS 66.1 = 14 °C

Min Metal Temp. w/o impact per UCS-66, Curve B 6 °C

Min Metal Temp. at Required thickness (UCS 66.1) -8 °C

Note: UCS-66(b)-(c) was considered in the flange MDMT calculation.

Per ASME VIII-2, 4.16.2.3, Perform Calculations in the New Condition:

Flange Stress Analysis Results: N./mm²

	Operating		Gasket Seating	
	Actual	Allowed	Actual	Allowed
Longitudinal Hub	100.90	222.00	158.09	222.00
Radial Flange	10.60	148.00	16.61	148.00
Tangential Flange	67.87	148.00	106.34	148.00
Maximum Average	84.39	148.00	132.22	148.00
Bolting	103.38	172.00	29.70	172.00

Minimum Req'd Thickness, ambient, no Corrosion: 191.770 mm

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Internal Pressure Calculations: Step: 5 9:45am Sep 9,2026

Internal Pressure Results Summary:

Element Thickness, Pressure, Diameter and Allowable Stress :

From	To	Int. Press + Liq. Hd KPa.	Nominal Thickness mm	Total Corr Allowance mm	Element Diameter mm	Allowable Stress(SE) N./mm ²	Element#
CH COVER		5900	178	3	1460	147	1
CH Flange		5900	213	3	1100	147	2
Channel shell		5910.8	32	3	1100	124.95	3
Shell		1090.8	18	3	1100	117.3	4
Shell head		1090.8	14	3	1100	116.45	5

Element Required Thickness and MAWP :

From	To	Design Pressure KPa.	M.A.W.P. Corroded KPa.	M.A.P. New & Cold KPa.	Minimum Thickness mm	Required Thickness mm	Element#
CH COVER		5900	6332	7292	170	165.306	1
CH Flange		5900	7473	8320	205	200.101	2
Channel shell		5900	6342	7072	32	29.9248	3
Shell		1080	3120	3765	18	8.17178	4
Shell head		1080	1260	1916	9	8.14808	5

Warning: Shell Side MAWP governed by a standard Flange!

Summary of Heat Exchanger Maximum Allowable Working Pressures :

Note:

For Exchanger designs, the following values include MAWPs that consider the tubesheet, tubes, tube/tubesheet joint etc. These values were determined by iteration. Review the tubesheet analysis report for more information.

Shell Side MAWP = 1115 KPa.
 Shell Side MAPnc = 1916 KPa.
 Channel Side MAWP = 6218 KPa.
 Channel Side MAPnc = 7072 KPa.

Elements Suitable for Design Internal Pressure.

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Element: Channel shell

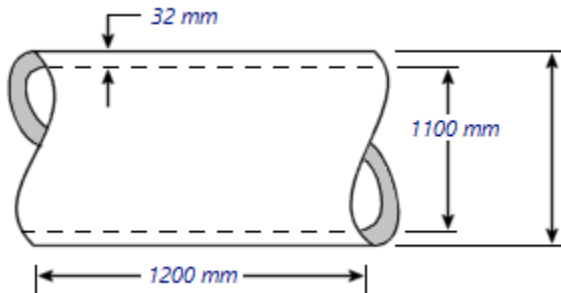
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Internal Pressure Calculation Results:

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Internal Pressure Results for: Channel shell

Cylindrical Shell Design Information:			
Design Pressure P	5910.812 KPa.	Int. Design Temperature	430 °C
Material	SA-387 11	Ext. Design Temperature	200 °C
Length	1200.000 mm	External Pressure Chart	CS-3
		UNS Number	K11789
Int. Corr. All. c	3.0000 mm	Allowable Stress (ope) S	147.0 N./mm ²
Ext. Corr. All. ce	0.0000 mm	Allowable Stress (amb) Sa	148.0 N./mm ²
Inside Diameter Di	1100.000 mm	Actual Stress	136.8 N./mm ²
Spec. Min. Thk t,e, or sa	32.0000 mm	Spec. Nominal Thk.	32.000 mm
Surface Area	43881.8 cm ²	SG of Contents	1.000
Element Volume	1140602.5 cm ³	Weight of Contents	1139.9 Kg
Empty Weight	1058.4 Kg	Operating Weight	2198.3 Kg
		Comp. All. Ope. Stress	83.6 N./mm ²

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	85 %	Long. Joint Eff.	E or v 85 %

Required Thickness due to Internal Pressure [tr]:

$$\begin{aligned}
 &= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)} \\
 &= (5911 \cdot 553) / (147 \cdot 0.85 - 0.6 \cdot 5911) \\
 &= 26.9248 + 3.0000 = 29.9248 \text{ mm}
 \end{aligned}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 10.812 KPa.

$$\begin{aligned}
 &= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)} \\
 &= (147 \cdot 0.85 \cdot 29) / (553 + 0.6 \cdot 29) \\
 &= 6352 - 10.8 = 6342 \text{ KPa.}
 \end{aligned}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$\begin{aligned}
 &= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)} \\
 &= (148 \cdot 0.85 \cdot 32) / (550 + 0.6 \cdot 32) \\
 &= 7072 \text{ KPa.}
 \end{aligned}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$\begin{aligned}
 &= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t) \\
 &= (5911 \cdot (553 + 0.6 \cdot 29)) / (0.85 \cdot 29) \\
 &= 136.784 \text{ N./mm}^2
 \end{aligned}$$



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Element: Channel shell

Step: 6

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% Elongation per Table UG-79-1 ($50 \cdot t_{nom} / R_f \cdot (1 - R_f / R_o)$) 2.827 %

Minimum Design Metal Temperature Results:

Govern. thk, $t_g = 32$, $t_r = 28.2$, $c = 3$ mm, $E^* = 0.85$

Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.83$, Temp. Reduction = 10 °C

Min Metal Temp. w/o impact per UCS-66, Curve A 25 °C

Min Metal Temp. at Required thickness (UCS 66.1) 16 °C

Note:

Post Weld Heat Treatment is required for this Element/Joint and it was specified as being heat treated.

MDMT Design Information:			
MDMT Curve	A	Press. at MDMT	6218.147 KPa.
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-12 °C
Governing Thk.	32.000 mm	Basic MDMT	25 °C
Stress/Thk. Ratio	0.826	Temperature Difference	10 °C
UCS-66(c) red.	No	Computed Min. Temp.	16 °C
		Joint Efficiency E^*	85 %

EXTERNAL PRESSURE RESULTS for: Channel shell

Cylindrical Shell From 30 to Tsht Ext. Chart: CS-2 at 200 °C

Elastic Modulus from Chart: CS-2 at 200 °C: 193602528 KPa.

Results for Maximum Allowable Ext. Pressure					MAEP	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
29.000	1164.00	1199.97	40.14	1.0309	0.0051885	107.46

$$MAEP = (4 \cdot B) / (3 \cdot (Do/t)) = (4 \cdot 107) / (3 \cdot 40.1) = 3569 \text{ KPa.}$$

Results for Required Thickness					Tca	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
3.806	1164.00	1199.97	305.86	1.0309	0.0002401	23.24

$$MAEP = (4 \cdot B) / (3 \cdot (Do/t)) = (4 \cdot 23.2) / (3 \cdot 306) = 101 \text{ KPa.}$$

Results for Maximum Stiffened Length					Slen	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
29.000	1164.00	34620.78	40.14	29.7429	0.0006852	66.33

$$MAEP = (4 \cdot B) / (3 \cdot (Do/t)) = (4 \cdot 66.3) / (3 \cdot 40.1) = 2203 \text{ KPa.}$$



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FileName : 175-C

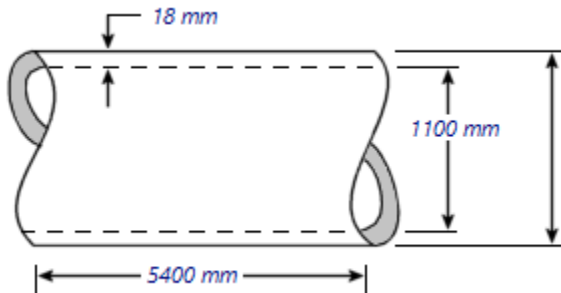
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Element: Shell

Step: 7 9:45am Sep 9,2026

Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025



Internal Pressure Results for: Shell

Cylindrical Shell Design Information:			
Design Pressure P	1090.812 KPa.	Int. Design Temperature	275 °C
Material	SA-516 70	Ext. Design Temperature	25 °C
Length	5400.000 mm	External Pressure Chart	CS-2
		UNS Number	K02700
Int. Corr. All. c	3.0000 mm	Allowable Stress (ope) S	138.0 N./mm ²
Ext. Corr. All. ce	0.0000 mm	Allowable Stress (amb) Sa	138.0 N./mm ²
Inside Diameter Di	1100.000 mm	Actual Stress	48.1 N./mm ²
Spec. Min. Thk t,e, or sa	18.0000 mm	Spec. Nominal Thk.	18.000 mm
Surface Area	192717.9 cm ²	SG of Contents	1.000
Element Volume	4381930.0 cm ³	Weight of Contents	4379.3 Kg
Empty Weight	2646.0 Kg	Operating Weight	7025.3 Kg
		Comp. All. Ope. Stress	101.1 N./mm ²

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	85 %	Long. Joint Eff.	E or v 85 %

Required Thickness due to Internal Pressure [tr]:

$$\begin{aligned}
 &= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)} \\
 &= (1091 \cdot 553) / (138 \cdot 0.85 - 0.6 \cdot 1091) \\
 &= 5.1718 + 3.0000 = 8.1718 \text{ mm}
 \end{aligned}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 10.812 KPa.

$$\begin{aligned}
 &= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)} \\
 &= (138 \cdot 0.85 \cdot 15) / (553 + 0.6 \cdot 15) \\
 &= 3131 - 10.8 = 3120 \text{ KPa.}
 \end{aligned}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$\begin{aligned}
 &= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)} \\
 &= (138 \cdot 0.85 \cdot 18) / (550 + 0.6 \cdot 18) \\
 &= 3765 \text{ KPa.}
 \end{aligned}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$\begin{aligned}
 &= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t) \\
 &= (1091 \cdot (553 + 0.6 \cdot 15)) / (0.85 \cdot 15) \\
 &= 48.084 \text{ N./mm}^2
 \end{aligned}$$



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Element: Shell

Step: 7

9:45am

Sep 9,2026

% Elongation per Table UG-79-1 ($50 \cdot t_{nom} / R_f \cdot (1 - R_f / R_o)$) 1.610 %**Minimum Design Metal Temperature Results:**Govrn. thk, $t_g = 18$, $t_r = 5.29$, $c = 3$ mm, $E^* = 0.85$ Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.3$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -11 °C

Min Metal Temp. at Required thickness (UCS 66.1) -104 °C

Min Metal Temp. w/o impact per UG-20(f) -29 °C

MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	1115.044 KPa.
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-12 °C
Governing Thk.	18.000 mm	Basic MDMT	-11 °C
Stress/Thk. Ratio	0.300	Temperature Difference	93 °C
UCS-66(c) red.	No	Computed Min. Temp.	-104 °C
		MDMT per UG-20(f)	-29 °C
		Joint Efficiency E^*	85 %

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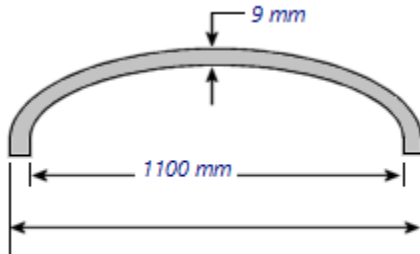
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Element: Shell head

Step: 8 9:45am Sep 9,2026

Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025



Internal Pressure Results for: Shell head

Elliptical Head Design Information:			
Design Pressure P	1090.812 KPa.	Int. Design Temperature	275 °C
Material	SA-516 70	Ext. Design Temperature	25 °C
Length	50.000 mm	External Pressure Chart	CS-2
		UNS Number	K02700
Int. Corr. All. c	3.0000 mm	Allowable Stress (ope) S	137.0 N./mm ²
Ext. Corr. All. ce	0.0000 mm	Allowable Stress (amb) Sa	138.0 N./mm ²
Inside Diameter Di	1100.000 mm	Actual Stress	117.6 N./mm ²
Spec. Min. Thk t,e, or sa	9.0000 mm	Spec. Nominal Thk.	14.000 mm
Surface Area	15564.3 cm ²	SG of Contents	1.000
Element Volume	221783.8 cm ³	Weight of Contents	221.6 Kg
Empty Weight	180.2 Kg	Operating Weight	401.8 Kg
		Comp. All. Ope. Stress	101.1 N./mm ²
		Aspect Ratio	2.00

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	85 %	Long. Joint Eff. E or v	85 %

Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 13.750 mm

Head inner surface maximum deviation inside the specified shape, 0.625% of D: 6.875 mm

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot D \cdot K_{cor}) / (2 \cdot S \cdot E - 0.2 \cdot P) \text{ Appendix 1-4(c)}$$

$$= (1091 \cdot 1106 \cdot 0.99) / (2 \cdot 137 \cdot 0.85 - 0.2 \cdot 1091)$$

$$= 5.1481 + 3.0000 = 8.1481 \text{ mm}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 10.812 KPa.

$$= (2 \cdot S \cdot E \cdot t) / (K_{cor} \cdot D + 0.2 \cdot t) \text{ per Appendix 1-4 (c)}$$

$$= (2 \cdot 137 \cdot 0.85 \cdot 6) / (0.99 \cdot 1106 + 0.2 \cdot 6)$$

$$= 1271 - 10.8 = 1260 \text{ KPa.}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (2 \cdot S \cdot E \cdot t) / (K \cdot D + 0.2 \cdot t) \text{ per Appendix 1-4 (c)}$$

$$= (2 \cdot 138 \cdot 0.85 \cdot 9) / (1 \cdot 1100 + 0.2 \cdot 9)$$

$$= 1916 \text{ KPa.}$$



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Element: Shell head

Step: 8

9:45am

Sep 9, 2026

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (K_{cor} \cdot D + 0.2 \cdot t)) / (2 \cdot E \cdot t)$$

$$= (1091 \cdot (0.99 \cdot 1106 + 0.2 \cdot 6)) / (2 \cdot 0.85 \cdot 6)$$

$$= 117.565 \text{ N./mm}^2$$

Straight Flange Required Thickness:

$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) + c \text{ per UG-27 (c)(1)}$$

$$= (1091 \cdot 553) / (137 \cdot 0.85 - 0.6 \cdot 1091) + 3$$

$$= 8.210 \text{ mm}$$

Straight Flange Maximum Allowable Working Pressure:

[Less Operating Hydrostatic Head Pressure of 10.812 KPa.](#)

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (137 \cdot 0.85 \cdot 11) / (553 + 0.6 \cdot 11)$$

$$= 2289 - 10.8 = 2278 \text{ KPa.}$$

Factor K, corroded condition [Kcor]:

$$= (2 + (\text{Inside Diameter} / (2 \cdot \text{Inside Head Depth})))^2 / 6$$

$$= (2 + (1106 / (2 \cdot 278)))^2 / 6$$

$$= 0.992825$$

Percent Elong. per UCS-79, VIII-1-01-57 $(75 \cdot t_{nom} / R_f) \cdot (1 - R_f / R_o)$ 5.412 %

[Note: Please Check Requirements of UCS-79 as Elongation is > 5%.](#)

MDMT Calculations in the Knuckle Portion:

Govrn. thk, tg = 9, tr = 5.22, c = 3 mm, E* = 0.85

Thickness Ratio = $tr \cdot E* / (tg - c) = 0.74$, Temp. Reduction = 14 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -29 °C

Min Metal Temp. at Required thickness (UCS 66.1) -43 °C

MDMT Calculations in the Head Straight Flange:

Govrn. thk, tg = 14, tr = 5.29, c = 3 mm, E* = 0.85

Thickness Ratio = $tr \cdot E* / (tg - c) = 0.41$, Temp. Reduction = 48 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -18 °C

Min Metal Temp. at Required thickness (UCS 66.1) -48 °C

Min Metal Temp. w/o impact per UG-20(f) -29 °C

MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	1115.044 KPa.
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-12 °C
Governing Thk.	9.000 mm	Basic MDMT	-29 °C
Stress/Thk. Ratio	0.740	Temperature Difference	14 °C
UCS-66(c) red.	No	Computed Min. Temp.	-43 °C
		MDMT per UG-20(f)	-29 °C
		Joint Efficiency E*	85 %

[Note: Temperature values displayed above are for the location with the highest basic MDMT](#)

Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 13.750 mm

Head inner surface maximum deviation inside the specified shape, 0.625% of D: 6.875 mm

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Shop Pressure Test Results: Step: 9 9:45am Sep 9, 2026

Hydrostatic Test Pressure Results:

Exchanger Shell Side Hydrostatic Test Pressures:

UG99(b)	= 1.30 * M.A.W.P. * Sa/S	1449.558 KPa.
UG99b(b)(35)	= 1.30 * Design Pres * Sa/S	1404.000 KPa.
UG99(c)	= 1.30 * M.A.P. - Head(Hyd)	2476.714 KPa.
UG100	= 1.10 * M.A.W.P. * Sa/S	1226.549 KPa.
PED	= max(1.43*DP, 1.25*DP*ratio)	1544.400 KPa.
PED&UG99(b)(35)	= max(1.43*DP, 1.3*DP*ratio)	1544.400 KPa.
App 27-4	= M.A.W.P.	1115.044 KPa.

Exchanger Channel Side Hydrostatic Test Pressures:

UG99(b)	= 1.30 * M.A.W.P. * Sa/S	8138.184 KPa.
UG99b(b)(35)	= 1.30 * Design Pres * Sa/S	7721.800 KPa.
UG99(c)	= 1.30 * M.A.P. - Head(Hyd)	9179.087 KPa.
UG100	= 1.10 * M.A.W.P. * Sa/S	6886.156 KPa.
PED	= max(1.43*DP, 1.25*DP*ratio)	8437.000 KPa.
PED&UG99(b)(35)	= max(1.43*DP, 1.3*DP*ratio)	8437.000 KPa.
App 27-4	= M.A.W.P.	6218.147 KPa.

Horizontal Test performed per: UG-99c

Please note that Nozzle, Shell, Head, Flange, etc MAWPs are all considered when determining the hydrotest pressure for those test types that are based on the MAWP of the vessel.

Stresses on Elements due to Test Pressure (N./mm² & KPa.):

From To	Stress	Allowable	Ratio	Pressure
Channel shell	192.3	279.0	0.689	9189.87
Shell	91.2	234.0	0.390	2487.50
Shell head	179.1	234.0	0.766	2487.50

Stress ratios for Nozzle and Pad Materials (N./mm²):

Description	Pad/Nozzle	Ambient	Operating	Ratio
N3- Tube inlet	Nozzle	118.00	98.98	1.192
N3- Tube inlet	Pad	148.00	147.00	1.007
Tube outlet	Nozzle	118.00	98.98	1.192
Tube outlet	Pad	148.00	147.00	1.007
Shell inlet-N1	Nozzle	118.00	118.00	1.000
Shell inlet-N1	Pad	138.00	137.00	1.007
Shell Inlet	Nozzle	118.00	118.00	1.000
Shell Inlet	Pad	138.00	137.00	1.007
Vent	Nozzle	138.00	132.50	1.041
Drain	Nozzle	138.00	132.50	1.041

Minimum

1.000

Stress ratios for Pressurized Vessel Elements (N./mm²):

Description	Ambient	Operating	Ratio
CH COVER	148.00	147.00	1.007
CH Flange	148.00	147.00	1.007
Channel shell	148.00	147.00	1.007
Shell	138.00	138.00	1.000
Shell head	138.00	137.00	1.007



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Shop Pressure Test Results: Step: 9 9:45am Sep 9,2026

Minimum 1.000

Stress ratios for Exchanger Materials (N./mm²):

Description	Ambient	Operating	Ratio
Tube Material	118.00	98.98	1.192
Tubesheet Material	148.00	147.00	1.007

Minimum 1.007

Hoop Stress in Nozzle Wall during Pressure Test (N./mm²):

Description	Ambient	Operating	Ratio
N3- Tube inlet	31.39	184.50	0.170
Tube outlet	26.96	184.50	0.146
Shell inlet-N1	23.13	216.00	0.107
Shell Inlet	23.13	216.00	0.107
Vent	6.19	225.00	0.027
Drain	6.19	225.00	0.027

Elements Suitable for Test Pressure.

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Field Pressure Test Results: Step: 10 9:45am Sep 9, 2026

Hydrostatic Test Pressure Results:

Exchanger Shell Side Hydrostatic Test Pressures:

UG99(b)	= 1.30 * M.A.W.P. * Sa/S	1449.558 KPa.
UG99b(b)(35)	= 1.30 * Design Pres * Sa/S	1404.000 KPa.
UG99(c)	= 1.30 * M.A.P. - Head(Hyd)	2476.714 KPa.
UG100	= 1.10 * M.A.W.P. * Sa/S	1226.549 KPa.
PED	= max(1.43*DP, 1.25*DP*ratio)	1544.400 KPa.
PED&UG99(b)(35)	= max(1.43*DP, 1.3*DP*ratio)	1544.400 KPa.
App 27-4	= M.A.W.P.	1115.044 KPa.

Exchanger Channel Side Hydrostatic Test Pressures:

UG99(b)	= 1.30 * M.A.W.P. * Sa/S	8138.184 KPa.
UG99b(b)(35)	= 1.30 * Design Pres * Sa/S	7721.800 KPa.
UG99(c)	= 1.30 * M.A.P. - Head(Hyd)	9179.087 KPa.
UG100	= 1.10 * M.A.W.P. * Sa/S	6886.156 KPa.
PED	= max(1.43*DP, 1.25*DP*ratio)	8437.000 KPa.
PED&UG99(b)(35)	= max(1.43*DP, 1.3*DP*ratio)	8437.000 KPa.
App 27-4	= M.A.W.P.	6218.147 KPa.

UG-99(b)(35), Test Pressure Calculation [Shell Side]:

= Test Factor • Design Pressure • Stress Ratio

= 1.3 • 1080 • 1

= 1404.000 KPa.

UG-99(b) Note 35, Test Pressure Calculation [Channel Side]:

= Test Factor • Design Pressure • Stress Ratio

= 1.3 • 5900 • 1.01

= 7721.800 KPa.

Horizontal Test performed per: UG-99b (Note 35)

Please note that Nozzle, Shell, Head, Flange, etc MAWPs are all considered when determining the hydrotest pressure for those test types that are based on the MAWP of the vessel.

Stresses on Elements due to Test Pressure (N./mm² & KPa.):

From To	Stress	Allowable	Ratio	Pressure
Channel shell	178.9	279.0	0.641	7732.58
Shell	62.4	234.0	0.267	1414.78
Shell head	152.5	234.0	0.652	1414.78

Stress ratios for Nozzle and Pad Materials (N./mm²):

Description	Pad/Nozzle	Ambient	Operating	Ratio
N3- Tube inlet	Nozzle	118.00	98.98	1.192
N3- Tube inlet	Pad	148.00	147.00	1.007
Tube outlet	Nozzle	118.00	98.98	1.192
Tube outlet	Pad	148.00	147.00	1.007
Shell inlet-N1	Nozzle	118.00	118.00	1.000
Shell inlet-N1	Pad	138.00	137.00	1.007
Shell Inlet	Nozzle	118.00	118.00	1.000
Shell Inlet	Pad	138.00	137.00	1.007
Vent	Nozzle	138.00	132.50	1.041
Drain	Nozzle	138.00	132.50	1.041

Minimum

1.000



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Field Pressure Test Results: Step: 10 9:45am Sep 9,2026

Stress ratios for Pressurized Vessel Elements (N./mm²):

Description	Ambient	Operating	Ratio
CH COVER	148.00	147.00	1.007
CH Flange	148.00	147.00	1.007
Channel shell	148.00	147.00	1.007
Shell	138.00	138.00	1.000
Shell head	138.00	137.00	1.007

Minimum 1.000

Stress ratios for Exchanger Materials (N./mm²):

Description	Ambient	Operating	Ratio
Tube Material	118.00	98.98	1.192
Tubesheet Material	148.00	147.00	1.007

Minimum 1.007

Hoop Stress in Nozzle Wall during Pressure Test (N./mm²):

Description	Ambient	Operating	Ratio
N3- Tube inlet	26.41	184.50	0.143
Tube outlet	22.68	184.50	0.123
Shell inlet-N1	13.16	216.00	0.061
Shell Inlet	13.16	216.00	0.061
Vent	3.52	225.00	0.016
Drain	3.52	225.00	0.016

Elements Suitable for Test Pressure.

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ASME TS Calc: Pressure Test

Case: 1 9:45a Sep 9, 2026

Input Echo, Tubesheet Number 1, Description: Tubesheet

Shell Data:

Main Shell Description: Shell

Shell Test Pressure		1404.00 KPa.
Shell Thickness	ts	18.0000 mm
Shell Internal Corrosion Allowance	cas	0.0000 mm
Shell External Corrosion Allowance	casext	0.0000 mm
Inside Diameter of Shell	Ds	1100.000 mm
Shell Temperature for Internal Pressure	Ts	21.11 °C
Shell Material		SA-516 70

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Shell Material UNS Number		K02700
Shell Allowable Stress at Temperature	Ss	138.00 N./mm ²
Shell Allowable Stress at Ambient		138.00 N./mm ²

Channel Data:

Channel Description: Channel shell

Channel Type:		Cylinder
Channel Test Pressure		7721.80 KPa.
Channel Thickness	tc	32.0000 mm
Channel Internal Corrosion Allowance	cac	0.0000 mm
Channel External Corrosion Allowance	cacext	0.0000 mm
Inside Diameter of Channel	Dc	1100.000 mm
Channel Design Temperature	TEMPC	21.11 °C
Channel Flange Matl		SA-387 11

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Channel Material UNS Number		K11789
Channel Allowable Stress at Temperature	Sc	148.00 N./mm ²
Channel Allowable Stress at Ambient		148.00 N./mm ²

Tube Data:

Number of Tube Holes	Nt	446
Tube Wall Thickness	et	2.1100 mm
Tube Outside Diameter	D	19.0500 mm
Total Straight Tube Length	Lt	5200.00 mm
Straight Tube Length (bet. inner tubsht faces) L		5030.00 mm
Design Temperature of the Tubes		21.11 °C
Tube Material		SA-213 T11
Tube Material UNS Number		K11597
Is this a Welded Tube		No
Tube Material Specification used	Smls. tube	
Tube Allowable Stress at Temperature		118.00 N./mm ²
Tube Allowable Stress At Ambient		118.00 N./mm ²
Tube Yield Stress At design Temperature	Syt	205.00 N./mm ²
Tube Pitch (Center to Center Spacing)	P	23.8125 mm
Tube Layout Pattern		Triangular

Fillet Weld Leg	af	2.1100 mm
Groove Weld Leg	ag	2.1100 mm
Tube-Tubesheet Joint Weld Type		Full Strength
Method for Tube-Tubesheet Jt. Allowable		UW-20.3
Is the Tube Inset		No

Radius to Outermost Tube Hole Center	ro	528.475 mm
Largest Center-to-Center Tube Distance	U1	57.1500 mm
Length of Expanded Portion of Tube	ltx	0.0000 mm
Tube-side pass partition groove depth	hg	8.0000 mm
Area of Tube Layout	Ap	2685.8 cm ²
Perimeter of Tube Layout	Cp	2543.2 mm



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ASME TS Calc: Pressure Test

Case: 1 9:45a Sep 9,2026

Tubesheet Data:

Tubesheet TYPE: U-tube, Integral both sides, Conf. a

Tubesheet Design Metal Temperature T 21.11 °C
Tubesheet Material SA-336 F11

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

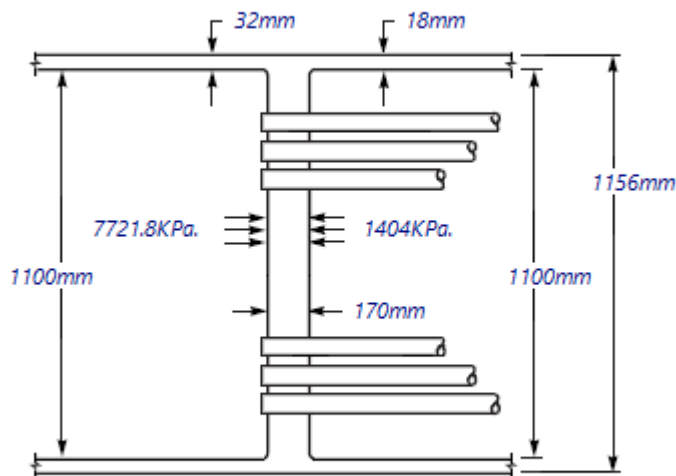
Tubesheet Material UNS Number K11572
Tubesheet Allowable Stress at Temperature S 148.00 N./mm²
Tubesheet Allowable Stress at Ambient Tt 148.00 N./mm²
Thickness of Tubesheet h 170.0000 mm
Tubesheet Corr. Allowance (Shell side) Cats 0.0000 mm
Tubesheet Corr. Allowance (Channel side) Catc 0.0000 mm
Tubesheet Outside Diameter A 1156.000 mm

Area of the Untubed Lanes AL 419.1 cm²

Junction Stress Reduction option Perform Plastic Calculation
Perform Differential Pressure Design No
Run Multiple Load Cases Yes

Tubesheet Integral with Both
Tubesheet Extended as Flange No

Tubesheet Integral w/Shell and Channel



Configuration a:

ASME TubeSheet Results per Part UHX/4.18, 2025

Elasticity/Expansion Material Properties :

Shell - TM-1 Carbon Steels with C<= 0.3%

Elastic Mod. at Design Temperature 21.1 °C 0.20270E+09 KPa.
Elastic Mod. at Ambient Temperature 21.1 °C 0.20270E+09 KPa.

Channel - TM-1 Material Group C



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ASME TS Calc: Pressure Test

Case: 1 9:45a Sep 9,2026

Elastic Mod. at Design Temperature	21.1 °C	0.20408E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20408E+09 KPa.

Tubes - TM-1 Material Group C

Elastic Mod. at Tubsht. Design Temp.	21.1 °C	0.20408E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20408E+09 KPa.

TubeSheet - TM-1 Material Group C

Elastic Mod. at Design Temperature	21.1 °C	0.20408E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20408E+09 KPa.

Tube Required Thickness under Internal Pressure (Tubeside pressure):

Thickness Due to Internal Pressure:

= $(P \cdot (D/2 - CAE)) / (S \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)

= $(7722 \cdot (19/2 - 0)) / (118 \cdot 1 + 0.4 \cdot 7722)$

= $0.6075 + 0.0000 = 0.6075$ mm

Tube Required Thickness under External Pressure (Shellside pressure) :

External Pressure Chart	CS-2	at	21.11 °C
Elastic Modulus for Material			199943392.00 KPa.

Results for Max. Allowable External Pressure (Emawp):

TCA	ODCA	SLEN	D/T	L/D	Factor A	B
2.1100	19.05	5558.48	9.03	50.0000	0.0134948	122.73

EMAWP = $(2.167 / (D/T) - 0.0833) \cdot B = 19233$ KPa.

Results for Req'd Thickness for Ext. Pressure (Tca):

TCA	ODCA	SLEN	D/T	L/D	Factor A	B
0.4045	19.05	5558.48	47.09	50.0000	0.0004960	49.59

EMAWP = $(4 \cdot B) / (3 \cdot (D/T)) = (4 \cdot 49.6) / (3 \cdot 47.1) = 1404$ KPa.

Summary of Tube Required Thickness Results:

Total Required Thickness including Corrosion all.	0.6075 mm
Allowable Internal Pressure at Corroded thickness	28678.85 KPa.
Required Internal Design Pressure	7721.80 KPa.
Allowable External Pressure at Corroded thickness	19233.22 KPa.
Required External Design Pressure	1404.00 KPa.
Required Thickness due to Shell Side pressure	0.4045 mm

U-Tube Tubesheet results per ASME Div. 2 Part 4.18.7 2025

Results for 3 Load Cases:

Case#	--Reqd. Thk. + CA		---- Tubesheet Stresses				Case Type	Pass/Fail
	Tbsht	Extnsn	Bend	Allwd	Shear	Allwd		
D1uc	117.869	...	297	589	24	236	Ps+Pt D1	Ok
D2uc	55.272	...	56	589	4	236	Ps+Pt D2	Ok
D3uc	105.606	...	241	589	20	236	Ps+Pt D3	Ok
Max:	117.8686	... mm		0.504		0.102	(Str. Ratio)	

Load Case Definitions:

[Ps & Pt]:

Shell-side and Tube-side Design or Operating Pressures derived from Psd,min Ptd,max, Psox,min, Ptox,max etc. per the Load Case Tables

[c]:



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ASME TS Calc: Pressure Test

Case: 1 9:45a Sep 9,2026

With or Without Corrosion Allowance

[D1, D2, D3]:

Design Load Cases using the Maximum and Minimum Design Pressures

[D4]:

Design Load Case using the Minimum (Vacuum) Pressures (if specified)

Shell and Channel Stress Summary:

-- Integral Cyl. Req'd. Thk.	Shell	Channel	Shell Stress	Stress Allwd	Channel Stress	Stress Allwd	Pass Fail
D1uc	...	...	...	370	...	442	Ok
D2uc	...	...	...	370	...	442	Ok
D3uc	...	...	...	370	...	442	Ok
Max	...	...		...		...	

Note:

Because there was no net pressure differential across the tubesheet due to the specified pressures psd,min and ptd,min; load case D4 was not processed as the resulting tubesheet bending stress would be zero.

Summary of Thickness Comparisons for 3 Load Cases:

Thickness (mm)	Required	Actual	P/F
Tubesheet thickness :	117.869	170.000	Ok
Tube thickness :	0.607	2.110	Ok
Tube-tubesheet fillet weld leg :	1.281	2.110	Ok
Tube-tubesheet groove weld leg :	1.281	2.110	Ok

Given Shell Thickness: 18.0000 mm

Given Channel Thickness: 32.0000 mm

Min Shell length of thk, (18.000) adj. to tubesheet: 253.282 mm

Min Channel length of thk, (32.000) adj. to tubesheet: 337.710 mm

Note: This is a full strength Tube to Tubesheet Joint.

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External Pressure Calculations: Step: 11 9:45am Sep 9,2026

External Pressure Calculation Results Summary:

ASME Code, Section VIII Division 1, 2025

External Pressure Calculations:

From	To	Section Length mm	Outside Diameter mm	Corroded Thickness mm	Factor A	Factor B N./mm ²
10	20	No Calc	...	167	No Calc	No Calc
20	30	No Calc	...	202	No Calc	No Calc
30	Tsht	1199.97	1164	29	0.0051885	107.457
Tsht	40	No Calc	...	29	No Calc	No Calc
40	50	5541.7	1136	15	0.00037196	37.1879
50	60	No Calc	1118	6	0.00074538	74.5211

External Pressure Calculations:

From	To	External Actual T. mm	External Required T. mm	External Design Pressure KPa.	External M.A.W.P. KPa.
10	20	170	141.58	101.3	No Calc
20	30	205	200.101	101.3	No Calc
30	Tsht	32	6.80561	101.3	3569
Tsht	40	32	No Calc	101.3	No Calc
40	50	18	No Calc	...	655
50	60	9	No Calc	...	444

Minimum

444

External Pressure Calculations:

From	To	Actual Length Bet. Stiffeners mm	Allowable Length Bet. Stiffeners mm	Ring Inertia Required cm ⁴	Ring Inertia Available cm ⁴
10	20	No Calc	No Calc	No Calc	No Calc
20	30	No Calc	No Calc	No Calc	No Calc
30	Tsht	1199.97	34620.8	No Calc	No Calc
Tsht	40	No Calc	No Calc	No Calc	No Calc
40	50	5541.7	No Calc	No Calc	No Calc
50	60	No Calc	No Calc	No Calc	No Calc

Elements Suitable for External Pressure.

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Flg Calc [Ext P]: New Flange

Step: 12

9:45am

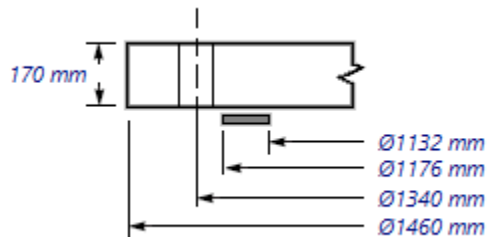
Sep 9, 2026

Flange Input Data Values

Description: New Flange :

CH COVER

Flange Type	TEMA Channel Cover		
Design Pressure	P	-101.30	KPa.
Design Temperature		200	°C
Internal Corrosion Allowance	ci	3.0000	mm
External Corrosion Allowance	ce	0.0000	mm
Use Corrosion Allowance in Thickness Calcs.		Yes	
Flange Outside Diameter	A	1460.000	mm
Flange Thickness	t	170.0000	mm
Flange Material	SA-336 F11		
Flange Material UNS number	K11572		
Flange Allowable Stress At Temperature	Sfo	148.00	N./mm ²
Flange Allowable Stress At Ambient	Sfa	148.00	N./mm ²
Bolt Material	SA-193 B16		
Bolt Allowable Stress At Temperature	Sb	172.00	N./mm ²
Bolt Allowable Stress At Ambient	Sa	172.00	N./mm ²
Perimeter along the Center of the Bolts	L	4209.734	mm
Youngs Modulus for Blind Flange Matl.	Sy	176184.59	N./mm ²
Allowed Channel Cover Deflection	dfAllow	1.3750	mm
Diameter of Bolt Circle	C	1340.000	mm
Nominal Bolt Diameter	a	57.1500	mm
Type of Threads	TEMA Thread Series		
Number of Bolts	32		
Flange Face Outside Diameter	Fod	1187.000	mm
Flange Face Inside Diameter	Fid	1100.000	mm
Flange Facing Sketch	Sketch 1a		
Gasket Outside Diameter	Go	1176.000	mm
Gasket Inside Diameter	Gi	1132.000	mm
Gasket Factor	m	3.0000	
Gasket Design Seating Stress	y	68.95	N./mm ²
Gasket Thickness	tg	4.5000	mm
Raised face depth	Fd	0.0000	mm



Bolted Flange Calculation Results:

Gasket Contact Width,	$N = (Go - Gi)/2$	22.000 mm
Basic Gasket Width,	$bo = N/2$	11.000 mm
Effective Gasket Width,	$b = Cb \cdot \sqrt{bo}$	8.358 mm
Gasket Reaction Diameter,	$G = Go - 2 \cdot b$	1159.285 mm



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Flg Calc [Ext P]: New Flange

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Basic Flange and Bolt Loads:

Hydrostatic End Load due to Pressure [H or Wo,1]:

$$= \pi/4 \cdot G^2 \cdot P$$

$$= \pi/4 \cdot 1159^2 \cdot -101$$

$$= -10903.757 \text{ Kgf}$$

Contact Load on Gasket Surfaces [Hp or Wo,2]:

$$= 2 \cdot b \cdot \pi \cdot G \cdot m \cdot P$$

$$= 2 \cdot 8.36 \cdot \pi \cdot 1159 \cdot 3 \cdot -101$$

$$= -1886.604 \text{ Kgf}$$

Design Bolt Load for the Operating Condition [Wo]:

$$= \max(W_{o,1} + W_{o,2} + H'p, 0)$$

$$= \max(-10904 + -1887 + 0, 0)$$

$$= 0.000 \text{ Kgf}$$

Design Bolt Load for the Gasket Seating Condition [Wg]:

$$= y \cdot b \cdot \pi \cdot G + y_{Part} \cdot b_{Part} \cdot l_p$$

$$= 68.9 \cdot 8.36 \cdot \pi \cdot 1159 + 0 \cdot 0 \cdot 0$$

$$= 214007.469 \text{ Kgf}$$

Required Bolt Area [Am]:

$$= \max((W_o + F_a + 4Me/G) / S_b, W_g / S_a)$$

$$= \max((0 + 0 + 4 \cdot 0 / 1159) / 172, 214007 / 172)$$

$$= 122.022 \text{ cm}^2$$

Bolting Information for TEMA Imperial Thread Series (Non Mandatory):

	Minimum	Actual	Maximum
Bolt Area:	122.022	706.682	
Radial Distance between Bolts and Edge:	57.150	60.000	
Circ. Spacing between the Bolts:	120.650	131.343	...

Min. Gasket Contact Width (Brownell Young) [Not an ASME Calc] [Nmin]:

$$= A_b \cdot S_a / (y \cdot \pi (G_o + G_i))$$

$$= 707 \cdot 172 / (68.9 \cdot \pi (1176 + 1132))$$

$$= 24.312 \text{ mm [Note: Exceeds actual gasket width, 22.000]}$$

Flange Design Bolt Load, Gasket Seating [W or Wgs]:

$$= S_a \cdot A_b$$

$$= 172 \cdot 707$$

$$= 1239413.50 \text{ Kgf}$$

Moment Arm Calculations:

Distance to Gasket Load Reaction [hg]:

$$= (C - G) / 2$$

$$= (1340 - 1159) / 2$$

$$= 90.3576 \text{ mm}$$

Flange Stresses

Tangential Flange Stress, Flat Head (4.6.2.3), Operating [STo]:

$$= 1.9 \cdot W_{m1} \cdot hG \cdot B_{sc} / (t^2 \cdot G) + C \cdot Z \cdot P_{eq} \cdot G^2 / t^2$$

$$= 1.9 \cdot 0 \cdot 90.4 \cdot 1 / (167^2 \cdot 1159) + 0.3 \cdot 1 \cdot 101 \cdot 1159^2 / 167^2$$

$$= 1.46 \text{ N./mm}^2$$

Tangential Flange Stress, Flat Head (4.6.2.4), Seating [STa]:

$$= 1.9 \cdot W \cdot hG \cdot B_{sc} / (t^2 \cdot G)$$

$$= 1.9 \cdot 1239413 \cdot 90.4 \cdot 1 / (167^2 \cdot 1159)$$

$$= 64.54 \text{ N./mm}^2$$



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Flg Calc [Ext P]: New Flange

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Bolt Stress, Operating [BSO]:

= $Wm1 / Ab$

= 0/707

= 0.00 N./mm²

Bolt Stress, Seating [BSa]:

= ($Wm2 / Ab$)

= (214007/707)

= 29.70 N./mm²

Flange Stress Analysis Results: N./mm²

	Actual	Operating Allowed	Gasket Actual	Seating Allowed
Tangential Flange	1.46	148.00	64.54	148.00
Bolting	0.00	172.00	29.70	172.00

TEMA Channel Cover Deflection Calculations:

Actual deflection at center per TEMA formula 0.7158 mm

Allowed deflection at center of Cover 1.3750 mm

Reqd. Blind Flange Thickness at Center 113.283 mm

Reqd. Blind Flange Thickness at Gasket 113.283 mm

Min. Required Flange Thickness per TEMA 134.343 mm

Stress Check: OK for External Pressure -101.3 KPa.

Estimated Finished Weight of Flange at given Thk. 2205.8 kg.

Estimated Unfinished Weight of Forging at given Thk 2205.8 kg.

Per ASME VIII-2, 4.16.2.3, Perform Calculations in the New Condition:

Flange Stress Analysis Results: N./mm²

	Actual	Operating Allowed	Gasket Actual	Seating Allowed
Longitudinal Hub	100.90	222.00	158.09	222.00
Radial Flange	10.60	148.00	16.61	148.00
Tangential Flange	67.87	148.00	106.34	148.00
Maximum Average	84.39	148.00	132.22	148.00
Bolting	103.38	172.00	29.70	172.00

Minimum Req'd Thickness, ambient, no Corrosion: 191.770 mm

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Flg Calc [Ext P]: New Flange

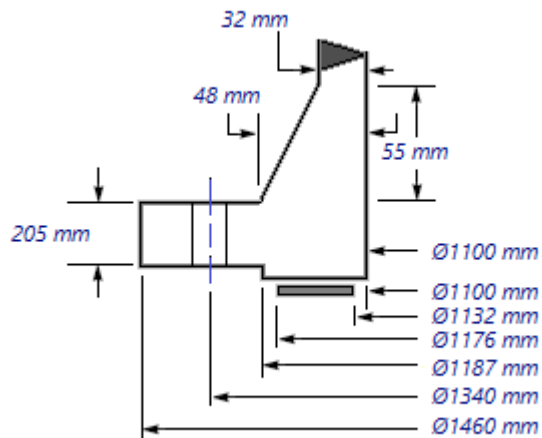
Step: 13 9:45am Sep 9,2026

Flange Input Data Values

Description: New Flange :

CH Flange

Flange Type	Integral Weld Neck		
Design Pressure	P	-101.30	KPa.
Design Temperature		200	°C
Internal Corrosion Allowance	ci	3.0000	mm
External Corrosion Allowance	ce	0.0000	mm
Use Corrosion Allowance in Thickness Calcs.		Yes	
Flange Inside Diameter	B	1100.000	mm
Flange Outside Diameter	A	1460.000	mm
Flange Thickness	t	205.0000	mm
Thickness of Hub at Small End	go	32.0000	mm
Thickness of Hub at Large End	gl	48.0000	mm
Length of Hub	h	55.0000	mm
Flange Material	SA-336 F11		
Flange Material UNS number	K11572		
Flange Allowable Stress At Temperature	Sfo	148.00	N./mm ²
Flange Allowable Stress At Ambient	Sfa	148.00	N./mm ²
Bolt Material	SA-193 B16		
Bolt Allowable Stress At Temperature	Sb	172.00	N./mm ²
Bolt Allowable Stress At Ambient	Sa	172.00	N./mm ²
Diameter of Bolt Circle	C	1340.000	mm
Nominal Bolt Diameter	a	57.1500	mm
Type of Threads	TEMA Thread Series		
Number of Bolts	32		
Flange Face Outside Diameter	Fod	1187.000	mm
Flange Face Inside Diameter	Fid	1100.000	mm
Flange Facing Sketch	Sketch 1a		
Gasket Outside Diameter	Go	1176.000	mm
Gasket Inside Diameter	Gi	1132.000	mm
Gasket Factor	m	3.0000	
Gasket Design Seating Stress	y	68.95	N./mm ²
Gasket Thickness	tg	4.5000	mm
Raised face depth	Fd	0.0000	mm





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Flg Calc [Ext P]: New Flange

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Bolted Flange Calculation Results:

Corroded Flange Thickness, $t_c = T - c_i$	202.000 mm
Corroded Flange ID, $B_{cor} = B + 2 \cdot c_i$	1106.000 mm
Corroded Large Hub, $g1_{cor} = g1 - c_i$	45.000 mm
Corroded Small Hub, $g0_{cor} = g0 - c_i$	29.000 mm
Code R Dimension, $R = ((C - B_{cor})/2) - g1_{cor}$	72.000 mm
Gasket Contact Width, $N = (G_o - G_i)/2$	22.000 mm
Basic Gasket Width, $b_o = N/2$	11.000 mm
Effective Gasket Width, $b = C_b \cdot \sqrt{b_o}$	8.358 mm
Gasket Reaction Diameter, $G = G_o - 2 \cdot b$	1159.285 mm

Basic Flange and Bolt Loads:

Hydrostatic End Load due to Pressure [H or Wo,1]:

$$\begin{aligned}
 &= \pi/4 \cdot G^2 \cdot P \\
 &= \pi/4 \cdot 1159^2 \cdot -101 \\
 &= -10903.757 \text{ Kg}
 \end{aligned}$$

Contact Load on Gasket Surfaces [Hp or Wo,2]:

$$\begin{aligned}
 &= 2 \cdot b \cdot \pi \cdot G \cdot m \cdot P \\
 &= 2 \cdot 8.36 \cdot \pi \cdot 1159 \cdot 3 \cdot -101 \\
 &= -1886.604 \text{ Kg}
 \end{aligned}$$

Hydrostatic End Load at Flange ID [Hd]:

$$\begin{aligned}
 &= \pi/4 \cdot B_{cor}^2 \cdot P \\
 &= \pi/4 \cdot 1106^2 \cdot -101 \\
 &= -9924.443 \text{ Kg}
 \end{aligned}$$

Pressure Force on Flange Face [Ht]:

$$\begin{aligned}
 &= H - H_d \\
 &= -10904 - -9924 \\
 &= -979.314 \text{ Kg}
 \end{aligned}$$

Design Bolt Load for the Operating Condition [Wo]:

$$\begin{aligned}
 &= \max(W_{o,1} + W_{o,2} + H'p, 0) \\
 &= \max(-10904 + -1887 + 0, 0) \\
 &= 0.000 \text{ Kg}
 \end{aligned}$$

Design Bolt Load for the Gasket Seating Condition [Wg]:

$$\begin{aligned}
 &= y \cdot b \cdot \pi \cdot G + y_{Part} \cdot b_{Part} \cdot l_p \\
 &= 68.9 \cdot 8.36 \cdot \pi \cdot 1159 + 0 \cdot 0 \cdot 0 \\
 &= 214007.469 \text{ Kg}
 \end{aligned}$$

Required Bolt Area [Am]:

$$\begin{aligned}
 &= \max((W_o + F_a + 4Me/G) / S_b, W_g / S_a) \\
 &= \max((0 + 0 + 4 \cdot 0 / 1159) / 172, 214007 / 172) \\
 &= 122.022 \text{ cm}^2
 \end{aligned}$$

ASME Maximum Circumferential Spacing between Bolts, Table 4.16.11 [Bsmax]:

$$\begin{aligned}
 &= 2a + 6t / (m + 0.5) \\
 &= 2 \cdot 57.1 + 6 \cdot 202 / (3 + 0.5) \\
 &= 460.586 \text{ mm}
 \end{aligned}$$

Actual Circumferential Bolt Spacing [Bs]:

$$\begin{aligned}
 &= C \cdot \sin(\pi / n) \\
 &= 1340 \cdot \sin(\pi / 32) \\
 &= 131.343 \text{ mm}
 \end{aligned}$$

ASME Moment Multiplier for Bolt Spacing per Table 4.16.11 [Bsc]:

$$\begin{aligned}
 &= \max(\sqrt{ B_s / (2a + t) }, 1) \\
 &= \max(\sqrt{ 131 / (2 \cdot 57.1 + 202) }, 1) \\
 &= 1.0000
 \end{aligned}$$



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Flg Calc [Ext P]: New Flange

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Bolting Information for TEMA Imperial Thread Series (Non Mandatory):

	Minimum	Actual	Maximum
Bolt Area:	122.022	706.682	
Radial Distance between Hub and Bolts:	69.850	72.000	
Radial Distance between Bolts and Edge:	57.150	60.000	
Circ. Spacing between the Bolts:	120.650	131.343	460.586

Min. Gasket Contact Width (Brownell Young) [Not an ASME Calc] [Nmin]:

$$= A_b \cdot S_a / (y \cdot \pi(G_o + G_i))$$

$$= 707 \cdot 172 / (68.9 \cdot \pi(1176 + 1132))$$

$$= 24.312 \text{ mm [Note: Exceeds actual gasket width, 22.000]}$$

Recommended Minimum Gasket Contact Width per Table 4.16.2

Sheet Gaskets including laminated sheet gaskets with or without a metal core: 19 mm

Preformed comp. gaskets incl. spiral wound, jacketed, & solid flat metal gaskets: 16 mm

Flange Design Bolt Load, Gasket Seating [W or Wgs]:

$$= S_a \cdot A_b$$

$$= 172 \cdot 707$$

$$= 1239413.50 \text{ Kg}$$

Moment Arm Calculations:

Distance to Gasket Load Reaction [hg]:

$$= (C - G) / 2$$

$$= (1340 - 1159) / 2$$

$$= 90.3576 \text{ mm}$$

Distance to Face Pressure Reaction [ht]:

$$= 0.5 \cdot (C - B) / 2 + h_g$$

$$= 0.5 \cdot (1340 - 1106) / 2 + 90.4$$

$$= 103.6788 \text{ mm}$$

Distance to End Pressure Reaction [hd]:

$$= (C - B - g_1) / 2$$

$$= (1340 - 1106 - 45) / 2$$

$$= 94.5000 \text{ mm}$$

Summary of Moments for External Pressure: (Kg-m.)

Loading	Force	Distance	Bolt Corr	Moment
End Pressure, Md	-9924.	4.1424	1.0000	-41.
Face Pressure, Mt	-979.	13.3212	1.0000	-13.
Gasket Seating, Matm	1239414.	90.3576	1.0000	111993.

Total Moment for Operation, Mo -54. Kg-m.

Total Moment for Gasket seating, Mg 111993. Kg-m.

Effective Hub Length, ho = sqrt(Bcor*goCor) 179.092 mm

Hub Ratio, h/h0 = HL / H0 0.307

Thickness Ratio, g1/g0 = (g1Cor/goCor) 1.552

Flange Factors for Integral Flange:

Factor F per 4.16.5 0.878

Factor V per 4.16.5 0.358

Factor f per 4.16.5 1.231

Stress factor from Table 4.16.4 [K]:

$$= A / B$$

$$= 1460 / 1106$$

$$= 1.320$$



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Flg Calc [Ext P]: New Flange

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Stress factor from Table 4.16.4 [Y]:

$$= (0.66845 + (5.7169(K^2 \cdot \log_{10}(K))) / (K^2 - 1))(K - 1)$$

$$= (0.66845 + (5.7169(1.32^2 \cdot \log_{10}(1.32)) / (1.32^2 - 1))(1.32 - 1)$$

$$= 7.143$$

Stress factor from Table 4.16.4 [T]:

$$= (K^2(1 + 8.55246 \cdot \log_{10}(K)) - 1) / ((1.0472 + 1.9448 \cdot K^2) \cdot (K - 1))$$

$$= (1.32^2(1 + 8.55246 \cdot \log_{10}(1.32)) - 1) / ((1.0472 + 1.9448 \cdot 1.32^2) \cdot (1.32 - 1))$$

$$= 1.789$$

Stress factor from Table 4.16.4 [U]:

$$= (K^2(1 + 8.55246 \cdot \log_{10}(K)) - 1) / (1.36136(K^2 - 1) \cdot (K - 1))$$

$$= (1.32^2(1 + 8.55246 \cdot \log_{10}(1.32)) - 1) / (1.36136(1.32^2 - 1) \cdot (1.32 - 1))$$

$$= 7.850$$

Stress factor from Table 4.16.4 [Z]:

$$= (K^2 + 1) / (K^2 - 1)$$

$$= (1.32^2 + 1) / (1.32^2 - 1)$$

$$= 3.693$$

Stress factor from Table 4.16.4 for integral flanges [d]:

$$= U \cdot g_0^2 \cdot h_0 / V$$

$$= 7.85 \cdot 29^2 \cdot 179 / 0.36$$

$$= 3300758.500 \text{ mm}^3$$

Stress factor from Table 4.16.4 for integral flanges [e]:

$$= F / h_0$$

$$= 0.88 / 179$$

$$= 0.005 \text{ 1/mm}$$

Stress factor from Table 4.16.4 [L]:

$$= (t \cdot e + 1) / T + t^3/d$$

$$= (202 \cdot 0.0049 + 1) / 1.79 + 202^3 / 3300758$$

$$= 3.610$$

Flange Stresses

Longitudinal Hub Stress, Operating [SHo]:

$$= (f \cdot Mop / Bcor) / (L \cdot g_1^2)$$

$$= (1.23 \cdot 54.2 / 1106) / (3.61 \cdot 45^2)$$

$$= 0.08 \text{ N./mm}^2$$

Longitudinal Hub Stress, Seating [SHa]:

$$= (f \cdot Matm / Bcor) / (L \cdot g_1^2)$$

$$= (1.23 \cdot 111993 / 1106) / (3.61 \cdot 45^2)$$

$$= 167.24 \text{ N./mm}^2$$

Radial Flange Stress, Operating [SRo]:

$$= (\beta \cdot Mop / Bcor) / (L \cdot t^2)$$

$$= (2.32 \cdot 54.2 / 1106) / (3.61 \cdot 202^2)$$

$$= 0.01 \text{ N./mm}^2$$

Radial Flange Stress, Seating [SRa]:

$$= (\beta \cdot Matm / Bcor) / (L \cdot t^2)$$

$$= (2.32 \cdot 111993 / 1106) / (3.61 \cdot 202^2)$$

$$= 15.65 \text{ N./mm}^2$$

Tangential Flange Stress, Operating [STo]:

$$= (Y \cdot Mo / (t^2 \cdot Bcor)) - Z \cdot SRo$$

$$= (7.14 \cdot 54.2 / (202^2 \cdot 1106)) + 3.69 \cdot 0.0076$$

$$= 0.06 \text{ N./mm}^2$$

Tangential Flange Stress, Seating [STa]:

$$= (y \cdot Matm / (t^2 \cdot Bcor)) - Z \cdot SRa$$



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Flg Calc [Ext P]: New Flange

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$$= (7.14 \cdot 111993 / (202^2 \cdot 1106)) - 3.69 \cdot 15.6$$

$$= 116.05 \text{ N./mm}^2$$

Average Flange Stress, Operating [SAo]:

$$= (S_{Ho} + \max(S_{Ro}, S_{To})) / 2$$

$$= (0.081 + \max(0.0076, 0.056)) / 2$$

$$= 0.07 \text{ N./mm}^2$$

Average Flange Stress, Seating [SAa]:

$$= (S_{Ha} + \max(S_{Ra}, S_{Ta})) / 2$$

$$= (167 + \max(15.6, 116)) / 2$$

$$= 141.65 \text{ N./mm}^2$$

Bolt Stress, Operating [BSo]:

$$= W_{m1} / A_b$$

$$= 0 / 707$$

$$= 0.00 \text{ N./mm}^2$$

Bolt Stress, Seating [BSa]:

$$= (W_{m2} / A_b)$$

$$= (214007 / 707)$$

$$= 29.70 \text{ N./mm}^2$$

Flange Stress Analysis Results: N./mm²

	Actual	Operating Allowed	Gasket Actual	Seating Allowed
Longitudinal Hub	0.08	222.00	167.24	222.00
Radial Flange	0.01	148.00	15.65	148.00
Tangential Flange	0.06	148.00	116.05	148.00
Maximum Average	0.07	148.00	141.65	148.00
Bolting	0.00	172.00	29.70	172.00

Minimum Required Flange Thickness 200.101 mm
 Stress Check: OK for External Pressure -101.3 KPa.
 Estimated Finished Weight of Flange at given Thk. 1211.1 kg.
 Estimated Unfinished Weight of Forging at given Thk 1458.6 kg.

Flange rigidity based on required thickness [ASME]:

Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:

$$= 52.14 \cdot M_a / B_{sc} \cdot C_{nvfac} \cdot V / (\lambda \cdot E_{amb} \cdot g_o^2 \cdot h_o \cdot K_i)$$

$$= 52.14 \cdot 111993 / 1 \cdot 9807 \cdot 0.36 / (3.42 \cdot 204092 \cdot 29^2 \cdot 179 \cdot 0.3)$$

$$= 0.650 \text{ (should be } \leq 1)$$

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:

$$= 52.14 \cdot M_o / B_{sc} \cdot C_{nvfac} \cdot V / (\lambda \cdot E_{op} \cdot g_o^2 \cdot h_o \cdot K_i)$$

$$= 52.14 \cdot 54.2 / 1 \cdot 9807 \cdot 0.36 / (3.42 \cdot 193337 \cdot 29^2 \cdot 179 \cdot 0.3)$$

$$= 0.000 \text{ (should be } \leq 1)$$

Flange rigidity based on given thickness [ASME]:

Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:

$$= 52.14 \cdot M_a / B_{sc} \cdot C_{nvfac} \cdot V / (\lambda \cdot E_{amb} \cdot g_o^2 \cdot h_o \cdot K_i)$$

$$= 52.14 \cdot 111993 / 1 \cdot 9807 \cdot 0.36 / (3.61 \cdot 204092 \cdot 29^2 \cdot 179 \cdot 0.3)$$

$$= 0.616 \text{ (should be } \leq 1)$$

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:

$$= 52.14 \cdot M_o / B_{sc} \cdot C_{nvfac} \cdot V / (\lambda \cdot E_{op} \cdot g_o^2 \cdot h_o \cdot K_i)$$

$$= 52.14 \cdot 54.2 / 1 \cdot 9807 \cdot 0.36 / (3.61 \cdot 193337 \cdot 29^2 \cdot 179 \cdot 0.3)$$

$$= 0.000 \text{ (should be } \leq 1)$$

Per ASME VIII-2, 4.16.2.3, Perform Calculations in the New Condition:



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Flg Calc [Ext P]: New Flange

Step: 13

9:45am

Sep 9, 2026

Flange Stress Analysis Results: N./mm²

	Actual	Operating Allowed	Gasket Actual	Seating Allowed
Longitudinal Hub	100.90	222.00	158.09	222.00
Radial Flange	10.60	148.00	16.61	148.00
Tangential Flange	67.87	148.00	106.34	148.00
Maximum Average	84.39	148.00	132.22	148.00
Bolting	103.38	172.00	29.70	172.00

Minimum Req'd Thickness, ambient, no Corrosion: 191.770 mm

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Element and Detail Weights:

Step: 14

9:45am

Sep 9, 2026

Element and Detail Weights:

From	To	Element Metal Wgt. kg.	Element ID Volume cm ³	Corroded Metal Wgt. kg.	Corroded ID Volume cm ³	Extra due Misc % kg.
10	20	2309.65	...	2309.65	...	230.965
20	30	1256	256949	1234.4	257521	125.6
30	40	1058.42	1140602	961.734	1153079	105.842
40	50	2645.99	4381930	2210.91	4438076	264.599
50	60	180.152	221784	141.548	225171	18.0152
Total		7450	6001265.50	6858	6073847.50	745



For elements specified as shell side elements, the volume(s) shown above for those elements, reflects the displacement of the tubes.

Weight of Details:

From	Type	Weight of Detail kg.	X Offset, Dtl. Cent. mm	Y Offset, Dtl. Cent. mm	Z Offset, Dtl. Cent. mm	Description
10	Ins1	121.941	76	...	...	Ins: 10
20	Ins1	58.4705	91	...	...	Ins: 20
30	Liqd	1139.91	600	...	...	Liquid: 20
30	Ins1	261.86	600	...	...	Ins: 40
30	Noz1	30.1719	400	592.17	...	N3- Tube inlet
30	Noz1	22.3943	500	-581.115	...	Tube outlet
30	Wght	260	500	...	...	Pass Partition
40	Sad1	223.427	1050	758.5	...	Left Saddle
40	Sad1	223.427	4150	758.5	...	Right saddle
40	Liqd	4379.26	2700	...	...	Liquid: 40
40	Ins1	738.946	2700	...	...	Ins: 40
40	Noz1	238.998	4800	-804	...	Shell inlet-N1
40	Noz1	238.998	1000	777.813	...	Shell Inlet
40	Noz1	8.66791	2000	575.4	...	Vent
40	Noz1	8.66791	1500	-575.4	...	Drain
40	Wght	250	1050	...	-1000	Fire proofing
40	Wght	250	4150	...	-1000	Fire proofing
50	Liqd	221.648	141.667	...	...	Liquid: 50
50	Ins1	99.9614	156.25	...	...	Ins: 40
30	FTsh	1336.93	1285	...	...	Tubesheet
30	Tube	2479.66	3800	...	...	Tubes
40	Baf1	170.683	1950	...	...	Baffles
40	Baf1	171.108	3900	...	...	Baffles

Total Weight of Each Detail Type:

Saddles	446.9
Liquid	5740.8
Insulation	1281.2
Nozzles	547.9
Weights	760.0
Exchanger Components	4158.4
Liquid in Tubes	467.8

Sum of the Detail Weights 13402.9 kg.

Weight Summation Results: (kg.)



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Element and Detail Weights:

Step: 14

9:45am

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	Fabricated	Shop Test	Shipping	Erected	Empty	Operating
Main Elements	8195.2	8195.2	8195.2	8195.2	8195.2	8195.2
Saddles	446.9	446.9	446.9	446.9	446.9	446.9
Nozzles	547.9	547.9	547.9	547.9	547.9	547.9
Wld Weights	260.0	260.0	260.0	260.0	260.0	260.0
Exchanger	4158.4	4158.4	4158.4	4158.4	4158.4	4158.4
Insulation	...	...	...	...	1281.2	1281.2
Empty Weights	...	...	...	500.0	500.0	...
Ope Weights	...	...	...	...	...	500.0
Ope. Liquid	...	...	...	...	...	5740.8
Tube Ope Lqd	...	...	...	...	...	467.8
Test Liquid	...	5997.6	...	...	...	...
Tube Tst Lqd	...	467.8	...	...	...	...
Totals	13608.4	20073.8	13608.4	14108.4	15389.5	21598.2

Field Installation Options:

* Insulation installed after lifting.

Miscellaneous Weight Percent: 10.0 %

Note that the above value for the miscellaneous weight percent has been applied to the shells/heads/flange/tubesheets/tubes etc. in the weight calculations for metallic components.

Weight Summary:

Fabricated Wt.	- Bare weight without removable internals	13608.4 kg.
Shop Test Wt.	- Fabricated weight + water (full)	20073.8 kg.
Shipping Wt.	- Fab. weight + removable intls.+ shipping app.	13608.4 kg.
Erected Wt.	- Fab. wt + or - loose items (trays,platforms etc.)	14108.4 kg.
Ope. Wt. no Liq	- Fab. weight + internals. + details + weights	15389.5 kg.
Operating Wt.	- Empty weight + operating liq. uncorroded	21598.2 kg.
Oper. Wt. + CA	- Corr wt. + operating liquid	20947.0 kg.
Field Test Wt.	- Empty weight + test medium (full)	21355.0 kg.

Exchanger Tube Data

Volume of Exchanger tubes :	468092.8 cm ³
Weight of Ope Liq in tubes :	467.8 kg.
Weight of Water in tubes :	467.8 kg.

Surface Areas of Elements:

From	To	Outside Surface Area cm ²	Inside Surface Area cm ²
10	20	24905.9	...
20	30	19074.5	9261.42
30	40	43881.8	41469
40	50	192718	186611
50	60	15564.3	14844.1

Total	296144.4	252185.1 cm ²
Total	318.8	271.4 ft ²



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Nozzle Flange MAWP:

Step: 15

9:45am

Sep 9, 2026

Nozzle Flange MAWP Results: (KPa. & °C)

Nozzle Description	Flange Rating		Des. Temp	Class	Grade/Group	Equiv. Press	Max Pressure		
	Ope.	Ambient					UG-44	50%	DNV
Shell Inlet	1115.0	1960.0	275	150	GR 1.1	1657.6	795.5	1115.0	14.9
Vent	1115.0	1960.0	275	150	GR 1.1	...	...	...	...
Drain	1115.0	1960.0	275	150	GR 1.1	...	...	...	...

Shellside Flange Rating

Lowest Flange Pressure Rating was (Ope)[ShellSide]: 1115.044 KPa.

Lowest Flange Pressure Rating was (Amb)[ShellSide]: 1960.006 KPa.

Channelside Flange Rating**Selected Method for Derating ASME B16.5 Flange MAWP: None Selected**

It appears that the ASME UG-44(b) method of pressure de-rating was not selected. If external loads (forces and bending moments) are also required to be evaluated on standard flanged joints with welding neck flanges as mentioned in UG-44(a)(2), (a)(9) and (a)(10) for vessels constructed per the ASME VIII-1 2019 Code Edition onward, they may be evaluated per the requirements specified in UG-44(b).

You can set the year of construction in Tools->Enter U1 Form Information.

Flange Pressure Ratings are per ASME B16.5 2025 Metric Edition

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Wind Load Calculation:

Step: 16 9:45am Sep 9,2026

Wind Load Calculation Results:

Input Values:

Wind Design Code	ASCE/SEI 7-16
Wind Load Reduction Scale Factor	1.100
Basic Wind Speed [V]	97.2 Km/hr
Surface Roughness Category	C: Open Terrain
Type of Surface	Moderately Smooth
Base Elevation	300 mm
Percent Wind for Hydrotest	50.0
Using User defined Wind Press. Vs Elev.	N
Height of Hill or Escarpment H or Hh	0 mm
Distance Upwind of Crest Lh	0 mm
Distance from Crest to the Vessel x	0 mm
Type of Terrain (Hill, Escarpment)	Flat
Damping Factor (Beta) for Wind (Ope)	0.0100
Damping Factor (Beta) for Wind (Empty)	0.0000
Damping Factor (Beta) for Wind (Filled)	0.0000
Site Elevation above Sea Level	0 mm

Wind Analysis Results

The importance factor is not used in the calculation of wind pressure vs. elevation for the selected wind code.

Static Gust-Effect Factor, Operating Case [G]:

$$= \min(0.85, 0.925((1 + 1.7 \cdot gQ \cdot I_{zbar} \cdot Q) / (1 + 1.7 \cdot gV \cdot I_{zbar})))$$

$$= \min(0.85, 0.925((1 + 1.7 \cdot 3.4 \cdot 0.23 \cdot 0.98) / (1 + 1.7 \cdot 3.4 \cdot 0.23)))$$

$$= \min(0.85, 0.91)$$

$$= 0.850$$

Natural Frequency of Vessel (Operating)	33.000 Hz
Natural Frequency of Vessel (Empty)	33.000 Hz
Natural Frequency of Vessel (Test)	33.000 Hz

User Entered Importance Factor is	1.000
Force Coefficient [Cf]	0.580
Structure Height to Diameter ratio	5.802

This is classified as a rigid structure. Static analysis performed.

Sample Calculation for the First Element:

The ASCE code performs calculations in Imperial Units. The wind pressure is therefore computed in these units.

Value of [α] and [Zg]:

Exposure Category: C from Table 26.11-1
 $\alpha = 9.5 : Z_g = 274320 \text{ mm}$

Effective Height [z]:

$$= \text{Centroid Height} + \text{Vessel Base Elevation}$$

$$= 1000 + 300 = 1300 \text{ mm}$$

$$= 4.27 \text{ ft. Imperial Units}$$

Velocity Pressure coefficient evaluated at height z [Kz]:

$$\text{Because } z (4.27 \text{ ft.}) < 15 \text{ ft.}$$

$$= 2.01(15 / Z_g)^{2/\alpha}$$

$$= 2.01(15/900)^{2/9.5}$$

$$= 0.85$$

Type of Hill: No Hill



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Wind Load Calculation:

Step: 16

9:45am

Sep 9, 2026

Wind Directionality Factor [Kd]:

= 1, per Table 26.6-1

As there is No Hill Present: [Kzt]:

K1 = 0, K2 = 0, K3 = 0

Topographical Factor [Kzt]:

= (1 + K1 • K2 • K3)²

= (1 + 0 • 0 • 0)²

= 1

Velocity Pressure evaluated at height z, Imperial Units [qz]:

= max(16, 0.00256 • Kz • Kzt • Kd • Ke • V(mph)²)

= max(16, 0.00256 • 0.85 • 1 • 1 • 1 • 60.4²)

= 16 psf [78.1] Kgs/m²

Force on the first element [F]:

= qz • G • Cf • WindArea

= 16 • 0.85 • 0.58 • 3.79

= 29.9 lbs. [13.6] Kgf

Element	Hgt (z) mm	K1	K2	K3	Kz	Kzt	qz Kgs/m ²
CH COVER	1300.0	0.000	0.000	0.000	0.849	1.000	78.120
CH Flange	1300.0	0.000	0.000	0.000	0.849	1.000	78.120
Channel shell	1300.0	0.000	0.000	0.000	0.849	1.000	78.120
Shell	1300.0	0.000	0.000	0.000	0.849	1.000	78.120
Shell head	1300.0	0.000	0.000	0.000	0.849	1.000	78.120

Wind Loads on Masses/Equipment/Piping

ID	Wind Area cm ²	Elevation mm	Pressure Kgs/m ²	Force Kgf
Pass Partitio	0.00	1300.00	78.12	0.00
Fire proofing	0.00	1300.00	78.12	0.00
Fire proofing	0.00	1300.00	78.12	0.00

Wind Load Calculation:

From	To	Wind Height mm	Wind Diameter mm	Wind Area cm ²	Wind Pressure Kgs/m ²	Element Wind Load Kgf
10	20	1300	2073.88	3525.6	78.12	14.9367
20	30	1300	1572	4087.2	78.12	17.316
30	40	1300	1756.8	21081.6	78.12	89.3152
40	50	1300	1603.2	86572.8	78.12	366.778
50	60	1300	1581.6	4765.79	78.12	20.191

The Wind Loads calculated and printed in the Wind Load calculation report have been factored by the input scalar/load reduction factor of: 1.100.

Be sure the wind speed is in accordance with the specified wind design code.

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Earthquake Load Calculation: Step: 17 9:45am Sep 9,2026

Earthquake Load Calculation Results:

Earthquake Load Calculation:

Input Values:

Seismic Design Code		ASCE/SEI 7-16
Seismic Load Reduction Scale Factor		1.000
Importance Factor		1.250
Table Value Fa		1.600
Table Value Fv		2.400
Max. Mapped Res. Acceleration	[Ss]	0.147
Max. Eff. Ground Acceleration	[S]	0.065
Force Modification Factor R		3.000
Site Class		D
Component Elevation Ratio	z/h	0.000
Amplification Factor	Ap	1.000
Force Factor		0.000
Consider Vertical Acceleration		Yes
Minimum Acceleration Multiplier		0.000
User Value of Sds (used if > 0)		0.000
User Value of Sd1 (used if > 0)		0.000
Moment Reduction Factor Tau		1.000

Seismic Analysis Results:

$$\begin{aligned} SMS &= Fa * Ss = 1.6 * 0.15 = 0.24 \\ SM1 &= Fv * S1 = 2.4 * 0.065 = 0.16 \\ SDS &= 2/3 * Sms = 2/3 * 0.24 = 0.16 \\ SD1 &= 2/3 * Sm1 = 2/3 * 0.16 = 0.1 \end{aligned}$$

Check Approximate Fundamental Period from 12.8-7 [Ta]:
 $= Ct \cdot h_n^x$ where $Ct = 0.020$, $x = 0.75$ and h_n = Structural Height (ft.)
 $= 0.020(5.680.75)$
 $= 0.074$ seconds

The Coefficient C_u from Table 12.8-1 is : 1.700

Fundamental Period (1/Frequency) [T]:
 $= (1/\text{Natural Frequency}) = (1/33)$
 $= 0.030$

Check the Value of T which is the smaller of $C_u \cdot T_a$ and T:
 $= \text{Minimum Value of } (1.7 \cdot 0.074, 0.03) \text{ per 12.8.2}$
 $= 0.030$
 As the time period is < 0.06 second, use section 15.4.2.

Base Shear per equation 15.4-5, [V]:
 $= 0.3 \cdot Sds \cdot W \cdot I$
 $= 0.3 \cdot 0.16 \cdot 21151 \cdot 1.25$
 $= 1243.697$ Kg

Vertical load per 12.4-4, [Ev]:
 $= 0.2 \cdot Sds \cdot W$
 $= 0.2 \cdot 0.16 \cdot 21151$
 $= 663.3$ Kg

Earthquake Load Calculation:

From	To	Earthquake Height mm	Earthquake Weight Kgf	Element Ope Load Kgf
10	20	730	3021.61	225.285



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Earthquake Load Calculation:

Step: 17

9:45am

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20	30	550	3021.61	169.735
30	40	550	3021.61	169.735
40	Sad1	550	3021.61	169.735
Sad1	50	550	3021.61	169.735
40	50	550	3021.61	169.735
50	60	550	3021.61	169.735

The Earthquake Weight does not include the weight of the saddles.

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Center of Gravity Calculation: Step: 18 9:45am Sep 9,2026

Shop/Field Installation Options :

Insulation is installed in the Field after being lifted.

Note : The CG is computed from the first Element From Node

Center of Gravity of the Saddles	4409.000 mm
Center of Gravity of the Liquid	3929.704 mm
Center of Gravity of the Insulation	3407.234 mm
Center of Gravity of the Nozzles	4305.406 mm
Center of Gravity of the Added Weights (Operating)	3221.895 mm
Center of Gravity of the Added Weights (Empty)	3221.895 mm
Center of Gravity of the Tubesheet(s)	1724.000 mm
Center of Gravity of the Tubes	4239.000 mm
Center of Gravity of the Baffles	4735.210 mm
Center of Gravity of Bare Shell New and Cold	2005.075 mm
Center of Gravity of Bare Shell Corroded	1834.964 mm
Vessel CG in the Operating Condition	3053.586 mm
Vessel CG in the Fabricated (Shop/Empty) Condition	2668.233 mm
Vessel CG in the Test Condition	3026.725 mm

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Saddle Calcs: Road Transport:

Step: 19

9:45am

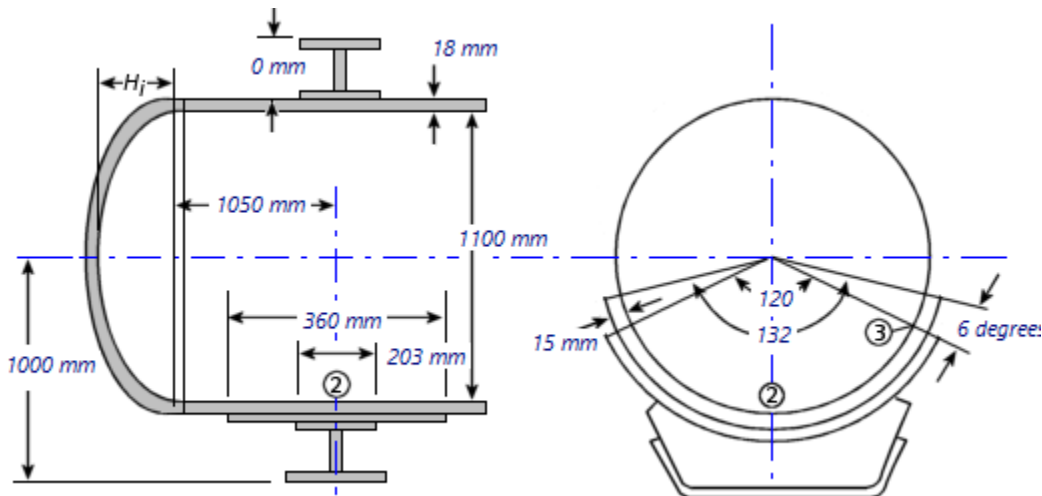
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ASME VIII Division 2 Horizontal Vessel Analysis, Left Saddle:

Horizontal Vessel Stress Calculations : Road Transport

Input and Calculated Values:

Vessel Mean Radius	Rm	559.00	mm
Shell Thickness used in this Case	t	18.000	mm
Stiffened Vessel Length per 4.15.6	L	5450.00	mm
Distance from Saddle to Vessel tangent	a1 or a	1050.00	mm
Saddle Width	b1 or b	203.00	mm
Saddle Bearing Angle	delta or theta	120.00	degrees
Wear Plate Width	b2 or b1	360.00	mm
Wear Plate Bearing Angle	delta2 or theta1	132.00	degrees
Wear Plate Thickness	e2 or tr	15.0	mm
Wear Plate Allowable Stress	fw or Sr	138.00	N./mm ²
Shell Allowable Stress used in Calculation		138.00	N./mm ²
Head Allowable Stress used in Calculation		147.00	N./mm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	1000.00	mm
Coefficient of Friction	mu	0.00	



Results for Horizontal Vessel Analysis:

Saddle Force Q for this Transportation Case		16388.98	Kgf
Shipping Longitudinal Acceleration	gx	0.10	g
Shipping Transverse Acceleration	gz	0.10	g
Shipping Vertical Acceleration	gy	0.05	g
Vertical Acceleration Acting with Transverse	gw	1.00	g
Internal Pressure during Transport		0.00	KPa.
Transportation Wind Load Scalar Factor	wlsf	1.00	
Transportation Wind Velocity	wv	92.7	Km/hr



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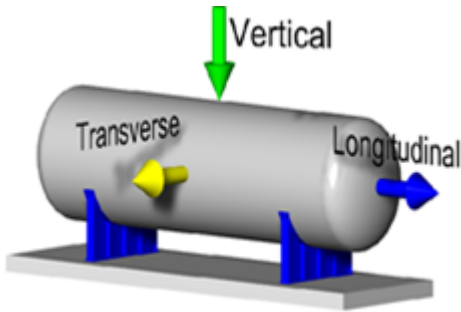
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Saddle Calcs: Road Transport:

Step: 19

9:45am

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Horizontal Vessel Analysis Results:	Actual N./mm ²	Allowable N./mm ²
Long. Stress at Top of Midspan	-3.10	-234.00
Long. Stress at Top of Midspan	3.10	138.00
Long. Stress at Bottom of Midspan	3.10	138.00
Long. Stress at Top of Saddles	14.81	138.00
Long. Stress at Bottom of Saddles	-8.21	-234.00
Long. Stress at Bottom of Saddles	8.21	138.00
Tangential Shear in Shell	11.49	110.40
Circ. Stress at Horn of Saddle	15.09	207.00
Circ. Compressive Stress in Shell	1.03	138.00

Intermediate Results: Saddle Reaction Q due to Road Transport Load:

Transverse Force due to Wind during Transport [F_{tw}]:

$$\begin{aligned}
 &= \text{Transverse Area} \cdot 0.00256 \cdot wv^2 \cdot wlsf \\
 &= 129 \cdot 0.00256 \cdot 57.6^2 \cdot 1 \\
 &= 1097.5 \text{ lbf} \\
 &= 497.816 \text{ Kg}
 \end{aligned}$$

Longitudinal Force due to Wind during Transport [F_{lw}]:

$$\begin{aligned}
 &= \pi \cdot D^2/4 \cdot 0.00256 \cdot wv^2 \cdot wlsf \cdot \text{Wind Dia. Mult} \\
 &= \pi \cdot 4.71^2/4 \cdot 0.00256 \cdot 57.6^2 \cdot 1 \cdot 1.2 \\
 &= 177.7 \text{ lbf} \\
 &= 80.6 \text{ Kg}
 \end{aligned}$$

Transverse Force due to Lateral Acceleration [F_t]:

$$\begin{aligned}
 &= g_z \cdot \text{Shipping Weight} \cdot g_w + F_{tw} \\
 &= 0.1 \cdot 13162 \cdot 1 + 498 \\
 &= 1814.0 \text{ Kg}
 \end{aligned}$$

Longitudinal Force due to Axial Acceleration [F_l]:

$$\begin{aligned}
 &= g_x \cdot \text{Shipping Weight} \cdot g_w + F_{lw} \\
 &= 0.1 \cdot 13162 \cdot 1 + 80.6 \\
 &= 1396.8 \text{ Kg}
 \end{aligned}$$

Saddle Reaction Force due to Transverse Load [F_{wt}]:

$$\begin{aligned}
 &= F_t / \text{Num of Saddles} \cdot B / E \\
 &= 1814/2 \cdot 1000/1010 \\
 &= 898.2 \text{ Kg}
 \end{aligned}$$

Saddle Reaction Force due to Longitudinal Load [F_{wl}]:

$$\begin{aligned}
 &= F_l \cdot B / L_s \\
 &= 1397 \cdot 1000/3100 \\
 &= 450.6 \text{ Kg}
 \end{aligned}$$

Saddle Reaction Force due to Vertical Transportation Acceleration [F_{rv}]:

$$\begin{aligned}
 &= \text{Saddle Load} \cdot g_y \\
 &= 14753 \cdot 0.05
 \end{aligned}$$



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Saddle Calcs: Road Transport: Step: 19 9:45am Sep 9, 2026

= 737.656 Kgf

Load Combination Results due to Transport Loading [Q]:

= Saddle Load + Frv + max(Fwl, Fwt)

= 14753 + 738 + max(451, 898)

= 16389.0 Kgf

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight) 16612.40 Kgf

Transverse Shear Load 906.98 Kgf

Longitudinal Shear Load 1396.75 Kgf

Note: Wear Plate is Welded to the Shell, k = 0.1

Saddle Dimension [E]:

= min(2(ShellID/2 + t + WearPadThickness)sin(θ/2), 2•Rm)

= min(2(1100/2 + 18 + 15)sin(120/2), 2•559)

= 1009.786 mm

The Computed K values from Table 4.15.1:

K1 = 0.1066 K2 = 1.1707 K3 = 0.8799 K4 = 0.4011

K5 = 0.7603 K6 = 0.0529 K7 = 0.0529 K8 = 0.3405

K9 = 0.2711 K10 = 0.0581 K1* = 0.1923 K6p = 0.0434

K7p = 0.0434

The suffix 'p' denotes the values for a wear plate if it exists.

Note: Dimension a is greater than or equal to Rm/2.

Moment per Equation 4.15.1 [M1]:

= -Q•a [1 - (1 - a/L + (Rm²-hm²)/(2a•L))/(1+(4hm)/3L)]

= -16389•1050[1-(1-1050/5450+(559²-0²)/

(2•1050•5450))/(1+(4•0)/(3•5450))]

= -2845.6 Kg-m.

Moment per Equation 4.15.2 [M2]:

= Q•L/4((1+2(Rm²-hm²)/(L²))/(1+(4hm²)/(3L))-4a/L)

= 16389•5450/4((1+2(559²-0²)/(5450²))/(1+(4•0)/

(3•5450))-4•1050/5450)

= 5591.5 Kg-m.

Longitudinal Stress at Top of Shell (4.15.4) [σ1]:

= P • Rm/(2t) - M2/(π•Rm²t)

= 0 • 559/(2•18) - 5592/(π•559²•18)

= -3.10 N./mm²

Longitudinal Stress at Bottom of Shell (4.15.5) [σ2]:

= P • Rm/(2t) + M2/(π • Rm² • t)

= 0 • 559/(2 • 18) + 5592/(π • 559² • 18)

= 3.10 N./mm²

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ*3]:

= P • Rm/(2t) - M1/(K1•π•Rm²t)

= 0•559/(2•18)-2846/(0.11•π•559²•18)

= 14.81 N./mm²

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ*4]:

= P • Rm/(2t) + M1/(K1* • π • Rm² • t)

= 0•559/(2•18)+2846/(0.19•π•559²•18)

= -8.21 N./mm²

Maximum Shear Force in the Saddle (4.15.3) [T]:

= Q(L-2a)/(L+(4•hm/3))

= 16389 (5450 - 2 • 1050)/(5450 + (4 • 0/3))

= 10074.0 Kgf



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Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$= K_2 \cdot T / (R_m \cdot t)$$

$$= 1.17 \cdot 10074 / (559 \cdot 18)$$

$$= 11.49 \text{ N./mm}^2$$

Decay Length (4.15.20) [x_1, x_2]:

$$= 0.78 \cdot \sqrt{(R_m \cdot t)}$$

$$= 0.78 \cdot \sqrt{(559 \cdot 18)}$$

$$= 78.241 \text{ mm}$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{(R_m \cdot t)}, 2a)$$

$$= \min(203 + 1.56 \cdot \sqrt{(559 \cdot 18)}, 2 \cdot 1050)$$

$$= 359.48 \text{ mm}$$

Wear Plate/Shell Stress ratio (4.15.26) [η]:

$$= 1.0000 \text{ Materials are the same, transport case}$$

Circumferential Stress at Saddle Base with Wear Plate (4.15.25) [σ_6, r]:

$$= -K_5 \cdot Q \cdot k / (B_1(t + \eta \cdot tr))$$

$$= -0.76 \cdot 16389 \cdot 0.1 / (359(18 + 1 \cdot 15))$$

$$= -1.03 \text{ N./mm}^2$$

Circ. Comp. Stress at Horn of Saddle, $L \geq 8R_m$ (4.15.28) [σ_7, r]:

$$= -Q / (4(t + \eta \cdot tr)b_1) - 3 \cdot K_7 \cdot Q / (2(t + \eta \cdot tr)^2)$$

$$= -16389 / (4(18 + 1 \cdot 15)359) -$$

$$3 \cdot 0.053 \cdot 16389 / (2(18 + 1 \cdot 15)^2)$$

$$= -15.09 \text{ N./mm}^2$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	1150.0000	mm
Baseplate Thickness	Bpthk	20.0000	mm
Baseplate Width	Bpwid	392.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	3	
Rib Thickness	Ribtk	10.0000	mm
Web Thickness	Webtk	10.0000	mm
Web Location	Webloc	Center	
Saddle Yield Stress	Sy	260.0	N./mm ²
Height of Web at Center	Hw, c	306.0	mm
Friction Coefficient	mu	0.000	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})t^3) / 12$$

$$= ((1.56 \cdot \sqrt{(550 \cdot 18)} + 360)18^3) / 12$$

$$= 25.040 \text{ cm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}}(C_1 - Y)^2$$

$$= 25 + 92.7(173 - 9)^2$$

$$= 25061.312 \text{ cm}^4$$

	B mm	D mm	Y mm	A cm ²	AY mm ³	I + AD ⁽²⁾ cm ⁴
Shell	515.2	18.0	9.0	92.7	83465.3	0.251E+05
Wearplate	360.0	15.0	25.5	54.0	137700.0	0.118E+05
Web	10.0	397.0	231.5	39.7	919055.2	0.656E+04
BasePlate	392.0	20.0	440.0	78.4	3449600.8	0.558E+05
Totals	...	...	...	264.8	4589821.5	0.992E+05

Distance to Centroid [C1]:

$$= AY / A$$



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$$= 4589821/265$$

$$= 173.306 \text{ mm}$$

Angle [β]:

$$= 180 - \text{Saddle Angle}/2$$

$$= 180 - 120/2$$

$$= 120.0$$

Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta))$$

$$= (1 + \cos(120) - 0.5 \cdot \sin(120)^2) / (\pi - 2.09 + \sin(120) \cos(120))$$

$$= 0.2035$$

Saddle Splitting Force [Fh]:

$$= K1 \cdot Q$$

$$= 0.2 \cdot 16389$$

$$= 3335.5125 \text{ Kgf}$$

Tension Stress, St = (Fh/As) = 1.9007 N./mm²

Allowed Stress, Sa = 0.6 * Yield Str = 155.9972 N./mm²

Saddle Splitting Dimension [d]:

$$= B - R \cdot \sin(\text{theta}/2) / (\text{theta}/2 \text{ in radians})$$

$$= 1000 - 550 \cdot \sin(120/2) / 1.05$$

$$= 545.154 \text{ mm}$$

Bending Moment, M = Fh * d = 1818.4026 Kg-m.

Bending Stress, Sb = (M * C1 / I) = 3.1149 N./mm²

Allowed Stress, Sa = 2/3 * Yield Str = 173.3302 N./mm²

Minimum Thickness of Baseplate per Moss:

$$= \sqrt{(3(Q + \text{SaddleWt}) \text{BasePlateWidth}) / (4 \cdot \text{BasePlateLength} \cdot \text{AllStress})}$$

$$= \sqrt{(3(16389 + 223)392) / (4 \cdot 1150 \cdot 173)}$$

$$= 15.501 \text{ mm}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$$= 2 \cdot \cos(90 - \text{Saddle Angle}/2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$$

$$= 2 \cdot \cos(90 - 120/2) (550 + 18 + 15)$$

$$= 1009.786 \text{ mm}$$

Distance between Ribs [e]:

$$= \text{Web Length} / (\text{Nr ribs} - 1)$$

$$= 1010 / (3 - 1)$$

$$= 504.893 \text{ mm}$$

Baseplate Pressure Area [Ap]:

$$= e \cdot \text{Bpwid} / 2$$

$$= 505 \cdot 392 / 2$$

$$= 989.590 \text{ cm}^2$$

Bearing Pressure [Bp]:

$$= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth})$$

$$= 16612 / (1150 \cdot 392)$$

$$= 3.685 \text{ Kgf/cm}^2$$

Axial Load [P]:

$$= Ap \cdot Bp$$

$$= 990 \cdot 3.69$$

$$= 3646.732 \text{ Kgf}$$

Area of the Rib and Web [Ar]:

$$= \text{Rib Area} + \text{Web Area}$$



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$$= 19.3 + 25.2$$

$$= 44.545 \text{ cm}^2$$

Compressive Stress [Sc]:

$$= P/A_r$$

$$= 3647/44.5$$

$$= 8.029 \text{ N./mm}^2$$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	10.0	203.0	...	20.3	...	697.

Rib dimension [D]:

$$= \text{Saddle Width} - \text{Web Thickness}$$

$$= 203 - 10$$

$$= 193.000 \text{ mm}$$

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0/44.5$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 203/2$$

$$= 101.500 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{(\text{Total Inertia} / \text{Total Area})}$$

$$= \sqrt{(697/44.5)}$$

$$= 39.560 \text{ mm}$$

Length of Outer Rib [L]:

$$= \text{Saddle Height} - \cos(\theta/2) (\text{radius} + \text{shlthk} + \text{wpdthk}) - \text{bpthk}$$

$$= 1000 - \cos(120/2) (550 + 18 + 15) - 20$$

$$= 688.500 \text{ mm}$$

Intermediate Term [Cc]:

$$= \sqrt{(2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress})}$$

$$= \sqrt{(2 \cdot \pi^2 \cdot 199943008/260)}$$

$$= 123.211$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 688/39.6$$

$$= 17.404$$

Bending Moment [Rm]:

$$= FI / (2 \cdot \text{Bplen}) \cdot e \cdot L / 2$$

$$= 1397 / (2 \cdot 1150) \cdot 505 \cdot 688/2$$

$$= 105.554 \text{ Kg-m.}$$

Compressive Allowable, $KL/r < C_c$ ($17.4 < 123$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (17.4)^2 / (2 \cdot 123^2)) 260 / (5/3 + 3 \cdot (17.4) / (8 \cdot 123) - (17.4^3) / (8 \cdot 123^3))$$

$$= 149.7 \text{ N./mm}^2$$

AISC Unity Check of Outside Ribs (must be ≤ 1)

$$= Sc/Sca + (Rm \cdot C1 / I) / Sba$$

$$= 8.03/150 + (106 \cdot 102/6971188) / 173$$



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= 0.141

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	10.0	193.0	0.0	19.3	0.0	697.
Web	504.9	10.0	0.0	50.5	0.0	4.21
Totals	...	...	...	69.8	...	701.

Distance to Centroid from Datum [ytot]:

= AY / A

= 0/69.8

= 0.000 mm

Distance to Centroid [C1]:

= Saddle Width / 2

= 203/2

= 101.500 mm

Length of Inner Rib [L]:

= Saddle Height - Outside Radius - Bpthk

= 1000 - cos(583/2)(20 + 0 + 0) - 0

= 397.000 mm

Radius of Gyration [r]:

= $\sqrt{\text{Total Inertia} / \text{Total Area}}$

= $\sqrt{701/69.8}$

= 31.699 mm

Slenderness ratio [KL/r]:

= KL/r

= 1 • 397/31.7

= 12.524

Unit Force [Force,u]:

= FI / (2 • Baseplate Length)

= 1397/(2 • 1150)

= 0.607 Kg/mm

Moment at base of inner Rib [Mbase,c]:

= Unit Force • e • L

= 0.61 • 505 • 397

= 121.728 Kg-m.

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

= Bending Moment / Section Modulus

= 122/69088

= 17.279 N./mm²

Compressive Allowable, KL/r < Cc (12.5 < 123) per AISC E2-1 [Sca]:

= $(1 - (KL/r)^2 / (2 \cdot Cc^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot Cc) - (KL/r)^3 / (8 \cdot Cc^3))$

= $(1 - (12.5)^2 / (2 \cdot 123^2)) 260 /$

$(5/3 + 3 \cdot (12.5) / (8 \cdot 123) - (12.5^3) / (8 \cdot 123^3))$

= 151.7 N./mm²

AISC Unity Check of Inside Ribs (must be ≤ 1)

= Sc/Sca + (Mbase,c • C1/I)/Sba

= 8.3/152 + (122 • 102/701)/173

= 0.154

ASME VIII Division 2 Horizontal Vessel Analysis, Right Saddle:

Input and Calculated Values:



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Vessel Mean Radius	Rm	559.00	mm
Shell Thickness used in this Case	t	18.000	mm
Stiffened Vessel Length per 4.15.6	L	5450.00	mm
Distance from Saddle to Vessel tangent	a1 or a	1050.00	mm
Saddle Width	b1 or b	203.00	mm
Saddle Bearing Angle	delta or theta	120.00	degrees
Wear Plate Width	b2 or b1	360.00	mm
Wear Plate Bearing Angle	delta2 or theta1	132.00	degrees
Wear Plate Thickness	e2 or tr	15.0	mm
Wear Plate Allowable Stress	fw or Sr	138.00	N./mm ²
Inside or Mean Depth of Head	Hi or hm	279.50	mm
Shell Allowable Stress used in Calculation		138.00	N./mm ²
Head Allowable Stress used in Calculation		137.00	N./mm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	1000.00	mm
Coefficient of Friction	mu	0.40	

Results for Horizontal Vessel Analysis:

Saddle Force Q for this Transportation Case		1575.20	Kgf
Shipping Longitudinal Acceleration	gx	0.10	g
Shipping Transverse Acceleration	gz	0.10	g
Shipping Vertical Acceleration	gy	0.05	g
Vertical Acceleration Acting with Transverse	gw	1.00	g
Internal Pressure during Transport		0.00	KPa.
Transportation Wind Load Scalar Factor	wlsf	1.00	
Transportation Wind Velocity	wv	92.7	Km/hr

Horizontal Vessel Analysis Results:	Actual N./mm ²	Allowable N./mm ²
Long. Stress at Top of Midspan	-0.21	-234.00
Long. Stress at Top of Midspan	0.21	138.00
Long. Stress at Bottom of Midspan	0.21	138.00
Long. Stress at Top of Saddles	1.94	138.00
Long. Stress at Bottom of Saddles	-1.07	-234.00
Long. Stress at Bottom of Saddles	1.07	138.00
Tangential Shear in Shell	1.03	110.40
Circ. Stress at Horn of Saddle	1.45	207.00
Circ. Compressive Stress in Shell	0.10	138.00

Intermediate Results: Saddle Reaction Q due to Road Transport Load:

Transverse Force due to Wind during Transport [Ftw]:

$$\begin{aligned}
 &= \text{Transverse Area} \cdot 0.00256 \cdot wv^2 \cdot wlsf \\
 &= 129 \cdot 0.00256 \cdot 57.6^2 \cdot 1 \\
 &= 1097.5 \text{ lbf} \\
 &= 497.816 \text{ Kgf}
 \end{aligned}$$

Longitudinal Force due to Wind during Transport [Flw]:

$$\begin{aligned}
 &= \pi \cdot D^2/4 \cdot 0.00256 \cdot wv^2 \cdot wlsf \cdot \text{Wind Dia. Mult} \\
 &= \pi \cdot 4.71^2/4 \cdot 0.00256 \cdot 57.6^2 \cdot 1 \cdot 1.2 \\
 &= 177.7 \text{ lbf} \\
 &= 80.6 \text{ Kgf}
 \end{aligned}$$

Transverse Force due to Lateral Acceleration [Ft]:



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$$\begin{aligned}
 &= gz \cdot \text{Shipping Weight} \cdot gw + Ftw \\
 &= 0.1 \cdot 13162 \cdot 1 + 498 \\
 &= 1814.0 \text{ Kgf}
 \end{aligned}$$

Longitudinal Force due to Axial Acceleration [FI]:

$$\begin{aligned}
 &= gx \cdot \text{Shipping Weight} \cdot gw + Flw \\
 &= 0.1 \cdot 13162 \cdot 1 + 80.6 \\
 &= 1396.8 \text{ Kgf}
 \end{aligned}$$

Saddle Reaction Force due to Transverse Load [Fwt]:

$$\begin{aligned}
 &= Ft/\text{Num of Saddles} \cdot B / E \\
 &= 1814/2 \cdot 1000/1010 \\
 &= 898.2 \text{ Kgf}
 \end{aligned}$$

Saddle Reaction Force due to Longitudinal Load [Fwl]:

$$\begin{aligned}
 &= FI \cdot B / Ls \\
 &= 1397 \cdot 1000/3100 \\
 &= 450.6 \text{ Kgf}
 \end{aligned}$$

Saddle Reaction Force due to Vertical Transportation Acceleration [Frv]:

$$\begin{aligned}
 &= \text{Saddle Load} \cdot gy \\
 &= 645 \cdot 0.05 \\
 &= 32.238 \text{ Kgf}
 \end{aligned}$$

Load Combination Results due to Transport Loading [Q]:

$$\begin{aligned}
 &= \text{Saddle Load} + Frv + \max(Fwl, Fwt) \\
 &= 645 + 32.2 + \max(451, 898) \\
 &= 1575.2 \text{ Kgf}
 \end{aligned}$$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	1798.62	Kgf
Transverse Shear Load	906.98	Kgf
Longitudinal Shear Load	1396.75	Kgf

Note: Wear Plate is Welded to the Shell, $k = 0.1$

Saddle Dimension [E]:

$$\begin{aligned}
 &= \min(2(\text{ShellID}/2 + t + \text{WearPadThickness})\sin(\theta/2), 2 \cdot Rm) \\
 &= \min(2(1100/2 + 18 + 15)\sin(120/2), 2 \cdot 559) \\
 &= 1009.786 \text{ mm}
 \end{aligned}$$

The Computed K values from Table 4.15.1:

K1 = 0.1066	K2 = 1.1707	K3 = 0.8799	K4 = 0.4011
K5 = 0.7603	K6 = 0.0529	K7 = 0.0529	K8 = 0.3405
K9 = 0.2711	K10 = 0.0581	K1* = 0.1923	K6p = 0.0434
K7p = 0.0434			

The suffix 'p' denotes the values for a wear plate if it exists.

Note: Dimension a is greater than or equal to $Rm/2$.

Moment per Equation 4.15.1 [M1]:

$$\begin{aligned}
 &= -Q \cdot a [1 - (1 - a/L + (Rm^2 - hm^2)/(2a \cdot L))/(1 + (4hm)/3L)] \\
 &= -1575 \cdot 1050 [1 - (1 - 1050/5450 + (559^2 - 280^2)/(2 \cdot 1050 \cdot 5450))/(1 + (4 \cdot 280)/(3 \cdot 5450))] \\
 &= -372.4 \text{ Kg-m.}
 \end{aligned}$$

Moment per Equation 4.15.2 [M2]:

$$\begin{aligned}
 &= Q \cdot L/4((1 + 2(Rm^2 - hm^2)/(L^2))/(1 + (4hm^2)/(3L)) - 4a/L) \\
 &= 1575 \cdot 5450/4((1 + 2(559^2 - 280^2)/(5450^2))/(1 + (4 \cdot 280)/(3 \cdot 5450)) - 4 \cdot 1050/5450) \\
 &= 386.6 \text{ Kg-m.}
 \end{aligned}$$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

$$= P \cdot Rm/(2t) - M2/(\pi \cdot Rm^2 t)$$



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$$= 0 \cdot 559 / (2 \cdot 18) - 387 / (\pi \cdot 559^2 \cdot 18)$$

$$= -0.21 \text{ N./mm}^2$$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

$$= P \cdot R_m / (2t) + M_2 / (\pi \cdot R_m^2 \cdot t)$$

$$= 0 \cdot 559 / (2 \cdot 18) + 387 / (\pi \cdot 559^2 \cdot 18)$$

$$= 0.21 \text{ N./mm}^2$$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ_{*3}]:

$$= P \cdot R_m / (2t) - M_1 / (K_1 \cdot \pi \cdot R_m^2 \cdot t)$$

$$= 0 \cdot 559 / (2 \cdot 18) - 372 / (0.11 \cdot \pi \cdot 559^2 \cdot 18)$$

$$= 1.94 \text{ N./mm}^2$$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ_{*4}]:

$$= P \cdot R_m / (2t) + M_1 / (K_1 \cdot \pi \cdot R_m^2 \cdot t)$$

$$= 0 \cdot 559 / (2 \cdot 18) + 372 / (0.19 \cdot \pi \cdot 559^2 \cdot 18)$$

$$= -1.07 \text{ N./mm}^2$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L - 2a) / (L + (4 \cdot h_m / 3))$$

$$= 1575 (5450 - 2 \cdot 1050) / (5450 + (4 \cdot 280 / 3))$$

$$= 906.3 \text{ Kg}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$= K_2 \cdot T / (R_m \cdot t)$$

$$= 1.17 \cdot 906 / (559 \cdot 18)$$

$$= 1.03 \text{ N./mm}^2$$

Decay Length (4.15.20) [x_1, x_2]:

$$= 0.78 \cdot \sqrt{(R_m \cdot t)}$$

$$= 0.78 \cdot \sqrt{(559 \cdot 18)}$$

$$= 78.241 \text{ mm}$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{(R_m \cdot t)}, 2a)$$

$$= \min(203 + 1.56 \cdot \sqrt{(559 \cdot 18)}, 2 \cdot 1050)$$

$$= 359.48 \text{ mm}$$

Wear Plate/Shell Stress ratio (4.15.26) [η]:

$$= 1.0000 \text{ Materials are the same, transport case}$$

Circumferential Stress at Saddle Base with Wear Plate (4.15.25) [σ_6, r]:

$$= -K_5 \cdot Q \cdot k / (B_1(t + \eta \cdot tr))$$

$$= -0.76 \cdot 1575 \cdot 0.1 / (359(18 + 1 \cdot 15))$$

$$= -0.10 \text{ N./mm}^2$$

Circ. Comp. Stress at Horn of Saddle, $L \geq 8R_m$ (4.15.28) [σ_7, r]:

$$= -Q / (4(t + \eta \cdot tr)b_1) - 3 \cdot K_7 \cdot Q / (2(t + \eta \cdot tr)^2)$$

$$= -1575 / (4(18 + 1 \cdot 15)359) -$$

$$3 \cdot 0.053 \cdot 1575 / (2(18 + 1 \cdot 15)^2)$$

$$= -1.45 \text{ N./mm}^2$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	1150.0000	mm
Baseplate Thickness	Bpthk	20.0000	mm
Baseplate Width	Bpwid	392.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	3	
Rib Thickness	Ribtk	10.0000	mm
Web Thickness	Webtk	10.0000	mm
Web Location	Webloc	Center	
Saddle Yield Stress	Sy	260.0	N./mm ²
Height of Web at Center	Hw,c	306.0	mm
Friction Coefficient	mu	0.400	



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Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})^3) / 12$$

$$= ((1.56 \cdot \sqrt{(550 \cdot 18)} + 360)^3) / 12$$

$$= 25.040 \text{ cm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}} (C1 - Y)^2$$

$$= 25 + 92.7 (173 - 9)^2$$

$$= 25061.312 \text{ cm}^4$$

	B mm	D mm	Y mm	A cm ²	AY mm ³	I + AD ² cm ⁴
Shell	515.2	18.0	9.0	92.7	83465.3	0.251E+05
Wearplate	360.0	15.0	25.5	54.0	137700.0	0.118E+05
Web	10.0	397.0	231.5	39.7	919055.2	0.656E+04
BasePlate	392.0	20.0	440.0	78.4	3449600.8	0.558E+05
Totals	...	...	...	264.8	4589821.5	0.992E+05

Distance to Centroid [C1]:

$$= AY / A$$

$$= 4589821 / 265$$

$$= 173.306 \text{ mm}$$

Angle [β]:

$$= 180 - \text{Saddle Angle} / 2$$

$$= 180 - 120 / 2$$

$$= 120.0$$

Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta))$$

$$= (1 + \cos(120) - 0.5 \cdot \sin(120)^2) / (\pi - 2.09 + \sin(120) \cos(120))$$

$$= 0.2035$$

Saddle Splitting Force [Fh]:

$$= K1 \cdot Q$$

$$= 0.2 \cdot 1575$$

$$= 320.5868 \text{ Kgf}$$

$$\text{Tension Stress, } St = (Fh / As) = 0.1827 \text{ N./mm}^2$$

$$\text{Allowed Stress, } Sa = 0.6 \cdot \text{Yield Str} = 155.9972 \text{ N./mm}^2$$

Saddle Splitting Dimension [d]:

$$= B - R \cdot \sin(\text{theta} / 2) / (\text{theta} / 2 \text{ in radians})$$

$$= 1000 - 550 \cdot \sin(120 / 2) / 1.05$$

$$= 545.154 \text{ mm}$$

$$\text{Bending Moment, } M = Fh \cdot d = 174.7725 \text{ Kg-m.}$$

$$\text{Bending Stress, } Sb = (M \cdot C1 / I) = 0.2994 \text{ N./mm}^2$$

$$\text{Allowed Stress, } Sa = 2/3 \cdot \text{Yield Str} = 173.3302 \text{ N./mm}^2$$

Minimum Thickness of Baseplate per Moss:

$$= \sqrt{(3(Q + \text{SaddleWt}) \cdot \text{BasePlateWidth}) / (4 \cdot \text{BasePlateLength} \cdot \text{AllStress})}$$

$$= \sqrt{(3(1575 + 223) \cdot 392) / (4 \cdot 1150 \cdot 173)}$$

$$= 5.101 \text{ mm}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$$= 2 \cdot \cos(90 - \text{Saddle Angle} / 2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$$

$$= 2 \cdot \cos(90 - 120 / 2) (550 + 18 + 15)$$

$$= 1009.786 \text{ mm}$$



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Distance between Ribs [e]:

$$= \text{Web Length} / (\text{Nribs} - 1)$$

$$= 1010 / (3 - 1)$$

$$= 504.893 \text{ mm}$$

Baseplate Pressure Area [Ap]:

$$= e \cdot \text{Bpwid} / 2$$

$$= 505 \cdot 392 / 2$$

$$= 989.590 \text{ cm}^2$$

Bearing Pressure [Bp]:

$$= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth})$$

$$= 1799 / (1150 \cdot 392)$$

$$= 0.399 \text{ Kg/cm}^2$$

Axial Load [P]:

$$= \text{Ap} \cdot \text{Bp}$$

$$= 990 \cdot 0.4$$

$$= 394.831 \text{ Kg}$$

Area of the Rib and Web [Ar]:

$$= \text{Rib Area} + \text{Web Area}$$

$$= 19.3 + 25.2$$

$$= 44.545 \text{ cm}^2$$

Compressive Stress [Sc]:

$$= P / \text{Ar}$$

$$= 395 / 44.5$$

$$= 0.869 \text{ N./mm}^2$$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	10.0	203.0	...	20.3	...	697.

Rib dimension [D]:

$$= \text{Saddle Width} - \text{Web Thickness}$$

$$= 203 - 10$$

$$= 193.000 \text{ mm}$$

Distance to Centroid from Datum [ytot]:

$$= \text{AY} / \text{A}$$

$$= 0 / 44.5$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 203 / 2$$

$$= 101.500 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{(\text{Total Inertia} / \text{Total Area})}$$

$$= \sqrt{(697 / 44.5)}$$

$$= 39.560 \text{ mm}$$

Length of Outer Rib [L]:

$$= \text{Saddle Height} - \cos(\text{theta}/2) (\text{radius} + \text{shlthk} + \text{wpdthk}) - \text{bpthk}$$

$$= 1000 - \cos(120/2) (550 + 18 + 15) - 20$$

$$= 688.500 \text{ mm}$$

Intermediate Term [Cc]:



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$$= \sqrt{(2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress})}$$

$$= \sqrt{(2 \cdot \pi^2 \cdot 199943008 / 260)}$$

$$= 123.211$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 688 / 39.6$$

$$= 17.404$$

Bending Moment [Rm]:

$$= FI / (2 \cdot B_{plen}) \cdot e \cdot L / 2$$

$$= 1397 / (2 \cdot 1150) \cdot 505 \cdot 688 / 2$$

$$= 105.554 \text{ Kg-m.}$$

Compressive Allowable, $KL/r < C_c$ ($17.4 < 123$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (17.4)^2 / (2 \cdot 123^2)) 260 /$$

$$(5/3 + 3 \cdot (17.4) / (8 \cdot 123) - (17.4^3) / (8 \cdot 123^3))$$

$$= 149.7 \text{ N./mm}^2$$

AISC Unity Check of Outside Ribs (must be ≤ 1)

$$= S_c / S_{ca} + (R_m \cdot C_1 / I) / S_{ba}$$

$$= 0.87 / 150 + (106 \cdot 102 / 6971188) / 173$$

$$= 0.093$$

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	10.0	193.0	0.0	19.3	0.0	697.
Web	504.9	10.0	0.0	50.5	0.0	4.21
Totals	...	...	...	69.8	...	701.

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0 / 69.8$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 203 / 2$$

$$= 101.500 \text{ mm}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \text{Outside Radius} - B_{pthk}$$

$$= 1000 - \cos(583/2) (20 + 0 + 0) - 0$$

$$= 397.000 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{(\text{Total Inertia} / \text{Total Area})}$$

$$= \sqrt{(701 / 69.8)}$$

$$= 31.699 \text{ mm}$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 397 / 31.7$$

$$= 12.524$$

Unit Force [Force,u]:

$$= FI / (2 \cdot \text{Baseplate Length})$$

$$= 1397 / (2 \cdot 1150)$$

$$= 0.607 \text{ Kg/mm}$$

Moment at base of inner Rib [Mbase,c]:



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$$\begin{aligned}
 &= \text{Unit Force} \cdot e \cdot L \\
 &= 0.61 \cdot 505 \cdot 397 \\
 &= 121.728 \text{ Kg-m.}
 \end{aligned}$$

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

$$\begin{aligned}
 &= \text{Bending Moment} / \text{Section Modulus} \\
 &= 122/69088 \\
 &= 17.279 \text{ N./mm}^2
 \end{aligned}$$

Compressive Allowable, $KL/r < C_c$ ($12.5 < 123$) per AISC E2-1 [Sca]:

$$\begin{aligned}
 &= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3)) \\
 &= (1 - (12.5)^2 / (2 \cdot 123^2)) 260 / \\
 &\quad (5/3 + 3 \cdot (12.5) / (8 \cdot 123) - (12.5^3) / (8 \cdot 123^3)) \\
 &= 151.7 \text{ N./mm}^2
 \end{aligned}$$

AISC Unity Check of Inside Ribs (must be ≤ 1)

$$\begin{aligned}
 &= S_c / S_{ca} + (M_{base,c} \cdot C_{1/I}) / S_{ba} \\
 &= 0.9/152 + (122 \cdot 102/701) / 173 \\
 &= 0.106
 \end{aligned}$$

Review notes about nozzle loadings and supports in the Warnings report.

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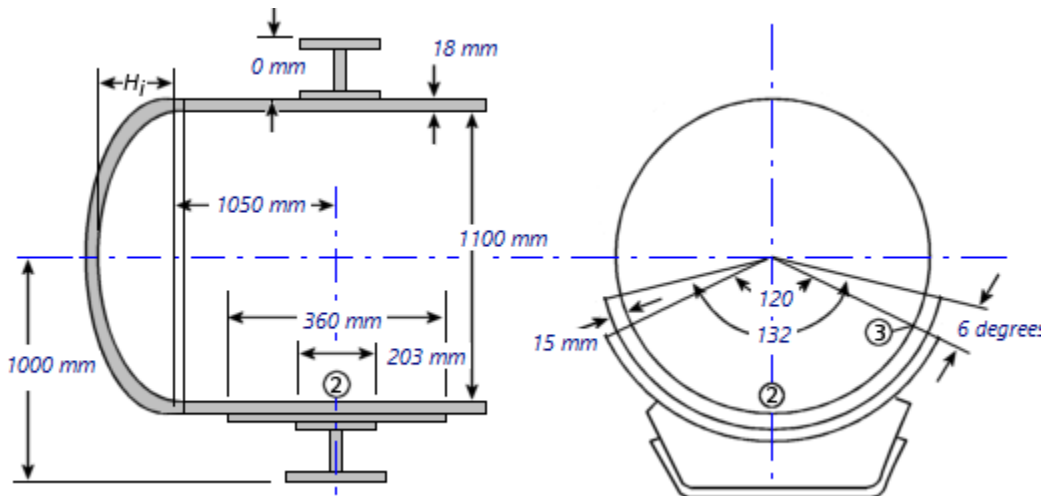
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ASME VIII Division 2 Horizontal Vessel Analysis, Left Saddle:

Horizontal Vessel Stress Calculations : Operating Case

Input and Calculated Values:

Vessel Mean Radius	Rm	560.50	mm
Shell Thickness used in this Case	t	18.000	mm
Stiffened Vessel Length per 4.15.6	L	5450.00	mm
Distance from Saddle to Vessel tangent	a1 or a	1050.00	mm
Saddle Width	b1 or b	203.00	mm
Saddle Bearing Angle	delta or theta	120.00	degrees
Wear Plate Width	b2 or b1	360.00	mm
Wear Plate Bearing Angle	delta2 or theta1	132.00	degrees
Wear Plate Thickness	e2 or tr	15.0	mm
Wear Plate Allowable Stress	fw or Sr	138.00	N./mm ²
Shell Allowable Stress used in Calculation		138.00	N./mm ²
Head Allowable Stress used in Calculation		147.00	N./mm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	1000.00	mm
Coefficient of Friction	mu	0.00	



Results for Horizontal Vessel Analysis:

Saddle Force Q, Operating Case	22004.41	Kgf
Pressure used in Analysis	P 1085.391	KPa.

Horizontal Vessel Analysis Results:	Actual N./mm ²	Allowable N./mm ²
Long. Stress at Top of Midspan	15.30	138.00
Long. Stress at Bottom of Midspan	25.26	138.00
Long. Stress at Top of Saddles	44.00	138.00
Long. Stress at Bottom of Saddles	7.13	138.00
Tangential Shear in Shell	18.47	110.40
Circ. Stress at Horn of Saddle	24.21	172.50
Circ. Compressive Stress in Shell	1.58	138.00



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Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:

$$= \text{Ftr}(\text{Ft/Num of Saddles} + \text{Z Force Load}) \cdot B / E$$

$$= 3 (509/2 + 0) \cdot 1000/1010$$

$$= 755.4 \text{ Kgf}$$

Longitudinal Saddle Reaction Force [Fwl]:

$$= \max(\text{Fl,wind}; \text{Sum of X Forces}) \cdot B / \text{Ls}$$

$$= \max(200, 0) \cdot 1000/3100$$

$$= 64.6 \text{ Kgf}$$

Saddle Reaction Force due to Earthquake Fl [Fsl]:

$$= \max(\text{Fl,eq}; \text{Sum of X Forces}) \cdot B / \text{Ls}$$

$$= \max(1244, 0) \cdot 1000/3100$$

$$= 401.2 \text{ Kgf}$$

Saddle Reaction Force due to Earthquake Ft [Fst]:

$$= \text{Ftr}(\text{Ft/Num of Saddles} + \text{Z Force Load}) \cdot B / E$$

$$= 3 (1244/2 + 0) \cdot 1000/1010$$

$$= 1847.5 \text{ Kgf}$$

Load Combination Results for Q + Wind or Seismic [Q]:

$$= \text{Saddle Load} + \max(\text{Fwl}, \text{Fwt}, \text{Fsl}, \text{Fst})$$

$$= 20157 + \max(64.6, 755, 401, 1847)$$

$$= 22004.4 \text{ Kgf}$$

Longitudinal Wind Force [Fl]:

$$= \text{WindScalar} \cdot \text{WindPress}(\text{Platform Area} + (\pi/4(\text{OD} \cdot \text{WindDiaMult})^2))$$

$$= 1.1 \cdot 766 (0 + \pi/4(1.44 \cdot 1.2)^2)$$

$$= 1965.311 \text{ N}$$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	22227.83	Kgf
Transverse Shear Load	621.85	Kgf
Longitudinal Shear Load	1243.70	Kgf

Note: Wear Plate is Welded to the Shell, k = 0.1

Saddle Dimension [E]:

$$= \min(2(\text{ShellID}/2 + t + \text{WearPadThickness}) \sin(\theta/2), 2 \cdot \text{Rm})$$

$$= \min(2(1100/2 + 18 + 15) \sin(120/2), 2 \cdot 560)$$

$$= 1009.786 \text{ mm}$$

The Computed K values from Table 4.15.1:

K1 = 0.1066	K2 = 1.1707	K3 = 0.8799	K4 = 0.4011
K5 = 0.7603	K6 = 0.0529	K7 = 0.0529	K8 = 0.3405
K9 = 0.2711	K10 = 0.0581	K1* = 0.1923	K6p = 0.0434
K7p = 0.0434			

The suffix 'p' denotes the values for a wear plate if it exists.

Note: Dimension a is greater than or equal to Rm/2.

Moment per Equation 4.15.1 [M1]:

$$= -Q \cdot a [1 - (1 - a/L + (Rm^2 - hm^2)/(2a \cdot L)) / (1 + (4hm)/3L)]$$

$$= -22004 \cdot 1050 [1 - (1 - 1050/5450 + (560^2 - 0^2) / (2 \cdot 1050 \cdot 5450)) / (1 + (4 \cdot 0) / (3 \cdot 5450))]$$

$$= -3817.2 \text{ Kg-m.}$$

Moment per Equation 4.15.2 [M2]:

$$= Q \cdot L / 4 ((1 + 2(Rm^2 - hm^2)/(L^2)) / (1 + (4hm^2)/(3L)) - 4a/L)$$

$$= 22004 \cdot 5450 / 4 ((1 + 2(560^2 - 0^2)/(5450^2)) / (1 + (4 \cdot 0) / (3 \cdot 5450)) - 4 \cdot 1050 / 5450)$$



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$$(3 \cdot 5450) - 4 \cdot 1050 / 5450 \\ = 7510.7 \text{ Kg-m.}$$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

$$= P \cdot R_m / (2t) - M_2 / (\pi \cdot R_m^2 \cdot t) \\ = 1085 \cdot 560 / (2 \cdot 15) - 7511 / (\pi \cdot 560^2 \cdot 15) \\ = 15.30 \text{ N./mm}^2$$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

$$= P \cdot R_m / (2t) + M_2 / (\pi \cdot R_m^2 \cdot t) \\ = 1085 \cdot 560 / (2 \cdot 15) + 7511 / (\pi \cdot 560^2 \cdot 15) \\ = 25.26 \text{ N./mm}^2$$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ_{*3}]:

$$= P \cdot R_m / (2t) - M_1 / (K_1 \cdot \pi \cdot R_m^2 \cdot t) \\ = 1085 \cdot 560 / (2 \cdot 15) - 3817 / (0.11 \cdot \pi \cdot 560^2 \cdot 15) \\ = 44.00 \text{ N./mm}^2$$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ_{*4}]:

$$= P \cdot R_m / (2t) + M_1 / (K_1 \cdot \pi \cdot R_m^2 \cdot t) \\ = 1085 \cdot 560 / (2 \cdot 15) + 3817 / (0.19 \cdot \pi \cdot 560^2 \cdot 15) \\ = 7.13 \text{ N./mm}^2$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L - 2a) / (L + (4 \cdot h_m / 3)) \\ = 22004 (5450 - 2 \cdot 1050) / (5450 + (4 \cdot 0 / 3)) \\ = 13525.6 \text{ Kgf}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$= K_2 \cdot T / (R_m \cdot t) \\ = 1.17 \cdot 13526 / (560 \cdot 15) \\ = 18.47 \text{ N./mm}^2$$

Decay Length (4.15.20) [x_1, x_2]:

$$= 0.78 \cdot \sqrt{R_m \cdot t} \\ = 0.78 \cdot \sqrt{560 \cdot 15} \\ = 71.520 \text{ mm}$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{R_m \cdot t}, 2a) \\ = \min(203 + 1.56 \cdot \sqrt{560 \cdot 15}, 2 \cdot 1050) \\ = 346.04 \text{ mm}$$

Wear Plate/Shell Stress ratio (4.15.26) [η]:

$$= \min(S_r / S, 1) \\ = \min(138 / 138, 1) \\ = 1.0000$$

Circumferential Stress at Saddle Base with Wear Plate (4.15.25) [σ_6, r]:

$$= -K_5 \cdot Q \cdot k / (B_1(t + \eta \cdot tr)) \\ = -0.76 \cdot 22004 \cdot 0.1 / (346(15 + 1 \cdot 15)) \\ = -1.58 \text{ N./mm}^2$$

Circ. Comp. Stress at Horn of Saddle, $L \geq 8R_m$ (4.15.28) [σ_7, r]:

$$= -Q / (4(t + \eta \cdot tr)b_1) - 3 \cdot K_7 \cdot Q / (2(t + \eta \cdot tr)^2) \\ = -22004 / (4(15 + 1 \cdot 15)346) - 3 \cdot 0.053 \cdot 22004 / (2(15 + 1 \cdot 15)^2) \\ = -24.21 \text{ N./mm}^2$$

Distance between Saddle Supports [L_s]:

$$= 3100.0 \text{ mm}$$

Free Un-Restrained Thermal Expansion between the Saddles [Exp]:

$$= \alpha \cdot L_s (\text{Design Temperature} - \text{Ambient Temperature}) \\ = 0.13139\text{E-}04 \cdot 3100 (275 - 21.1)$$



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= 10.341 mm

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	1150.0000	mm
Baseplate Thickness	Bpthk	20.0000	mm
Baseplate Width	Bpwid	392.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	3	
Rib Thickness	Ribtk	10.0000	mm
Web Thickness	Webtk	10.0000	mm
Web Location	Webloc	Center	
Saddle Design Temperature		21.1	°C
Saddle Yield Stress	Sy	262.0	N./mm ²
Height of Web at Center	Hw,c	306.0	mm
Friction Coefficient	mu	0.000	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})t^3)/12$$

$$= ((1.56 \cdot \sqrt{(553 \cdot 15)} + 360)15^3)/12$$

$$= 14.121 \text{ cm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}}(C1 - Y)^2$$

$$= 14.1 + 75.3(182 - 7.5)^2$$

$$= 23035.172 \text{ cm}^4$$

	B mm	D mm	Y mm	A cm ²	AY mm ³	I + AD ² (2) cm ⁴
Shell	502.1	15.0	7.5	75.3	56484.0	0.230E+05
Wearplate	360.0	15.0	22.5	54.0	121500.0	0.138E+05
Web	10.0	397.0	228.5	39.7	907145.3	0.606E+04
BasePlate	392.0	20.0	437.0	78.4	3426080.5	0.509E+05
Totals	...	...	...	247.4	4511210.0	0.938E+05

Distance to Centroid [C1]:

$$= AY / A$$

$$= 4511210/247$$

$$= 182.336 \text{ mm}$$

Angle [β]:

$$= 180 - \text{Saddle Angle}/2$$

$$= 180 - 120/2$$

$$= 120.0$$

Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta))$$

$$= (1 + \cos(120) - 0.5 \cdot \sin(120)^2) / (\pi - 2.09 + \sin(120) \cos(120))$$

$$= 0.2035$$

Saddle Splitting Force [Fh]:

$$= K1 \cdot Q$$

$$= 0.2 \cdot 22004$$

$$= 4478.3740 \text{ Kgf}$$

$$\text{Tension Stress, } St = (Fh/As) = 2.5519 \text{ N./mm}^2$$

$$\text{Allowed Stress, } Sa = 0.6 \cdot \text{Yield Str} = 157.1972 \text{ N./mm}^2$$

Saddle Splitting Dimension [d]:

$$= B - R \cdot \sin(\text{theta}/2) / (\text{theta}/2 \text{ in radians})$$

$$= 1000 - 553 \cdot \sin(120/2) / 1.05$$

$$= 542.673 \text{ mm}$$

$$\text{Bending Moment, } M = Fh \cdot d = 2430.3391 \text{ Kg-m.}$$



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Bending Stress, $S_b = (M \cdot C_1 / I) = 4.6343 \text{ N./mm}^2$
 Allowed Stress, $S_a = 2/3 \cdot \text{Yield Str} = 174.6635 \text{ N./mm}^2$

Minimum Thickness of Baseplate per Moss:

$= \sqrt{(3(Q + \text{SaddleWt}) \cdot \text{BasePlateWidth}) / (4 \cdot \text{BasePlateLength} \cdot \text{AllStress}))}$
 $= \sqrt{(3(22004 + 223) \cdot 392) / (4 \cdot 1150 \cdot 175)}$
 $= 17.862 \text{ mm}$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:
 $= 2 \cdot \cos(90 - \text{Saddle Angle}/2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$
 $= 2 \cdot \cos(90 - 120/2) (550 + 18 + 15)$
 $= 1009.786 \text{ mm}$

Distance between Ribs [e]:
 $= \text{Web Length} / (\text{Nr ribs} - 1)$
 $= 1010 / (3 - 1)$
 $= 504.893 \text{ mm}$

Baseplate Pressure Area [Ap]:
 $= e \cdot \text{Bpwid} / 2$
 $= 505 \cdot 392 / 2$
 $= 989.590 \text{ cm}^2$

Bearing Pressure [Bp]:
 $= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth})$
 $= 22228 / (1150 \cdot 392)$
 $= 4.931 \text{ Kg/cm}^2$

Axial Load [P]:
 $= A_p \cdot B_p$
 $= 990 \cdot 4.93$
 $= 4879.423 \text{ Kg}$

Area of the Rib and Web [Ar]:
 $= \text{Rib Area} + \text{Web Area}$
 $= 19.3 + 25.2$
 $= 44.545 \text{ cm}^2$

Compressive Stress [Sc]:
 $= P / A_r$
 $= 4879 / 44.5$
 $= 10.742 \text{ N./mm}^2$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	10.0	203.0	...	20.3	...	697.

Rib dimension [D]:
 $= \text{Saddle Width} - \text{Web Thickness}$
 $= 203 - 10$
 $= 193.000 \text{ mm}$

Distance to Centroid from Datum [ytot]:
 $= AY / A$
 $= 0 / 44.5$
 $= 0.000 \text{ mm}$

Distance to Centroid [C1]:



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= Saddle Width / 2
= 203/2
= 101.500 mm

Radius of Gyration [r]:
= $\sqrt{\text{Total Inertia} / \text{Total Area}}$
= $\sqrt{697/44.5}$
= 39.560 mm

Length of Outer Rib [L]:
= Saddle Height - $\cos(\theta/2)$ (radius + shlthk + wpdthk) - bpthk
= 1000 - $\cos(120/2)$ (550 + 18 + 15) - 20
= 688.500 mm

Intermediate Term [Cc]:
= $\sqrt{2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress}}$
= $\sqrt{2 \cdot \pi^2 \cdot 199943008/262}$
= 122.739

Slenderness ratio [KL/r]:
= KL/r
= 1 • 688/39.6
= 17.404

Bending Moment [Rm]:
= FI / (2 • Bplen) • e • L / 2
= 1244 / (2 • 1150) • 505 • 688/2
= 93.987 Kg-m.

Compressive Allowable, $KL/r < Cc$ (17.4 < 123) per AISC E2-1 [Sca]:
= $(1 - (KL/r)^2 / (2 \cdot Cc^2)) Fy / (5/3 + 3 \cdot (KL/r) / (8 \cdot Cc) - (KL/r)^3 / (8 \cdot Cc^3))$
= $(1 - (17.4)^2 / (2 \cdot 123^2)) 262 / (5/3 + 3 \cdot (17.4) / (8 \cdot 123) - (17.4)^3 / (8 \cdot 123^3))$
= 150.8 N./mm²

AISC Unity Check of Outside Ribs (must be ≤ 1)
= $Sc/Sca + (Rm \cdot C1 / I) / Sba$
= 10.7/151 + (94 • 102/6971188)/175
= 0.148

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	10.0	193.0	0.0	19.3	0.0	697.
Web	504.9	10.0	0.0	50.5	0.0	4.21
Totals	...	...	...	69.8	...	701.

Distance to Centroid from Datum [ytot]:
= AY / A
= 0/69.8
= 0.000 mm

Distance to Centroid [C1]:
= Saddle Width / 2
= 203/2
= 101.500 mm

Length of Inner Rib [L]:
= Saddle Height - Outside Radius - Bpthk
= 1000 - $\cos(583/2)$ (20 + 0 + 0) - 0
= 397.000 mm

Radius of Gyration [r]:



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$$= \sqrt{\text{Total Inertia} / \text{Total Area}}$$

$$= \sqrt{701/69.8}$$

$$= 31.699 \text{ mm}$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 397/31.7$$

$$= 12.524$$

Unit Force [Force,u]:

$$= FI / (2 \cdot \text{Baseplate Length})$$

$$= 1244 / (2 \cdot 1150)$$

$$= 0.541 \text{ Kg/mm}$$

Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 0.54 \cdot 505 \cdot 397$$

$$= 108.389 \text{ Kg-m.}$$

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

$$= \text{Bending Moment} / \text{Section Modulus}$$

$$= 108/69088$$

$$= 15.385 \text{ N./mm}^2$$

Compressive Allowable, $KL/r < C_c$ ($12.5 < 123$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (12.5)^2 / (2 \cdot 123^2)) 262 / (5/3 + 3 \cdot (12.5) / (8 \cdot 123) - (12.5)^3 / (8 \cdot 123^3))$$

$$= 152.9 \text{ N./mm}^2$$

AISC Unity Check of Inside Ribs (must be ≤ 1)

$$= S_c / S_{ca} + (M_{base,c} \cdot C_{1/I}) / S_{ba}$$

$$= 11.1/153 + (108 \cdot 102/701) / 175$$

$$= 0.161$$

Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate	Nbolts	4
Total Number of Bolts in Tension/Baseplate	Nbt	2
Bolt Design Temperature		21.1 °C
Bolt Material Specification		SA-307 A
Bolt Allowable Stress	Stba	138.00 N./mm ²
Bolt Corrosion Allowance	Bca	1.5 mm
Distance from Bolts to Edge	Edgedis	50.0 mm
User Defined Area in a Single Bolt	UserBoltArea	4.0400 cm ²
BasePlate Allowable Stress	S	138.00 N./mm ²
Area Available in a Single Bolt	BltArea	3.0418 cm ²
Saddle Load Q0 (Weight)	Q0	20380.4 Kg
Saddle Load QL (Wind/Seismic contribution)	QL	401.2 Kg
Maximum Transverse Force	Ft	621.8 Kg
Maximum Longitudinal Force	F1	1243.7 Kg
Saddle Bolted to Steel Foundation		No

Friction force due to thermal growth, [Ff]:

$$= 752.4 \text{ Kg}$$

Shear Stress in a Single Bolt, Longitudinal Direction [τ_b ,l]:

$$= \max(F_f, F_1) / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 1244 / (3.04 \cdot 4)$$

$$= 10.0 \text{ N./mm}^2. \text{ Must be less than } 82.8 \text{ N./mm}^2.$$

Shear Stress in a Single Bolt, Transverse Direction [τ_b ,t]:

$$= F_t / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 622 / (3.04 \cdot 4)$$

$$= 5.0 \text{ N./mm}^2. \text{ Must be less than } 82.8 \text{ N./mm}^2.$$



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Bolt Area Calculation per Dennis R. Moss

Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:
= 0.0 (QO > QL --> No Uplift in Longitudinal direction)

Bolt Area due to Shear Load [Bltarears]:
= FI / (BoltShearAllowable • Nbolts)
= 1244/(82.8 • 4)
= 0.3683 cm²

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:
= B • Ft + Sum of X Moments
= 1000 • 622 + 0
= 621.86 Kg-m.

Eccentricity (e):
= Rmom / QO
= 622/20380
= 30.51 mm < Bplen/6 --> No Uplift in Transverse direction

Bolt Area due to Transverse Load [Bltareart]:
= 0 (No Uplift)

Required Area of a Single Bolt [Bltarear]:
= max[Bltarearl, Bltarears, Bltareart]
= max[0, 0.37, 0]
= 0.3683 cm²

MDMT Calculations for: Left Saddle, Saddle to Wearplate Junction, Curve: B

Govrn. thk, tg = 15, c = 3 mm, E* = 0.85
Thickness Ratio = tr * E*/(tg - c) = 0.3, Temp. Reduction = 78 °C
Attachment governing, Attachment stress = Shell stress per Code

Min Metal Temp. w/o impact per UCS-66, Curve B	-16 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

ASME VIII Division 2 Horizontal Vessel Analysis, Right Saddle:

Input and Calculated Values:

Vessel Mean Radius	Rm	560.50	mm
Shell Thickness used in this Case	t	18.000	mm
Stiffened Vessel Length per 4.15.6	L	5450.00	mm
Distance from Saddle to Vessel tangent	a1 or a	1050.00	mm
Saddle Width	b1 or b	203.00	mm
Saddle Bearing Angle	delta or theta	120.00	degrees
Wear Plate Width	b2 or b1	360.00	mm
Wear Plate Bearing Angle	delta2 or theta1	132.00	degrees
Wear Plate Thickness	e2 or tr	15.0	mm
Wear Plate Allowable Stress	fw or Sr	138.00	N./mm ²
Inside or Mean Depth of Head	Hi or hm	281.00	mm
Shell Allowable Stress used in Calculation		138.00	N./mm ²
Head Allowable Stress used in Calculation		137.00	N./mm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	



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Distance from Saddle Base to Centerline B 1000.00 mm
Coefficient of Friction mu 0.40

Results for Horizontal Vessel Analysis:

Saddle Force Q, Operating Case 3505.14 Kgf
Pressure used in Analysis P 1085.391 KPa.

Horizontal Vessel Analysis Results:	Actual N./mm ²	Allowable N./mm ²
Long. Stress at Top of Midspan	19.71	138.00
Long. Stress at Bottom of Midspan	20.85	138.00
Long. Stress at Top of Saddles	25.43	138.00
Long. Stress at Bottom of Saddles	17.42	138.00
Tangential Shear in Shell	2.75	110.40
Circ. Stress at Horn of Saddle	3.86	172.50
Circ. Compressive Stress in Shell	0.25	138.00

Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:
= Ftr(Ft/Num of Saddles + Z Force Load) • B / E
= 3 (509/2 + 0) • 1000/1010
= 755.4 Kgf

Longitudinal Saddle Reaction Force [Fwl]:
= max(Fl,wind; Sum of X Forces) • B / Ls
= max(200, 0) • 1000/3100
= 64.6 Kgf

Saddle Reaction Force due to Earthquake Fl [Fsl]:
= max(Fl,eq; Sum of X Forces) • B / Ls
= max(1244, 0) • 1000/3100
= 401.2 Kgf

Saddle Reaction Force due to Earthquake Ft [Fst]:
= Ftr(Ft/Num of Saddles + Z Force Load) • B / E
= 3 (1244/2 + 0) • 1000/1010
= 1847.5 Kgf

Load Combination Results for Q + Wind or Seismic [Q]:
= Saddle Load + max(Fwl, Fwt, Fsl, Fst)
= 1658 + max(64.6, 755, 401, 1847)
= 3505.1 Kgf

Longitudinal Wind Force [Fl]:
= WindScalar • WindPress(Platform Area + ($\pi/4$ (OD • WindDiaMult)²))
= 1.1 • 766 (0 + $\pi/4$ (1.44 • 1.2)²)
= 1965.311 N

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight) 3728.56 Kgf
Transverse Shear Load 621.85 Kgf
Longitudinal Shear Load 1243.70 Kgf

Note: Wear Plate is Welded to the Shell, k = 0.1

Saddle Dimension [E]:
= min(2(ShellID/2 + t + WearPadThickness)sin(θ/2), 2•Rm)
= min(2(1100/2 + 18 + 15)sin(120/2), 2•560)
= 1009.786 mm



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The Computed K values from Table 4.15.1:

$K1 = 0.1066$ $K2 = 1.1707$ $K3 = 0.8799$ $K4 = 0.4011$
 $K5 = 0.7603$ $K6 = 0.0529$ $K7 = 0.0529$ $K8 = 0.3405$
 $K9 = 0.2711$ $K10 = 0.0581$ $K1^* = 0.1923$ $K6p = 0.0434$
 $K7p = 0.0434$

The suffix 'p' denotes the values for a wear plate if it exists.

Note: Dimension a is greater than or equal to $Rm/2$.

Moment per Equation 4.15.1 [M1]:

$$\begin{aligned}
 &= -Q \cdot a [1 - (1 - a/L + (Rm^2 - hm^2)/(2a \cdot L))/(1 + (4hm)/3L)] \\
 &= -3505 \cdot 1050 [1 - (1 - 1050/5450 + (560^2 - 281^2)/(2 \cdot 1050 \cdot 5450))/(1 + (4 \cdot 281)/(3 \cdot 5450))] \\
 &= -829.4 \text{ Kg-m.}
 \end{aligned}$$

Moment per Equation 4.15.2 [M2]:

$$\begin{aligned}
 &= Q \cdot L/4 [(1 + 2(Rm^2 - hm^2)/(L^2))/(1 + (4hm^2)/(3L)) - 4a/L] \\
 &= 3505 \cdot 5450/4 [(1 + 2(560^2 - 281^2)/(5450^2))/(1 + (4 \cdot 281)/(3 \cdot 5450)) - 4 \cdot 1050/5450] \\
 &= 858.9 \text{ Kg-m.}
 \end{aligned}$$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

$$\begin{aligned}
 &= P \cdot Rm/(2t) - M2/(\pi \cdot Rm^2 t) \\
 &= 1085 \cdot 560/(2 \cdot 15) - 859/(\pi \cdot 560^2 \cdot 15) \\
 &= 19.71 \text{ N./mm}^2
 \end{aligned}$$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

$$\begin{aligned}
 &= P \cdot Rm/(2t) + M2/(\pi \cdot Rm^2 \cdot t) \\
 &= 1085 \cdot 560/(2 \cdot 15) + 859/(\pi \cdot 560^2 \cdot 15) \\
 &= 20.85 \text{ N./mm}^2
 \end{aligned}$$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ_3]:

$$\begin{aligned}
 &= P \cdot Rm/(2t) - M1/(K1 \cdot \pi \cdot Rm^2 t) \\
 &= 1085 \cdot 560/(2 \cdot 15) - 829/(0.11 \cdot \pi \cdot 560^2 \cdot 15) \\
 &= 25.43 \text{ N./mm}^2
 \end{aligned}$$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ_4]:

$$\begin{aligned}
 &= P \cdot Rm/(2t) + M1/(K1^* \cdot \pi \cdot Rm^2 \cdot t) \\
 &= 1085 \cdot 560/(2 \cdot 15) + 829/(0.19 \cdot \pi \cdot 560^2 \cdot 15) \\
 &= 17.42 \text{ N./mm}^2
 \end{aligned}$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$\begin{aligned}
 &= Q(L - 2a)/(L + (4 \cdot hm/3)) \\
 &= 3505 (5450 - 2 \cdot 1050)/(5450 + (4 \cdot 281/3)) \\
 &= 2015.9 \text{ Kg}
 \end{aligned}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$\begin{aligned}
 &= K2 \cdot T / (Rm \cdot t) \\
 &= 1.17 \cdot 2016/(560 \cdot 15) \\
 &= 2.75 \text{ N./mm}^2
 \end{aligned}$$

Decay Length (4.15.20) [x_1, x_2]:

$$\begin{aligned}
 &= 0.78 \cdot \sqrt{(Rm \cdot t)} \\
 &= 0.78 \cdot \sqrt{(560 \cdot 15)} \\
 &= 71.520 \text{ mm}
 \end{aligned}$$

Effective reinforcing plate width (4.15.24) [B1]:

$$\begin{aligned}
 &= \min(b + 1.56 \cdot \sqrt{(Rm \cdot t)}, 2a) \\
 &= \min(203 + 1.56 \cdot \sqrt{(560 \cdot 15)}, 2 \cdot 1050) \\
 &= 346.04 \text{ mm}
 \end{aligned}$$

Wear Plate/Shell Stress ratio (4.15.26) [η]:

$$= \min(Sr/S, 1)$$



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$$= \min(138/138, 1)$$

$$= 1.0000$$

Circumferential Stress at Saddle Base with Wear Plate (4.15.25) $[\sigma_6, r]$:

$$= -K5 \cdot Q \cdot k / (B1(t + \eta \cdot tr))$$

$$= -0.76 \cdot 3505 \cdot 0.1 / (346(15 + 1 \cdot 15))$$

$$= -0.25 \text{ N./mm}^2$$

Circ. Comp. Stress at Horn of Saddle, $L \geq 8R_m$ (4.15.28) $[\sigma_7, r]$:

$$= -Q / (4(t + \eta \cdot tr)b1) - 3 \cdot K7 \cdot Q / (2(t + \eta \cdot tr)^2)$$

$$= -3505 / (4(15 + 1 \cdot 15)346) -$$

$$3 \cdot 0.053 \cdot 3505 / (2(15 + 1 \cdot 15)^2)$$

$$= -3.86 \text{ N./mm}^2$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	1150.0000	mm
Baseplate Thickness	Bpthk	20.0000	mm
Baseplate Width	Bpwid	392.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	3	
Rib Thickness	Ribtk	10.0000	mm
Web Thickness	Webtk	10.0000	mm
Web Location	Webloc	Center	
Saddle Design Temperature		21.1	°C
Saddle Yield Stress	Sy	262.0	N./mm ²
Height of Web at Center	Hw,c	306.0	mm
Friction Coefficient	mu	0.400	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})t^3) / 12$$

$$= ((1.56 \cdot \sqrt{(553 \cdot 15)} + 360)15^3) / 12$$

$$= 14.121 \text{ cm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}}(C1 - Y)^2$$

$$= 14.1 + 75.3(182 - 7.5)^2$$

$$= 23035.172 \text{ cm}^4$$

	B mm	D mm	Y mm	A cm ²	AY mm ³	I + AD ⁽²⁾ cm ⁴
Shell	502.1	15.0	7.5	75.3	56484.0	0.230E+05
Wearplate	360.0	15.0	22.5	54.0	121500.0	0.138E+05
Web	10.0	397.0	228.5	39.7	907145.3	0.606E+04
BasePlate	392.0	20.0	437.0	78.4	3426080.5	0.509E+05
Totals	...	...	...	247.4	4511210.0	0.938E+05

Distance to Centroid [C1]:

$$= AY / A$$

$$= 4511210/247$$

$$= 182.336 \text{ mm}$$

Angle [β]:

$$= 180 - \text{Saddle Angle}/2$$

$$= 180 - 120/2$$

$$= 120.0$$

Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta)\cos(\beta))$$

$$= (1 + \cos(120) - 0.5 \cdot \sin(120)^2) / (\pi - 2.09 + \sin(120)\cos(120))$$

$$= 0.2035$$

Saddle Splitting Force [Fh]:



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$$= K1 \cdot Q$$

$$= 0.2 \cdot 3505$$

$$= 713.3715 \text{ Kgf}$$

$$\text{Tension Stress, } St = (Fh/As) = 0.4065 \text{ N./mm}^2$$

$$\text{Allowed Stress, } Sa = 0.6 \cdot \text{Yield Str} = 157.1972 \text{ N./mm}^2$$

Saddle Splitting Dimension [d]:

$$= B - R \cdot \sin(\theta/2) / (\theta/2 \text{ in radians})$$

$$= 1000 - 553 \cdot \sin(120/2) / 1.05$$

$$= 542.673 \text{ mm}$$

$$\text{Bending Moment, } M = Fh \cdot d = 387.1349 \text{ Kg-m.}$$

$$\text{Bending Stress, } Sb = (M \cdot C1 / I) = 0.7382 \text{ N./mm}^2$$

$$\text{Allowed Stress, } Sa = 2/3 \cdot \text{Yield Str} = 174.6635 \text{ N./mm}^2$$

Minimum Thickness of Baseplate per Moss:

$$= \sqrt{3(Q + \text{SaddleWt}) \text{BasePlateWidth} / (4 \cdot \text{BasePlateLength} \cdot \text{AllStress})}$$

$$= \sqrt{3(3505 + 223)392 / (4 \cdot 1150 \cdot 175)}$$

$$= 7.316 \text{ mm}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$$= 2 \cdot \cos(90 - \text{Saddle Angle}/2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$$

$$= 2 \cdot \cos(90 - 120/2) (550 + 18 + 15)$$

$$= 1009.786 \text{ mm}$$

Distance between Ribs [e]:

$$= \text{Web Length} / (\text{Nr ribs} - 1)$$

$$= 1010 / (3 - 1)$$

$$= 504.893 \text{ mm}$$

Baseplate Pressure Area [Ap]:

$$= e \cdot \text{Bpwid} / 2$$

$$= 505 \cdot 392 / 2$$

$$= 989.590 \text{ cm}^2$$

Bearing Pressure [Bp]:

$$= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth})$$

$$= 3729 / (1150 \cdot 392)$$

$$= 0.827 \text{ Kgf/cm}^2$$

Axial Load [P]:

$$= Ap \cdot Bp$$

$$= 990 \cdot 0.83$$

$$= 818.489 \text{ Kgf}$$

Area of the Rib and Web [Ar]:

$$= \text{Rib Area} + \text{Web Area}$$

$$= 19.3 + 25.2$$

$$= 44.545 \text{ cm}^2$$

Compressive Stress [Sc]:

$$= P/Ar$$

$$= 818/44.5$$

$$= 1.802 \text{ N./mm}^2$$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

B	D	Y	A	AY	Io
---	---	---	---	----	----



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Rib+Web	10.0	203.0	...	20.3	...	697.
---------	------	-------	-----	------	-----	------

Rib dimension [D]:

= Saddle Width - Web Thickness
= 203 - 10
= 193.000 mm

Distance to Centroid from Datum [ytot]:

= AY / A
= 0/44.5
= 0.000 mm

Distance to Centroid [C1]:

= Saddle Width / 2
= 203/2
= 101.500 mm

Radius of Gyration [r]:

= $\sqrt{\text{Total Inertia} / \text{Total Area}}$
= $\sqrt{697/44.5}$
= 39.560 mm

Length of Outer Rib [L]:

= Saddle Height - $\cos(\text{theta}/2)$ (radius + shlthk + wpdthk) - bpthk
= 1000 - $\cos(120/2)$ (550 + 18 + 15) - 20
= 688.500 mm

Intermediate Term [Cc]:

= $\sqrt{(2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress})}$
= $\sqrt{(2 \cdot \pi^2 \cdot 199943008/262)}$
= 122.739

Slenderness ratio [KL/r]:

= KL/r
= 1 • 688/39.6
= 17.404

Bending Moment [Rm]:

= FI / (2 • Bplen) • e • L / 2
= 1244 / (2 • 1150) • 505 • 688/2
= 93.987 Kg-m.

Compressive Allowable, KL/r < Cc (17.4 < 123) per AISC E2-1 [Sca]:

= $(1 - (KL/r)^2 / (2 \cdot Cc^2)) Fy / (5/3 + 3 \cdot (KL/r) / (8 \cdot Cc) - (KL/r)^3 / (8 \cdot Cc^3))$
= $(1 - (17.4)^2 / (2 \cdot 123^2)) 262 /$
 $(5/3 + 3 \cdot (17.4) / (8 \cdot 123) - (17.4^3) / (8 \cdot 123^3))$
= 150.8 N./mm²

AISC Unity Check of Outside Ribs (must be ≤ 1)

= Sc/Sca + (Rm • C1 / I) / Sba
= 1.8/151 + (94 • 102/6971188) / 175
= 0.089

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	10.0	193.0	0.0	19.3	0.0	697.
Web	504.9	10.0	0.0	50.5	0.0	4.21
Totals	...	...	...	69.8	...	701.

Distance to Centroid from Datum [ytot]:

= AY / A
= 0/69.8



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= 0.000 mm

Distance to Centroid [C1]:

= Saddle Width / 2

= 203/2

= 101.500 mm

Length of Inner Rib [L]:

= Saddle Height - Outside Radius - Bpthk

= 1000 - cos(583/2)(20 + 0 + 0) - 0

= 397.000 mm

Radius of Gyration [r]:

= $\sqrt{\text{Total Inertia} / \text{Total Area}}$

= $\sqrt{(701/69.8)}$

= 31.699 mm

Slenderness ratio [KL/r]:

= KL/r

= 1 • 397/31.7

= 12.524

Unit Force [Force,u]:

= FI / (2 • Baseplate Length)

= 1244/(2 • 1150)

= 0.541 Kg/mm

Moment at base of inner Rib [Mbase,c]:

= Unit Force • e • L

= 0.54 • 505 • 397

= 108.389 Kg-m.

Bending Stress due to Transverse Force and Weight Load [$\sigma_{B,base,c}$]:

= Bending Moment / Section Modulus

= 108/69088

= 15.385 N./mm²

Compressive Allowable, KL/r < Cc (12.5 < 123) per AISC E2-1 [Sca]:

= $(1 - (KL/r)^2 / (2 \cdot Cc^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot Cc) - (KL/r)^3 / (8 \cdot Cc^3))$

= $(1 - (12.5)^2 / (2 \cdot 123^2)) 262 /$

$(5/3 + 3 \cdot (12.5) / (8 \cdot 123) - (12.5^3) / (8 \cdot 123^3))$

= 152.9 N./mm²

AISC Unity Check of Inside Ribs (must be ≤ 1)

= Sc/Sca + (Mbase,c • C1/I)/Sba

= 1.86/153 + (108 • 102/701)/175

= 0.100

Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate	Nbolts	4
Total Number of Bolts in Tension/Baseplate	Nbt	2
Bolt Design Temperature		21.1 °C
Bolt Material Specification		SA-307 A
Bolt Allowable Stress	Stba	138.00 N./mm ²
Bolt Corrosion Allowance	Bca	1.5 mm
Distance from Bolts to Edge	Edgedis	50.0 mm
User Defined Area in a Single Bolt	UserBoltArea	4.0400 cm ²
BasePlate Allowable Stress	S	138.00 N./mm ²
Area Available in a Single Bolt	BltArea	3.0428 cm ²
Saddle Load Q0 (Weight)	Q0	1881.1 Kg
Saddle Load QL (Wind/Seismic contribution)	QL	401.2 Kg
Maximum Transverse Force	Ft	621.8 Kg
Maximum Longitudinal Force	F1	1243.7 Kg
Saddle Bolted to Steel Foundation	No	



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Shear Stress in a Single Bolt, Longitudinal Direction [τ_{b,l}]:

$$= F_l / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 1244 / (3.04 \cdot 4)$$

$$= 10.0 \text{ N./mm}^2. \text{ Must be less than } 82.8 \text{ N./mm}^2.$$

Shear Stress in a Single Bolt, Transverse Direction [τ_{b,t}]:

$$= F_t / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 622 / (3.04 \cdot 4)$$

$$= 5.0 \text{ N./mm}^2. \text{ Must be less than } 82.8 \text{ N./mm}^2.$$

Bolt Area Calculation per Dennis R. Moss

Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:

$$= 0.0 \text{ (QO > QL --> No Uplift in Longitudinal direction)}$$

Bolt Area due to Shear Load [Bltarears]:

$$= F_l / (\text{BoltShearAllowable} \cdot \text{Nbols})$$

$$= 1244 / (82.8 \cdot 4)$$

$$= 0.3683 \text{ cm}^2$$

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:

$$= B \cdot F_t + \text{Sum of X Moments}$$

$$= 1000 \cdot 622 + 0$$

$$= 621.86 \text{ Kg-m.}$$

Eccentricity (e):

$$= R_{mom} / QO$$

$$= 622 / 1881$$

$$= 330.58 \text{ mm} > B_{plen} / 6 \text{ --> Uplift in Transverse direction}$$

f = B_{plen} / 2 - Edgedis

$$= 1150 / 2 - 50$$

$$= 524.99 \text{ mm}$$

Modular Ratio Of Steel/Concrete (n1):

$$= E_s / E_c$$

$$= 203402 / 21526$$

$$= 9.45$$

K1 = 3 (e - 0.5 • B_{plen})

$$= 3 (331 - 0.5 \cdot 1150)$$

$$= -733.27 \text{ mm}$$

K2 = 6 • n1 • A_t / B_{pwid} • (f + e)

$$= 6 \cdot 9.45 \cdot 6.09 / 392 \cdot (525 + 331)$$

$$= 75301.65 \text{ mm}^2$$

K3 = -K2 • (0.5 • B_{plen} + f)

$$= -75302 \cdot (0.5 \cdot 1150 + 525)$$

$$= -82830870.66 \text{ mm}^3$$

Iteratively Solving for the Effective Bearing Length:

$$Y^3 + K_1 \cdot Y^2 + K_2 \cdot Y + K_3 = 0$$

$$Y^3 + -733 \cdot Y^2 + 75302 \cdot Y + -82830872 = 0$$

$$Y = 774.20 \text{ mm}$$

Num = (B_{plen} / 2 - Y / 3 - e)

$$= (1150 / 2 - 774 / 3 - 331)$$

$$= -13.64$$

Denom = (B_{plen} / 2 - Y / 3 + f)

$$= (1150 / 2 - 774 / 3 + 525)$$



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= 841.92

Total Bolt Tension Force [Tforce]:

= - QO • Num / Denom

= - 1881 • -13.6/842

= 30.48 Kgf

Bolt Area Required due to Transverse Load [Bltareart]:

= Tforce / (Stba • Nbt)

= 30.5/(138 • 2)

= 0.0108 cm²

Required Area of a Single Bolt [Bltarear]:

= max[Bltarearl, Bltarears, Bltareart]

= max[0, 0.37, 0.011]

= 0.3683 cm²

Baseplate Thickness Calculation per D. Moss:

Bearing Pressure (fc)

= 2(QO + Tforce) / (Y • Bpwid)

= 2(1881 + 30.5)/(774 • 392)

= 123.53 KPa.

Distance from Baseplate Edge to the Web [ADIST]:

= (Bplen - WebIngh) / 2

= (1150 - 1099)/2

= 25.4000 mm

Overturning Moment due To Bolt Tension [Mt]:

= Tforce • Adist

= 30.5 • 25.4

= 0.77 Kg-m.

Equivalent Bearing Pressure (f1):

= fc • (Y - Adist) / Y

= 124 • (774 - 25.4)/774

= 119.48 KPa.

Overturning Moment due to Bearing Pressure [Mc]:

= (Adist² • Bpwid / 6) • (f1 + 2 • fc)

= (25.4² • 392/6) • (119 + 2 • 124)

= 1.58 Kg-m.

Baseplate Required Thickness [Treq]:

= $\sqrt{(6 \cdot \max(Mt, Mc) / (Bpwid \cdot Sba))}$

= $\sqrt{(6 \cdot \max(0.77, 1.58) / (392 \cdot 207))}$

= 1.0689 mm

MDMT Calculations for: Right saddle, Saddle to Wearplate Junction, Curve: B

Govrn. thk, tg = 15, c = 3 mm, E* = 0.85

Thickness Ratio = tr • E*/(tg - c) = 0.3, Temp. Reduction = 78 °C

Attachment governing, Attachment stress = Shell stress per Code

Min Metal Temp. w/o impact per UCS-66, Curve B -16 °C

Min Metal Temp. at Required thickness (UCS 66.1) -104 °C

Min Metal Temp. w/o impact per UG-20(f) -29 °C

Review notes about nozzle loadings and supports in the Warnings report.

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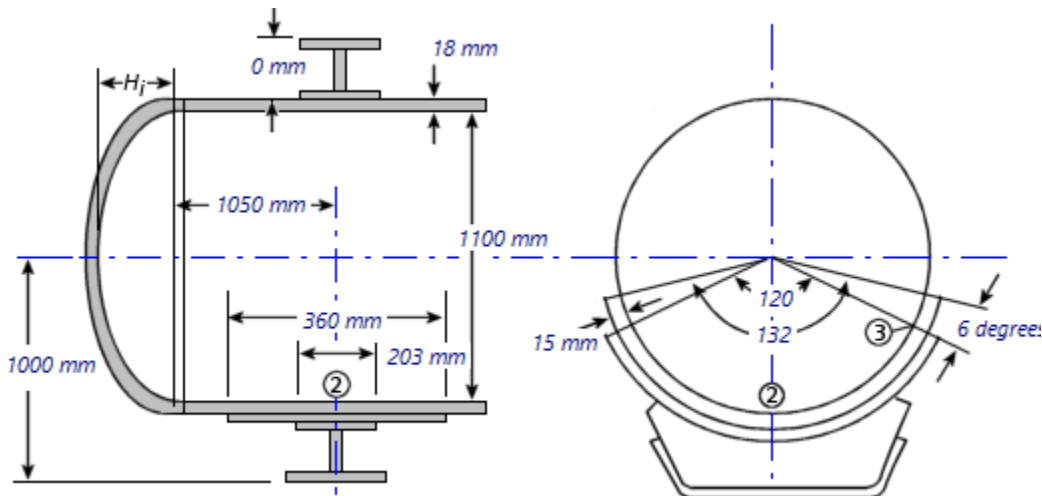
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ASME VIII Division 2 Horizontal Vessel Analysis, Left Saddle:

Horizontal Vessel Stress Calculations : Test Case

Input and Calculated Values:

Vessel Mean Radius	Rm	559.00	mm
Shell Thickness used in this Case	t	18.000	mm
Stiffened Vessel Length per 4.15.6	L	5450.00	mm
Distance from Saddle to Vessel tangent	a1 or a	1050.00	mm
Saddle Width	b1 or b	203.00	mm
Saddle Bearing Angle	delta or theta	120.00	degrees
Wear Plate Width	b2 or b1	360.00	mm
Wear Plate Bearing Angle	delta2 or theta1	132.00	degrees
Wear Plate Thickness	e2 or tr	15.0	mm
Wear Plate Allowable Stress	fw or Sr	138.00	N./mm ²
Shell Allowable Stress used in Calculation		234.00	N./mm ²
Head Allowable Stress used in Calculation		234.00	N./mm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	1000.00	mm
Coefficient of Friction	mu	0.00	



Results for Horizontal Vessel Analysis:

Saddle Force Q, Test Case, no Ext. Forces	19614.91	Kgf
Pressure used in Analysis	P	1409.479 KPa.

Horizontal Vessel Analysis Results:	Actual N./mm ²	Allowable N./mm ²
Long. Stress at Top of Midspan	18.17	234.00
Long. Stress at Bottom of Midspan	25.60	234.00
Long. Stress at Top of Saddles	39.62	234.00
Long. Stress at Bottom of Saddles	12.06	234.00
Tangential Shear in Shell	13.76	138.58
Circ. Stress at Horn of Saddle	18.06	350.99
Circ. Compressive Stress in Shell	1.23	234.00



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Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:

$$= \text{Ftr}(\text{Ft/Num of Saddles} + \text{Z Force Load}) \cdot B / E$$

$$= 3 (254/2 + 0) \cdot 1000/1010$$

$$= 377.7 \text{ Kgf}$$

Longitudinal Saddle Reaction Force [Fwl]:

$$= \max(\text{Fl,wind}; \text{Sum of X Forces}) \cdot B / \text{Ls}$$

$$= \max(100, 0) \cdot 1000/3100$$

$$= 32.3 \text{ Kgf}$$

Load Combination Results for Q + Wind or Seismic [Q]:

$$= \text{Saddle Load} + \max(\text{Fwl}, \text{Fwt}, \text{Fsl}, \text{Fst})$$

$$= 19237 + \max(32.3, 378, 0, 0)$$

$$= 19614.9 \text{ Kgf}$$

Longitudinal Wind Force [Fl]:

$$= \text{WindScalar} \cdot \text{WindPress}(\text{Platform Area} + (\pi/4(\text{OD} \cdot \text{WindDiaMult})^2))$$

$$= 1.1 \cdot 766 (0 + \pi/4(1.44 \cdot 1.2)^2)$$

$$= 1965.311 \text{ N}$$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight) 19838.33 Kgf

Transverse Shear Load 127.13 Kgf

Longitudinal Shear Load 100.20 Kgf

Note: Wear Plate is Welded to the Shell, $k = 0.1$

Saddle Dimension [E]:

$$= \min(2(\text{ShellID}/2 + t + \text{WearPadThickness})\sin(\theta/2), 2 \cdot R_m)$$

$$= \min(2(1100/2 + 18 + 15)\sin(120/2), 2 \cdot 559)$$

$$= 1009.786 \text{ mm}$$

The Computed K values from Table 4.15.1:

K1 = 0.1066 K2 = 1.1707 K3 = 0.8799 K4 = 0.4011

K5 = 0.7603 K6 = 0.0529 K7 = 0.0529 K8 = 0.3405

K9 = 0.2711 K10 = 0.0581 K1* = 0.1923 K6p = 0.0434

K7p = 0.0434

The suffix 'p' denotes the values for a wear plate if it exists.

Note: Dimension a is greater than or equal to $R_m/2$.

Moment per Equation 4.15.1 [M1]:

$$= -Q \cdot a [1 - (1 - a/L + (R_m^2 - h_m^2)/(2a \cdot L))/(1 + (4h_m)/3L)]$$

$$= -19615 \cdot 1050 [1 - (1 - 1050/5450 + (559^2 - 0^2)/(2 \cdot 1050 \cdot 5450))/(1 + (4 \cdot 0)/(3 \cdot 5450))]$$

$$= -3405.7 \text{ Kg-m.}$$

Moment per Equation 4.15.2 [M2]:

$$= Q \cdot L/4 ((1 + 2(R_m^2 - h_m^2)/(L^2))/(1 + (4h_m^2)/(3L)) - 4a/L)$$

$$= 19615 \cdot 5450/4 ((1 + 2(559^2 - 0^2)/(5450^2))/(1 + (4 \cdot 0)/(3 \cdot 5450)) - 4 \cdot 1050/5450)$$

$$= 6692.1 \text{ Kg-m.}$$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

$$= P \cdot R_m/(2t) - M2/(\pi \cdot R_m^2 t)$$

$$= 1409 \cdot 559/(2 \cdot 18) - 6692/(\pi \cdot 559^2 \cdot 18)$$

$$= 18.17 \text{ N/mm}^2$$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

$$= P \cdot R_m/(2t) + M2/(\pi \cdot R_m^2 \cdot t)$$



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$$= 1409 \cdot 559 / (2 \cdot 18) + 6692 / (\pi \cdot 559^2 \cdot 18)$$

$$= 25.60 \text{ N./mm}^2$$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ_{*3}]:

$$= P \cdot R_m / (2t) - M1 / (K1 \cdot \pi \cdot R_m^2 t)$$

$$= 1409 \cdot 559 / (2 \cdot 18) - 3406 / (0.11 \cdot \pi \cdot 559^2 \cdot 18)$$

$$= 39.62 \text{ N./mm}^2$$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ_{*4}]:

$$= P \cdot R_m / (2t) + M1 / (K1 \cdot \pi \cdot R_m^2 \cdot t)$$

$$= 1409 \cdot 559 / (2 \cdot 18) + 3406 / (0.19 \cdot \pi \cdot 559^2 \cdot 18)$$

$$= 12.06 \text{ N./mm}^2$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L - 2a) / (L + (4 \cdot h_m / 3))$$

$$= 19615 \cdot (5450 - 2 \cdot 1050) / (5450 + (4 \cdot 0 / 3))$$

$$= 12056.9 \text{ Kg}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$= K2 \cdot T / (R_m \cdot t)$$

$$= 1.17 \cdot 12057 / (559 \cdot 18)$$

$$= 13.76 \text{ N./mm}^2$$

Decay Length (4.15.20) [x_1, x_2]:

$$= 0.78 \cdot \sqrt{(R_m \cdot t)}$$

$$= 0.78 \cdot \sqrt{(559 \cdot 18)}$$

$$= 78.241 \text{ mm}$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{(R_m \cdot t)}, 2a)$$

$$= \min(203 + 1.56 \cdot \sqrt{(559 \cdot 18)}, 2 \cdot 1050)$$

$$= 359.48 \text{ mm}$$

Wear Plate/Shell Stress ratio (4.15.26) [eta]:

$$= 1.0000 \text{ Materials are the same, test case}$$

Circumferential Stress at Saddle Base with Wear Plate (4.15.25) [σ_6, r]:

$$= -K5 \cdot Q \cdot k / (B1(t + \eta \cdot tr))$$

$$= -0.76 \cdot 19615 \cdot 0.1 / (359(18 + 1 \cdot 15))$$

$$= -1.23 \text{ N./mm}^2$$

Circ. Comp. Stress at Horn of Saddle, $L \geq 8R_m$ (4.15.28) [σ_7, r]:

$$= -Q / (4(t + \eta \cdot tr)b1) - 3 \cdot K7 \cdot Q / (2(t + \eta \cdot tr)^2)$$

$$= -19615 / (4(18 + 1 \cdot 15)359) -$$

$$3 \cdot 0.053 \cdot 19615 / (2(18 + 1 \cdot 15)^2)$$

$$= -18.06 \text{ N./mm}^2$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	1150.0000	mm
Baseplate Thickness	Bpthk	20.0000	mm
Baseplate Width	Bpwid	392.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	3	
Rib Thickness	Ribtk	10.0000	mm
Web Thickness	Webtk	10.0000	mm
Web Location	Webloc	Center	
Saddle Yield Stress	Sy	260.0	N./mm ²
Height of Web at Center	Hw,c	306.0	mm
Friction Coefficient	mu	0.000	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})^3) / 12$$

$$= ((1.56 \cdot \sqrt{(550 \cdot 18)} + 360)^3) / 12$$

$$= 25.040 \text{ cm}^4$$



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Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}}(C1 - Y)^2$$

$$= 25 + 92.7 (173 - 9)^2$$

$$= 25061.312 \text{ cm}^4$$

	B mm	D mm	Y mm	A cm ²	AY mm ³	I + AD ² cm ⁴
Shell	515.2	18.0	9.0	92.7	83465.3	0.251E+05
Wearplate	360.0	15.0	25.5	54.0	137700.0	0.118E+05
Web	10.0	397.0	231.5	39.7	919055.2	0.656E+04
BasePlate	392.0	20.0	440.0	78.4	3449600.8	0.558E+05
Totals	...	...	...	264.8	4589821.5	0.992E+05

Distance to Centroid [C1]:

$$= AY / A$$

$$= 4589821/265$$

$$= 173.306 \text{ mm}$$

Angle [β]:

$$= 180 - \text{Saddle Angle}/2$$

$$= 180 - 120/2$$

$$= 120.0$$

Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta))$$

$$= (1 + \cos(120) - 0.5 \sin(120)^2) / (\pi - 2.09 + \sin(120) \cos(120))$$

$$= 0.2035$$

Saddle Splitting Force [Fh]:

$$= K1 \cdot Q$$

$$= 0.2 \cdot 19615$$

$$= 3992.0591 \text{ Kg}$$

$$\text{Tension Stress, } St = (Fh / As) = 2.2748 \text{ N./mm}^2$$

$$\text{Allowed Stress, } Sa = 0.6 \cdot \text{Yield Str} = 155.9972 \text{ N./mm}^2$$

Saddle Splitting Dimension [d]:

$$= B - R \cdot \sin(\theta/2) / (\theta/2 \text{ in radians})$$

$$= 1000 - 550 \cdot \sin(120/2) / 1.05$$

$$= 545.154 \text{ mm}$$

$$\text{Bending Moment, } M = Fh \cdot d = 2176.3284 \text{ Kg-m.}$$

$$\text{Bending Stress, } Sb = (M \cdot C1 / I) = 3.7280 \text{ N./mm}^2$$

$$\text{Allowed Stress, } Sa = 2/3 \cdot \text{Yield Str} = 173.3302 \text{ N./mm}^2$$

Minimum Thickness of Baseplate per Moss:

$$= \sqrt[3]{(Q + \text{SaddleWt}) \cdot \text{BasePlateWidth} / (4 \cdot \text{BasePlateLength} \cdot \text{AllStress})}$$

$$= \sqrt[3]{(3(19615 + 223) \cdot 392) / (4 \cdot 1150 \cdot 173)}$$

$$= 16.940 \text{ mm}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$$= 2 \cdot \cos(90 - \text{Saddle Angle}/2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$$

$$= 2 \cdot \cos(90 - 120/2) (550 + 18 + 15)$$

$$= 1009.786 \text{ mm}$$

Distance between Ribs [e]:

$$= \text{Web Length} / (N_{\text{ribs}} - 1)$$

$$= 1010 / (3 - 1)$$

$$= 504.893 \text{ mm}$$



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Baseplate Pressure Area [Ap]:

$$= e \cdot Bpwid / 2$$

$$= 505 \cdot 392 / 2$$

$$= 989.590 \text{ cm}^2$$

Bearing Pressure [Bp]:

$$= (Q + SaddleWt) / (BasePlateLength \cdot BasePlateWidth)$$

$$= 19838 / (1150 \cdot 392)$$

$$= 4.401 \text{ Kg/cm}^2$$

Axial Load [P]:

$$= Ap \cdot Bp$$

$$= 990 \cdot 4.4$$

$$= 4354.883 \text{ Kg}$$

Area of the Rib and Web [Ar]:

$$= Rib \text{ Area} + Web \text{ Area}$$

$$= 19.3 + 25.2$$

$$= 44.545 \text{ cm}^2$$

Compressive Stress [Sc]:

$$= P / Ar$$

$$= 4355 / 44.5$$

$$= 9.588 \text{ N./mm}^2$$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	10.0	203.0	...	20.3	...	697.

Rib dimension [D]:

$$= Saddle \text{ Width} - Web \text{ Thickness}$$

$$= 203 - 10$$

$$= 193.000 \text{ mm}$$

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0 / 44.5$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= Saddle \text{ Width} / 2$$

$$= 203 / 2$$

$$= 101.500 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{ (Total \text{ Inertia} / Total \text{ Area}) }$$

$$= \sqrt{ (697 / 44.5) }$$

$$= 39.560 \text{ mm}$$

Length of Outer Rib [L]:

$$= Saddle \text{ Height} - \cos(\theta / 2) (radius + shlthk + wpdthk) - bpthk$$

$$= 1000 - \cos(120 / 2) (550 + 18 + 15) - 20$$

$$= 688.500 \text{ mm}$$

Intermediate Term [Cc]:

$$= \sqrt{ (2 \cdot \pi^2 \cdot Elastic \text{ Modulus} / Yield \text{ Stress}) }$$

$$= \sqrt{ (2 \cdot \pi^2 \cdot 199943008 / 260) }$$

$$= 123.211$$

Slenderness ratio [KL/r]:

$$= KL / r$$



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$$= 1 \cdot 688 / 39.6$$

$$= 17.404$$

Bending Moment [Rm]:

$$= F_l / (2 \cdot B_{pln}) \cdot e \cdot L / 2$$

$$= 100 / (2 \cdot 1150) \cdot 505 \cdot 688 / 2$$

$$= 7.573 \text{ Kg-m.}$$

Compressive Allowable, $KL/r < C_c$ ($17.4 < 123$) per AISC E2-1 [Sca]:

$$= (1 - (Kl/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (Kl/r) / (8 \cdot C_c) - (Kl/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (17.4)^2 / (2 \cdot 123^2)) 260 /$$

$$(5/3 + 3 \cdot (17.4) / (8 \cdot 123) - (17.4^3) / (8 \cdot 123^3))$$

$$= 149.7 \text{ N/mm}^2$$

AISC Unity Check of Outside Ribs (must be ≤ 1)

$$= S_c / S_{ca} + (R_m \cdot C_1 / I) / S_{ba}$$

$$= 9.59 / 150 + (7.57 \cdot 102 / 6971188) / 173$$

$$= 0.070$$

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	10.0	193.0	0.0	19.3	0.0	697.
Web	504.9	10.0	0.0	50.5	0.0	4.21
Totals	...	...	...	69.8	...	701.

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0 / 69.8$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 203 / 2$$

$$= 101.500 \text{ mm}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \text{Outside Radius} - B_{pthk}$$

$$= 1000 - \cos(583/2) (20 + 0 + 0) - 0$$

$$= 397.000 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{(\text{Total Inertia} / \text{Total Area})}$$

$$= \sqrt{(701 / 69.8)}$$

$$= 31.699 \text{ mm}$$

Slenderness ratio [KL/r]:

$$= KL / r$$

$$= 1 \cdot 397 / 31.7$$

$$= 12.524$$

Unit Force [Force,u]:

$$= F_l / (2 \cdot \text{Baseplate Length})$$

$$= 100 / (2 \cdot 1150)$$

$$= 0.044 \text{ Kg/mm}$$

Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 0.044 \cdot 505 \cdot 397$$

$$= 8.733 \text{ Kg-m.}$$

Bending Stress due to Transverse Force and Weight Load [$\sigma_{B,base,c}$]:

$$= \text{Bending Moment} / \text{Section Modulus}$$



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$$= 8.73/69088$$

$$= 1.240 \text{ N./mm}^2$$

Compressive Allowable, $KL/r < C_c$ ($12.5 < 123$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (12.5)^2 / (2 \cdot 123^2)) 260 / (5/3 + 3 \cdot (12.5) / (8 \cdot 123) - (12.5^3) / (8 \cdot 123^3))$$

$$= 151.7 \text{ N./mm}^2$$

AISC Unity Check of Inside Ribs (must be ≤ 1)

$$= S_c / S_{ca} + (M_{base,c} \cdot C1/I) / S_{ba}$$

$$= 9.91/152 + (8.73 \cdot 102/701) / 173$$

$$= 0.072$$

Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate	Nbolts	4
Total Number of Bolts in Tension/Baseplate	Nbt	2
Bolt Material Specification		SA-307 A
Bolt Allowable Stress	Stba	48.30 N./mm ²
Bolt Corrosion Allowance	Bca	1.5 mm
Distance from Bolts to Edge	Edgedis	50.0 mm
User Defined Area in a Single Bolt	UserBoltArea	4.0400 cm ²
BasePlate Allowable Stress	S	138.00 N./mm ²
Area Available in a Single Bolt	BltArea	3.0418 cm ²
Saddle Load QO (Weight)	QO	19460.6 Kg
Saddle Load QL (Wind/Seismic contribution)	QL	32.3 Kg
Maximum Transverse Force	Ft	127.1 Kg
Maximum Longitudinal Force	F1	100.2 Kg
Saddle Bolted to Steel Foundation		No

Shear Stress in a Single Bolt, Longitudinal Direction [tb,l]:

$$= F1 / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 100 / (3.04 \cdot 4)$$

$$= 0.8 \text{ N./mm}^2. \text{ Must be less than } 29.0 \text{ N./mm}^2.$$

Shear Stress in a Single Bolt, Transverse Direction [tb,t]:

$$= Ft / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 127 / (3.04 \cdot 4)$$

$$= 1.0 \text{ N./mm}^2. \text{ Must be less than } 29.0 \text{ N./mm}^2.$$

Bolt Area Calculation per Dennis R. Moss

Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:

$$= 0.0 \text{ (} QO > QL \text{ --> No Uplift in Longitudinal direction)}$$

Bolt Area due to Shear Load [Bltarears]:

$$= F1 / (\text{BoltShearAllowable} \cdot \text{Nbolts})$$

$$= 100 / (29 \cdot 4)$$

$$= 0.0848 \text{ cm}^2$$

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:

$$= B \cdot Ft + \text{Sum of X Moments}$$

$$= 1000 \cdot 127 + 0$$

$$= 127.14 \text{ Kg-m.}$$

Eccentricity (e):

$$= Rmom / QO$$

$$= 127/19461$$

$$= 6.53 \text{ mm} < B_{plen}/6 \text{ --> No Uplift in Transverse direction}$$

Bolt Area due to Transverse Load [Bltareart]:

$$= 0 \text{ (No Uplift)}$$



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Required Area of a Single Bolt [Bltarear]:

= max[Bltarearl, Bltarears, Bltareart]

= max[0, 0.085, 0]

= 0.0848 cm²

ASME VIII Division 2 Horizontal Vessel Analysis, Right Saddle:

Input and Calculated Values:

Vessel Mean Radius	Rm	559.00	mm
Shell Thickness used in this Case	t	18.000	mm
Stiffened Vessel Length per 4.15.6	L	5450.00	mm
Distance from Saddle to Vessel tangent	a1 or a	1050.00	mm
Saddle Width	b1 or b	203.00	mm
Saddle Bearing Angle	delta or theta	120.00	degrees
Wear Plate Width	b2 or b1	360.00	mm
Wear Plate Bearing Angle	delta2 or theta1	132.00	degrees
Wear Plate Thickness	e2 or tr	15.0	mm
Wear Plate Allowable Stress	fw or Sr	138.00	N./mm ²
Inside or Mean Depth of Head	Hi or hm	279.50	mm
Shell Allowable Stress used in Calculation		234.00	N./mm ²
Head Allowable Stress used in Calculation		234.00	N./mm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	1000.00	mm
Coefficient of Friction	mu	0.40	

Results for Horizontal Vessel Analysis:

Saddle Force Q, Test Case, no Ext. Forces		1267.43	Kgf
Pressure used in Analysis	P	1409.479	KPa.

Horizontal Vessel Analysis Results:	Actual N./mm ²	Allowable N./mm ²
Long. Stress at Top of Midspan	21.71	234.00
Long. Stress at Bottom of Midspan	22.06	234.00
Long. Stress at Top of Saddles	23.45	234.00
Long. Stress at Bottom of Saddles	21.02	234.00
Tangential Shear in Shell	0.83	138.58
Circ. Stress at Horn of Saddle	1.17	350.99
Circ. Compressive Stress in Shell	0.08	234.00

Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:

= Ftr(Ft/Num of Saddles + Z Force Load) • B / E

= 3 (254/2 + 0) • 1000/1010

= 377.7 Kgf

Longitudinal Saddle Reaction Force [Fwl]:

= max(Fl,wind; Sum of X Forces) • B / Ls

= max(100, 0) • 1000/3100

= 32.3 Kgf

Load Combination Results for Q + Wind or Seismic [Q]:



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= Saddle Load + max(Fwl, Fwt, Fsl, Fst)
 = 890 + max(32.3, 378, 0, 0)
 = 1267.4 Kgf

Longitudinal Wind Force [FI]:
 = WindScalar • WindPress(Platform Area + ($\pi/4$ (OD • WindDiaMult)²))
 = 1.1 • 766 (0 + $\pi/4$ (1.44 • 1.2)²)
 = 1965.311 N

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	1490.85	Kgf
Transverse Shear Load	127.13	Kgf
Longitudinal Shear Load	100.20	Kgf

Note: Wear Plate is Welded to the Shell, k = 0.1

Saddle Dimension [E]:
 = min(2(ShellID/2 + t + WearPadThickness)sin($\theta/2$), 2•Rm)
 = min(2(1100/2 + 18 + 15)sin(120/2), 2•559)
 = 1009.786 mm

The Computed K values from Table 4.15.1:

K1 = 0.1066	K2 = 1.1707	K3 = 0.8799	K4 = 0.4011
K5 = 0.7603	K6 = 0.0529	K7 = 0.0529	K8 = 0.3405
K9 = 0.2711	K10 = 0.0581	K1* = 0.1923	K6p = 0.0434
K7p = 0.0434			

The suffix 'p' denotes the values for a wear plate if it exists.

Note: Dimension a is greater than or equal to Rm/2.

Moment per Equation 4.15.1 [M1]:
 = -Q•a [1 - (1 - a/L + (Rm²-hm²)/(2a•L))/(1+(4hm)/3L)]
 = -1267•1050 [1 - (1 - 1050/5450 + (559²-280²)/(2•1050•5450))/(1+(4•280)/(3•5450))]
 = -299.7 Kg-m.

Moment per Equation 4.15.2 [M2]:
 = Q•L/4 ((1+2(Rm²-hm²)/(L²))/(1+(4hm²)/(3L)) - 4a/L)
 = 1267•5450/4 ((1+2(559²-280²)/(5450²))/(1+(4•280)/(3•5450)) - 4•1050/5450)
 = 311.1 Kg-m.

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:
 = P • Rm/(2t) - M2/(π •Rm²t)
 = 1409 • 559/(2•18) - 311/(π •559²•18)
 = 21.71 N./mm²

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:
 = P • Rm/(2t) + M2/(π • Rm² • t)
 = 1409 • 559/(2 • 18) + 311/(π • 559² • 18)
 = 22.06 N./mm²

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ_3]:
 = P • Rm/(2t) - M1/(K1• π •Rm²t)
 = 1409•559/(2•18)--300/(0.11• π •559²•18)
 = 23.45 N./mm²

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ_4]:
 = P • Rm/(2t) + M1/(K1* • π • Rm² • t)
 = 1409•559/(2•18)+-300/(0.19• π •559²•18)
 = 21.02 N./mm²

Maximum Shear Force in the Saddle (4.15.3) [T]:
 = Q(L-2a)/(L+(4•hm/3))



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$$= 1267 (5450 - 2 \cdot 1050) / (5450 + (4 \cdot 280/3))$$

$$= 729.2 \text{ Kgf}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$= K_2 \cdot T / (R_m \cdot t)$$

$$= 1.17 \cdot 729 / (559 \cdot 18)$$

$$= 0.83 \text{ N./mm}^2$$

Decay Length (4.15.20) [x_1, x_2]:

$$= 0.78 \cdot \sqrt{(R_m \cdot t)}$$

$$= 0.78 \cdot \sqrt{(559 \cdot 18)}$$

$$= 78.241 \text{ mm}$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{(R_m \cdot t)}, 2a)$$

$$= \min(203 + 1.56 \cdot \sqrt{(559 \cdot 18)}, 2 \cdot 1050)$$

$$= 359.48 \text{ mm}$$

Wear Plate/Shell Stress ratio (4.15.26) [η]:

$$= 1.0000 \text{ Materials are the same, test case}$$

Circumferential Stress at Saddle Base with Wear Plate (4.15.25) [σ_6, r]:

$$= -K_5 \cdot Q \cdot k / (B_1(t + \eta \cdot tr))$$

$$= -0.76 \cdot 1267 \cdot 0.1 / (359(18 + 1 \cdot 15))$$

$$= -0.08 \text{ N./mm}^2$$

Circ. Comp. Stress at Horn of Saddle, $L \geq 8R_m$ (4.15.28) [σ_7, r]:

$$= -Q / (4(t + \eta \cdot tr)b_1) - 3 \cdot K_7 \cdot Q / (2(t + \eta \cdot tr)^2)$$

$$= -1267 / (4(18 + 1 \cdot 15)359) - 3 \cdot 0.053 \cdot 1267 / (2(18 + 1 \cdot 15)^2)$$

$$= -1.17 \text{ N./mm}^2$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	1150.0000	mm
Baseplate Thickness	Bpthk	20.0000	mm
Baseplate Width	Bpwid	392.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	3	
Rib Thickness	Ribtk	10.0000	mm
Web Thickness	Webtk	10.0000	mm
Web Location	Webloc	Center	
Saddle Yield Stress	Sy	260.0	N./mm ²
Height of Web at Center	Hw, c	306.0	mm
Friction Coefficient	mu	0.400	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})t^3) / 12$$

$$= ((1.56 \cdot \sqrt{(550 \cdot 18)} + 360)18^3) / 12$$

$$= 25.040 \text{ cm}^4$$

Resolved Inertia for the Shell [I, shell]:

$$= \text{ShellInertia} + A_{\text{Shell}}(C_1 - Y)^2$$

$$= 25 + 92.7(173 - 9)^2$$

$$= 25061.312 \text{ cm}^4$$

	B mm	D mm	Y mm	A cm ²	AY mm ³	I + AD ² cm ⁴
Shell	515.2	18.0	9.0	92.7	83465.3	0.251E+05
Wearplate	360.0	15.0	25.5	54.0	137700.0	0.118E+05
Web	10.0	397.0	231.5	39.7	919055.2	0.656E+04
BasePlate	392.0	20.0	440.0	78.4	3449600.8	0.558E+05
Totals	...	...	...	264.8	4589821.5	0.992E+05



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Distance to Centroid [C1]:

$$= AY / A$$

$$= 4589821/265$$

$$= 173.306 \text{ mm}$$

Angle [β]:

$$= 180 - \text{Saddle Angle}/2$$

$$= 180 - 120/2$$

$$= 120.0$$

Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta))$$

$$= (1 + \cos(120) - 0.5 \cdot \sin(120)^2) / (\pi - 2.09 + \sin(120) \cos(120))$$

$$= 0.2035$$

Saddle Splitting Force [Fh]:

$$= K1 \cdot Q$$

$$= 0.2 \cdot 1267$$

$$= 257.9485 \text{ Kgf}$$

$$\text{Tension Stress, } St = (Fh / As) = 0.1470 \text{ N./mm}^2$$

$$\text{Allowed Stress, } Sa = 0.6 \cdot \text{Yield Str} = 155.9972 \text{ N./mm}^2$$

Saddle Splitting Dimension [d]:

$$= B - R \cdot \sin(\theta/2) / (\theta/2 \text{ in radians})$$

$$= 1000 - 550 \cdot \sin(120/2) / 1.05$$

$$= 545.154 \text{ mm}$$

$$\text{Bending Moment, } M = Fh \cdot d = 140.6244 \text{ Kg-m.}$$

$$\text{Bending Stress, } Sb = (M \cdot C1 / I) = 0.2409 \text{ N./mm}^2$$

$$\text{Allowed Stress, } Sa = 2/3 \cdot \text{Yield Str} = 173.3302 \text{ N./mm}^2$$

Minimum Thickness of Baseplate per Moss:

$$= \sqrt{(3(Q + \text{SaddleWt}) \cdot \text{BasePlateWidth}) / (4 \cdot \text{BasePlateLength} \cdot \text{AllStress})}$$

$$= \sqrt{(3(1267 + 223) \cdot 392) / (4 \cdot 1150 \cdot 173)}$$

$$= 4.644 \text{ mm}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$$= 2 \cdot \cos(90 - \text{Saddle Angle}/2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$$

$$= 2 \cdot \cos(90 - 120/2) (550 + 18 + 15)$$

$$= 1009.786 \text{ mm}$$

Distance between Ribs [e]:

$$= \text{Web Length} / (\text{Nr ribs} - 1)$$

$$= 1010 / (3 - 1)$$

$$= 504.893 \text{ mm}$$

Baseplate Pressure Area [Ap]:

$$= e \cdot \text{Bpwid} / 2$$

$$= 505 \cdot 392 / 2$$

$$= 989.590 \text{ cm}^2$$

Bearing Pressure [Bp]:

$$= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth})$$

$$= 1491 / (1150 \cdot 392)$$

$$= 0.331 \text{ Kg/cm}^2$$

Axial Load [P]:

$$= Ap \cdot Bp$$

$$= 990 \cdot 0.33$$

$$= 327.270 \text{ Kgf}$$



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Area of the Rib and Web [Ar]:

= Rib Area + Web Area

= 19.3 + 25.2

= 44.545 cm²

Compressive Stress [Sc]:

= P/Ar

= 327/44.5

= 0.721 N./mm²

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	10.0	203.0	...	20.3	...	697.

Rib dimension [D]:

= Saddle Width - Web Thickness

= 203 - 10

= 193.000 mm

Distance to Centroid from Datum [ytot]:

= AY / A

= 0/44.5

= 0.000 mm

Distance to Centroid [C1]:

= Saddle Width / 2

= 203/2

= 101.500 mm

Radius of Gyration [r]:

= $\sqrt{\text{Total Inertia} / \text{Total Area}}$

= $\sqrt{697/44.5}$

= 39.560 mm

Length of Outer Rib [L]:

= Saddle Height - cos(theta/2)(radius + shlthk + wpdthk) - bpthk

= 1000 - cos(120/2)(550 + 18 + 15) - 20

= 688.500 mm

Intermediate Term [Cc]:

= $\sqrt{2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress}}$

= $\sqrt{2 \cdot \pi^2 \cdot 199943008/260}$

= 123.211

Slenderness ratio [KL/r]:

= KL/r

= 1 • 688/39.6

= 17.404

Bending Moment [Rm]:

= FI / (2 • Bplen) • e • L / 2

= 100 / (2 • 1150) • 505 • 688/2

= 7.573 Kg-m.

Compressive Allowable, KL/r < Cc (17.4 < 123) per AISC E2-1 [Sca]:

= $(1 - (KL/r)^2 / (2 \cdot Cc^2)) Fy / (5/3 + 3 \cdot (KL/r) / (8 \cdot Cc) - (KL/r)^3 / (8 \cdot Cc^3))$

= $(1 - (17.4)^2 / (2 \cdot 123^2)) 260 /$

$(5/3 + 3 \cdot (17.4) / (8 \cdot 123) - (17.4^3) / (8 \cdot 123^3))$

= 149.7 N./mm²



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AISC Unity Check of Outside Ribs (must be ≤ 1)

$$= Sc/Sca + (Rm \cdot C1 / I) / Sba$$

$$= 0.72/150 + (7.57 \cdot 102/6971188) / 173$$

$$= 0.011$$

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	10.0	193.0	0.0	19.3	0.0	697.
Web	504.9	10.0	0.0	50.5	0.0	4.21
Totals	...	...	...	69.8	...	701.

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0/69.8$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 203/2$$

$$= 101.500 \text{ mm}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \text{Outside Radius} - Bpthk$$

$$= 1000 - \cos(583/2) (20 + 0 + 0) - 0$$

$$= 397.000 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{ (\text{Total Inertia} / \text{Total Area}) }$$

$$= \sqrt{ (701/69.8) }$$

$$= 31.699 \text{ mm}$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 397/31.7$$

$$= 12.524$$

Unit Force [Force,u]:

$$= FI / (2 \cdot \text{Baseplate Length})$$

$$= 100 / (2 \cdot 1150)$$

$$= 0.044 \text{ Kg/mm}$$

Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 0.044 \cdot 505 \cdot 397$$

$$= 8.733 \text{ Kg-m.}$$

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

$$= \text{Bending Moment} / \text{Section Modulus}$$

$$= 8.73/69088$$

$$= 1.240 \text{ N./mm}^2$$

Compressive Allowable, $KL/r < Cc$ ($12.5 < 123$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot Cc^2)) Fy / (5/3 + 3 \cdot (KL/r) / (8 \cdot Cc) - (KL/r)^3 / (8 \cdot Cc^3))$$

$$= (1 - (12.5)^2 / (2 \cdot 123^2)) 260 /$$

$$(5/3 + 3 \cdot (12.5) / (8 \cdot 123) - (12.5^3) / (8 \cdot 123^3))$$

$$= 151.7 \text{ N./mm}^2$$

AISC Unity Check of Inside Ribs (must be ≤ 1)

$$= Sc/Sca + (Mbase,c \cdot C1 / I) / Sba$$

$$= 0.75/152 + (8.73 \cdot 102/701) / 173$$

$$= 0.012$$



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Saddle Calcs: Test Case: Step: 21 9:45am Sep 9, 2026

Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate	Nbolts	4
Total Number of Bolts in Tension/Baseplate	Nbt	2
Bolt Material Specification		SA-307 A
Bolt Allowable Stress	Stba	48.30 N./mm ²
Bolt Corrosion Allowance	Bca	1.5 mm
Distance from Bolts to Edge	Edgedis	50.0 mm
User Defined Area in a Single Bolt	UserBoltArea	4.0400 cm ²
BasePlate Allowable Stress	S	138.00 N./mm ²
Area Available in a Single Bolt	BltArea	3.0428 cm ²
Saddle Load QO (Weight)	QO	1113.1 Kg
Saddle Load QL (Wind/Seismic contribution)	QL	32.3 Kg
Maximum Transverse Force	Ft	127.1 Kg
Maximum Longitudinal Force	Fl	100.2 Kg
Saddle Bolted to Steel Foundation		No

Shear Stress in a Single Bolt, Longitudinal Direction [tb,l]:

$$= Fl / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 100 / (3.04 \cdot 4)$$

$$= 0.8 \text{ N./mm}^2. \text{ Must be less than } 29.0 \text{ N./mm}^2.$$

Shear Stress in a Single Bolt, Transverse Direction [tb,t]:

$$= Ft / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 127 / (3.04 \cdot 4)$$

$$= 1.0 \text{ N./mm}^2. \text{ Must be less than } 29.0 \text{ N./mm}^2.$$

Bolt Area Calculation per Dennis R. Moss

Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:

$$= 0.0 \text{ (QO > QL --> No Uplift in Longitudinal direction)}$$

Bolt Area due to Shear Load [Bltarears]:

$$= Fl / (\text{BoltShearAllowable} \cdot \text{Nbolts})$$

$$= 100 / (29 \cdot 4)$$

$$= 0.0848 \text{ cm}^2$$

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:

$$= B \cdot Ft + \text{Sum of X Moments}$$

$$= 1000 \cdot 127 + 0$$

$$= 127.14 \text{ Kg-m.}$$

Eccentricity (e):

$$= Rmom / QO$$

$$= 127 / 1113$$

$$= 114.21 \text{ mm} < Bplen/6 \text{ --> No Uplift in Transverse direction}$$

Bolt Area due to Transverse Load [Bltareart]:

$$= 0 \text{ (No Uplift)}$$

Required Area of a Single Bolt [Bltarear]:

$$= \max[\text{Bltarearl}, \text{Bltarears}, \text{Bltareart}]$$

$$= \max[0, 0.085, 0]$$

$$= 0.0848 \text{ cm}^2$$

Review notes about nozzle loadings and supports in the Warnings report.

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Nozzle Calcs.: N3- Tube inlet

Nozl: 13

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Input, Nozzle Desc: N3- Tube inlet

From: 30

Pressure for Reinforcement Calculations	P	6218.147	KPa.
Temperature for Internal Pressure	Temp	430	°C
Design External Pressure	Pext	101.30	KPa.
Temperature for External Pressure	Tempex	200	°C
Parent Material		SA-387 11	
Parent Allowable Stress at Temperature	Sv	147.00	N./mm ²
Parent Allowable Stress At Ambient	Sva	148.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	1100.00	mm
Design Length of Section	L	1199.9701	mm
Shell Finished (Minimum) Thickness	t	32.0000	mm
Shell Internal Corrosion Allowance	c	3.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		839.00	mm
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-335 P11	
Material UNS Number		K11597	
Material Specification/Type		Smls. pipe	
Allowable Stress at Temperature	Sn	98.98	N./mm ²
Allowable Stress At Ambient	Sna	118.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		0.00	deg
Diameter		4.0000	in.
Size and Thickness Basis		Minimum	
Nominal Thickness		XXS	
Flange Material		SA-336 F11	
Flange Type		Weld Neck Flange	
Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	250.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	15.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	26.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Pad Material		SA-387 11	
Pad Allowable Stress at Temperature	Sp	147.00	N./mm ²
Pad Allowable Stress At Ambient	Spa	148.00	N./mm ²
Diameter of Pad along vessel surface	Dp	214.3000	mm
Thickness of Pad	te	28.0000	mm
Weld leg size between Pad and Shell	Wp	15.0000	mm
Groove weld depth between Pad and Nozzle	Wgpn	28.0000	mm
Reinforcing Pad Width		50.0000	mm

The Pressure Design option was MAWP + static head (to the nozzle).



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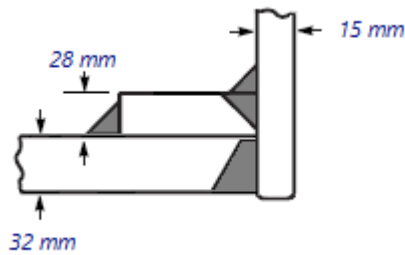
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Nozzle Calcs.: N3- Tube inlet

Noz1: 13

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Reinforcement CALCULATION, Description: N3- Tube inlet

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	3.320 in.
Actual Thickness Used in Calculation	0.590 in.

Note:

Post Weld Heat Treatment is required for this nozzle and it was specified as being heat treated.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]

= $P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)

= $6218 \cdot 553 / (147 \cdot 1 - 0.6 \cdot 6218)$

= 24.0019 mm

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]

= $P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)

= $6218 \cdot 45.2 / (99 \cdot 1 - 0.6 \cdot 6218)$

= 2.9490 mm

Required Nozzle thickness under External Pressure per UG-28 : 0.5092 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	180.6814 mm
--	----	-------------

Parallel to Vessel Wall, opening length	d	90.3407 mm
---	---	------------

Normal to Vessel Wall (Thickness Limit), pad side Tlwp		57.9491 mm
--	--	------------

The Pad diameter is greater than the Diameter Limit. The excess will not be considered.

Weld Strength Reduction Factor [fr1]:

= $\min(1, S_n / S_v)$

= $\min(1, 99 / 147)$

= 0.673

Weld Strength Reduction Factor [fr2]:

= $\min(1, S_n / S_v)$

= $\min(1, 99 / 147)$

= 0.673

Weld Strength Reduction Factor [fr4]:

= $\min(1, S_p / S_v)$

= $\min(1, 147 / 147)$

= 1.000

Weld Strength Reduction Factor [fr3]:

= $\min(fr2, fr4)$

= $\min(0.67, 1)$

= 0.673



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Nozzle Calcs.: N3- Tube inlet

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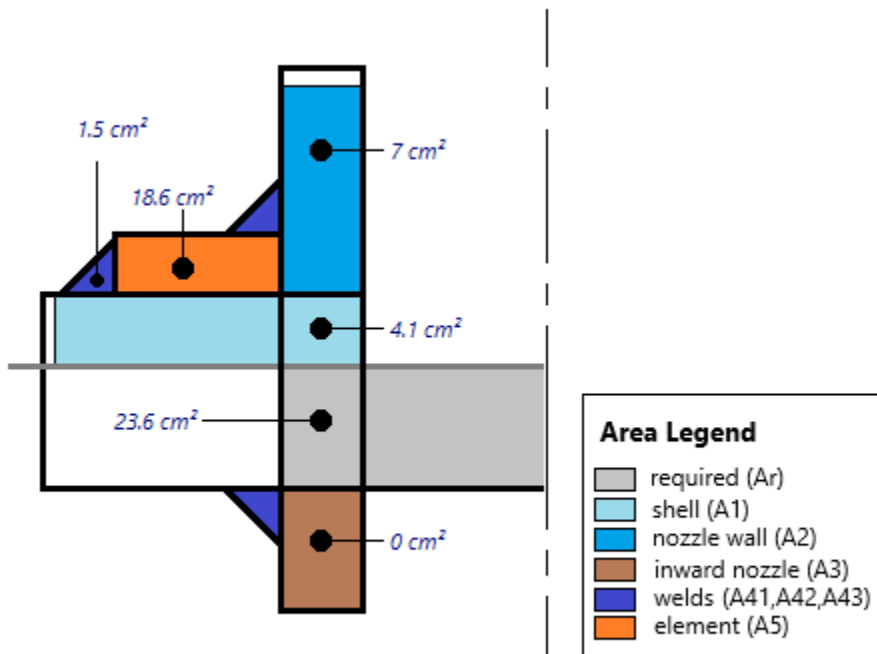


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		MAWP	External	Mapnc
Area Required	Ar	23.562	1.868	NA
Area in Shell	A1	4.124	20.789	NA
Area in Nozzle Wall	A2	7.047	8.951	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	1.515	1.515	NA
Area in Element	A5	18.587	18.587	NA
TOTAL AREA AVAILABLE	Atot	31.273	49.842	NA

The MAWP Case Governs the Analysis.

Nozzle Angle Used in Area Calculations

90.00 degs.

The area available without a pad is Insufficient.
The area available with the given pad is Sufficient.

SELECTION OF POSSIBLE REINFORCING PADS: Diameter Thickness
Based on given Pad Thickness: 153.1418 28.0000 mm
Based on given Pad Diameter: 214.3000 16.3837 mm
Based on the Estimated Diameter Limit: 179.0939 16.7851 mm

Area Required [A]:
= (d • tr • F + 2 • tn • tr • F • (1 - fr1)) UG-37(c)
= (90.3 • 24 • 1 + 2 • 12 • 24 • 1 • (1 - 0.67))
= 23.562 cm²

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
= d (E1 • t - F • tr) - 2 • tn (E1 • t - F • tr) • (1 - fr1)
= 90.3 (1 • 29 - 1 • 24) - 2 • 12
(1 • 29 - 1 • 24) • (1 - 0.67)



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Nozzle Calcs.: N3- Tube inlet Nozl: 13 9:45am Sep 9, 2026

$$= 4.124 \text{ cm}^2$$

Area Available in Nozzle Wall Projecting Outward [A2]:

$$\begin{aligned} &= (2 \cdot Tlwp) \cdot (tn - trn) \cdot fr2 \\ &= (2 \cdot 57.9) \cdot (12 - 2.95) \cdot 0.67 \\ &= 7.047 \text{ cm}^2 \end{aligned}$$

Area Available in Welds [A41 + A42 + A43]:

$$\begin{aligned} &= Wo^2 \cdot fr3 + (Wi-can/0.707)^2 \cdot fr2 + Wp^2 \cdot fr4 \\ &= 15^2 \cdot 0.67 + (0)^2 \cdot 0.67 + 0^2 \cdot 1 \\ &= 1.515 \text{ cm}^2 \end{aligned}$$

Area Available in Element [A5]:

$$\begin{aligned} &= (\min(Dp, DL) - (\text{Nozzle OD})) \cdot (\min(tp, Tlwp, te)) \cdot fr4 \\ &= (181 - 114) \cdot 28 \cdot 1 \\ &= 18.587 \text{ cm}^2 \end{aligned}$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 5.9490 mm
Wall Thickness per UG16(b),	tr16b = 5.5000 mm
Wall Thickness, shell/head, internal pressure	trb1 = 27.0019 mm
Wall Thickness	tb1 = max(trb1, tr16b) = 27.0019 mm
Wall Thickness, shell/head, external pressure	trb2 = 3.3812 mm
Wall Thickness	tb2 = max(trb2, tr16b) = 5.5000 mm
Wall Thickness per table UG-45	tb3 = 8.2578 mm

Determine Nozzle Thickness candidate [tb]:

$$\begin{aligned} &= \min[tb3, \max(tb1, tb2)] \\ &= \min[8.26, \max(27, 5.5)] \\ &= 8.2578 \text{ mm} \end{aligned}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$\begin{aligned} &= \max(ta, tb) \\ &= \max(5.95, 8.26) \\ &= 8.2578 \text{ mm} \end{aligned}$$

Available Nozzle Neck Thickness = 14.9797 mm -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	: 39.7,	Allowable	: 99.0 N./mm ²	Passed
Expansion	: 0.0,	Allowable	: 231.6 N./mm ²	Passed
Occasional	: 10.4,	Allowable	: 131.6 N./mm ²	Passed
Shear	: 20.4,	Allowable	: 69.3 N./mm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, Curve: B

Govrn. thk, tg = 15, tr = 2.46, c = 3 mm, E* = 1
Thickness Ratio = tr * E*/(tg - c) = 0.21, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-16 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Nozzle Neck to Pad Weld for the Nozzle, Curve: B

Govrn. thk, tg = 15, tr = 2.46, c = 3 mm, E* = 1
Thickness Ratio = tr * E*/(tg - c) = 0.21, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-16 °C
--	--------



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Nozzle Calcs.: N3- Tube inlet Nozl: 13 9:45am Sep 9, 2026

Min Metal Temp. at Required thickness (UCS 66.1) -104 °C

Nozzle Neck to Pad Weld for Reinforcement pad, Curve: A

Govrn. thk, $t_g = 15$, $t_r = 2.46$, $c = 3$ mm, $E^* = 1$

Thickness Ratio = $t_r * E^* / (t_g - c) = 0.21$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A 4 °C

Min Metal Temp. at Required thickness (UCS 66.1) -104 °C

Shell to Pad Weld Junction at Pad OD, Curve: A

Govrn. thk, $t_g = 28$, $c = 3$ mm, $E^* = 1$

Thickness Ratio = $t_r * E^* / (t_g - c) = 0.82$, Temp. Reduction = 10 °C

Attachment governing, Attachment stress = Shell stress per Code

Min Metal Temp. w/o impact per UCS-66, Curve A 22 °C

Min Metal Temp. at Required thickness (UCS 66.1) 12 °C

Nozzle-Shell/Head Weld (UCS-66(a)1(b)), min(Curve:B, Curve:A)

Govrn. thk, $t_g = 15$, $t_r = 2.46$, $c = 3$ mm, $E^* = 1$

Thickness Ratio = $t_r * E^* / (t_g - c) = 0.21$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A 4 °C

Min Metal Temp. at Required thickness (UCS 66.1) -104 °C

Gov. MDMT of the Nozzle : -104 °C

Gov. MDMT of the Reinforcement Pad : 12 °C

Gov. MDMT of the nozzle to shell joint welded assembly : 12 °C

Weld Size Calculations, Description: N3- Tube inlet

Intermediate Calc. for nozzle/shell Welds T_{min} 11.9797 mm

Intermediate Calc. for pad/shell Welds T_{minPad} 19.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$6.0000 = \text{Min per Code}$	$10.6050 = 0.7 * W_o$ mm
Pad Weld	$9.5000 = 0.5 * T_{minPad}$	$10.6050 = 0.7 * W_p$ mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:

$$= \max(0, (A - A_1 + 2 \cdot t_n \cdot f_r \cdot (E_1 \cdot t - t_r)) S_v)$$

$$= \max(0, (23.6 - 4.12 + 2 \cdot 12 \cdot 0.67 \cdot (1 \cdot 29 - 24)) 147)$$

$$= 30346.22 \text{ Kg}$$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:

$$= (A_2 + A_5 + A_4 - (W_i - \text{Can} / .707)^2 \cdot f_r \cdot 2) \cdot S_v$$

$$= (7.05 + 18.6 + 1.51 - 0 \cdot 0.67) \cdot 147$$

$$= 40696.48 \text{ Kg}$$

Weld Load [W2]:

$$= (A_2 + A_3 + A_4 + (2 \cdot t_n \cdot t \cdot f_r)) \cdot S_v$$

$$= (7.05 + 0 + 1.51 + (4.68)) \cdot 147$$

$$= 19847.79 \text{ Kg}$$

Weld Load [W3]:

$$= (A_2 + A_3 + A_4 + A_5 + (2 \cdot t_n \cdot t \cdot f_r)) \cdot S$$

$$= (7.05 + 0 + 1.51 + 18.6 + (4.68)) \cdot 147$$

$$= 47709.41 \text{ Kg}$$



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Nozzle Calcs.: N3- Tube inlet

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Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$= (\pi/2) \cdot Dlo \cdot Wo \cdot 0.49 \cdot Snw$$

$$= (3.14/2.0) \cdot 114 \cdot 15 \cdot 0.49 \cdot 99$$

$$= 13319. \text{ Kg}$$

Shear, Pad Element Weld [Spew]:

$$= (\pi/2) \cdot DP \cdot WP \cdot 0.49 \cdot SEW$$

$$= (3.14/2.0) \cdot 214 \cdot 15 \cdot 0.49 \cdot 147$$

$$= 37088. \text{ Kg}$$

Shear, Nozzle Wall [Snw]:

$$= (\pi \cdot (Dlr + Dlo) / 4) \cdot (Thk - Can) \cdot 0.7 \cdot Sn$$

$$= (3.14 \cdot 51.2) \cdot (15 - 3) \cdot 0.7 \cdot 99$$

$$= 13604. \text{ Kg}$$

Tension, Pad Groove Weld [Tpgw]:

$$= (\pi/2) \cdot Dlo \cdot Wgpn \cdot 0.74 \cdot Seg$$

$$= (3.14/2) \cdot 114 \cdot 28 \cdot 0.74 \cdot 147$$

$$= 55765. \text{ Kg}$$

Tension, Shell Groove Weld [Tngw]:

$$= (\pi/2) \cdot Dlo \cdot Wgnvi \cdot 0.74 \cdot Sng$$

$$= (3.14/2.0) \cdot 114 \cdot (26 - 0) \cdot 0.74 \cdot 147$$

$$= 51781. \text{ Kg}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (SPEW + SNW) = (37088 + 13604) = 50691 \text{ Kg}$$

Failure Path [Path 2-2]:

$$= (Sonw + Tpgw + Tngw + Sinw)$$

$$= (13319 + 55765 + 51781 + 0) = 120865 \text{ Kg}$$

Failure Path [Path 3-3]:

$$= (Spew + Tngw + Sinw)$$

$$= (37088 + 51781 + 0) = 88869 \text{ Kg}$$

Summary of Failure Path Calculations:

Path 1-1 = 50691	Kgf, must exceed W = 30346 Kg	or W1 = 40696 Kg
Path 2-2 = 120865	Kgf, must exceed W = 30346 Kg	or W2 = 19847 Kg
Path 3-3 = 88869	Kgf, must exceed W = 30346 Kg	or W3 = 47709 Kg

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 6342 KPa.

Note: The MAWP of this junction was limited by the parent Shell/Head.

Nozzle is O.K. for the External Pressure: 101 KPa.

The Drop for this Nozzle is : 2.9772 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 284.9772 mm

This nozzle or its diameter limit intersects weld seam # [1] on this element. The distance between the nozzle and seam is, 0.00 mm , while the distance from the center of the nozzle to the edge of the diameter limit is: 90.34 mm

Input Echo, WRC297 Item 1, Description: N3- Tube inlet:

Diameter Basis for Cylindrical Shell

ID



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Nozzle Calcs.: N3- Tube inlet Nozl: 13 9:45am Sep 9, 2026

Shell Internal Corrosion Allowance		3.0000	mm
Shell External Corrosion Allowance		0.0000	mm
Shell Diameter		1100.000	mm
Shell Thickness		32.0000	mm
Shell Stress Concentration Factor		1.000	
Vessel Material		SA-387 11	
Vessel Cold S.I. Allowable	Smc	148.00	N./mm ²
Vessel Hot S.I. Allowable	Smh	147.00	N./mm ²
Diameter Basis for Nozzle	ID		
Nozzle Corrosion Allowance		3.0000	mm
Nozzle Diameter		84.341	mm
Nozzle Thickness		14.9797	mm
Nozzle Stress Concentration Factor		1.000	
Nozzle Material		SA-335 P11	
Nozzle Cold S.I. Allowable	SNmc	118.00	N./mm ²
Nozzle Hot S.I. Allowable	SNmh	98.98	N./mm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter of Reinforcing Pad	214.300	mm
Thickness of Reinforcing Pad	28.0000	mm

Note:

External Forces and Moments in WRC 107/537 Convention:
These loads are assumed to be SUSained loads.

Design Internal Pressure	Dp	6218.15	KPa.
Radial Load	P	-917.00	Kgf
Circumferential Shear	Vc	663.00	Kgf
Longitudinal Shear	Vl	663.00	Kgf
Circumferential Moment	Mc	204.00	Kg-m.
Longitudinal Moment	Ml	204.00	Kg-m.
Torsional Moment	Mt	235.00	Kg-m.

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

Warning - The ratio, Dn/Tv, (3.572) is < 5
check the Limitations of WRC bulletin 297.

Warning - The ratio, Dn/Tn (7.630) is < 10
check the Limitations of WRC bulletin 297.

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{R_{\text{mean}}(t - ca)}$$

$$= 114.3 + 2 \cdot 1.65 \cdot \sqrt{581.5(32.0 - 3.0)}$$

$$= 542.836 \text{ mm}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	1163.000	mm
Nozzle Outside Diameter	(d) =	114.300	mm
Vessel Thickness used	(T) =	57.000	mm
Nozzle Thickness used	(t) =	11.980	mm
T / t	=	4.758	
d / t	=	10.000	

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 0.444$$

Nr/P	=	0.188
Mr/P	=	0.121



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MØ/P = 0.259
 NØ/P = 0.254
 MrD/Mc = 0.121
 NrDL/Mc = 0.160
 MØd/Mc = 0.289
 NØDL/Mc = -0.107
 MrD/Ml = 0.120
 NrDL/Ml = 0.162
 MØD/Ml = 0.282
 NØDL/Ml = -0.048

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	0	0	0	0
Outplane Bending (P)	4	-4	4	-4
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	-9	9	9	-9
Normal Pressure Stress	57	63	57	63
Outplane Stress Summary	53	69	71	50

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	0	0	0	0
Inplane Bending (P)	2	-2	2	-2
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	-3	3	3	-3
Inplane Pressure Stress	28	28	28	28
Inplane Stress Summary	26	30	35	24

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	1	1	1	1
Shear Stress Summary	2	2	1	1

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Two * Max Shear Stress	53	69	71	50

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Membrane (P)	0	0	0	0



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Outplane Bending (P)	4	-4	4	-4
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	-9	9	9	-9
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	28	28	28	28

Outplane Stress Summary	24	35	42	15
-------------------------	----	----	----	----

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses parallel to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside

Inplane Membrane (P)	0	0	0	0
Inplane Bending (P)	2	-2	2	-2
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	-3	3	3	-3
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	57	63	57	63

Inplane Stress Summary	55	65	64	59
------------------------	----	----	----	----

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Shear stress normal to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside

Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Torsional Shear (Mt)	1	1	1	1

Shear Stress Summary	1	1	2	2
----------------------	---	---	---	---

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stress Intensities)	Left Outside	Left Inside	Right Outside	Right Inside

Two * Max Shear Stress	55	65	64	59
------------------------	----	----	----	----

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stresses in the hoop direction)	Top Outside	Top Inside	Bottom Outside	Bottom Inside

Hoop Membrane (P)	0	0	0	0
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	24	32	24	32

Hoop Stress Summary	25	33	25	32
---------------------	----	----	----	----

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stresses Normal to pipe cross-section)	Top Outside	Top Inside	Bottom Outside	Bottom Inside

Axial Membrane (P)	2	2	2	2
Axial Bending (P)	10	-10	10	-10
Axial Membrane (Mc)	0	0	0	0



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Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-22	-22	22	22
Axial Bending (ML)	-28	28	28	-28
Axial Pressure Stress	12	12	12	12

Axial Stress Summary	-25	10	75	-1
----------------------	-----	----	----	----

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Shear stress)	Outside	Inside	Outside	Inside

Shear due to (Vc)	3	3	-3	-3
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	9	9	9	9

Shear Stress Summary	12	12	6	6
----------------------	----	----	---	---

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stress Intensities	Outside	Inside	Outside	Inside

Two * Max Shear Stress	57	38	76	36
------------------------	----	----	----	----

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses in the	Left	Left	Right	Right
hoop direction)	Outside	Inside	Outside	Inside

Hoop Membrane (P)	0	0	0	0
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	24	32	24	32

Hoop Stress Summary	26	33	24	32
---------------------	----	----	----	----

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses Normal to	Left	Left	Right	Right
pipe cross-section)	Outside	Inside	Outside	Inside

Axial Membrane (P)	2	2	2	2
Axial Bending (P)	10	-10	10	-10
Axial Membrane (Mc)	-22	-22	22	22
Axial Bending (Mc)	-30	30	30	-30
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	12	12	12	12

Axial Stress Summary	-27	12	77	-3
----------------------	-----	----	----	----

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Shear stress)	Outside	Inside	Outside	Inside

Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	-3	-3	3	3
Shear due to Torsion	9	9	9	9

Shear Stress Summary	6	6	12	12
----------------------	---	---	----	----

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
-----------------------	----	----	----	----



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(Stress Intensities	Outside	Inside	Outside	Inside
Two * Max Shear Stress	55	35	80	43

Stress Computations at the Edge of the Reinforcing Pad:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(\text{Rmean}(t - ca))}$$

$$= 114.3 + 2 \cdot 1.65 \cdot \sqrt{567.5(32.0 - 3.0)}$$

$$= 537.646 \text{ mm}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	1135.000 mm
Nozzle Outside Diameter	(d) =	214.300 mm
Vessel Thickness used	(T) =	29.000 mm
Nozzle Thickness used	(t) =	61.980 mm
T / t	=	0.468
d / t	=	10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 1.181$$

Note: Re-pad thickness is added to nozzle thickness to simulate the Re-pad.

Nr/P	=	0.096
Mr/P	=	0.202
M0/P	=	0.059
N0/P	=	0.123
MrD/Mc	=	0.305
NrDL/Mc	=	0.092
M0d/Mc	=	0.095
N0DL/Mc	=	0.048
MrD/Ml	=	0.174
NrDL/Ml	=	0.085
M0D/Ml	=	0.056
N0DL/Ml	=	0.127

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	1	1	1	1
Outplane Bending (P)	3	-3	3	-3
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-1	-1	1	1
Outplane Bending (ML)	-3	3	3	-3
Normal Pressure Stress	115	121	115	121
Outplane Stress Summary	115	121	125	116

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	1	1	1	1
Inplane Bending (P)	12	-12	12	-12
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	-11	11	11	-11
Inplane Pressure Stress	57	57	57	57



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Inplane Stress Summary	59	56	84	35
------------------------	----	----	----	----

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Shear stress normal to longitudinal plane)	Top Outside	Top Inside	Bottom Outside	Bottom Inside

Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	1	1	1	1

Shear Stress Summary	1	1	0	0
----------------------	---	---	---	---

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stress Intensities)	Top Outside	Top Inside	Bottom Outside	Bottom Inside

Two * Max Shear Stress	115	121	125	116
------------------------	-----	-----	-----	-----

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses Normal to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside

Outplane Membrane (P)	1	1	1	1
Outplane Bending (P)	3	-3	3	-3
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	-6	6	6	-6
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	57	57	57	57

Outplane Stress Summary	56	61	69	49
-------------------------	----	----	----	----

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses parallel to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside

Inplane Membrane (P)	1	1	1	1
Inplane Bending (P)	12	-12	12	-12
Inplane Membrane (Mc)	-1	-1	1	1
Inplane Bending (Mc)	-20	20	20	-20
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	115	121	115	121

Inplane Stress Summary	108	129	150	90
------------------------	-----	-----	-----	----

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Shear stress normal to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside

Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Torsional Shear (Mt)	1	1	1	1

Shear Stress Summary	0	0	1	1
----------------------	---	---	---	---

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stress Intensities)	Left	Left	Right	Right



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	Outside	Inside	Outside	Inside
Two * Max Shear Stress	108	129	150	90

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	57	63	57	63	57	63	57	63
Circ. Pl (SUS)	0	0	0	0	0	0	0	0
Circ. Q (SUS)	-5	5	13	-13	-1	1	5	-5
Long. Pm (SUS)	28	28	28	28	28	28	28	28
Long. Pl (SUS)	0	0	0	0	0	0	0	0
Long. Q (SUS)	-1	1	5	-5	-5	5	13	-13
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	1	1	1	1	1	1	1	1
Pm (SUS)	57.0	63.0	57.0	63.0	57.0	63.0	57.0	63.0
Pm+Pl (SUS)	57.0	63.0	57.0	63.0	57.0	63.0	57.0	63.0
Pm+Pl+Q (Total)	52.0	68.0	70.0	50.0	56.0	64.0	62.0	58.0

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	63.00	147.00	Passed
Pm+Pl (SUS)	63.00	220.51	Passed
Pm+Pl+Q (TOTAL)	70.03	441.01	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Reinforcing Pad Edge (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	115	121	115	121	115	121	115	121
Circ. Pl (SUS)	0	0	2	2	0	0	2	2
Circ. Q (SUS)	0	0	6	-6	-8	8	32	-32
Long. Pm (SUS)	57	57	57	57	57	57	57	57
Long. Pl (SUS)	1	1	1	1	1	1	1	1
Long. Q (SUS)	1	-1	23	-23	-3	3	9	-9
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	1	1	1	1	1	1	1	1
Pm (SUS)	115.0	121.0	115.0	121.0	115.0	121.0	115.0	121.0
Pm+Pl (SUS)	115.0	121.0	117.0	123.0	115.0	121.0	117.0	123.0



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Pm+Pl+Q (Total)	115.0	121.0	123.0	117.0	107.0	129.0	149.0	91.0
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Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	121.00	147.00	Passed
Pm+Pl (SUS)	123.00	220.51	Passed
Pm+Pl+Q (TOTAL)	149.01	441.01	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	24	32	24	32	24	32	24	32
Circ. Pl (SUS)	0	0	0	0	0	0	0	0
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	12	12	12	12	12	12	12	12
Long. Pl (SUS)	-20	-20	24	24	-20	-20	24	24
Long. Q (SUS)	-18	18	38	-38	-20	20	40	-40
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	3	3	-3	-3	-3	-3	3	3
Shear Q (SUS)	9	9	9	9	9	9	9	9
Pm (SUS)	24.0	32.0	24.0	32.0	24.0	32.0	24.0	32.0
Pm+Pl (SUS)	32.6	40.4	36.7	37.6	32.6	40.4	36.7	37.6
Pm+Pl+Q (Total)	55.5	37.3	74.7	36.1	53.4	33.7	78.6	43.3

Nozzle Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	32.00	98.98	Passed
Pm+Pl (SUS)	40.45	148.47	Passed
Pm+Pl+Q (TOTAL)	78.64	296.95	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

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Input, Nozzle Desc: Tube outlet

From: 30

Pressure for Reinforcement Calculations	P	6228.929	KPa.
Temperature for Internal Pressure	Temp	430	°C
Design External Pressure	Pext	101.30	KPa.
Temperature for External Pressure	Tempex	200	°C
Parent Material		SA-387 11	
Parent Allowable Stress at Temperature	Sv	147.00	N./mm ²
Parent Allowable Stress At Ambient	Sva	148.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	1100.00	mm
Design Length of Section	L	1199.9701	mm
Shell Finished (Minimum) Thickness	t	32.0000	mm
Shell Internal Corrosion Allowance	c	3.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		939.00	mm
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-335 P11	
Material UNS Number		K11597	
Material Specification/Type		Smls. pipe	
Allowable Stress at Temperature	Sn	98.98	N./mm ²
Allowable Stress At Ambient	Sna	118.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		180.00	deg
Diameter		3.0000	in.
Size and Thickness Basis		Minimum	
Nominal Thickness		XXS	
Flange Material		SA-182 F11	
Flange Type		Weld Neck Flange	
Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	250.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	15.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	26.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Pad Material		SA-387 11	
Pad Allowable Stress at Temperature	Sp	147.00	N./mm ²
Pad Allowable Stress At Ambient	Spa	148.00	N./mm ²
Diameter of Pad along vessel surface	Dp	214.3000	mm
Thickness of Pad	te	28.0000	mm
Weld leg size between Pad and Shell	Wp	15.0000	mm
Groove weld depth between Pad and Nozzle	Wgpn	28.0000	mm
Reinforcing Pad Width		62.7000	mm

The Pressure Design option was MAWP + static head (to the nozzle).



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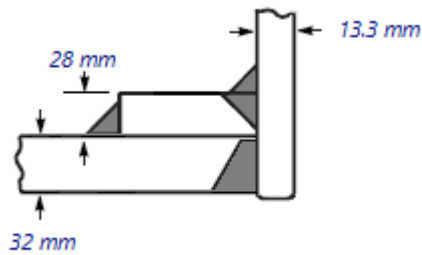
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Reinforcement CALCULATION, Description: Tube outlet

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	2.450 in.
Actual Thickness Used in Calculation	0.525 in.

Note:

Post Weld Heat Treatment is required for this nozzle and it was specified as being heat treated.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 6229 \cdot 553 / (147 \cdot 1 - 0.6 \cdot 6229)$
 $= 24.0446 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 6229 \cdot 34.1 / (99 \cdot 1 - 0.6 \cdot 6229)$
 $= 2.2312 \text{ mm}$

Required Nozzle thickness under External Pressure per UG-28 : 0.4421 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	146.9000 mm
Parallel to Vessel Wall	Rn+tn+t	73.4500 mm
Normal to Vessel Wall (Thickness Limit), pad side Tlwp		53.8375 mm

The Pad diameter is greater than the Diameter Limit. The excess will not be considered.

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 99 / 147)$
 $= 0.673$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 99 / 147)$
 $= 0.673$

Weld Strength Reduction Factor [fr4]:
 $= \min(1, S_p / S_v)$
 $= \min(1, 147 / 147)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(0.67, 1)$
 $= 0.673$



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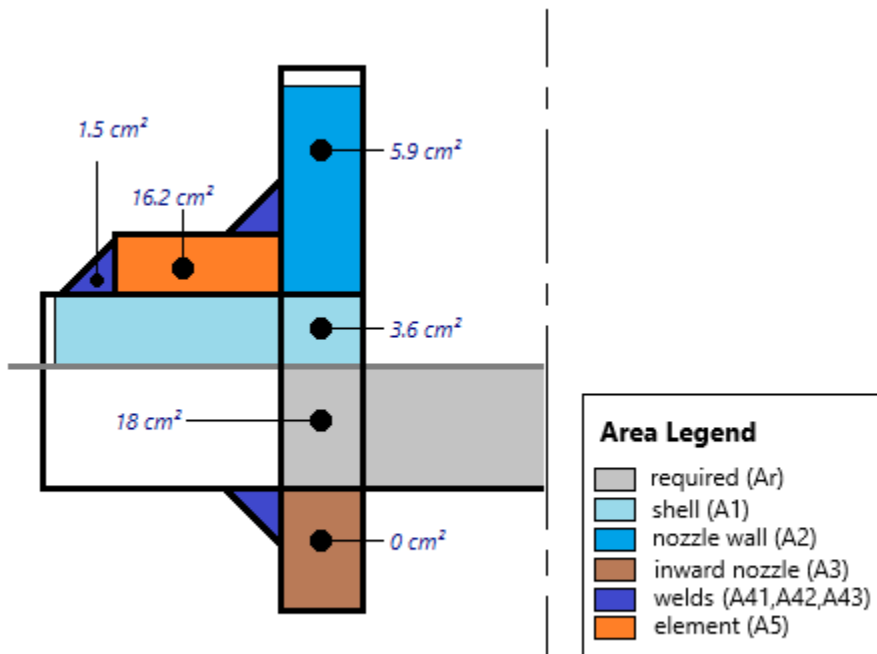


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		MAWP	External	Mapnc
Area Required	Ar	18.029	1.427	NA
Area in Shell	A1	3.564	18.119	NA
Area in Nozzle Wall	A2	5.875	7.172	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	1.515	1.515	NA
Area in Element	A5	16.240	16.240	NA
TOTAL AREA AVAILABLE	Atot	27.194	43.047	NA

The MAWP Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Insufficient.
The area available with the given pad is Sufficient.

SELECTION OF POSSIBLE REINFORCING PADS: Diameter Thickness

Based on given Pad Thickness: 114.1685 28.0000 mm

Based on given Pad Diameter: 214.3000 12.1986 mm

Based on Shell or Nozzle Thickness: 141.9572 13.3350 mm

Area Required [A]:

$$= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$$

$$= (68.2 \cdot 24 \cdot 1 + 2 \cdot 10.3 \cdot 24 \cdot 1 \cdot (1 - 0.67))$$

$$= 18.029 \text{ cm}^2$$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

$$= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$$

$$= 78.7(1 \cdot 29 - 1 \cdot 24) - 2 \cdot 10.3$$

$$(1 \cdot 29 - 1 \cdot 24) \cdot (1 - 0.67)$$



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$$= 3.564 \text{ cm}^2$$

Area Available in Nozzle Wall Projecting Outward [A2]:

$$= (2 \cdot T_{lwp}) \cdot (t_n - t_{rn}) \cdot f_{r2}$$

$$= (2 \cdot 53.8) \cdot (10.3 - 2.23) \cdot 0.67$$

$$= 5.875 \text{ cm}^2$$

Area Available in Welds [A41 + A42 + A43]:

$$= W_o^2 \cdot f_{r3} + (W_i - c_{an}/0.707)^2 \cdot f_{r2} + W_p^2 \cdot f_{r4}$$

$$= 15^2 \cdot 0.67 + (0)^2 \cdot 0.67 + 0^2 \cdot 1$$

$$= 1.515 \text{ cm}^2$$

Area Available in Element [A5]:

$$= (\min(D_p, D_L) - (\text{Nozzle OD})) \cdot (\min(t_p, T_{lwp}, t_e)) \cdot f_{r4}$$

$$= (147 - 88.9) \cdot 28 \cdot 1$$

$$= 16.240 \text{ cm}^2$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures $t_a = 5.2312 \text{ mm}$

Wall Thickness per UG16(b), $tr_{16b} = 5.5000 \text{ mm}$

Wall Thickness, shell/head, internal pressure $tr_{b1} = 27.0446 \text{ mm}$

Wall Thickness $tb_1 = \max(tr_{b1}, tr_{16b}) = 27.0446 \text{ mm}$

Wall Thickness, shell/head, external pressure $tr_{b2} = 3.3812 \text{ mm}$

Wall Thickness $tb_2 = \max(tr_{b2}, tr_{16b}) = 5.5000 \text{ mm}$

Wall Thickness per table UG-45 $tb_3 = 7.8000 \text{ mm}$

Determine Nozzle Thickness candidate [tb]:

$$= \min[tb_3, \max(tb_1, tb_2)]$$

$$= \min[7.8, \max(27, 5.5)]$$

$$= 7.8000 \text{ mm}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= \max(t_a, t_b)$$

$$= \max(5.23, 7.8)$$

$$= 7.8000 \text{ mm}$$

Available Nozzle Neck Thickness = 13.3350 mm -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	:	46.4,	Allowable	:	99.0 N./mm ²	Passed
Expansion	:	0.0,	Allowable	:	224.8 N./mm ²	Passed
Occasional	:	8.9,	Allowable	:	131.6 N./mm ²	Passed
Shear	:	26.1,	Allowable	:	69.3 N./mm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, Curve: B

Govrn. thk, $t_g = 13.3$, $t_r = 1.86$, $c = 3 \text{ mm}$, $E^* = 1$

Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.18$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -20 °C

Min Metal Temp. at Required thickness (UCS 66.1) -104 °C

Nozzle Neck to Pad Weld for the Nozzle, Curve: B

Govrn. thk, $t_g = 13.3$, $t_r = 1.86$, $c = 3 \text{ mm}$, $E^* = 1$

Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.18$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -20 °C



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Min Metal Temp. at Required thickness (UCS 66.1) -104 °C

Nozzle Neck to Pad Weld for Reinforcement pad, Curve: A

Govrn. thk, $t_g = 13.3$, $t_r = 1.86$, $c = 3$ mm, $E^* = 1$

Thickness Ratio = $t_r * E^* / (t_g - c) = 0.18$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A 0 °C

Min Metal Temp. at Required thickness (UCS 66.1) -104 °C

Shell to Pad Weld Junction at Pad OD, Curve: A

Govrn. thk, $t_g = 28$, $c = 3$ mm, $E^* = 1$

Thickness Ratio = $t_r * E^* / (t_g - c) = 0.82$, Temp. Reduction = 10 °C

Attachment governing, Attachment stress = Shell stress per Code

Min Metal Temp. w/o impact per UCS-66, Curve A 22 °C

Min Metal Temp. at Required thickness (UCS 66.1) 12 °C

Nozzle-Shell/Head Weld (UCS-66(a)1(b)), min(Curve:B, Curve:A)

Govrn. thk, $t_g = 13.3$, $t_r = 1.86$, $c = 3$ mm, $E^* = 1$

Thickness Ratio = $t_r * E^* / (t_g - c) = 0.18$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A 0 °C

Min Metal Temp. at Required thickness (UCS 66.1) -104 °C

Gov. MDMT of the Nozzle : -104 °C

Gov. MDMT of the Reinforcement Pad : 12 °C

Gov. MDMT of the nozzle to shell joint welded assembly : 12 °C

Weld Size Calculations, Description: Tube outlet

Intermediate Calc. for nozzle/shell Welds T_{min} 10.3350 mm

Intermediate Calc. for pad/shell Welds T_{minPad} 19.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$6.0000 = \text{Min per Code}$	$10.6050 = 0.7 * W_o$ mm
Pad Weld	$9.5000 = 0.5 * T_{minPad}$	$10.6050 = 0.7 * W_p$ mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:

$$= \max(0, (A - A1 + 2 \cdot t_n \cdot f_r1 \cdot (E1 \cdot t - t_r)) \cdot S_v)$$

$$= \max(0, (18 - 3.56 + 2 \cdot 10.3 \cdot 0.67 \cdot (1 \cdot 29 - 24)) \cdot 147)$$

$$= 22717.50 \text{ Kg}$$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:

$$= (A2 + A5 + A4 - (W_i - \text{Can} / .707)^2 \cdot f_r2) \cdot S_v$$

$$= (5.88 + 16.2 + 1.51 - 0 \cdot 0.67) \cdot 147$$

$$= 35421.77 \text{ Kg}$$

Weld Load [W2]:

$$= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot f_r1)) \cdot S_v$$

$$= (5.88 + 0 + 1.51 + (4.04)) \cdot 147$$

$$= 17128.13 \text{ Kg}$$

Weld Load [W3]:

$$= (A2 + A3 + A4 + A5 + (2 \cdot t_n \cdot t \cdot f_r1)) \cdot S$$

$$= (5.88 + 0 + 1.51 + 16.2 + (4.04)) \cdot 147$$

$$= 41471.91 \text{ Kg}$$



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Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$$

$$= (3.14/2.0) \cdot 88.9 \cdot 15 \cdot 0.49 \cdot 99$$

$$= 10359. \text{ Kg}$$

Shear, Pad Element Weld [Spew]:

$$= (\pi/2) \cdot D_P \cdot W_P \cdot 0.49 \cdot S_{EW}$$

$$= (3.14/2.0) \cdot 214 \cdot 15 \cdot 0.49 \cdot 147$$

$$= 37088. \text{ Kg}$$

Shear, Nozzle Wall [Snw]:

$$= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - Can) \cdot 0.7 \cdot S_n$$

$$= (3.14 \cdot 39.3) \cdot (13.3 - 3) \cdot 0.7 \cdot 99$$

$$= 9011. \text{ Kg}$$

Tension, Pad Groove Weld [Tpgw]:

$$= (\pi/2) \cdot D_{lo} \cdot W_{gpn} \cdot 0.74 \cdot S_{eg}$$

$$= (3.14/2) \cdot 88.9 \cdot 28 \cdot 0.74 \cdot 147$$

$$= 43372. \text{ Kg}$$

Tension, Shell Groove Weld [Tngw]:

$$= (\pi/2) \cdot D_{lo} \cdot W_{gvi} \cdot 0.74 \cdot S_{ng}$$

$$= (3.14/2.0) \cdot 88.9 \cdot (26 - 0) \cdot 0.74 \cdot 147$$

$$= 40274. \text{ Kg}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (S_{PEW} + S_{NW}) = (37088 + 9011) = 46099 \text{ Kg}$$

Failure Path [Path 2-2]:

$$= (S_{onw} + T_{pgw} + T_{ngw} + S_{inw})$$

$$= (10359 + 43372 + 40274 + 0) = 94006 \text{ Kg}$$

Failure Path [Path 3-3]:

$$= (S_{pew} + T_{ngw} + S_{inw})$$

$$= (37088 + 40274 + 0) = 77362 \text{ Kg}$$

Summary of Failure Path Calculations:

Path 1-1 = 46099	Kgf, must exceed W = 22717 Kg	or W1 = 35421 Kg
Path 2-2 = 94006	Kgf, must exceed W = 22717 Kg	or W2 = 17128 Kg
Path 3-3 = 77362	Kgf, must exceed W = 22717 Kg	or W3 = 41471 Kg

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 6342 KPa.

Note: The MAWP of this junction was limited by the parent Shell/Head.

Nozzle is O.K. for the External Pressure: 101 KPa.

The Drop for this Nozzle is : 1.7991 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 283.7991 mm

Input Echo, WRC297 Item 1, Description: Tube outlet :

Diameter Basis for Cylindrical Shell	ID
Shell Internal Corrosion Allowance	3.0000 mm
Shell External Corrosion Allowance	0.0000 mm
Shell Diameter	1100.000 mm
Shell Thickness	32.0000 mm



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Nozzle Calcs.: Tube outlet Nozl: 14 9:45am Sep 9, 2026

Shell Stress Concentration Factor		1.000	
Vessel Material		SA-387 11	
Vessel Cold S.I. Allowable	Smc	148.00	N./mm ²
Vessel Hot S.I. Allowable	Smh	147.00	N./mm ²
Diameter Basis for Nozzle		ID	
Nozzle Corrosion Allowance		3.0000	mm
Nozzle Diameter		62.230	mm
Nozzle Thickness		13.3350	mm
Nozzle Stress Concentration Factor		1.000	
Nozzle Material		SA-335 P11	
Nozzle Cold S.I. Allowable	SNmc	118.00	N./mm ²
Nozzle Hot S.I. Allowable	SNmh	98.98	N./mm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter of Reinforcing Pad	214.300	mm
Thickness of Reinforcing Pad	28.0000	mm

Note:

External Forces and Moments in WRC 107/537 Convention:

These loads are assumed to be SUSained loads.

Design Internal Pressure	Dp	6228.93	KPa.
Radial Load	P	-734.00	Kgf
Circumferential Shear	Vc	530.00	Kgf
Longitudinal Shear	Vl	530.00	Kgf
Circumferential Moment	Mc	131.00	Kg-m.
Longitudinal Moment	Ml	131.00	Kg-m.
Torsional Moment	Mt	150.00	Kg-m.

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

Warning - The ratio, Dn/Tv, (2.778) is < 5
check the Limitations of WRC bulletin 297.

Warning - The ratio, Dn/Tn (6.667) is < 10
check the Limitations of WRC bulletin 297.

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{R_{\text{mean}}(t - c_a)}$$

$$= 88.9 + 2 \cdot 1.65 \cdot \sqrt{581.5(32.0 - 3.0)}$$

$$= 517.436 \text{ mm}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	1163.000	mm
Nozzle Outside Diameter	(d) =	88.900	mm
Vessel Thickness used	(T) =	57.000	mm
Nozzle Thickness used	(t) =	10.335	mm
T / t	=	5.515	
d / t	=	10.000	

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 0.345$$

Nr/P	=	0.186
Mr/P	=	0.114
M0/P	=	0.271
N0/P	=	0.262
MrD/Mc	=	0.112
NrDL/Mc	=	0.157



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Nozzle Calcs.: Tube outlet

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MØd/Mc = 0.299
 NØDL/Mc = -0.118
 MrD/Ml = 0.111
 NrDL/Ml = 0.159
 MØD/Ml = 0.292
 NØDL/Ml = -0.074

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	0	0	0	0
Outplane Bending (P)	3	-3	3	-3
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	-7	7	7	-7
Normal Pressure Stress	57	63	57	63
Outplane Stress Summary	54	68	69	52

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	0	0	0	0
Inplane Bending (P)	1	-1	1	-1
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	-2	2	2	-2
Inplane Pressure Stress	28	28	28	28
Inplane Stress Summary	26	29	34	25

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	2	2	2	2
Shear Stress Summary	2	2	1	1

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Two * Max Shear Stress	54	68	69	52

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Membrane (P)	0	0	0	0
Outplane Bending (P)	3	-3	3	-3
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	-7	7	7	-7
Outplane Membrane (ML)	0	0	0	0



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Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	28	28	28	28

Outplane Stress Summary	25	34	40	17
-------------------------	----	----	----	----

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses parallel to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside

Inplane Membrane (P)	0	0	0	0
Inplane Bending (P)	1	-1	1	-1
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	-2	2	2	-2
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	57	63	57	63

Inplane Stress Summary	55	64	63	60
------------------------	----	----	----	----

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Shear stress normal to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside

Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Torsional Shear (Mt)	2	2	2	2

Shear Stress Summary	1	1	2	2
----------------------	---	---	---	---

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stress Intensities)	Left Outside	Left Inside	Right Outside	Right Inside

Two * Max Shear Stress	55	64	63	60
------------------------	----	----	----	----

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stresses in the hoop direction)	Top Outside	Top Inside	Bottom Outside	Bottom Inside

Hoop Membrane (P)	0	0	0	0
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	21	28	21	28

Hoop Stress Summary	22	29	21	29
---------------------	----	----	----	----

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stresses Normal to pipe cross-section)	Top Outside	Top Inside	Bottom Outside	Bottom Inside

Axial Membrane (P)	2	2	2	2
Axial Bending (P)	8	-8	8	-8
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-28	-28	28	28
Axial Bending (ML)	-25	25	25	-25
Axial Pressure Stress	10	10	10	10



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Axial Stress Summary	-32	2	76	7
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Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Shear stress)	Au Outside	Al Inside	Bu Outside	Bl Inside
--------------------------------------	---------------	--------------	---------------	--------------

Shear due to (Vc)	3	3	-3	-3
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	11	11	11	11

Shear Stress Summary	15	15	7	7
----------------------	----	----	---	---

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Outside	Al Inside	Bu Outside	Bl Inside
--	---------------	--------------	---------------	--------------

Two * Max Shear Stress	62	40	77	31
------------------------	----	----	----	----

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses in the hoop direction)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
--	-----------------------	----------------------	------------------------	-----------------------

Hoop Membrane (P)	0	0	0	0
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	21	28	21	28

Hoop Stress Summary	22	30	21	28
---------------------	----	----	----	----

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to pipe cross-section)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
---	-----------------------	----------------------	------------------------	-----------------------

Axial Membrane (P)	2	2	2	2
Axial Bending (P)	8	-8	8	-8
Axial Membrane (Mc)	-28	-28	28	28
Axial Bending (Mc)	-27	27	27	-27
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	10	10	10	10

Axial Stress Summary	-33	3	77	6
----------------------	-----	---	----	---

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Shear stress)	Cu Outside	Cl Inside	Du Outside	Dl Inside
---	---------------	--------------	---------------	--------------

Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	-3	-3	3	3
Shear due to Torsion	11	11	11	11

Shear Stress Summary	7	7	15	15
----------------------	---	---	----	----

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Outside	Cl Inside	Du Outside	Dl Inside
---	---------------	--------------	---------------	--------------

Two * Max Shear Stress	58	32	81	37
------------------------	----	----	----	----



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Nozzle Calcs.: Tube outlet

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Stress Computations at the Edge of the Reinforcing Pad:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{R_{\text{mean}}(t - c_a)}$$

$$= 88.9 + 2 \cdot 1.65 \cdot \sqrt{567.5(32.0 - 3.0)}$$

$$= 512.246 \text{ mm}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter (D) = 1135.000 mm

Nozzle Outside Diameter (d) = 214.300 mm

Vessel Thickness used (T) = 29.000 mm

Nozzle Thickness used (t) = 73.035 mm

T / t = 0.397

d / t = 10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 1.181$$

Note: Re-pad thickness is added to nozzle thickness to simulate the Re-pad.

Nr/P = 0.096

Mr/P = 0.203

M0/P = 0.059

N0/P = 0.119

MrD/Mc = 0.306

NrDL/Mc = 0.087

M0d/Mc = 0.094

N0DL/Mc = 0.049

MrD/Ml = 0.174

NrDL/Ml = 0.089

M0D/Ml = 0.055

N0DL/Ml = 0.123

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au	Al	Bu	B1
	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Outplane Membrane (P)	1	1	1	1
Outplane Bending (P)	3	-3	3	-3
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	-2	2	2	-2
Normal Pressure Stress	115	121	115	121
Outplane Stress Summary	116	121	123	118

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au	Al	Bu	B1
	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Inplane Membrane (P)	0	0	0	0
Inplane Bending (P)	10	-10	10	-10
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	-7	7	7	-7
Inplane Pressure Stress	57	57	57	57
Inplane Stress Summary	61	55	77	41

Vessel Stresses (N./mm²):



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LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)			Au Top Outside	Al Top Inside	Bu Bottom Outside	B1 Bottom Inside
Outplane Shear (Vc)			0	0	0	0
Outplane Shear (Vl)			0	0	0	0
Outplane Shear (Mt)			0	0	0	0
Shear Stress Summary			1	1	0	0

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stress Intensities)			Au Top Outside	Al Top Inside	Bu Bottom Outside	B1 Bottom Inside
Two * Max Shear Stress			116	121	123	118

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)			Cu Left Outside	Cl Left Inside	Du Right Outside	D1 Right Inside
Outplane Membrane (P)			1	1	1	1
Outplane Bending (P)			3	-3	3	-3
Outplane Membrane (Mc)			0	0	0	0
Outplane Bending (Mc)			-4	4	4	-4
Outplane Membrane (ML)			0	0	0	0
Outplane Bending (ML)			0	0	0	0
Normal Pressure Stress			57	57	57	57
Outplane Stress Summary			57	59	66	52

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses parallel to circumferential plane)			Cu Left Outside	Cl Left Inside	Du Right Outside	D1 Right Inside
Inplane Membrane (P)			0	0	0	0
Inplane Bending (P)			10	-10	10	-10
Inplane Membrane (Mc)			0	0	0	0
Inplane Bending (Mc)			-13	13	13	-13
Inplane Membrane (ML)			0	0	0	0
Inplane Bending (ML)			0	0	0	0
Inplane Pressure Stress			115	121	115	121
Inplane Stress Summary			113	124	140	99

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Shear stress normal to circumferential plane)			Cu Left Outside	Cl Left Inside	Du Right Outside	D1 Right Inside
Outplane Shear (Vc)			0	0	0	0
Outplane Shear (Vl)			0	0	0	0
Torsional Shear (Mt)			0	0	0	0
Shear Stress Summary			0	0	1	1

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)			Cu Left Outside	Cl Left Inside	Du Right Outside	D1 Right Inside
Two * Max Shear Stress			113	124	140	99



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Nozl: 14

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WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	57	63	57	63	57	63	57	63
Circ. Pl (SUS)	0	0	0	0	0	0	0	0
Circ. Q (SUS)	-4	4	10	-10	-1	1	3	-3
Long. Pm (SUS)	28	28	28	28	28	28	28	28
Long. Pl (SUS)	0	0	0	0	0	0	0	0
Long. Q (SUS)	-1	1	3	-3	-4	4	10	-10
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	2	2	2	2	2	2	2	2
Pm (SUS)	57.0	63.0	57.0	63.0	57.0	63.0	57.0	63.0
Pm+Pl (SUS)	57.0	63.0	57.0	63.0	57.0	63.0	57.0	63.0
Pm+Pl+Q (Total)	53.2	67.1	67.1	53.1	56.1	64.1	60.2	60.1

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	63.00	147.00	Passed
Pm+Pl (SUS)	63.00	220.51	Passed
Pm+Pl+Q (TOTAL)	67.11	441.01	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Reinforcing Pad Edge (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	115	121	115	121	115	121	115	121
Circ. Pl (SUS)	1	1	1	1	0	0	0	0
Circ. Q (SUS)	1	-1	5	-5	-3	3	23	-23
Long. Pm (SUS)	57	57	57	57	57	57	57	57
Long. Pl (SUS)	0	0	0	0	1	1	1	1
Long. Q (SUS)	3	-3	17	-17	-1	1	7	-7
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	0	0	0	0	0	0	0	0
Pm (SUS)	115.0	121.0	115.0	121.0	115.0	121.0	115.0	121.0
Pm+Pl (SUS)	116.0	122.0	116.0	122.0	115.0	121.0	115.0	121.0
Pm+Pl+Q (Total)	117.0	121.0	121.0	117.0	112.0	124.0	138.0	98.0

Vessel Stress Summation Comparison (N./mm²):



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Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	121.00	147.00	Passed
Pm+Pl (SUS)	122.00	220.51	Passed
Pm+Pl+Q (TOTAL)	138.00	441.01	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	21	28	21	28	21	28	21	28
Circ. Pl (SUS)	0	0	0	0	0	0	0	0
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	10	10	10	10	10	10	10	10
Long. Pl (SUS)	-26	-26	30	30	-26	-26	30	30
Long. Q (SUS)	-17	17	33	-33	-19	19	35	-35
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	3	3	-3	-3	-3	-3	3	3
Shear Q (SUS)	11	11	11	11	11	11	11	11
Pm (SUS)	21.0	28.0	21.0	28.0	21.0	28.0	21.0	28.0
Pm+Pl (SUS)	37.5	44.4	40.5	40.7	37.5	44.4	40.5	40.7
Pm+Pl+Q (Total)	60.8	38.9	74.2	30.7	58.2	30.3	78.4	36.2

Nozzle Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	28.00	98.98	Passed
Pm+Pl (SUS)	44.41	148.47	Passed
Pm+Pl+Q (TOTAL)	78.41	296.95	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

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Nozzle Calcs.: Shell inlet-N1

Nozl: 15

9:45am

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Input, Nozzle Desc: Shell inlet-N1

From: 40

Pressure for Reinforcement Calculations	P	1125.827	KPa.
Temperature for Internal Pressure	Temp	275	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm ²
Parent Allowable Stress At Ambient	Sva	138.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	1100.00	mm
Shell Finished (Minimum) Thickness	t	30.0000	mm
Shell Internal Corrosion Allowance	c	3.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		6609.00	mm
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-106 B	
Material UNS Number		K03006	
Material Specification/Type		Smls. pipe	
Allowable Stress at Temperature	Sn	118.00	N./mm ²
Allowable Stress At Ambient	Sna	118.00	N./mm ²
Diameter Basis (for tr calc only)		Outside	
Layout Angle		180.00	deg
Diameter		20.0000	in.
Size and Thickness Basis		Nominal	
Nominal Thickness		80	
Flange Material		SA-105	
Flange Type		Weld Neck Flange	
Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	250.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	20.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	14.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Pad Material		SA-516 70	
Pad Allowable Stress at Temperature	Sp	137.00	N./mm ²
Pad Allowable Stress At Ambient	Spa	138.00	N./mm ²
Diameter of Pad along vessel surface	Dp	808.0000	mm
Thickness of Pad	te	20.0000	mm
Weld leg size between Pad and Shell	Wp	10.0000	mm
Groove weld depth between Pad and Nozzle	Wgpn	20.0000	mm
Reinforcing Pad Width		150.0000	mm

The Pressure Design option was MAWP + static head (to the nozzle).



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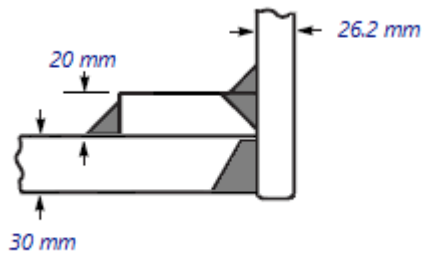
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Nozzle Calcs.: Shell inlet-N1

Nozl: 15

9:45am

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Reinforcement CALCULATION, Description: Shell inlet-N1

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation	20.000 in.
Actual Thickness Used in Calculation	1.031 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 1126 \cdot 553 / (138 \cdot 1 - 0.6 \cdot 1126)$
 $= 4.5340 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R_o / (S_n \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= 1126 \cdot 254 / (118 \cdot 1 + 0.4 \cdot 1126)$
 $= 2.4144 \text{ mm}$

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	923.2504 mm
Parallel to Vessel Wall, opening length	d	461.6252 mm
Normal to Vessel Wall (Thickness Limit), pad side Tlwp		67.5000 mm

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 118 / 138)$
 $= 0.855$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 118 / 138)$
 $= 0.855$

Weld Strength Reduction Factor [fr4]:
 $= \min(1, S_p / S_v)$
 $= \min(1, 137 / 138)$
 $= 0.993$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(0.86, 0.99)$
 $= 0.855$



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Nozzle Calcs.: Shell inlet-N1

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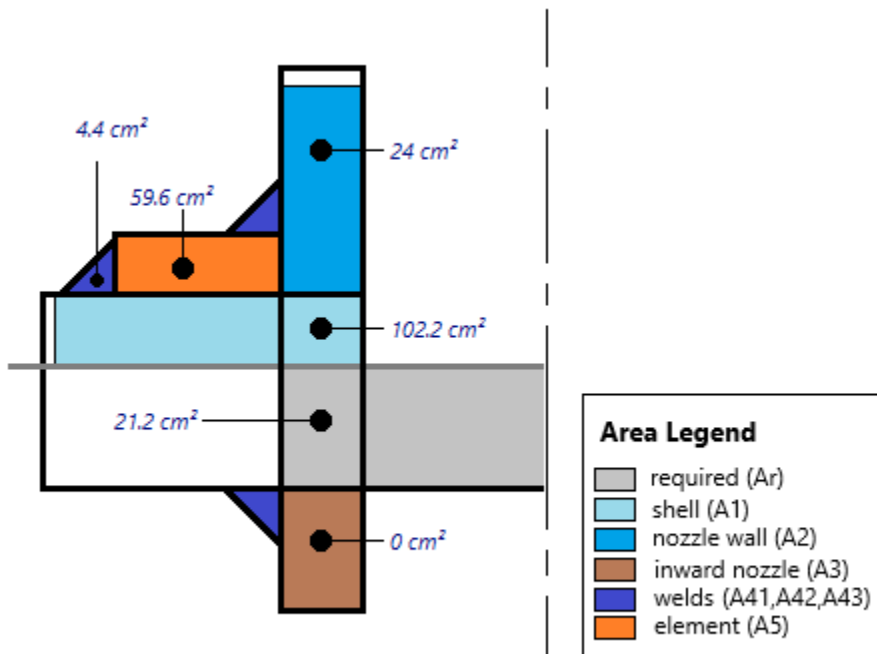


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		MAWP	External	Mapnc
Area Required	Ar	21.235	NA	NA
Area in Shell	A1	102.199	NA	NA
Area in Nozzle Wall	A2	23.979	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA
Area in Welds	A41+A42+A43	4.413	NA	NA
Area in Element	A5	59.566	NA	NA
TOTAL AREA AVAILABLE	Atot	190.157	NA	NA

The MAWP Case Governs the Analysis.

Nozzle Angle Used in Area Calculations

90.00 degs.

The area available without a pad is Sufficient.

The area available with the given pad is Sufficient.

Area Required [A]:

$$\begin{aligned}
 &= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)} \\
 &= (462 \cdot 4.53 \cdot 1 + 2 \cdot 23.2 \cdot 4.53 \cdot 1 \cdot (1 - 0.86)) \\
 &= 21.235 \text{ cm}^2
 \end{aligned}$$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

$$\begin{aligned}
 &= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1) \\
 &= 462(1 \cdot 27 - 1 \cdot 4.53) - 2 \cdot 23.2 \\
 &\quad (1 \cdot 27 - 1 \cdot 4.53) \cdot (1 - 0.86) \\
 &= 102.199 \text{ cm}^2
 \end{aligned}$$

Area Available in Nozzle Wall Projecting Outward [A2]:

$$\begin{aligned}
 &= (2 \cdot Tlwp) \cdot (tn - trn) \cdot fr2 \\
 &= (2 \cdot 67.5) \cdot (23.2 - 2.41) \cdot 0.86
 \end{aligned}$$



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$$= 23.979 \text{ cm}^2$$

Area Available in Welds [A41 + A42 + A43]:

$$= W_o^2 \cdot fr_3 + (W_i - can/0.707)^2 \cdot fr_2 + W_p^2 \cdot fr_4$$

$$= 20^2 \cdot 0.86 + (0)^2 \cdot 0.86 + 10^2 \cdot 0.99$$

$$= 4.413 \text{ cm}^2$$

Area Available in Element [A5]:

$$= (\min(D_p, D_L) - (\text{Nozzle OD})) \cdot (\min(t_p, t_{lp}, t_e)) \cdot fr_4$$

$$= (808 - 508) \cdot 20 \cdot 0.99$$

$$= 59.566 \text{ cm}^2$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures $t_a = 5.4144 \text{ mm}$

Wall Thickness per UG16(b), $tr_{16b} = 5.5000 \text{ mm}$

Wall Thickness, shell/head, internal pressure $tr_{b1} = 7.5340 \text{ mm}$

Wall Thickness $tb_1 = \max(tr_{b1}, tr_{16b}) = 7.5340 \text{ mm}$

Wall Thickness $tb_2 = \max(tr_{b2}, tr_{16b}) = 5.5000 \text{ mm}$

Wall Thickness per table UG-45 $tb_3 = 11.3312 \text{ mm}$

Determine Nozzle Thickness candidate [tb]:

$$= \min[tb_3, \max(tb_1, tb_2)]$$

$$= \min[11.3, \max(7.53, 5.5)]$$

$$= 7.5340 \text{ mm}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= \max(t_a, t_b)$$

$$= \max(5.41, 7.53)$$

$$= 7.5340 \text{ mm}$$

Available Nozzle Neck Thickness = $0.875 \cdot 26.187 = 22.914 \text{ mm} \rightarrow \text{OK}$

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	:	16.4,	Allowable	:	118.0 N./mm ²	Passed
Expansion	:	0.0,	Allowable	:	278.6 N./mm ²	Passed
Occasional	:	5.3,	Allowable	:	156.9 N./mm ²	Passed
Shear	:	8.5,	Allowable	:	82.6 N./mm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, min(Curve:B, Curve:A)

Govrn. thk, $t_g = 22.9$, $t_r = 2.41$, $c = 3 \text{ mm}$, $E^* = 1$

Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.12$, Temp. Reduction = 78°C

Min Metal Temp. w/o impact per UCS-66, Curve A 16°C

Min Metal Temp. at Required thickness (UCS 66.1) -104°C

Nozzle Neck to Pad Weld for the Nozzle, Curve: B

Govrn. thk, $t_g = 20$, $c = 3 \text{ mm}$, $E^* = 1$

Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.17$, Temp. Reduction = 78°C

Attachment governing, Attachment stress = Shell stress per Code

Min Metal Temp. w/o impact per UCS-66, Curve B -7°C

Min Metal Temp. at Required thickness (UCS 66.1) -104°C

Min Metal Temp. w/o impact per UG-20(f) -29°C

Nozzle Neck to Pad Weld for Reinforcement pad, Curve: B



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Nozzle Calcs.: Shell inlet-N1 Nozl: 15 9:45am Sep 9,2026

Govrn. thk, $t_g = 20$, $c = 3$ mm, $E^* = 1$
 Thickness Ratio = $tr * E^* / (t_g - c) = 0.17$, Temp. Reduction = 78 °C
 Attachment governing, Attachment stress = Shell stress per Code

Min Metal Temp. w/o impact per UCS-66, Curve B	-7 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Shell to Pad Weld Junction at Pad OD, Curve: B

Govrn. thk, $t_g = 20$, $c = 3$ mm, $E^* = 1$
 Thickness Ratio = $tr * E^* / (t_g - c) = 0.17$, Temp. Reduction = 78 °C
 Attachment governing, Attachment stress = Shell stress per Code

Min Metal Temp. w/o impact per UCS-66, Curve B	-7 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Nozzle-Shell/Head Weld (UCS-66(a)(b)), Curve: B

Govrn. thk, $t_g = 22.9$, $tr = 2.41$, $c = 3$ mm, $E^* = 1$
 Thickness Ratio = $tr * E^* / (t_g - c) = 0.12$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-3 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Gov. MDMT of the Nozzle	: -104 °C
Gov. MDMT of the Reinforcement Pad	: -104 °C
Gov. MDMT of the nozzle to shell joint welded assembly	: -104 °C

Weld Size Calculations, Description: Shell inlet-N1

Intermediate Calc. for nozzle/shell Welds	T_{min}	19.0000 mm
Intermediate Calc. for pad/shell Welds	T_{minPad}	19.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$6.0000 = \text{Min per Code}$	$14.1400 = 0.7 * W_o$ mm
Pad Weld	$9.5000 = 0.5 * T_{minPad}$	$7.0700 = 0.7 * W_p$ mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A - A1 + 2 \cdot t_n \cdot f_r1 \cdot (E1 \cdot t - tr)) \cdot Sv)$
 $= \max(0, (21.2 - 102 + 2 \cdot 23.2 \cdot 0.86 \cdot (1 \cdot 27 - 4.53)) \cdot 138)$
 $= \max(0, -101393.13) \text{ Kg}$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A2 + A5 + A4 - (W_i - Can / .707)^2 \cdot f_r2) \cdot Sv$
 $= (24 + 59.6 + 4.41 - 0 \cdot 0.86) \cdot 138$
 $= 123770.70 \text{ Kg}$

Weld Load [W2]:
 $= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot f_r1)) \cdot Sv$
 $= (24 + 0 + 3.42 + (10.7)) \cdot 138$
 $= 53621.31 \text{ Kg}$

Weld Load [W3]:
 $= (A2 + A3 + A4 + A5 + (2 \cdot t_n \cdot t \cdot f_r1)) \cdot S$
 $= (24 + 0 + 4.41 + 59.6 + (10.7)) \cdot 138$
 $= 138836.47 \text{ Kg}$



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Nozzle Calcs.: Shell inlet-N1

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Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$$

$$= (3.14/2.0) \cdot 508 \cdot 20 \cdot 0.49 \cdot 118$$

$$= 94093. \text{ Kg}$$

Shear, Pad Element Weld [Spew]:

$$= (\pi/2) \cdot D_P \cdot W_P \cdot 0.49 \cdot S_{EW}$$

$$= (3.14/2.0) \cdot 808 \cdot 10 \cdot 0.49 \cdot 137$$

$$= 86879. \text{ Kg}$$

Shear, Nozzle Wall [Snw]:

$$= (\pi \cdot (D_{lr} + D_{lo}) / 4) \cdot (Thk - Can) \cdot 0.7 \cdot S_n$$

$$= (3.14 \cdot 242) \cdot (26.2 - 3) \cdot 0.7 \cdot 118$$

$$= 148727. \text{ Kg}$$

Tension, Pad Groove Weld [Tpgw]:

$$= (\pi/2) \cdot D_{lo} \cdot W_{gpn} \cdot 0.74 \cdot S_{eg}$$

$$= (3.14/2) \cdot 508 \cdot 20 \cdot 0.74 \cdot 137$$

$$= 164980. \text{ Kg}$$

Tension, Shell Groove Weld [Tngw]:

$$= (\pi/2) \cdot D_{lo} \cdot W_{gvi} \cdot 0.74 \cdot S_{ng}$$

$$= (3.14/2.0) \cdot 508 \cdot (14 - 0) \cdot 0.74 \cdot 138$$

$$= 116328. \text{ Kg}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (S_{PEW} + S_{NW}) = (86879 + 148727) = 235606 \text{ Kg}$$

Failure Path [Path 2-2]:

$$= (S_{onw} + T_{pgw} + T_{ngw} + S_{inw})$$

$$= (94093 + 164980 + 116328 + 0) = 375401 \text{ Kg}$$

Failure Path [Path 3-3]:

$$= (S_{pew} + T_{ngw} + S_{inw})$$

$$= (86879 + 116328 + 0) = 203207 \text{ Kg}$$

Summary of Failure Path Calculations:

$$\text{Path 1-1} = 235605 \text{ Kg}, \text{ must exceed } W = 0 \text{ Kg}$$

$$\text{or } W_1 = 123770 \text{ Kg}$$

$$\text{Path 2-2} = 375401 \text{ Kg}, \text{ must exceed } W = 0 \text{ Kg}$$

$$\text{or } W_2 = 53621 \text{ Kg}$$

$$\text{Path 3-3} = 203207 \text{ Kg}, \text{ must exceed } W = 0 \text{ Kg}$$

$$\text{or } W_3 = 138836 \text{ Kg}$$

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 3120 KPa.

Note: The MAWP of this junction was limited by the parent Shell/Head.

The Drop for this Nozzle is : 62.1639 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 342.1639 mm

This nozzle or its diameter limit intersects weld seam # [2] on this element. The distance between the nozzle and seam is, 38.61 mm , while the distance from the center of the nozzle to the edge of the diameter limit is: 461.63 mm

Input Echo, WRC297 Item 1, Description: Shell inlet-N1:

Diameter Basis for Cylindrical Shell

ID

Shell Internal Corrosion Allowance

3.0000 mm

Shell External Corrosion Allowance

0.0000 mm



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Nozzle Calcs.: Shell inlet-N1 Nozl: 15 9:45am Sep 9,2026

Shell Diameter		1100.000	mm
Shell Thickness		30.0000	mm
Shell Stress Concentration Factor		1.000	
Vessel Material		SA-516 70	
Vessel Cold S.I. Allowable	Smc	138.00	N./mm ²
Vessel Hot S.I. Allowable	Smh	138.00	N./mm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter Basis for Nozzle		OD	
Nozzle Corrosion Allowance		3.0000	mm
Nozzle Diameter		508.000	mm
Nozzle Thickness		26.1874	mm
Nozzle Stress Concentration Factor		1.000	
Nozzle Material		SA-106 B	
Nozzle Cold S.I. Allowable	SNmc	118.00	N./mm ²
Nozzle Hot S.I. Allowable	SNmh	118.00	N./mm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter of Reinforcing Pad		808.000	mm
Thickness of Reinforcing Pad		20.0000	mm

Note:

External Forces and Moments in WRC 107/537 Convention:
These loads are assumed to be SUSained loads.

Design Internal Pressure	Dp	1125.83	KPa.
Radial Load	P	-3823.00	Kgf
Circumferential Shear	Vc	2548.00	Kgf
Longitudinal Shear	Vl	2548.00	Kgf
Circumferential Moment	Mc	3568.01	Kg-m.
Longitudinal Moment	Ml	3568.01	Kg-m.
Torsional Moment	Mt	5097.00	Kg-m.

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:
 $= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{R_{\text{mean}}(t - c_a)}$
 $= 508.0 + 2 \cdot 1.65 \cdot \sqrt{576.5(30.0 - 3.0)}$
 $= 919.714 \text{ mm}$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	1153.000	mm
Nozzle Outside Diameter	(d) =	508.000	mm
Vessel Thickness used	(T) =	47.000	mm
Nozzle Thickness used	(t) =	23.187	mm
T / t	=	2.027	
d / t	=	21.908	

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 2.182$$

Nr/P	=	0.071
Mr/P	=	0.076
M0/P	=	0.046
N0/P	=	0.149
MrD/Mc	=	0.192



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Nozzle Calcs.: Shell inlet-N1

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NrDL/Mc = 0.124
MØd/Mc = 0.133
NØDL/Mc = 0.161
MrD/Ml = 0.125
NrDL/Ml = 0.072
MØD/Ml = 0.081
NØDL/Ml = 0.260

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	2	2	2	2
Outplane Bending (P)	4	-4	4	-4
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-8	-8	8	8
Outplane Bending (ML)	-15	15	15	-15
Normal Pressure Stress	12	13	12	13
Outplane Stress Summary	-3	18	43	4

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	1	1	1	1
Inplane Bending (P)	7	-7	7	-7
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-2	-2	2	2
Inplane Bending (ML)	-23	23	23	-23
Inplane Pressure Stress	6	6	6	6
Inplane Stress Summary	-10	20	40	-21

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	2	2	2	2
Shear Stress Summary	3	3	1	1

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Two * Max Shear Stress	11	23	44	26

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Membrane (P)	2	2	2	2
Outplane Bending (P)	4	-4	4	-4
Outplane Membrane (Mc)	-5	-5	5	5
Outplane Bending (Mc)	-24	24	24	-24



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Nozzle Calcs.: Shell inlet-N1 Nozl: 15 9:45am Sep 9,2026

Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	6	6	6	6

Outplane Stress Summary	-16	24	43	-15
-------------------------	-----	----	----	-----

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses parallel to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside

Inplane Membrane (P)	1	1	1	1
Inplane Bending (P)	7	-7	7	-7
Inplane Membrane (Mc)	-3	-3	3	3
Inplane Bending (Mc)	-35	35	35	-35
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	12	13	12	13

Inplane Stress Summary	-18	39	61	-24
------------------------	-----	----	----	-----

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Shear stress normal to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside

Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Torsional Shear (Mt)	2	2	2	2

Shear Stress Summary	1	1	3	3
----------------------	---	---	---	---

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stress Intensities)	Left Outside	Left Inside	Right Outside	Right Inside

Two * Max Shear Stress	19	39	62	25
------------------------	----	----	----	----

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stresses in the hoop direction)	Top Outside	Top Inside	Bottom Outside	Bottom Inside

Hoop Membrane (P)	2	2	2	2
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-8	-8	8	8
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	12	14	12	14

Hoop Stress Summary	7	8	23	24
---------------------	---	---	----	----

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stresses Normal to pipe cross-section)	Top Outside	Top Inside	Bottom Outside	Bottom Inside

Axial Membrane (P)	1	1	1	1
Axial Bending (P)	17	-17	17	-17
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-8	-8	8	8
Axial Bending (ML)	-68	68	68	-68



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Nozzle Calcs.: Shell inlet-N1 Nozl: 15 9:45am Sep 9,2026

Axial Pressure Stress	6	6	6	6
-----------------------	---	---	---	---

Axial Stress Summary	-52	50	101	-69
----------------------	-----	----	-----	-----

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Shear stress)	Outside	Inside	Outside	Inside

Shear due to (Vc)	1	1	-1	-1
-------------------	---	---	----	----

Shear due to (Vl)	0	0	0	0
-------------------	---	---	---	---

Shear due to Torsion	5	5	5	5
----------------------	---	---	---	---

Shear Stress Summary	6	6	3	3
----------------------	---	---	---	---

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stress Intensities	Outside	Inside	Outside	Inside

Two * Max Shear Stress	60	51	101	94
------------------------	----	----	-----	----

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses in the	Left	Left	Right	Right
hoop direction)	Outside	Inside	Outside	Inside

Hoop Membrane (P)	2	2	2	2
--------------------	---	---	---	---

Hoop Bending (P)	0	0	0	0
-------------------	---	---	---	---

Hoop Membrane (Mc)	-5	-5	5	5
--------------------	----	----	---	---

Hoop Bending (Mc)	0	0	0	0
-------------------	---	---	---	---

Hoop Membrane (ML)	0	0	0	0
--------------------	---	---	---	---

Hoop Bending (ML)	0	0	0	0
-------------------	---	---	---	---

Hoop Pressure Stress	12	14	12	14
----------------------	----	----	----	----

Hoop Stress Summary	10	11	20	21
---------------------	----	----	----	----

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses Normal to	Left	Left	Right	Right
pipe cross-section)	Outside	Inside	Outside	Inside

Axial Membrane (P)	1	1	1	1
---------------------	---	---	---	---

Axial Bending (P)	17	-17	17	-17
--------------------	----	-----	----	-----

Axial Membrane (Mc)	-8	-8	8	8
---------------------	----	----	---	---

Axial Bending (Mc)	-99	99	99	-99
--------------------	-----	----	----	-----

Axial Membrane (ML)	0	0	0	0
---------------------	---	---	---	---

Axial Bending (ML)	0	0	0	0
--------------------	---	---	---	---

Axial Pressure Stress	6	6	6	6
-----------------------	---	---	---	---

Axial Stress Summary	-83	81	132	-100
----------------------	-----	----	-----	------

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Shear stress)	Outside	Inside	Outside	Inside

Shear due to (Vc)	0	0	0	0
-------------------	---	---	---	---

Shear due to (Vl)	-1	-1	1	1
-------------------	----	----	---	---

Shear due to Torsion	5	5	5	5
----------------------	---	---	---	---

Shear Stress Summary	3	3	6	6
----------------------	---	---	---	---

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stress Intensities	Outside	Inside	Outside	Inside

Two * Max Shear Stress	94	81	133	123
------------------------	----	----	-----	-----



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Nozzle Calcs.: Shell inlet-N1

Nozl: 15

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Stress Computations at the Edge of the Reinforcing Pad:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(\text{Rmean}(t - ca))}$$

$$= 508.0 + 2 \cdot 1.65 \cdot \sqrt{(566.5(30.0 - 3.0))}$$

$$= 916.127 \text{ mm}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter (D) = 1133.000 mm
 Nozzle Outside Diameter (d) = 808.000 mm
 Vessel Thickness used (T) = 27.000 mm
 Nozzle Thickness used (t) = 173.187 mm
 T / t = 0.156
 d / t = 10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 4.620$$

Note: Re-pad thickness is added to nozzle thickness to simulate the Re-pad.

Nr/P = 0.045
 Mr/P = 0.072
 M0/P = 0.022
 N0/P = 0.041
 MrD/Mc = 0.247
 NrDL/Mc = 0.092
 M0d/Mc = 0.074
 N0DL/Mc = 0.071
 MrD/Ml = 0.058
 NrDL/Ml = 0.045
 M0D/Ml = 0.018
 N0DL/Ml = 0.040

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	2	2	2	2
Outplane Bending (P)	6	-6	6	-6
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-2	-2	2	2
Outplane Bending (ML)	-6	6	6	-6
Normal Pressure Stress	22	23	22	23
Outplane Stress Summary	22	22	40	14

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	2	2	2	2
Inplane Bending (P)	22	-22	22	-22
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-2	-2	2	2
Inplane Bending (ML)	-20	20	20	-20
Inplane Pressure Stress	11	11	11	11
Inplane Stress Summary	12	9	59	-26



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Nozzle Calcs.: Shell inlet-N1

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Vessel Stresses (N./mm²):

LONGITUDINAL PLANE			Au	Al	Bu	B1
(Shear stress normal to longitudinal plane)			Top Outside	Top Inside	Bottom Outside	Bottom Inside
Outplane Shear (Vc)			0	0	0	0
Outplane Shear (Vl)			0	0	0	0
Outplane Shear (Mt)			1	1	1	1
Shear Stress Summary			2	2	1	1

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE			Au	Al	Bu	B1
(Stress Intensities)			Top Outside	Top Inside	Bottom Outside	Bottom Inside
Two * Max Shear Stress			23	23	59	41

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE			Cu	Cl	Du	D1
(Stresses Normal to circumferential plane)			Left Outside	Left Inside	Right Outside	Right Inside
Outplane Membrane (P)			2	2	2	2
Outplane Bending (P)			6	-6	6	-6
Outplane Membrane (Mc)			-4	-4	4	4
Outplane Bending (Mc)			-26	26	26	-26
Outplane Membrane (ML)			0	0	0	0
Outplane Bending (ML)			0	0	0	0
Normal Pressure Stress			11	11	11	11
Outplane Stress Summary			-10	28	50	-15

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE			Cu	Cl	Du	D1
(Stresses parallel to circumferential plane)			Left Outside	Left Inside	Right Outside	Right Inside
Inplane Membrane (P)			2	2	2	2
Inplane Bending (P)			22	-22	22	-22
Inplane Membrane (Mc)			-5	-5	5	5
Inplane Bending (Mc)			-88	88	88	-88
Inplane Membrane (ML)			0	0	0	0
Inplane Bending (ML)			0	0	0	0
Inplane Pressure Stress			22	23	22	23
Inplane Stress Summary			-46	86	140	-78

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE			Cu	Cl	Du	D1
(Shear stress normal to circumferential plane)			Left Outside	Left Inside	Right Outside	Right Inside
Outplane Shear (Vc)			0	0	0	0
Outplane Shear (Vl)			0	0	0	0
Torsional Shear (Mt)			1	1	1	1
Shear Stress Summary			1	1	2	2

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE			Cu	Cl	Du	D1
(Stress Intensities)			Left Outside	Left Inside	Right Outside	Right Inside
Two * Max Shear Stress			46	86	140	78



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WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	12	13	12	13	12	13	12	13
Circ. Pl (SUS)	-6	-6	10	10	-2	-2	4	4
Circ. Q (SUS)	-11	11	19	-19	-28	28	42	-42
Long. Pm (SUS)	6	6	6	6	6	6	6	6
Long. Pl (SUS)	-1	-1	3	3	-3	-3	7	7
Long. Q (SUS)	-16	16	30	-30	-20	20	28	-28
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	2	2	2	2	2	2	2	2
Pm (SUS)	12.0	13.0	12.0	13.0	12.0	13.0	12.0	13.0
Pm+Pl (SUS)	6.0	7.0	22.0	23.0	10.0	11.0	16.0	17.0
Pm+Pl+Q (Total)	11.6	22.0	42.2	25.3	19.6	39.2	58.2	25.4

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	13.00	138.00	Passed
Pm+Pl (SUS)	23.00	207.00	Passed
Pm+Pl+Q (TOTAL)	58.23	413.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Reinforcing Pad Edge (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	22	23	22	23	22	23	22	23
Circ. Pl (SUS)	0	0	4	4	-3	-3	7	7
Circ. Q (SUS)	0	0	12	-12	-66	66	110	-110
Long. Pm (SUS)	11	11	11	11	11	11	11	11
Long. Pl (SUS)	0	0	4	4	-2	-2	6	6
Long. Q (SUS)	2	-2	42	-42	-20	20	32	-32
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	1	1	1	1	1	1	1	1
Pm (SUS)	22.0	23.0	22.0	23.0	22.0	23.0	22.0	23.0
Pm+Pl (SUS)	22.0	23.0	26.0	27.0	19.0	20.0	29.0	30.0
Pm+Pl+Q (Total)	22.1	23.1	57.1	42.0	47.0	86.0	139.0	80.0



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Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	23.00	138.00	Passed
Pm+Pl (SUS)	30.00	207.00	Passed
Pm+Pl+Q (TOTAL)	139.01	413.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	12	14	12	14	12	14	12	14
Circ. Pl (SUS)	-6	-6	10	10	-3	-3	7	7
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	6	6	6	6	6	6	6	6
Long. Pl (SUS)	-7	-7	9	9	-7	-7	9	9
Long. Q (SUS)	-51	51	85	-85	-82	82	116	-116
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	1	1	-1	-1	-1	-1	1	1
Shear Q (SUS)	5	5	5	5	5	5	5	5
Pm (SUS)	12.0	14.0	12.0	14.0	12.0	14.0	12.0	14.0
Pm+Pl (SUS)	7.3	9.2	22.1	24.1	10.2	12.2	19.2	21.2
Pm+Pl+Q (Total)	59.2	50.8	100.2	94.3	92.3	81.2	131.3	122.6

Nozzle Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	14.00	118.00	Passed
Pm+Pl (SUS)	24.11	177.00	Passed
Pm+Pl+Q (TOTAL)	131.32	353.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

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Input, Nozzle Desc: Shell Inlet

From: 40

Pressure for Reinforcement Calculations	P	1115.044	KPa.
Temperature for Internal Pressure	Temp	275	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm ²
Parent Allowable Stress At Ambient	Sva	138.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	1100.00	mm
Shell Finished (Minimum) Thickness	t	18.0000	mm
Shell Internal Corrosion Allowance	c	3.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		2809.00	mm
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-106 B	
Material UNS Number		K03006	
Material Specification/Type		Smls. pipe	
Allowable Stress at Temperature	Sn	118.00	N./mm ²
Allowable Stress At Ambient	Sna	118.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		0.00	deg
Diameter		20.0000	in.
Size and Thickness Basis		Nominal	
Nominal Thickness		80	
Flange Material		SA-105	
Flange Type		Weld Neck Flange	
Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	250.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	16.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	14.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Pad Material		SA-516 70	
Pad Allowable Stress at Temperature	Sp	137.00	N./mm ²
Pad Allowable Stress At Ambient	Spa	138.00	N./mm ²
Diameter of Pad along vessel surface	Dp	808.0000	mm
Thickness of Pad	te	16.0000	mm
Weld leg size between Pad and Shell	Wp	15.0000	mm
Groove weld depth between Pad and Nozzle	Wgpn	6.0000	mm
Reinforcing Pad Width		150.0000	mm
Flange Class		150	
Flange Grade		GR 1.1	

The Pressure Design option was MAWP + static head (to the nozzle).



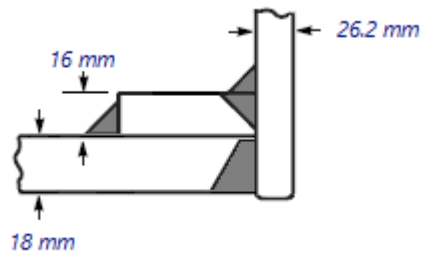
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Nozzle Calcs.: Shell Inlet

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Reinforcement CALCULATION, Description: Shell Inlet

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	17.938 in.
Actual Thickness Used in Calculation	1.031 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 1115 \cdot 553 / (138 \cdot 1 - 0.6 \cdot 1115)$
 $= 4.4904 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 1115 \cdot 231 / (118 \cdot 1 - 0.6 \cdot 1115)$
 $= 2.1937 \text{ mm}$

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	923.2504 mm
Parallel to Vessel Wall, opening length	d	461.6252 mm
Normal to Vessel Wall (Thickness Limit), pad side Tlwp		37.5000 mm

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 118 / 138)$
 $= 0.855$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 118 / 138)$
 $= 0.855$

Weld Strength Reduction Factor [fr4]:
 $= \min(1, S_p / S_v)$
 $= \min(1, 137 / 138)$
 $= 0.993$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(0.86, 0.99)$
 $= 0.855$



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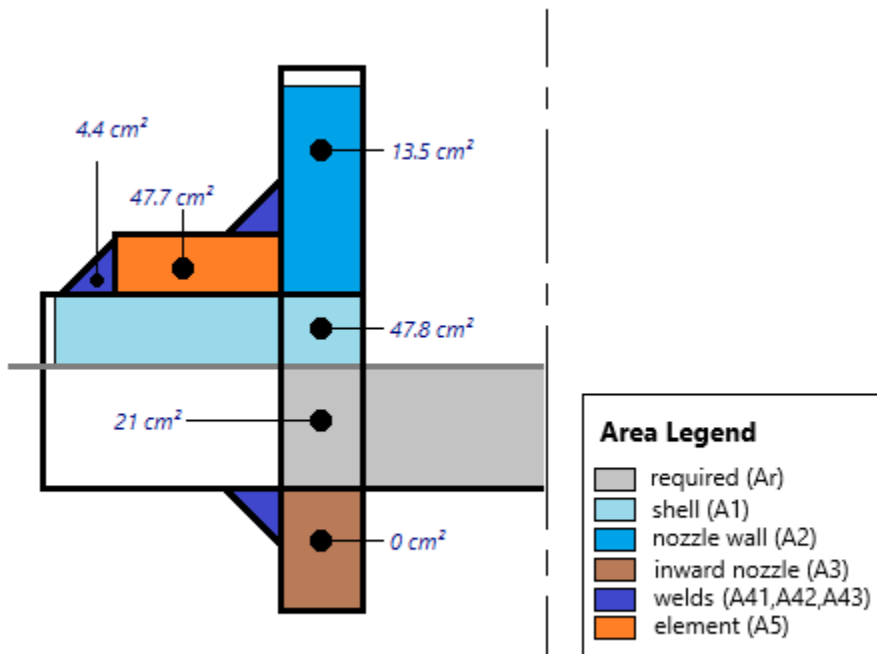


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		MAWP	External	Mapnc
Area Required	Ar	21.030	NA	NA
Area in Shell	A1	47.809	NA	NA
Area in Nozzle Wall	A2	13.463	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA
Area in Welds	A41+A42+A43	4.423	NA	NA
Area in Element	A5	47.653	NA	NA
TOTAL AREA AVAILABLE	Atot	113.347	NA	NA

The MAWP Case Governs the Analysis.

Nozzle Angle Used in Area Calculations

90.00 degs.

The area available without a pad is Sufficient.

The area available with the given pad is Sufficient.

Area Required [A]:

$$\begin{aligned}
 &= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)} \\
 &= (462 \cdot 4.49 \cdot 1 + 2 \cdot 23.2 \cdot 4.49 \cdot 1 \cdot (1 - 0.86)) \\
 &= 21.030 \text{ cm}^2
 \end{aligned}$$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

$$\begin{aligned}
 &= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1) \\
 &= 462(1 \cdot 15 - 1 \cdot 4.49) - 2 \cdot 23.2 \\
 &\quad (1 \cdot 15 - 1 \cdot 4.49) \cdot (1 - 0.86) \\
 &= 47.809 \text{ cm}^2
 \end{aligned}$$

Area Available in Nozzle Wall Projecting Outward [A2]:

$$\begin{aligned}
 &= (2 \cdot Tlwp) \cdot (tn - trn) \cdot fr2 \\
 &= (2 \cdot 37.5) \cdot (23.2 - 2.19) \cdot 0.86
 \end{aligned}$$



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$$= 13.463 \text{ cm}^2$$

Area Available in Welds [A41 + A42 + A43]:

$$= W_o^2 \cdot fr_3 + (W_i - can/0.707)^2 \cdot fr_2 + W_p^2 \cdot fr_4$$

$$= 16^2 \cdot 0.86 + (0)^2 \cdot 0.86 + 15^2 \cdot 0.99$$

$$= 4.423 \text{ cm}^2$$

Area Available in Element [A5]:

$$= (\min(D_p, D_L) - (\text{Nozzle OD})) \cdot (\min(t_p, t_{lwp}, t_e)) \cdot fr_4$$

$$= (808 - 508) \cdot 16 \cdot 0.99$$

$$= 47.653 \text{ cm}^2$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures $t_a = 5.1937 \text{ mm}$

Wall Thickness per UG16(b), $tr_{16b} = 5.5000 \text{ mm}$

Wall Thickness, shell/head, internal pressure $trb_1 = 7.4904 \text{ mm}$

Wall Thickness $tb_1 = \max(trb_1, tr_{16b}) = 7.4904 \text{ mm}$

Wall Thickness $tb_2 = \max(trb_2, tr_{16b}) = 5.5000 \text{ mm}$

Wall Thickness per table UG-45 $tb_3 = 11.3312 \text{ mm}$

Determine Nozzle Thickness candidate [tb]:

$$= \min[tb_3, \max(tb_1, tb_2)]$$

$$= \min[11.3, \max(7.49, 5.5)]$$

$$= 7.4904 \text{ mm}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= \max(t_a, t_b)$$

$$= \max(5.19, 7.49)$$

$$= 7.4904 \text{ mm}$$

Available Nozzle Neck Thickness = $0.875 \cdot 26.187 = 22.914 \text{ mm} \rightarrow \text{OK}$

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	:	16.3,	Allowable	:	118.0 N./mm ²	Passed
Expansion	:	0.0,	Allowable	:	278.7 N./mm ²	Passed
Occasional	:	5.3,	Allowable	:	156.9 N./mm ²	Passed
Shear	:	8.7,	Allowable	:	82.6 N./mm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, min(Curve:B, Curve:A)

Govrn. thk, $t_g = 22.9$, $t_r = 2.19$, $c = 3 \text{ mm}$, $E^* = 1$

Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.11$, Temp. Reduction = 78°C

Min Metal Temp. w/o impact per UCS-66, Curve A 16°C

Min Metal Temp. at Required thickness (UCS 66.1) -104°C

Nozzle Neck to Pad Weld for the Nozzle, Curve: B

Govrn. thk, $t_g = 16$, $c = 3 \text{ mm}$, $E^* = 1$

Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.3$, Temp. Reduction = 78°C

Attachment governing, Attachment stress = Shell stress per Code

Min Metal Temp. w/o impact per UCS-66, Curve B -14°C

Min Metal Temp. at Required thickness (UCS 66.1) -104°C

Min Metal Temp. w/o impact per UG-20(f) -29°C

Nozzle Neck to Pad Weld for Reinforcement pad, Curve: B



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Govrn. thk, $t_g = 16$, $c = 3$ mm, $E^* = 1$
 Thickness Ratio = $tr * E^* / (t_g - c) = 0.3$, Temp. Reduction = 78 °C
 Attachment governing, Attachment stress = Shell stress per Code

Min Metal Temp. w/o impact per UCS-66, Curve B	-14 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Shell to Pad Weld Junction at Pad OD, Curve: B

Govrn. thk, $t_g = 16$, $c = 3$ mm, $E^* = 1$
 Thickness Ratio = $tr * E^* / (t_g - c) = 0.3$, Temp. Reduction = 78 °C
 Attachment governing, Attachment stress = Shell stress per Code

Min Metal Temp. w/o impact per UCS-66, Curve B	-14 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Nozzle-Shell/Head Weld (UCS-66(a)(b)), Curve: B

Govrn. thk, $t_g = 18$, $tr = 4.49$, $c = 3$ mm, $E^* = 1$
 Thickness Ratio = $tr * E^* / (t_g - c) = 0.3$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-11 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Gov. MDMT of the Nozzle	: -104 °C
Gov. MDMT of the Reinforcement Pad	: -104 °C
Gov. MDMT of the nozzle to shell joint welded assembly	: -104 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-43 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = $1115.04 / 1960.01 = 0.569$

Weld Size Calculations, Description: Shell Inlet

Intermediate Calc. for nozzle/shell Welds	T_{min}	16.0000 mm
Intermediate Calc. for pad/shell Welds	T_{minPad}	15.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$11.2000 = 0.7 * t_{min}$	$11.3120 = 0.7 * W_o$ mm
Pad Weld	$7.5000 = 0.5 * T_{minPad}$	$10.6050 = 0.7 * W_p$ mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A - A1 + 2 * t_n * fr1 * (E1 * t - tr)) * Sv)$
 $= \max(0, (21 - 47.8 + 2 * 23.2 * 0.86 * (1 * 15 - 4.49)) * 138)$
 $= \max(0, -31816.97) \text{ Kgf}$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A2 + A5 + A4 - (Wi - Can / .707)^2 * fr2) * Sv$
 $= (13.5 + 47.7 + 4.42 - 0 * 0.86) * 138$
 $= 92223.05 \text{ Kgf}$

Weld Load [W2]:



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$$= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot fr1)) \cdot Sv$$

$$= (13.5 + 0 + 2.19 + (5.95)) \cdot 138$$

$$= 30395.19 \text{ Kgf}$$

Weld Load [W3]:

$$= (A2 + A3 + A4 + A5 + (2 \cdot t_n \cdot t \cdot fr1)) \cdot S$$

$$= (13.5 + 0 + 4.42 + 47.7 + (5.95)) \cdot 138$$

$$= 100592.92 \text{ Kgf}$$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$= (\pi/2) \cdot Dlo \cdot Wo \cdot 0.49 \cdot Snw$$

$$= (3.14/2.0) \cdot 508 \cdot 16 \cdot 0.49 \cdot 118$$

$$= 75274. \text{ Kgf}$$

Shear, Pad Element Weld [Spew]:

$$= (\pi/2) \cdot DP \cdot WP \cdot 0.49 \cdot SEW$$

$$= (3.14/2.0) \cdot 808 \cdot 15 \cdot 0.49 \cdot 137$$

$$= 130318. \text{ Kgf}$$

Shear, Nozzle Wall [Snw]:

$$= (\pi \cdot (Dlr + Dlo) / 4) \cdot (Thk - Can) \cdot 0.7 \cdot Sn$$

$$= (3.14 \cdot 242) \cdot (26.2 - 3) \cdot 0.7 \cdot 118$$

$$= 148727. \text{ Kgf}$$

Tension, Pad Groove Weld [Tpgw]:

$$= (\pi/2) \cdot Dlo \cdot Wgpn \cdot 0.74 \cdot Seg$$

$$= (3.14/2) \cdot 508 \cdot 6 \cdot 0.74 \cdot 137$$

$$= 49494. \text{ Kgf}$$

Tension, Shell Groove Weld [Tngw]:

$$= (\pi/2) \cdot Dlo \cdot Wgnvi \cdot 0.74 \cdot Sng$$

$$= (3.14/2.0) \cdot 508 \cdot (14 - 0) \cdot 0.74 \cdot 138$$

$$= 116328. \text{ Kgf}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (SPEW + SNW) = (130318 + 148727) = 279045 \text{ Kgf}$$

Failure Path [Path 2-2]:

$$= (Sonw + Tpgw + Tngw + Sinw)$$

$$= (75274 + 49494 + 116328 + 0) = 241097 \text{ Kgf}$$

Failure Path [Path 3-3]:

$$= (Spew + Tngw + Sinw)$$

$$= (130318 + 116328 + 0) = 246647 \text{ Kgf}$$

Summary of Failure Path Calculations:

Path 1-1 = 279045 Kgf, must exceed W = 0 Kgf	or W1 = 92223 Kgf
Path 2-2 = 241096 Kgf, must exceed W = 0 Kgf	or W2 = 30395 Kgf
Path 3-3 = 246646 Kgf, must exceed W = 0 Kgf	or W3 = 100592 Kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 3120 KPa.

Note: The MAWP of this junction was limited by the parent Shell/Head.

The Drop for this Nozzle is : 62.1639 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 330.1639 mm

This nozzle or its diameter limit intersects weld seam # [1] on this element. The



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distance between the nozzle and seam is, 434.32 mm , while the distance from the center of the nozzle to the edge of the diameter limit is: 461.63 mm

This nozzle or its diameter limit intersects weld seam # [3] on this element. The distance between the nozzle and seam is, 366.76 mm , while the distance from the center of the nozzle to the edge of the diameter limit is: 461.63 mm

Input Echo, WRC297 Item 1, Description: Shell Inlet :

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	3.0000	mm
Shell External Corrosion Allowance	0.0000	mm
Shell Diameter	1100.000	mm
Shell Thickness	18.0000	mm
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-516 70	
Vessel Cold S.I. Allowable	Smc 138.00	N./mm ²
Vessel Hot S.I. Allowable	Smh 138.00	N./mm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter Basis for Nozzle	ID	
Nozzle Corrosion Allowance	3.0000	mm
Nozzle Diameter	455.625	mm
Nozzle Thickness	26.1874	mm
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-106 B	
Nozzle Cold S.I. Allowable	SNmc 118.00	N./mm ²
Nozzle Hot S.I. Allowable	SNmh 118.00	N./mm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter of Reinforcing Pad	808.000	mm
Thickness of Reinforcing Pad	16.0000	mm

Note:

External Forces and Moments in WRC 107/537 Convention:
These loads are assumed to be SUSained loads.

Design Internal Pressure	Dp	1115.04	KPa.
Radial Load	P	-3823.00	Kgf
Circumferential Shear	Vc	2548.00	Kgf
Longitudinal Shear	Vl	2548.00	Kgf
Circumferential Moment	Mc	3568.01	Kg-m.
Longitudinal Moment	Ml	3568.01	Kg-m.
Torsional Moment	Mt	5199.00	Kg-m.

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{R_{\text{mean}}(t - ca)}$$

$$= 508.0 + 2 \cdot 1.65 \cdot \sqrt{568.5(18.0 - 3.0)}$$

$$= 812.737 \text{ mm}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	1137.000	mm
Nozzle Outside Diameter	(d) =	508.000	mm



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Vessel Thickness used (T) = 31.000 mm
 Nozzle Thickness used (t) = 23.187 mm
 T / t = 1.337
 d / t = 21.908

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 2.706$$

Nr/P = 0.057
 Mr/P = 0.093
 M0/P = 0.037
 N0/P = 0.125
 MrD/Mc = 0.224
 NrDL/Mc = 0.104
 M0d/Mc = 0.101
 N0DL/Mc = 0.126
 MrD/Ml = 0.109
 NrDL/Ml = 0.054
 M0D/Ml = 0.050
 N0DL/Ml = 0.184

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	4	4	4	4
Outplane Bending (P)	8	-8	8	-8
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-13	-13	13	13
Outplane Bending (ML)	-21	21	21	-21
Normal Pressure Stress	19	20	19	20
Outplane Stress Summary	-1	25	67	8

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	2	2	2	2
Inplane Bending (P)	21	-21	21	-21
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-3	-3	3	3
Inplane Bending (ML)	-46	46	46	-46
Inplane Pressure Stress	9	9	9	9
Inplane Stress Summary	-17	33	84	-52

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	1	1	-1	-1
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	4	4	4	4
Shear Stress Summary	5	5	3	3

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
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Two * Max Shear Stress	18	35	84	61
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu	Cl	Du	Dl
	Left	Left	Right	Right
	Outside	Inside	Outside	Inside
Outplane Membrane (P)	4	4	4	4
Outplane Bending (P)	8	-8	8	-8
Outplane Membrane (Mc)	-9	-9	9	9
Outplane Bending (Mc)	-43	43	43	-43
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	9	9	9	9
Outplane Stress Summary	-29	40	75	-28

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses parallel to circumferential plane)	Cu	Cl	Du	Dl
	Left	Left	Right	Right
	Outside	Inside	Outside	Inside
Inplane Membrane (P)	2	2	2	2
Inplane Bending (P)	21	-21	21	-21
Inplane Membrane (Mc)	-7	-7	7	7
Inplane Bending (Mc)	-96	96	96	-96
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	19	20	19	20
Inplane Stress Summary	-60	89	147	-88

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Shear stress normal to circumferential plane)	Cu	Cl	Du	Dl
	Left	Left	Right	Right
	Outside	Inside	Outside	Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-1	-1	1	1
Torsional Shear (Mt)	4	4	4	4
Shear Stress Summary	3	3	5	5

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu	Cl	Du	Dl
	Left	Left	Right	Right
	Outside	Inside	Outside	Inside
Two * Max Shear Stress	60	90	147	88

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses in the hoop direction)	Au	Al	Bu	Bl
	Top	Top	Bottom	Bottom
	Outside	Inside	Outside	Inside
Hoop Membrane (P)	4	4	4	4
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-13	-13	13	13
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	12	14	12	14
Hoop Stress Summary	4	5	30	32



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Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to pipe cross-section)	Au	Al	Bu	Bl
	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Axial Membrane (P)	1	1	1	1
Axial Bending (P)	26	-26	26	-26
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-8	-8	8	8
Axial Bending (ML)	-62	62	62	-62
Axial Pressure Stress	6	6	6	6
Axial Stress Summary	-37	34	105	-73

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Shear stress)	Au	Al	Bu	Bl
	Outside	Inside	Outside	Inside
Shear due to (Vc)	1	1	-1	-1
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	5	5	5	5
Shear Stress Summary	6	6	4	4

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stress Intensities)	Au	Al	Bu	Bl
	Outside	Inside	Outside	Inside
Two * Max Shear Stress	43	36	106	106

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses in the hoop direction)	Cu	Cl	Du	Dl
	Left Outside	Left Inside	Right Outside	Right Inside
Hoop Membrane (P)	4	4	4	4
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	-9	-9	9	9
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	12	14	12	14
Hoop Stress Summary	8	9	26	27

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to pipe cross-section)	Cu	Cl	Du	Dl
	Left Outside	Left Inside	Right Outside	Right Inside
Axial Membrane (P)	1	1	1	1
Axial Bending (P)	26	-26	26	-26
Axial Membrane (Mc)	-8	-8	8	8
Axial Bending (Mc)	-132	132	132	-132
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	6	6	6	6
Axial Stress Summary	-106	104	175	-143

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Shear stress)	Cu	Cl	Du	Dl
	Outside	Inside	Outside	Inside



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Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	-1	-1	1	1
Shear due to Torsion	5	5	5	5
Shear Stress Summary	4	4	6	6

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Outside	Cl Inside	Du Outside	Dl Inside
Two * Max Shear Stress	115	104	175	171

Stress Computations at the Edge of the Reinforcing Pad:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{R_{\text{mean}}(t - c_a)}$$

$$= 508.0 + 2 \cdot 1.65 \cdot \sqrt{560.5(18.0 - 3.0)}$$

$$= 810.585 \text{ mm}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	1121.000 mm
Nozzle Outside Diameter	(d) =	808.000 mm
Vessel Thickness used	(T) =	15.000 mm
Nozzle Thickness used	(t) =	173.187 mm
T / t	=	0.087
d / t	=	10.000

$$\lambda = [(d/D) \cdot \sqrt{D/T}] = 6.231$$

Note: Re-pad thickness is added to nozzle thickness to simulate the Re-pad.

Nr/P	=	0.045
Mr/P	=	0.070
M0/P	=	0.019
N0/P	=	0.028
MrD/Mc	=	0.239
NrDL/Mc	=	0.093
M0d/Mc	=	0.068
N0DL/Mc	=	0.052
MrD/Ml	=	0.042
NrDL/Ml	=	0.038
M0D/Ml	=	0.013
N0DL/Ml	=	0.024

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	4	4	4	4
Outplane Bending (P)	19	-19	19	-19
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-4	-4	4	4
Outplane Bending (ML)	-15	15	15	-15
Normal Pressure Stress	40	41	40	41
Outplane Stress Summary	44	37	84	16

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses parallel to	Au Top	Al Top	Bu Bottom	Bl Bottom
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longitudinal plane)	Outside	Inside	Outside	Inside
Inplane Membrane (P)	7	7	7	7
Inplane Bending (P)	70	-70	70	-70
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-7	-7	7	7
Inplane Bending (ML)	-48	48	48	-48
Inplane Pressure Stress	20	20	20	20
Inplane Stress Summary	42	-1	153	-83

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE		Au	Al	Bu	Bl
(Shear stress normal to longitudinal plane)		Top	Top	Bottom	Bottom
		Outside	Inside	Outside	Inside
Outplane Shear (Vc)		1	1	-1	-1
Outplane Shear (Vl)		0	0	0	0
Outplane Shear (Mt)		3	3	3	3
Shear Stress Summary		4	4	1	1

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE		Au	Al	Bu	Bl
(Stress Intensities		Top	Top	Bottom	Bottom
		Outside	Inside	Outside	Inside
Two * Max Shear Stress		48	40	153	99

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE		Cu	Cl	Du	Dl
(Stresses Normal to circumferential plane)		Left	Left	Right	Right
		Outside	Inside	Outside	Inside
Outplane Membrane (P)		4	4	4	4
Outplane Bending (P)		19	-19	19	-19
Outplane Membrane (Mc)		-9	-9	9	9
Outplane Bending (Mc)		-79	79	79	-79
Outplane Membrane (ML)		0	0	0	0
Outplane Bending (ML)		0	0	0	0
Normal Pressure Stress		20	20	20	20
Outplane Stress Summary		-44	74	133	-63

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE		Cu	Cl	Du	Dl
(Stresses parallel to circumferential plane)		Left	Left	Right	Right
		Outside	Inside	Outside	Inside
Inplane Membrane (P)		7	7	7	7
Inplane Bending (P)		70	-70	70	-70
Inplane Membrane (Mc)		-17	-17	17	17
Inplane Bending (Mc)		-276	276	276	-276
Inplane Membrane (ML)		0	0	0	0
Inplane Bending (ML)		0	0	0	0
Inplane Pressure Stress		40	41	40	41
Inplane Stress Summary		-175	237	412	-279

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE		Cu	Cl	Du	Dl
(Shear stress normal to circumferential plane)		Left	Left	Right	Right
		Outside	Inside	Outside	Inside



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Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-1	-1	1	1
Torsional Shear (Mt)	3	3	3	3

Shear Stress Summary	1	1	4	4
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Two * Max Shear Stress	175	237	412	279

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	19	20	19	20	19	20	19	20
Circ. Pl (SUS)	-9	-9	17	17	-5	-5	9	9
Circ. Q (SUS)	-13	13	29	-29	-75	75	117	-117
Long. Pm (SUS)	9	9	9	9	9	9	9	9
Long. Pl (SUS)	-1	-1	5	5	-5	-5	13	13
Long. Q (SUS)	-25	25	67	-67	-35	35	51	-51
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	1	1	-1	-1	-1	-1	1	1
Shear Q (SUS)	4	4	4	4	4	4	4	4
Pm (SUS)	19.0	20.0	19.0	20.0	19.0	20.0	19.0	20.0
Pm+Pl (SUS)	10.4	11.3	36.0	37.0	14.1	15.1	28.2	29.1
Pm+Pl+Q (Total)	18.6	35.2	81.5	61.3	61.3	90.2	145.3	88.4

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	20.00	138.00	Passed
Pm+Pl (SUS)	37.04	207.00	Passed
Pm+Pl+Q (TOTAL)	145.35	413.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Reinforcing Pad Edge (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	40	41	40	41	40	41	40	41
Circ. Pl (SUS)	0	0	8	8	-10	-10	24	24
Circ. Q (SUS)	4	-4	34	-34	-206	206	346	-346
Long. Pm (SUS)	20	20	20	20	20	20	20	20
Long. Pl (SUS)	0	0	14	14	-5	-5	13	13



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Long. Q (SUS)	22	-22	118	-118	-60	60	98	-98
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	1	1	-1	-1	-1	-1	1	1
Shear Q (SUS)	3	3	3	3	3	3	3	3
Pm (SUS)	40.0	41.0	40.0	41.0	40.0	41.0	40.0	41.0
Pm+Pl (SUS)	40.0	41.0	48.1	49.1	30.1	31.1	64.0	65.0
Pm+Pl+Q (Total)	47.1	39.8	152.1	99.1	176.0	237.0	410.1	281.1

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	41.00	138.00	Passed
Pm+Pl (SUS)	65.03	207.00	Passed
Pm+Pl+Q (TOTAL)	410.06	413.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	12	14	12	14	12	14	12	14
Circ. Pl (SUS)	-9	-9	17	17	-5	-5	13	13
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	6	6	6	6	6	6	6	6
Long. Pl (SUS)	-7	-7	9	9	-7	-7	9	9
Long. Q (SUS)	-36	36	88	-88	-106	106	158	-158
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	1	1	-1	-1	-1	-1	1	1
Shear Q (SUS)	5	5	5	5	5	5	5	5
Pm (SUS)	12.0	14.0	12.0	14.0	12.0	14.0	12.0	14.0
Pm+Pl (SUS)	4.5	6.3	29.1	31.1	8.2	10.2	25.1	27.1
Pm+Pl+Q (Total)	41.8	36.2	103.2	104.3	114.3	105.2	173.2	170.4

Nozzle Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	14.00	118.00	Passed
Pm+Pl (SUS)	31.06	177.00	Passed
Pm+Pl+Q (TOTAL)	173.24	353.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

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Nozzle Calcs.: Vent

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Input, Nozzle Desc: Vent

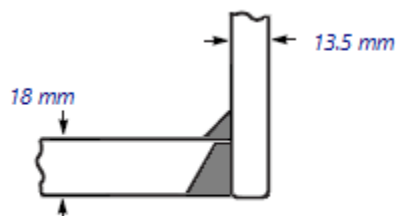
From: 40

Pressure for Reinforcement Calculations	P	1115.044	KPa.
Temperature for Internal Pressure	Temp	275	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm ²
Parent Allowable Stress At Ambient	Sva	138.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	1100.00	mm
Shell Finished (Minimum) Thickness	t	18.0000	mm
Shell Internal Corrosion Allowance	c	3.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		3809.00	mm
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-105	
Material UNS Number		K03504	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	132.50	N./mm ²
Allowable Stress At Ambient	Sna	138.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		0.00	deg
Diameter		2.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	13.4620	mm
Flange Material		SA-105	
Flange Type		Long Weld Neck	
Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	209.5500	mm
Weld leg size between Nozzle and Pad/Shell	Wo	20.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	14.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Flange Class		150	
Flange Grade		GR 1.1	

The Pressure Design option was MAWP + static head (to the nozzle).





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Nozzle Calcs.: Vent

Nozl: 17 9:45am Sep 9, 2026

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation 2.000 in.
Actual Thickness Used in Calculation 0.530 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
= $P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
= $1115 \cdot 553 / (138 \cdot 1 - 0.6 \cdot 1115)$
= 4.4904 mm

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
= $P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
= $1115 \cdot 28.4 / (133 \cdot 1 - 0.6 \cdot 1115)$
= 0.2402 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	113.6000	mm
Parallel to Vessel Wall, opening length	d	56.8000	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	26.1550	mm

*Taking a UG-36(c)(3)(a) exemption for nozzle: Vent.
This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation.*

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 3.2402	mm
Wall Thickness per UG16(b),	tr16b	= 5.5000	mm
Wall Thickness, shell/head, internal pressure	trb1	= 7.4904	mm
Wall Thickness	tb1 = max(trb1, tr16b)	= 7.4904	mm
Wall Thickness	tb2 = max(trb2, tr16b)	= 5.5000	mm
Wall Thickness per table UG-45	tb3	= 7.8000	mm

Determine Nozzle Thickness candidate [tb]:
= min[tb3, max(tb1, tb2)]
= min[7.8, max(7.49, 5.5)]
= 7.4904 mm

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
= max(ta, tb)
= max(3.24, 7.49)
= 7.4904 mm

Available Nozzle Neck Thickness = 13.4620 mm -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld (UCS-66(a)(1)(b)), min(Curve:A, Curve:B)

Govrn. thk, tg = 13.5, tr = 0.23, c = 3 mm, E* = 1
Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.022$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	1 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 °C
--	--------



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Nozzle Calcs.: Vent

Nozl: 17 9:45am

Sep 9, 2026

Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -43 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :

Design Pressure/Ambient Rating = $1115.04/1960.01 = 0.569$ Weld Size Calculations, Description: Vent

Intermediate Calc. for nozzle/shell Welds Tmin 10.4620 mm

Results Per UW-16.1:

Required Thickness Actual Thickness

Nozzle Weld $6.0000 = \text{Min per Code}$ $14.1400 = 0.7 * W_o$ mm

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 1.3747 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 228.9247 mm

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Nozzle Calcs.: Drain

Nozl: 18

9:45am

Sep 9, 2026

Input, Nozzle Desc: Drain

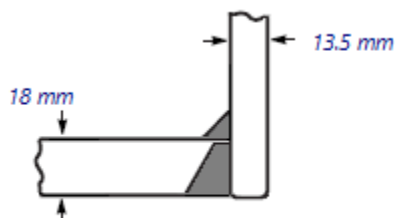
From: 40

Pressure for Reinforcement Calculations	P	1125.827	KPa.
Temperature for Internal Pressure	Temp	275	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm ²
Parent Allowable Stress At Ambient	Sva	138.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	1100.00	mm
Shell Finished (Minimum) Thickness	t	18.0000	mm
Shell Internal Corrosion Allowance	c	3.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		3309.00	mm
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-105	
Material UNS Number		K03504	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	132.50	N./mm ²
Allowable Stress At Ambient	Sna	138.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		180.00	deg
Diameter		2.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	13.4620	mm
Flange Material		SA-105	
Flange Type		Long Weld Neck	
Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	209.5500	mm
Weld leg size between Nozzle and Pad/Shell	Wo	20.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	14.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Flange Class		150	
Flange Grade		GR 1.1	

The Pressure Design option was MAWP + static head (to the nozzle).





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Nozzle Calcs.: Drain

Nozl: 18 9:45am Sep 9, 2026

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation 2.000 in.
Actual Thickness Used in Calculation 0.530 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
= $P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
= $1126 \cdot 553 / (138 \cdot 1 - 0.6 \cdot 1126)$
= 4.5340 mm

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
= $P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
= $1126 \cdot 28.4 / (133 \cdot 1 - 0.6 \cdot 1126)$
= 0.2426 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	113.6000	mm
Parallel to Vessel Wall, opening length	d	56.8000	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	26.1550	mm

*Taking a UG-36(c)(3)(a) exemption for nozzle: Drain.
This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation.*

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 3.2426	mm
Wall Thickness per UG16(b),	tr16b	= 5.5000	mm
Wall Thickness, shell/head, internal pressure	trb1	= 7.5340	mm
Wall Thickness	tb1 = max(trb1, tr16b)	= 7.5340	mm
Wall Thickness	tb2 = max(trb2, tr16b)	= 5.5000	mm
Wall Thickness per table UG-45	tb3	= 7.8000	mm

Determine Nozzle Thickness candidate [tb]:
= min[tb3, max(tb1, tb2)]
= min[7.8, max(7.53, 5.5)]
= 7.5340 mm

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
= max(ta, tb)
= max(3.24, 7.53)
= 7.5340 mm

Available Nozzle Neck Thickness = 13.4620 mm -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld (UCS-66(a)(1)(b)), min(Curve:A, Curve:B)

Govrn. thk, tg = 13.5, tr = 0.23, c = 3 mm, E* = 1
Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.022$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	1 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 °C
--	--------



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Nozzle Calcs.: Drain

Nozl: 18 9:45am

Sep 9, 2026

Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -42 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :

Design Pressure/Ambient Rating = $1125.83/1960.01 = 0.574$

Weld Size Calculations, Description: Drain

Intermediate Calc. for nozzle/shell Welds Tmin 10.4620 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$6.0000 = \text{Min per Code}$	$14.1400 = 0.7 * W_o$ mm

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 1.3747 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 228.9247 mm

This nozzle or its diameter limit intersects weld seam # [2] on this element. The distance between the nozzle and seam is, 38.61 mm , while the distance from the center of the nozzle to the edge of the diameter limit is: 56.80 mm

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ASME TS Calc: DESIGN 1

Case: 1 9:45a Sep 9,2026

Input Echo, Tubesheet Number 1, Description: Tubesheet

Shell Data:

Main Shell Description: Shell

Shell Maximum Design Pressure	Psd,max	1080.00 KPa.
Shell Maximum Operating Pressure	Psox,max	1080.00 KPa.
Shell Minimum Operating Pressure	Psox,min	0.00 KPa.
Shell Thickness	ts	18.0000 mm
Shell Internal Corrosion Allowance	cas	3.0000 mm
Shell External Corrosion Allowance	casext	0.0000 mm
Inside Diameter of Shell	Ds	1100.000 mm
Shell Temperature for Internal Pressure	Ts	275.00 °C
Shell Material		SA-516 70

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Shell Material UNS Number		K02700
Shell Allowable Stress at Temperature	Ss	137.00 N./mm ²
Shell Allowable Stress at Ambient		138.00 N./mm ²

Channel Data:

Channel Description: Channel shell

Channel Type:		Cylinder
Channel Maximum Design Pressure	Ptd,max	5900.00 KPa.
Channel Maximum Operating Pressure	Ptox,max	5900.00 KPa.
Channel Minimum Operating Pressure	Ptox,min	0.00 KPa.
Channel Thickness	tc	32.0000 mm
Channel Internal Corrosion Allowance	cac	3.0000 mm
Channel External Corrosion Allowance	cacext	0.0000 mm
Inside Diameter of Channel	Dc	1100.000 mm
Channel Design Temperature	TEMPC	430.00 °C
Channel Flange Matl		SA-387 11

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Channel Material UNS Number		K11789
Channel Allowable Stress at Temperature	Sc	147.00 N./mm ²
Channel Allowable Stress at Ambient		148.00 N./mm ²

Tube Data:

Number of Tube Holes	Nt	446
Tube Wall Thickness	et	2.1100 mm
Tube Outside Diameter	D	19.0500 mm
Total Straight Tube Length	Lt	5200.00 mm
Straight Tube Length (bet. inner tubsht faces) L		5030.00 mm
Design Temperature of the Tubes		430.00 °C
Tube Material		SA-213 T11
Tube Material UNS Number		K11597
Is this a Welded Tube		No
Tube Material Specification used		Smls. tube
Tube Allowable Stress at Temperature		98.98 N./mm ²
Tube Allowable Stress At Ambient		118.00 N./mm ²
Tube Yield Stress At design Temperature	Syt	148.20 N./mm ²
Tube Pitch (Center to Center Spacing)	P	23.8125 mm
Tube Layout Pattern		Triangular
Fillet Weld Leg	af	2.1100 mm
Groove Weld Leg	ag	2.1100 mm
Tube-Tubesheet Joint Weld Type		Full Strength
Method for Tube-Tubesheet Jt. Allowable		UW-20.3
Is the Tube Inset		No
Radius to Outermost Tube Hole Center	ro	528.475 mm
Largest Center-to-Center Tube Distance	U1	57.1500 mm



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ASME TS Calc: DESIGN 1

Case: 1 9:45a Sep 9, 2026

Length of Expanded Portion of Tube	ltx	0.0000 mm
Tube-side pass partition groove depth	hg	8.0000 mm
Area of Tube Layout	Ap	2685.8 cm ²
Perimeter of Tube Layout	Cp	2543.2 mm

Tubesheet Data:

Tubesheet TYPE: U-tube, Integral both sides, Conf. a

Tubesheet Design Metal Temperature	T	430.00 °C
Tubesheet Material		SA-336 F11

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

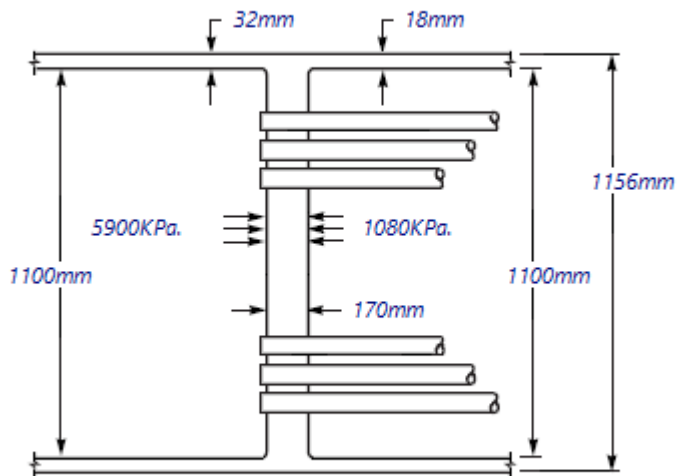
Tubesheet Material UNS Number		K11572
Tubesheet Allowable Stress at Temperature	S	147.00 N./mm ²
Tubesheet Allowable Stress at Ambient	Tt	148.00 N./mm ²
Thickness of Tubesheet	h	170.0000 mm
Tubesheet Corr. Allowance (Shell side)	Cats	3.0000 mm
Tubesheet Corr. Allowance (Channel side)	Catc	3.0000 mm
Tubesheet Outside Diameter	A	1156.000 mm

Area of the Untubed Lanes	AL	419.1 cm ²
---------------------------	----	-----------------------

Junction Stress Reduction option	Perform Plastic Calculation
Perform Differential Pressure Design	No
Run Multiple Load Cases	Yes

Tubesheet Integral with	Both
Tubesheet Extended as Flange	No

Tubesheet Integral w/Shell and Channel



Configuration a:

ASME TubeSheet Results per Part UHX/4.18, 2025

Elasticity/Expansion Material Properties :

Shell - TM-1 Carbon Steels with C<= 0.3%



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ASME TS Calc: DESIGN 1

Case: 1 9:45a Sep 9,2026

Elastic Mod. at Design Temperature	275.0 °C	0.18674E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20270E+09 KPa.

Channel - TM-1 Material Group C

Elastic Mod. at Design Temperature	430.0 °C	0.17617E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20408E+09 KPa.

Tubes - TM-1 Material Group C

Elastic Mod. at Tubsht. Design Temp.	430.0 °C	0.17617E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20408E+09 KPa.

TubeSheet - TM-1 Material Group C

Elastic Mod. at Design Temperature	430.0 °C	0.17617E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20408E+09 KPa.

Tube Required Thickness under Internal Pressure (Tubeside pressure):

Thickness Due to Internal Pressure:

$$= (P \cdot (D/2 - CAE)) / (S \cdot E + 0.4 \cdot P) \text{ per Appendix 1-1 (a)(1)}$$

$$= (5900 \cdot (19/2 - 0)) / (99 \cdot 1 + 0.4 \cdot 5900)$$

$$= 0.5546 + 0.0000 = 0.5546 \text{ mm}$$

Tube Required Thickness under External Pressure (Shellside pressure) :

External Pressure Chart CS-2 at 430.00 °C
Elastic Modulus for Material 156378400.00 KPa.

Results for Max. Allowable External Pressure (Emapw):

TCA	ODCA	SLEN	D/T	L/D	Factor A	B
2.1100	19.05	5558.48	9.03	50.0000	0.0134948	77.47

EMAWP = (2.167 / (D/T) - 0.0833) * B = 12140 KPa.

Results for Req'd Thickness for Ext. Pressure (Tca):

TCA	ODCA	SLEN	D/T	L/D	Factor A	B
0.4023	19.05	5558.48	47.35	50.0000	0.0004906	38.36

EMAWP = (4 * B) / (3 * (D/T)) = (4 * 38.4) / (3 * 47.4) = 1080 KPa.

Summary of Tube Required Thickness Results:

Total Required Thickness including Corrosion all.	0.5546 mm
Allowable Internal Pressure at Corroded thickness	24057.11 KPa.
Required Internal Design Pressure	5900.00 KPa.
Allowable External Pressure at Corroded thickness	12139.82 KPa.
Required External Design Pressure	1080.00 KPa.
Required Thickness due to Shell Side pressure	0.4023 mm

Detailed Results for load Case D1 corr. (Psd,min + Ptd,max)

Results for ASME U-tube Tubesheet Calculations for Configuration a:

Per Edition 2025, Original Thickness :

ASME Section VIII-2 Part 4.18.7.4 Step 1:

Equivalent outer tube limit circle diameter [Do]:

$$= 2 \cdot r_o + dt$$

$$= 2 \cdot 528 + 19$$

$$= 1076.000 \text{ mm}$$

Basic ligament efficiency for shear [μ]:



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ASME TS Calc: DESIGN 1

Case: 1 9:45a Sep 9, 2026

$$\begin{aligned}
 &= (p - dt) / p \\
 &= (23.8 - 19) / 23.8 \\
 &= 0.200
 \end{aligned}$$

ASME Section VIII-2 Part 4.18.7.4 Step 2:

Ratio [ps]:

$$\begin{aligned}
 &= D_s / D_o \text{ (Configurations a, b, c)} \\
 &= 1106 / 1076 \\
 &= 1.0279
 \end{aligned}$$

Ratio [pc]:

$$\begin{aligned}
 &= D_c / D_o \text{ (Configurations a, e, f)} \\
 &= 1106 / 1076 \\
 &= 1.0279
 \end{aligned}$$

ASME Section VIII-2 Part 4.18.7.4 Step 3:

Ratio [p]:

$$\begin{aligned}
 &= l_{tx} / h \\
 &= 0 / 164 \\
 &= 0.000 \text{ (must be } 0 \leq p \leq 1 \text{)}
 \end{aligned}$$

Effective tube hole Diameter [d*]:

$$\begin{aligned}
 &= \max(dt - 2t_t (E_t T / E) \cdot (S_t T / S) \cdot (p); dt - 2t_t) \\
 &= \max(19 - 2 \cdot 2.11 (176174368 / 176174368) \cdot (99 / 147) \cdot (0); 19 - 2 \cdot 2.11) \\
 &= 19.0500 \text{ mm}
 \end{aligned}$$

Effective tube pitch [p*]:

$$\begin{aligned}
 &= p / \sqrt{ (1 - 4 \cdot \min(A_l; 4 \cdot D_o \cdot p) / (\pi \cdot D_o^2)) } \\
 &= 23.8 / \sqrt{ (1 - 4 \cdot \min(419; 4 \cdot 1076 \cdot 23.8) / (\pi \cdot 1076^2)) } \\
 &= 24.3810 \text{ mm}
 \end{aligned}$$

Effective ligament efficiency for bending [μ*]:

$$\begin{aligned}
 &= (p^* - d^*) / p^* \\
 &= (24.4 - 19) / 24.4 \\
 &= 0.2187
 \end{aligned}$$

$h/p = 6.887139$; $\mu^* = 0.218653$; $E^* = 31303484 \text{ KPa}$.

$E^*/E = 0.177685$; $\nu^* = 0.432843$

Note: As h/p (6.887) is > 2 , data values for $h/p = 2$ were used.

ASME Section VIII-2 Part 4.18.7.4 Step 4:

Shell coefficient [βs]:

$$\begin{aligned}
 &= ((12 (1 - \nu_s^2))^{1/4}) / \sqrt{ (D_s + t_s) t_s } \\
 &= ((12 (1 - 0.3^2))^{1/4}) / \sqrt{ (1106 + 15) 15 } \\
 &= 0.0140 \text{ 1/mm}
 \end{aligned}$$

Shell coefficient [ks]:

$$\begin{aligned}
 &= \beta_s \cdot E_s \cdot t_s^3 / (6 (1 - \nu_s^2)) \\
 &= 0.014 \cdot 186735680 \cdot 15^3 / (6 (1 - 0.3^2)) \\
 &= 1618138240.0000 \text{ KPa} \cdot \text{mm}^2
 \end{aligned}$$

Shell coefficient [λs]:

$$\begin{aligned}
 &= 6 (D_s + t_s) \cdot k_s / h^3 \cdot (1 + h \cdot \beta_s + (h \cdot \beta_s)^2 / 2) \\
 &= 6 (1106 + 15) \cdot 1618138240 / 164^3 \cdot (1 + 164 \cdot 0.014 + (164 \cdot 0.014)^2 / 2) \\
 &= 14661119.0 \text{ KPa}
 \end{aligned}$$

Shell coefficient [δs]:

$$\begin{aligned}
 &= D_s^2 / (4 \cdot E_s \cdot t_s) \cdot (1 - D_s \cdot \nu_s / (2 (D_s + t_s))) \\
 &= 1106^2 / (4 \cdot 186735680 \cdot 15) \cdot (1 - 1106 \cdot 0.3 / (2 (1106 + 15))) \\
 &= 0.93014E-01 \text{ mm/N} \cdot \text{mm}^2
 \end{aligned}$$



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ASME TS Calc: DESIGN 1

Case: 1 9:45a Sep 9, 2026

Parameter [ws]:

$$= p_s \cdot k_s \cdot \beta_s \cdot \delta_s (1 + h \cdot \beta_s)$$

$$= 1.03 \cdot 1618138240 \cdot 0.014 \cdot 0.093014315 (1 + 164 \cdot 0.014)$$

$$= 7155.3599 \text{ mm}^2$$

Channel coefficient [β_c]:

$$= \left(\left(12(1 - v_c^2) \right)^{1/4} \right) / \sqrt{\left((D_c + t_c) t_c \right)}$$

$$= \left(\left(12(1 - 0.3^2) \right)^{1/4} \right) / \sqrt{\left((1106 + 29) 29 \right)}$$

$$= 0.0100 \text{ 1/mm}$$

Channel coefficient [kc]:

$$= \beta_c \cdot E_c \cdot t_c^3 / (6(1 - v_s^2))$$

$$= 0.01 \cdot 176174368 \cdot 29^3 / (6(1 - 0.3^2))$$

$$= 7885020160.0000 \text{ KPa} \cdot \text{mm}^2$$

Channel coefficients [CC, CqDelta, CqTheta, CmTheta, CmDelta]:

$$CqDelta = CqTheta = CmTheta = CmDelta = 1$$

Channel coefficient [λ_c]:

$$= 6(D_c + t_c / \cos(\theta)) \cdot k_c / h^3 \cdot (CmTheta + h \cdot \beta_c \cdot (Cm\Delta + CqTheta) / 2 + (h \cdot \beta_c)^2 / 2 \cdot Cq\Delta)$$

$$= 6(1106 + 29 / \cos(0)) \cdot 7.89E+09 / 164^3 \cdot (1 + 164 \cdot 0.01 \cdot (1 + 1) / 2 + (164 \cdot 0.01)^2 / 2 \cdot 1)$$

$$= 48613668.0 \text{ KPa}$$

Channel coefficient [δ_c]:

$$= D_c^2 / (4 \cdot E_c \cdot t_c) \cdot (1 - D_c \cdot v_c / (2(D_c + t_c)))$$

$$= 1106^2 / (4 \cdot 176174368 \cdot 29) \cdot (1 - 1106 \cdot 0.3 / (2(1106 + 29)))$$

$$= 0.51104E-01 \text{ mm/N./mm}^2$$

Channel coefficient [zetac]:

$$zetac = 0$$

Parameter [ω_c]:

$$= p_c \cdot k_c [\delta_c (CmTheta \cdot \beta_c + Cq\Delta \cdot h \cdot \beta_c^2) - \zeta_c (CmTheta + h \cdot CqTheta \cdot h \cdot \beta_c / 2)]$$

$$= 1.03 \cdot 7.89E+09 [0.051104259 (1 + 0.01 \cdot 1 \cdot 164) - 0.000 (1/164 + 1 \cdot 0.01/2)]$$

$$= 10970.4473 \text{ mm}^2$$

ASME Section VIII-2 Part 4.18.7.4 Step 5:

Diameter ratio [K]:

$$= A / D_o$$

$$= 1156 / 1076$$

$$= 1.0743$$

Coefficient [F]:

$$= (1 - v^*) / E^* \cdot (\lambda_s + \lambda_c + E \cdot \ln(K))$$

$$= (1 - 0.43) / 31303484 \cdot (14661119 + 48613668 + 176174368 \cdot \ln(1.07))$$

$$= 1.3753$$

ASME Section VIII-2 Part 4.18.7.4 Step 6:

Moment on tubesheet due to pressures (Ps, Pt) [Mts]:

$$= D_o^2 / 16 \cdot [(p_s - 1) \cdot (p_s^2 + 1) \cdot P_s - (p_c - 1) \cdot (p_c^2 + 1) \cdot P_t]$$

$$= 1076^2 / 16 \cdot [(1.03 - 1) \cdot (1.03^2 + 1) \cdot 0 - (1.03 - 1) \cdot (1.03^2 + 1) \cdot 5900]$$

$$= -24479592.0000 \text{ KPa} \cdot \text{mm}^2$$

Moment acting on unperforated tubesheet rim [M*]:

$$= Mts + \omega_c \cdot P_t - \omega_s \cdot P_s$$

$$= -24479592 + 10970 \cdot 5900 - 7155 \cdot 0$$



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$$= 40246052.0000 \text{ KPa} \cdot \text{mm}^2$$

ASME Section VIII-2 Part 4.18.7.4 Step 7:

Maximum bending moment acting on periphery of tubesheet [Mp]:

$$\begin{aligned} &= ((M \cdot) - D_o^2/32 \cdot F (P_s - P_t)) / (1 + F) \\ &= ((40246052) - 1076^2/32 \cdot 1.38 (0 - 5900)) / (1 + 1.38) \\ &= 140540608.0000 \text{ KPa} \cdot \text{mm}^2 \end{aligned}$$

Maximum bending moment acting on center of tubesheet [Mo]:

$$\begin{aligned} &= M_p + D_o^2/64 \cdot (3 + \nu) \cdot (P_s - P_t) \\ &= 140540608 + 1076^2/64 \cdot (3 + 0.43) \cdot (0 - 5900) \\ &= -225855168.0000 \text{ KPa} \cdot \text{mm}^2 \end{aligned}$$

Maximum bending moment acting on tubesheet [M]:

$$\begin{aligned} &= \max(| M_p | ; | M_o |) \\ &= \max(| 140540608 | ; | -225855168 |) \\ &= 225855168.0000 \text{ KPa} \cdot \text{mm}^2 \end{aligned}$$

ASME Section VIII-2 Part 4.18.7.4 Step 8:

Tubesheet Bending Stress at original thickness:

$$\begin{aligned} &= 6 \cdot M / (\mu \cdot (h - h_g')^2) \\ &= 6 \cdot 225855168 / (0.22 (164 - 5)^2) \\ &= 245.1636 \text{ N./mm}^2 \end{aligned}$$

Allowable Tubesheet Bending Stress [σ_{All}]:

$$\begin{aligned} &= 2 \cdot S \\ &= 2 \cdot 147 \\ &= 294.01 \text{ N./mm}^2 \end{aligned}$$

Tubesheet bending stress at final thickness [σ]:

$$\begin{aligned} &= 6 \cdot M / (\mu \cdot (h - h_g')^2) \\ &= 6 \cdot 220727728 / (0.22 (149 - 5)^2) \\ &= 293.6114 \text{ N./mm}^2 \end{aligned}$$

Required tubesheet thickness for Bending Stress [HreqB]:

$$\begin{aligned} &= h + C_{ats} + C_{atc} \\ &= 149 + 3 + 3 \\ &= 154.6323 \text{ mm} \end{aligned}$$

Required tubesheet thickness for given loadings (includes CA) [Hreq]:

$$\begin{aligned} &= \max(H_{reqB} ; H_{reqS}) \\ &= \max(155 ; 32.5) \\ &= 154.6323 \text{ mm} \end{aligned}$$

ASME Section VIII-2 Part 4.18.7.4 Step 9:

Absolute pressure differential [$| P_s - P_t |$]:

$$\begin{aligned} &= | P_s - P_t | \\ &= | 0 - 5900 | \\ &= 5900.000 \text{ KPa.} \end{aligned}$$

Shear Stress check [τ_{limit}]:

$$\begin{aligned} &= 4 \cdot \mu \cdot h / D_o \cdot \min(0.8 \cdot S ; 0.533 \cdot S_y) \\ &= 4 \cdot 0.2 \cdot 164 / 1076 \cdot \min(0.8 \cdot 147 ; 0.533 \cdot 223) \\ &= 14.340 \text{ N./mm}^2 \end{aligned}$$

$| P_s - P_t |$ is $\leq$ the Shear Stress limit (τ_{limit}). Therefore, the required thickness for shear is for information only.

Minimum required thickness for Shear [HreqS]:

$$\begin{aligned} &= 1 / (4 \cdot \mu \cdot \min(0.8 \cdot S ; 0.533 \cdot S_y)) \cdot (4 \cdot A_p / C_p) \cdot | P_s - P_t | + C_{ats} + C_{atc} \\ &= 1 / (4 \cdot 0.2 \cdot 118) \cdot (4 \cdot 2686 / 2543) \cdot | 0 - 5900 | + 6 \\ &= 32.4927 \text{ mm} \end{aligned}$$



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ASME Section VIII-2 Part 4.18.7.4 Step 10:

Note: For a given Shell thickness of: 15.000mm

Min. Shell len. adjacent to the tubesheet is: 231.844 mm

Axial Shell Membrane Stress [σ_s, m]:

$$= D_s^2 / (4 \cdot t_s (D_s + t_s)) \cdot P_s$$

$$= 1106^2 / (4 \cdot 15 (1106 + 15)) \cdot 0$$

$$= 0.0000 \text{ N./mm}^2$$

Axial Shell Bending Stress [σ_s, b]:

$$= 6 \cdot k_s / t_s^2 \cdot [\beta S \cdot P_s + 6(1 - \nu^*) / E^* \cdot$$

$$Do / h^3 \cdot (1 + h \cdot \beta S / 2) \cdot (M_p + Do^2 / 32 \cdot (P_s - P_t))]$$

$$= 6 \cdot 363780 / 15^2 \cdot [0.014 \cdot 0 + 6(1 - 0.43) / 31303484 \cdot$$

$$1076 / 164^3 \cdot (1 + 164 \cdot 0.014 / 2) \cdot (140540608 + 1076^2 / 32 \cdot$$

$$(0 - 5900))]$$

$$= -179.378 \text{ N./mm}^2$$

Shell Membrane + Bending Stress [σ_s]:

$$= | \sigma_{s,m} | + | \sigma_{s,b} | \text{ (Should be } \leq 1.5 \cdot S_s)$$

$$= 179.378 \text{ N./mm}^2 \text{ (Should be } \leq 205.50)$$

Note: For a given Channel thickness of: 29.000 mm

Min. Channel len. adjacent to the tubesheet is: 322.366mm

Axial Channel Membrane Stress [σ_c, m]:

$$= D_c^2 / (4 \cdot t_c (D_c + t_c) \cdot \cos(\theta)) \cdot P_t$$

$$= 1106^2 / (4 \cdot 29 (1106 + 29) \cdot \cos(0)) \cdot 5900$$

$$= 54.819 \text{ N./mm}^2$$

Axial Channel Bending Stress [σ_c, b]:

$$= 6 \cdot k_c / t_c^2 \cdot [(\delta_c \cdot C_m \Delta \cdot \beta_c - \zeta_c \cdot C_m \theta) \cdot P_t -$$

$$6((1 - \nu^*) / E^*) \cdot Do / h^3 \cdot (C_m \theta +$$

$$h \cdot \beta_c / 2 \cdot C_m \Delta) \cdot (M_p + Do^2 / 32 \cdot (P_s - P_t))]$$

$$= 6 \cdot 1772663 / 29^2 \cdot [(0.051104259 \cdot 1 \cdot 0.01 - 0.000 \cdot 1) \cdot 5900 -$$

$$6((1 - 0.43) / 31303484) \cdot 1076 / 164^3 \cdot (1 +$$

$$0.01 \cdot 164 / 2 \cdot 1) \cdot (140540608 + 1076^2 / 32 \cdot (0 - 5900))]$$

$$= 368.150 \text{ N./mm}^2$$

Channel Membrane + Bending Stress [σ_c]:

$$= | \sigma_{c,m} | + | \sigma_{c,b} | \text{ (Should be } \leq 1.5 \cdot S_c)$$

$$= 422.969 \text{ N./mm}^2 \text{ (Should be } \leq 220.51)$$

Step 11, The Cylinder-to-Tubesheet Juncture is Overstressed.

Performing a simplified Elastic-Plastic calculation

(option 3) to reduce the overstress condition.

Modify Es and/or Ec and recompute from Step 4 onwards.

Modified Elasticity [E_c]:

$$= E_c \cdot \sqrt{(1.5 \cdot S_c / \sigma_c)}$$

$$= 176174368 \cdot \sqrt{(1.5 \cdot 147 / 423)}$$

$$= 127203536.0 \text{ KPa.}$$

Tubesheet Bending Stress at Original Thk., after the Elastic-Plastic iteration:

$$= 6 \cdot M / (\mu^* (h - h'g)^2)$$

$$= 6 \cdot 234235392 / (0.22 (164 - 5)^2)$$

$$= 254.3 \text{ N./mm}^2 \text{ (Should be } \leq 294.0)$$



Note: The tubesheet is Not overstressed after the Elastic-Plastic iteration.

The design is acceptable. Recomputing tubesheet required thickness.



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Required Tubesheet Thk., for Bending Stress after the Elastic-Plastic iteration [HreqB]:
 $= h + Cats + Catc = 152 + 3 + 3 = 158 \text{ mm}$

Required Tubesheet Thk. after the Elastic-Plastic iteration (includes CA) [Hreq]:
 $= \max(HreqB; HreqS) = \max(158; 32.5) = 158 \text{ mm}$

Tube Metal Area [At]:

$$= \pi (d_o - t) t$$

$$= \pi (19 - 2.11) 2.11$$

$$= 1.123 \text{ cm}^2$$

Tube Weld Size Results per UW-20.3:

Tube Strength [Ft]:

$$= \pi \cdot t (d_o - t) S$$

$$= \pi \cdot 2.11 (19 - 2.11) 99$$

$$= 1133.373 \text{ Kg}$$

Groove Weld Strength [Fg]:

$$= 0.85 \cdot \pi \cdot a_g (d_o + 0.67 \cdot a_g) S_w \text{ (but not } > F_t)$$

$$= 0.85 \cdot \pi \cdot 2.11 (19 + 0.67 \cdot 2.11) 99$$

$$= 1133.3734 \text{ Kg}$$

Weld Strength Factor [fw]:

$$= S / S_w \text{ where } S_w = \min(S \text{ and } S_t)$$

$$= 99/99$$

$$= 1.000$$

Design Strength Ratio [fd]:

$$= 1 \text{ for full strength welds}$$

Fillet Weld Strength [Ff]:

$$= 0.55 \cdot \pi \cdot a_f (d_o + 0.67 \cdot a_f) S_w \text{ (but not } > F_t)$$

$$= 0.55 \cdot \pi \cdot 2.11 (19 + 0.67 \cdot 2.11) 99$$

$$= 753.0200 \text{ Kg}$$

Minimum Length of Weld [ar]:

$$= 2 \left(\sqrt{((0.75 \cdot d_o)^2 + 1.07 \cdot t \cdot (d_o - t) \cdot f_w \cdot f_d)} - 0.75 \cdot d_o \right)$$

$$= 2 \left(\sqrt{((0.75 \cdot 19)^2 + 1.07 \cdot 2.11 \cdot (19 - 2.11) \cdot 1 \cdot 1)} - 0.75 \cdot 19 \right)$$

$$= 2.562 \text{ mm}$$

Minimum Required Fillet Weld Leg [afr]: 1.2810 mm

Minimum Required Groove Weld Leg [agr]: 1.2810 mm

Tube-Tubesheet Jt allowable, 1133 is $\geq$ tube strength 1133 Kg

Note: This tube-tubesheet joint is a Full Strength joint

Stress/Force summary for loadcase D1 corr. (Psd,min + Ptd,max):

Stress Description	Actual	Allowable	Pass/Fail
Tubesheet bend. stress	254.3 <=	294.0 N./mm ²	Ok
Tubesheet shear stress	19.0 <=	117.6 N./mm ²	Ok
Stress in Shell at Tubesheet	179.4 <=	205.5 N./mm ²	Ok
Stress in Channel at Tubesheet	423.0 <=	446.0 N./mm ²	Ok

Thickness results for loadcase D1 corr. (Psd,min + Ptd,max):

Thickness (mm)	Required	Actual	P/F
Tubesheet Thickness :	157.731	170.000	Ok



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Tube-Tubesheet Fillet Weld Leg :	1.281	2.110	Ok
Tube-Tubesheet Groove Weld Leg :	1.281	2.110	Ok

Detailed Results for load Case D2 corr. (Psd,max + Ptd,min)

Results for ASME U-tube Tubesheet Calculations for Configuration a:

Per Edition 2025, Original Thickness :

ASME Section VIII-2 Part 4.18.7.4 Step 1:

Equivalent outer tube limit circle diameter [Do]:

$$= 2 \cdot r_o + dt$$

$$= 2 \cdot 528 + 19$$

$$= 1076.000 \text{ mm}$$

Basic ligament efficiency for shear [μ]:

$$= (p - dt) / p$$

$$= (23.8 - 19) / 23.8$$

$$= 0.200$$

ASME Section VIII-2 Part 4.18.7.4 Step 2:

Ratio [ρ_s]:

$$= D_s / D_o \text{ (Configurations a, b, c)}$$

$$= 1106 / 1076$$

$$= 1.0279$$

Ratio [ρ_c]:

$$= D_c / D_o \text{ (Configurations a, e, f)}$$

$$= 1106 / 1076$$

$$= 1.0279$$

ASME Section VIII-2 Part 4.18.7.4 Step 3:

Ratio [ρ]:

$$= l_{tx} / h$$

$$= 0 / 164$$

$$= 0.000 \text{ (must be } 0 \leq \rho \leq 1 \text{)}$$

Effective tube hole Diameter [d^*]:

$$= \max(dt - 2tt(E_t/E) \cdot (StT/S) \cdot (\rho); dt - 2tt)$$

$$= \max(19 - 2 \cdot 2.11 (176174368 / 176174368) \cdot (99/147) \cdot (0); 19 - 2 \cdot 2.11)$$

$$= 19.0500 \text{ mm}$$

Effective tube pitch [p^*]:

$$= p / \sqrt{ 1 - 4 \cdot \min(A_l; 4 \cdot D_o \cdot p) / (\pi \cdot D_o^2) }$$

$$= 23.8 / \sqrt{ 1 - 4 \cdot \min(419; 4 \cdot 1076 \cdot 23.8) / (\pi \cdot 1076^2) }$$

$$= 24.3810 \text{ mm}$$

Effective ligament efficiency for bending [μ^*]:

$$= (p^* - d^*) / p^*$$

$$= (24.4 - 19) / 24.4$$

$$= 0.2187$$

$h/p = 6.887139$; $\mu^* = 0.218653$; $E^* = 31303484 \text{ KPa}$.

$E^*/E = 0.177685$; $v^* = 0.432843$

Note: As h/p (6.887) is > 2 , data values for $h/p = 2$ were used.

ASME Section VIII-2 Part 4.18.7.4 Step 4:



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Shell coefficient [β_s]:

$$= \left(\left(12(1 - \nu_s^2) \right)^{1/4} / \sqrt{(D_s + t_s)t_s} \right)$$

$$= \left(\left(12(1 - 0.3^2) \right)^{1/4} / \sqrt{(1106 + 15)15} \right)$$

$$= 0.0140 \text{ 1/mm}$$

Shell coefficient [k_s]:

$$= \beta_s \cdot E_s \cdot t_s^3 / (6(1 - \nu_s^2))$$

$$= 0.014 \cdot 186735680 \cdot 15^3 / (6(1 - 0.3^2))$$

$$= 1618138240.0000 \text{ KPa} \cdot \text{mm}^2$$

Shell coefficient [λ_s]:

$$= 6(D_s + t_s) \cdot k_s / h^3 \cdot (1 + h \cdot \beta_s + (h \cdot \beta_s)^2 / 2)$$

$$= 6(1106 + 15) \cdot 1618138240 / 164^3 \cdot (1 + 164 \cdot 0.014 + (164 \cdot 0.014)^2 / 2)$$

$$= 14661119.0 \text{ KPa}$$

Shell coefficient [δ_s]:

$$= D_s^2 / (4 \cdot E_s \cdot t_s) \cdot (1 - D_s \cdot \nu_s / (2(D_s + t_s)))$$

$$= 1106^2 / (4 \cdot 186735680 \cdot 15) \cdot (1 - 1106 \cdot 0.3 / (2(1106 + 15)))$$

$$= 0.93014\text{E-}01 \text{ mm/N./mm}^2$$

Parameter [w_s]:

$$= p_s \cdot k_s \cdot \beta_s \cdot \delta_s (1 + h \cdot \beta_s)$$

$$= 1.03 \cdot 1618138240 \cdot 0.014 \cdot 0.93014315 (1 + 164 \cdot 0.014)$$

$$= 7155.3599 \text{ mm}^2$$

Channel coefficient [β_c]:

$$= \left(\left(12(1 - \nu_c^2) \right)^{1/4} / \sqrt{(D_c + t_c)t_c} \right)$$

$$= \left(\left(12(1 - 0.3^2) \right)^{1/4} / \sqrt{(1106 + 29)29} \right)$$

$$= 0.0100 \text{ 1/mm}$$

Channel coefficient [k_c]:

$$= \beta_c \cdot E_c \cdot t_c^3 / (6(1 - \nu_c^2))$$

$$= 0.01 \cdot 176174368 \cdot 29^3 / (6(1 - 0.3^2))$$

$$= 7885020160.0000 \text{ KPa} \cdot \text{mm}^2$$

Channel coefficients [CC, CqDelta, CqTheta, CmTheta, CmDelta]:

$$CqDelta = CqTheta = CmTheta = CmDelta = 1$$

Channel coefficient [λ_c]:

$$= 6(D_c + t_c / \cos(\theta)) \cdot k_c / h^3 \cdot (CmTheta + h \cdot \beta_c \cdot (CmDelta + CqTheta) / 2 + (h \cdot \beta_c)^2 / 2 \cdot CqDelta)$$

$$= 6(1106 + 29 / \cos(0)) \cdot 7.89\text{E}+09 / 164^3 \cdot (1 + 164 \cdot 0.01 \cdot (1 + 1) / 2 + (164 \cdot 0.01)^2 / 2 \cdot 1)$$

$$= 48613668.0 \text{ KPa}$$

Channel coefficient [δ_c]:

$$= D_c^2 / (4 \cdot E_c \cdot t_c) \cdot (1 - D_c \cdot \nu_c / (2(D_c + t_c)))$$

$$= 1106^2 / (4 \cdot 176174368 \cdot 29) \cdot (1 - 1106 \cdot 0.3 / (2(1106 + 29)))$$

$$= 0.51104\text{E-}01 \text{ mm/N./mm}^2$$

Channel coefficient [zetac]:

$$\text{zetac} = 0$$

Parameter [w_c]:

$$= p_c \cdot k_c [\delta_c (CmTheta \cdot \beta_c + CqDelta \cdot h \cdot \beta_c^2) - \zeta_c (CmTheta + h \cdot CqTheta \cdot h \cdot \beta_c / 2)]$$

$$= 1.03 \cdot 7.89\text{E}+09 [0.051104259 (1 + 0.01 \cdot 1 \cdot 164) - 0.000 (1/164 + 1 \cdot 0.01/2)]$$

$$= 10970.4473 \text{ mm}^2$$

ASME Section VIII-2 Part 4.18.7.4 Step 5:

Diameter ratio [K]:

$$= A / D_o$$



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$$= 1156/1076$$

$$= 1.0743$$

Coefficient [F]:

$$= (1 - \nu^*)/E^* \cdot (\lambda_s + \lambda_c + E \cdot \ln(K))$$

$$= (1 - 0.43)/31303484 \cdot (14661119 + 48613668 + 176174368 \cdot \ln(1.07))$$

$$= 1.3753$$

ASME Section VIII-2 Part 4.18.7.4 Step 6:

Moment on tubesheet due to pressures (Ps, Pt) [Mts]:

$$= Do^2/16 \cdot [(ps-1) \cdot (ps^2+1) \cdot Ps - (pc-1) \cdot (pc^2+1) \cdot Pt]$$

$$= 1076^2/16 \cdot [(1.03-1) \cdot (1.03^2+1) \cdot 1080 - (1.03-1) \cdot (1.03^2+1) \cdot 0]$$

$$= 4481010.5000 \text{ KPa} \cdot \text{mm}^2$$

Moment acting on unperforated tubesheet rim [M*]:

$$= Mts + \omega_c \cdot Pt - \omega_s \cdot Ps$$

$$= 4481010 + 10970 \cdot 0 - 7155 \cdot 1080$$

$$= -3246779.0000 \text{ KPa} \cdot \text{mm}^2$$

ASME Section VIII-2 Part 4.18.7.4 Step 7:

Maximum bending moment acting on periphery of tubesheet [Mp]:

$$= (M^*) - Do^2/32 \cdot F(Ps - Pt) / (1 + F)$$

$$= (-3246779) - 1076^2/32 \cdot 1.38(1080 - 0) / (1 + 1.38)$$

$$= -23991454.0000 \text{ KPa} \cdot \text{mm}^2$$

Maximum bending moment acting on center of tubesheet [Mo]:

$$= Mp + Do^2/64 \cdot (3 + \nu^*) \cdot (Ps - Pt)$$

$$= -23991454 + 1076^2/64 \cdot (3 + 0.43) \cdot (1080 - 0)$$

$$= 43077604.0000 \text{ KPa} \cdot \text{mm}^2$$

Maximum bending moment acting on tubesheet [M]:

$$= \max(|Mp|; |Mo|)$$

$$= \max(|-23991454|; |43077604|)$$

$$= 43077604.0000 \text{ KPa} \cdot \text{mm}^2$$

ASME Section VIII-2 Part 4.18.7.4 Step 8:

Tubesheet Bending Stress at original thickness:

$$= 6 \cdot M / (\mu^* (h - hg')^2)$$

$$= 6 \cdot 43077604 / (0.22 (164 - 5)^2)$$

$$= 46.7603 \text{ N./mm}^2$$

Allowable Tubesheet Bending Stress [σ_{All}]:

$$= 2 \cdot S$$

$$= 2 \cdot 147$$

$$= 294.01 \text{ N./mm}^2$$

Tubesheet bending stress at final thickness [σ]:

$$= 6 \cdot M / (\mu^* (h - hg')^2)$$

$$= 6 \cdot 35277460 / (0.22 (62.5 - 5)^2)$$

$$= 292.6969 \text{ N./mm}^2$$

Required tubesheet thickness for Bending Stress [HreqB]:

$$= h + Cats + Catc$$

$$= 62.5 + 3 + 3$$

$$= 68.5108 \text{ mm}$$

Required tubesheet thickness for given loadings (includes CA) [Hreq]:

$$= \max(HreqB; HreqS)$$

$$= \max(68.5; 10.8)$$

$$= 68.5108 \text{ mm}$$



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ASME Section VIII-2 Part 4.18.7.4 Step 9:

Absolute pressure differential [| Ps - Pt |]:

$$= | P_s - P_t |$$

$$= | 1080 - 0 |$$

$$= 1080.000 \text{ KPa.}$$

Shear Stress check [τ limit]:

$$= 4 \cdot \mu \cdot h / D_o \cdot \min(0.8 \cdot S; 0.533 \cdot S_y)$$

$$= 4 \cdot 0.2 \cdot 164 / 1076 \cdot \min(0.8 \cdot 147; 0.533 \cdot 223)$$

$$= 14.340 \text{ N./mm}^2$$

| Ps - Pt | is $\leq$ the Shear Stress limit (τ limit). Therefore, the required thickness for shear is for information only.

Minimum required thickness for Shear [H_{reqS}]:

$$= 1 / (4 \cdot \mu \cdot \min(0.8 \cdot S; 0.533 \cdot S_y)) \cdot (4 \cdot A_p / C_p) \cdot | P_s - P_t | + C_{ats} + C_{atc}$$

$$= 1 / (4 \cdot 0.2 \cdot 118) \cdot (4 \cdot 2686 / 2543) \cdot | 1080 - 0 | + 6$$

$$= 10.8495 \text{ mm}$$

ASME Section VIII-2 Part 4.18.7.4 Step 10:

Note: For a given Shell thickness of: 15.000mm

Min. Shell len. adjacent to the tubesheet is: 231.844 mm

Axial Shell Membrane Stress [$\sigma_{s,m}$]:

$$= D_s^2 / (4 \cdot t_s (D_s + t_s)) \cdot P_s$$

$$= 1106^2 / (4 \cdot 15 (1106 + 15)) \cdot 1080$$

$$= 19.6428 \text{ N./mm}^2$$

Axial Shell Bending Stress [$\sigma_{s,b}$]:

$$= 6 \cdot k_s / t_s^2 \cdot [\beta S \cdot P'_s + 6(1 - \nu^*) / E^* \cdot D_o / h^3 \cdot (1 + h \cdot \beta S / 2) \cdot (M_p + D_o^2 / 32 \cdot (P_s - P_t))]$$

$$= 6 \cdot 363780 / 15^2 \cdot [0.014 \cdot 0.1 + 6(1 - 0.43) / 31303484 \cdot 1076 / 164^3 \cdot (1 + 164 \cdot 0.014 / 2) \cdot (-23991454 + 1076^2 / 32 \cdot (1080 - 0))]$$

$$= 97.876 \text{ N./mm}^2$$

Shell Membrane + Bending Stress [σ_s]:

$$= | \sigma_{s,m} | + | \sigma_{s,b} | \text{ (Should be } \leq 1.5 \cdot S_s)$$

$$= 117.518 \text{ N./mm}^2 \text{ (Should be } \leq 205.50)$$

Note: For a given Channel thickness of: 29.000 mm

Min. Channel len. adjacent to the tubesheet is: 322.366mm

Axial Channel Membrane Stress [$\sigma_{c,m}$]:

$$= D_c^2 / (4 \cdot t_c (D_c + t_c)) \cdot \cos(\theta) \cdot P_t$$

$$= 1106^2 / (4 \cdot 29 (1106 + 29)) \cdot \cos(0) \cdot 0$$

$$= 0.000 \text{ N./mm}^2$$

Axial Channel Bending Stress [$\sigma_{c,b}$]:

$$= 6 \cdot k_c / t_c^2 \cdot [(\delta_c \cdot C_m \Delta \cdot \beta_c - \zeta_c \cdot C_m \Theta) \cdot P_t - 6((1 - \nu^*) / E^*) \cdot D_o / h^3 \cdot (C_m \Theta + h \cdot \beta_c / 2 \cdot C_m \Delta) \cdot (M_p + D_o^2 / 32 \cdot (P_s - P_t))]$$

$$= 6 \cdot 1772663 / 29^2 \cdot [(0.051104259 \cdot 1 \cdot 0.01 - 0.000 \cdot 1) \cdot 0 - 6((1 - 0.43) / 31303484) \cdot 1076 / 164^3 \cdot (1 + 0.01 \cdot 164 / 2 \cdot 1) \cdot (-23991454 + 1076^2 / 32 \cdot (1080 - 0))]$$

$$= -40.991 \text{ N./mm}^2$$

Channel Membrane + Bending Stress [σ_c]:

$$= | \sigma_{c,m} | + | \sigma_{c,b} | \text{ (Should be } \leq 1.5 \cdot S_c)$$

$$= 40.991 \text{ N./mm}^2 \text{ (Should be } \leq 220.51)$$

Tube Metal Area [At]:



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$$\begin{aligned}
 &= \pi (d_o - t) t \\
 &= \pi (19 - 2.11) 2.11 \\
 &= 1.123 \text{ cm}^2
 \end{aligned}$$

Tube Weld Size Results per UW-20.3:

Tube Strength [Ft]:

$$\begin{aligned}
 &= \pi \cdot t (d_o - t) S \\
 &= \pi \cdot 2.11 (19 - 2.11) 99 \\
 &= 1133.373 \text{ Kg}
 \end{aligned}$$

Groove Weld Strength [Fg]:

$$\begin{aligned}
 &= 0.85 \cdot \pi \cdot a_g (d_o + 0.67 \cdot a_g) S_w \text{ (but not } > F_t) \\
 &= 0.85 \cdot \pi \cdot 2.11 (19 + 0.67 \cdot 2.11) 99 \\
 &= 1133.3734 \text{ Kg}
 \end{aligned}$$

Weld Strength Factor [fw]:

$$\begin{aligned}
 &= S / S_w \text{ where } S_w = \min(S \text{ and } S_t) \\
 &= 99/99 \\
 &= 1.000
 \end{aligned}$$

Design Strength Ratio [fd]:

$$= 1 \text{ for full strength welds}$$

Fillet Weld Strength [Ff]:

$$\begin{aligned}
 &= 0.55 \cdot \pi \cdot a_f (d_o + 0.67 \cdot a_f) S_w \text{ (but not } > F_t) \\
 &= 0.55 \cdot \pi \cdot 2.11 (19 + 0.67 \cdot 2.11) 99 \\
 &= 753.0200 \text{ Kg}
 \end{aligned}$$

Minimum Length of Weld [ar]:

$$\begin{aligned}
 &= 2 \left(\sqrt{(0.75 \cdot d_o)^2 + 1.07 \cdot t \cdot (d_o - t) f_w \cdot f_d} - 0.75 \cdot d_o \right) \\
 &= 2 \left(\sqrt{(0.75 \cdot 19)^2 + 1.07 \cdot 2.11 \cdot (19 - 2.11) 1 \cdot 1} - 0.75 \cdot 19 \right) \\
 &= 2.562 \text{ mm}
 \end{aligned}$$

Minimum Required Fillet Weld Leg [afr]: 1.2810 mm

Minimum Required Groove Weld Leg [agr]: 1.2810 mm

Tube-Tubesheet Jt allowable, 1133 is $\geq$ tube strength 1133 Kg

Note: This tube-tubesheet joint is a Full Strength joint

Stress/Force summary for loadcase D2 corr. (Psd,max + Ptd,min):

Stress Description	Actual	Allowable	Pass/Fail
Tubesheet bend. stress	46.8 <=	294.0 N./mm ²	Ok
Tubesheet shear stress	3.5 <=	117.6 N./mm ²	Ok
Stress in Shell at Tubesheet	117.5 <=	205.5 N./mm ²	Ok
Stress in Channel at Tubesheet	41.0 <=	220.5 N./mm ²	Ok

Thickness results for loadcase D2 corr. (Psd,max + Ptd,min):

Thickness (mm)	Required	Actual	P/F
Tubesheet Thickness :	68.511	170.000	Ok
Tube-Tubesheet Fillet Weld Leg :	1.281	2.110	Ok
Tube-Tubesheet Groove Weld Leg :	1.281	2.110	Ok

Detailed Results for load Case D3 corr. (Psd,max + Ptd,max)

Results for ASME U-tube Tubesheet Calculations for Configuration a:



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Per Edition 2025, Original Thickness :

ASME Section VIII-2 Part 4.18.7.4 Step 1:

Equivalent outer tube limit circle diameter [Do]:

$$\begin{aligned} &= 2 \cdot r_o + dt \\ &= 2 \cdot 528 + 19 \\ &= 1076.000 \text{ mm} \end{aligned}$$

Basic ligament efficiency for shear [μ]:

$$\begin{aligned} &= (p - dt) / p \\ &= (23.8 - 19) / 23.8 \\ &= 0.200 \end{aligned}$$

ASME Section VIII-2 Part 4.18.7.4 Step 2:

Ratio [ρ_s]:

$$\begin{aligned} &= D_s / D_o \text{ (Configurations a, b, c)} \\ &= 1106 / 1076 \\ &= 1.0279 \end{aligned}$$

Ratio [ρ_c]:

$$\begin{aligned} &= D_c / D_o \text{ (Configurations a, e, f)} \\ &= 1106 / 1076 \\ &= 1.0279 \end{aligned}$$

ASME Section VIII-2 Part 4.18.7.4 Step 3:

Ratio [ρ]:

$$\begin{aligned} &= l_{tx} / h \\ &= 0 / 164 \\ &= 0.000 \text{ (must be } 0 \leq \rho \leq 1 \text{)} \end{aligned}$$

Effective tube hole Diameter [d^*]:

$$\begin{aligned} &= \max(dt - 2t_t(E_t T / E) \cdot (S_t T / S) \cdot (\rho); dt - 2t_t) \\ &= \max(19 - 2 \cdot 2.11 (176174368 / 176174368) \cdot (99 / 147) \cdot (0); 19 - 2 \cdot 2.11) \\ &= 19.0500 \text{ mm} \end{aligned}$$

Effective tube pitch [p^*]:

$$\begin{aligned} &= p / \sqrt{ (1 - 4 \cdot \min(A_l; 4 \cdot D_o \cdot p) / (\pi \cdot D_o^2)) } \\ &= 23.8 / \sqrt{ (1 - 4 \cdot \min(419; 4 \cdot 1076 \cdot 23.8) / (\pi \cdot 1076^2)) } \\ &= 24.3810 \text{ mm} \end{aligned}$$

Effective ligament efficiency for bending [μ^*]:

$$\begin{aligned} &= (p^* - d^*) / p^* \\ &= (24.4 - 19) / 24.4 \\ &= 0.2187 \end{aligned}$$

$h/p = 6.887139$; $\mu^* = 0.218653$; $E^* = 31303484 \text{ KPa}$.

$E^*/E = 0.177685$; $v^* = 0.432843$

Note: As h/p (6.887) is > 2 , data values for $h/p = 2$ were used.

ASME Section VIII-2 Part 4.18.7.4 Step 4:

Shell coefficient [β_s]:

$$\begin{aligned} &= ((12(1 - v_s^2))^{1/4}) / \sqrt{ ((D_s + t_s) t_s) } \\ &= ((12(1 - 0.3^2))^{1/4}) / \sqrt{ ((1106 + 15) 15) } \\ &= 0.0140 \text{ 1/mm} \end{aligned}$$

Shell coefficient [k_s]:

$$= \beta_s \cdot E_s \cdot t_s^3 / (6(1 - v_s^2))$$



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$$= 0.014 \cdot 186735680 \cdot 15^3 / (6(1 - 0.3^2))$$

$$= 1618138240.0000 \text{ KPa} \cdot \text{mm}^2$$

Shell coefficient [λ_s]:

$$= 6(D_s + t_s) \cdot k_s / h^3 \cdot (1 + h \cdot \beta_s + (h \cdot \beta_s)^2 / 2)$$

$$= 6(1106 + 15) \cdot 1618138240 / 164^3 \cdot (1 + 164 \cdot 0.014 + (164 \cdot 0.014)^2 / 2)$$

$$= 14661119.0 \text{ KPa}$$

Shell coefficient [δ_s]:

$$= D_s^2 / (4 \cdot E_s \cdot t_s) \cdot (1 - D_s \cdot v_s / (2(D_s + t_s)))$$

$$= 1106^2 / (4 \cdot 186735680 \cdot 15) \cdot (1 - 1106 \cdot 0.3 / (2(1106 + 15)))$$

$$= 0.93014\text{E-}01 \text{ mm/N./mm}^2$$

Parameter [ω_s]:

$$= \rho_s \cdot k_s \cdot \beta_s \cdot \delta_s (1 + h \cdot \beta_s)$$

$$= 1.03 \cdot 1618138240 \cdot 0.014 \cdot 0.093014315 (1 + 164 \cdot 0.014)$$

$$= 7155.3599 \text{ mm}^2$$

Channel coefficient [β_c]:

$$= ((12(1 - v_c^2))^{1/4}) / \sqrt{(D_c + t_c) t_c}$$

$$= ((12(1 - 0.3^2))^{1/4}) / ((1106 + 29) 29)$$

$$= 0.0100 \text{ 1/mm}$$

Channel coefficient [k_c]:

$$= \beta_c \cdot E_c \cdot t_c^3 / (6(1 - v_s^2))$$

$$= 0.01 \cdot 176174368 \cdot 29^3 / (6(1 - 0.3^2))$$

$$= 7885020160.0000 \text{ KPa} \cdot \text{mm}^2$$

Channel coefficients [CC, CqDelta, CqTheta, CmTheta, CmDelta]:

$$\text{CqDelta} = \text{CqTheta} = \text{CmTheta} = \text{CmDelta} = 1$$

Channel coefficient [λ_c]:

$$= 6(D_c + t_c / \cos(\theta)) \cdot k_c / h^3 \cdot (C_m\Theta + h \cdot \beta_c \cdot (C_m\Delta + C_q\Theta) / 2 + (h \cdot \beta_c)^2 / 2 \cdot C_q\Delta)$$

$$= 6(1106 + 29 / \cos(0)) \cdot 7.89\text{E}+09 / 164^3 \cdot (1 + 164 \cdot 0.01 \cdot (1 + 1) / 2 + (164 \cdot 0.01)^2 / 2 \cdot 1)$$

$$= 48613668.0 \text{ KPa}$$

Channel coefficient [δ_c]:

$$= D_c^2 / (4 \cdot E_c \cdot t_c) \cdot (1 - D_c \cdot v_c / (2(D_c + t_c)))$$

$$= 1106^2 / (4 \cdot 176174368 \cdot 29) \cdot (1 - 1106 \cdot 0.3 / (2(1106 + 29)))$$

$$= 0.51104\text{E-}01 \text{ mm/N./mm}^2$$

Channel coefficient [zetac]:

$$\text{zetac} = 0$$

Parameter [ω_c]:

$$= \rho_c \cdot k_c [\delta_c (C_m\Theta + \beta_c + C_q\Delta \cdot h \cdot \beta_c^2) - \zeta_c (C_m\Theta + h \cdot C_q\Theta \cdot h \cdot \beta_c / 2)]$$

$$= 1.03 \cdot 7.89\text{E}+09 [0.051104259 (1 + 0.01 \cdot 1 \cdot 164) - 0.000 (1/164 + 1 \cdot 0.01/2)]$$

$$= 10970.4473 \text{ mm}^2$$

ASME Section VIII-2 Part 4.18.7.4 Step 5:

Diameter ratio [K]:

$$= A / D_o$$

$$= 1156 / 1076$$

$$= 1.0743$$

Coefficient [F]:

$$= (1 - v^*) / E^* \cdot (\lambda_s + \lambda_c + E \cdot \ln(K))$$

$$= (1 - 0.43) / 31303484 \cdot (14661119 + 48613668 + 176174368 \cdot \ln(1.07))$$

$$= 1.3753$$



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ASME Section VIII-2 Part 4.18.7.4 Step 6:

Moment on tubesheet due to pressures (Ps, Pt) [Mts]:

$$= \frac{D_o^2}{16} \cdot [(p_s - 1) \cdot (p_s^2 + 1) \cdot P_s - (p_c - 1) \cdot (p_c^2 + 1) \cdot P_t]$$

$$= \frac{1076^2}{16} \cdot [(1.03 - 1) \cdot (1.03^2 + 1) \cdot 1080 - (1.03 - 1) \cdot (1.03^2 + 1) \cdot 5900]$$

$$= -19998580.0000 \text{ KPa} \cdot \text{mm}^2$$

Moment acting on unperforated tubesheet rim [M*]:

$$= M_{ts} + \omega_c \cdot P_t - \omega_s \cdot P_s$$

$$= -19998580 + 10970 \cdot 5900 - 7155 \cdot 1080$$

$$= 36999276.0000 \text{ KPa} \cdot \text{mm}^2$$

ASME Section VIII-2 Part 4.18.7.4 Step 7:

Maximum bending moment acting on periphery of tubesheet [Mp]:

$$= ((M_{ts}) - \frac{D_o^2}{32} \cdot F (P_s - P_t)) / (1 + F)$$

$$= ((36999276) - \frac{1076^2}{32} \cdot 1.38 (1080 - 5900)) / (1 + 1.38)$$

$$= 116549160.0000 \text{ KPa} \cdot \text{mm}^2$$

Maximum bending moment acting on center of tubesheet [Mo]:

$$= M_p + \frac{D_o^2}{64} \cdot (3 + \nu^*) \cdot (P_s - P_t)$$

$$= 116549160 + \frac{1076^2}{64} \cdot (3 + 0.43) \cdot (1080 - 5900)$$

$$= -182777568.0000 \text{ KPa} \cdot \text{mm}^2$$

Maximum bending moment acting on tubesheet [M]:

$$= \max(|M_p| ; |M_o|)$$

$$= \max(|116549160| ; |-182777568|)$$

$$= 182777568.0000 \text{ KPa} \cdot \text{mm}^2$$

ASME Section VIII-2 Part 4.18.7.4 Step 8:

Tubesheet Bending Stress at original thickness:

$$= 6 \cdot M / (\mu^* (h - h_g')^2)$$

$$= 6 \cdot 182777568 / (0.22 (164 - 5)^2)$$

$$= 198.4033 \text{ N./mm}^2$$

Allowable Tubesheet Bending Stress [σ_{All}]:

$$= 2 \cdot S$$

$$= 2 \cdot 147$$

$$= 294.01 \text{ N./mm}^2$$

Tubesheet bending stress at final thickness [σ]:

$$= 6 \cdot M / (\mu^* (h - h_g')^2)$$

$$= 6 \cdot 173625520 / (0.22 (133 - 5)^2)$$

$$= 293.0731 \text{ N./mm}^2$$

Required tubesheet thickness for Bending Stress [HreqB]:

$$= h + C_{ats} + C_{atc}$$

$$= 133 + 3 + 3$$

$$= 138.5055 \text{ mm}$$

Required tubesheet thickness for given loadings (includes CA) [Hreq]:

$$= \max(H_{reqB}; H_{reqS})$$

$$= \max(139 ; 27.6)$$

$$= 138.5055 \text{ mm}$$

ASME Section VIII-2 Part 4.18.7.4 Step 9:

Absolute pressure differential [| Ps - Pt |]:

$$= | P_s - P_t |$$

$$= | 1080 - 5900 |$$

$$= 4820.000 \text{ KPa}$$



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Shear Stress check [τ limit]:

$$= 4 \cdot \mu \cdot h / D_o \cdot \min(0.8 \cdot S; 0.533 \cdot S_y)$$

$$= 4 \cdot 0.2 \cdot 164 / 1076 \cdot \min(0.8 \cdot 147; 0.533 \cdot 223)$$

$$= 14.340 \text{ N./mm}^2$$

$|P_s - P_t|$ is $\leq$ the Shear Stress limit (τ limit). Therefore, the required thickness for shear is for information only.

Minimum required thickness for Shear [H_{reqS}]:

$$= 1 / (4 \cdot \mu \cdot \min(0.8 \cdot S; 0.533 \cdot S_y)) \cdot (4 \cdot A_p / C_p) \cdot |P_s - P_t| + C_{ats} + C_{atc}$$

$$= 1 / (4 \cdot 0.2 \cdot 118) \cdot (4 \cdot 2686 / 2543) \cdot |1080 - 5900| + 6$$

$$= 27.6432 \text{ mm}$$

ASME Section VIII-2 Part 4.18.7.4 Step 10:

Note: For a given Shell thickness of: 15.000mm

Min. Shell len. adjacent to the tubesheet is: 231.844 mm

Axial Shell Membrane Stress [σ_s, m]:

$$= D_s^2 / (4 \cdot t_s (D_s + t_s)) \cdot P_s$$

$$= 1106^2 / (4 \cdot 15 (1106 + 15)) \cdot 1080$$

$$= 19.6428 \text{ N./mm}^2$$

Axial Shell Bending Stress [σ_s, b]:

$$= 6 \cdot k_s / t_s^2 \cdot [\beta S \cdot P_s + 6(1 - \nu^*) / E^* \cdot D_o / h^3 \cdot (1 + h \cdot \beta S / 2) \cdot (M_p + D_o^2 / 32 \cdot (P_s - P_t))]$$

$$= 6 \cdot 363780 / 15^2 \cdot [0.014 \cdot 0.1 + 6(1 - 0.43) / 31303484 \cdot 1076 / 164^3 \cdot (1 + 164 \cdot 0.014 / 2) \cdot (116549160 + 1076^2 / 32 \cdot (1080 - 5900))]$$

$$= -81.502 \text{ N./mm}^2$$

Shell Membrane + Bending Stress [σ_s]:

$$= |\sigma_{s,m}| + |\sigma_{s,b}| \text{ (Should be } \leq 1.5 \cdot S_s)$$

$$= 101.145 \text{ N./mm}^2 \text{ (Should be } \leq 205.50)$$

Note: For a given Channel thickness of: 29.000 mm

Min. Channel len. adjacent to the tubesheet is: 322.366mm

Axial Channel Membrane Stress [σ_c, m]:

$$= D_c^2 / (4 \cdot t_c (D_c + t_c) \cdot \cos(\theta)) \cdot P_t$$

$$= 1106^2 / (4 \cdot 29 (1106 + 29) \cdot \cos(0)) \cdot 5900$$

$$= 54.819 \text{ N./mm}^2$$

Axial Channel Bending Stress [σ_c, b]:

$$= 6 \cdot k_c / t_c^2 \cdot [(\delta_c \cdot C_m \Delta + \beta_c - \zeta_c \cdot C_m \theta) \cdot P_t - 6((1 - \nu^*) / E^*) \cdot D_o / h^3 \cdot (C_m \theta + h \cdot \beta_c / 2 \cdot C_m \Delta) \cdot (M_p + D_o^2 / 32 \cdot (P_s - P_t))]$$

$$= 6 \cdot 1772663 / 29^2 \cdot [(0.051104259 \cdot 1 \cdot 0.01 - 0.000 \cdot 1) \cdot 5900 - 6((1 - 0.43) / 31303484) \cdot 1076 / 164^3 \cdot (1 + 0.01 \cdot 164 / 2 \cdot 1) \cdot (116549160 + 1076^2 / 32 \cdot (1080 - 5900))]$$

$$= 327.159 \text{ N./mm}^2$$

Channel Membrane + Bending Stress [σ_c]:

$$= |\sigma_{c,m}| + |\sigma_{c,b}| \text{ (Should be } \leq 1.5 \cdot S_c)$$

$$= 381.979 \text{ N./mm}^2 \text{ (Should be } \leq 220.51)$$

Step 11, The Cylinder-to-Tubesheet Juncture is Overstressed.

Performing a simplified Elastic-Plastic calculation

(option 3) to reduce the overstress condition.

Modify E_s and/or E_c and recompute from Step 4 onwards.

Modified Elasticity [E_c]:

$$= E_c \cdot \sqrt{(1.5 \cdot S_c / \sigma_c)}$$

$$= 176174368 \cdot \sqrt{(1.5 \cdot 147 / 382)}$$



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= 133854848.0 KPa.

Tubesheet Bending Stress at Original Thk., after the Elastic-Plastic iteration:

= $6 \cdot M / (\mu \cdot (h - h'g)^2)$

= $6 \cdot 188433392 / (0.22 \cdot (164 - 5)^2)$

= 204.5 N./mm² (Should be ≤ 294.0)



Note: The tubesheet is Not overstressed after the Elastic-Plastic iteration.

The design is acceptable. Recomputing tubesheet required thickness.

Required Tubesheet Thk., for Bending Stress after the Elastic-Plastic iteration [HreqB]:

= $h + Cats + Catc = 135 + 3 + 3 = 141$ mm

Required Tubesheet Thk. after the Elastic-Plastic iteration (includes CA) [Hreq]:

= $\max(HreqB; HreqS) = \max(141; 27.6) = 141$ mm

Tube Metal Area [At]:

= $\pi (do - t) t$

= $\pi (19 - 2.11) 2.11$

= 1.123 cm²

Tube Weld Size Results per UW-20.3:

Tube Strength [Ft]:

= $\pi \cdot t (do - t) S$

= $\pi \cdot 2.11 (19 - 2.11) 99$

= 1133.373 Kgf

Groove Weld Strength [Fg]:

= $0.85 \cdot \pi \cdot ag (do + 0.67 \cdot ag) Sw$ (but not > Ft)

= $0.85 \cdot \pi \cdot 2.11 (19 + 0.67 \cdot 2.11) 99$

= 1133.3734 Kgf

Weld Strength Factor [fw]:

= S / Sw where $Sw = \min(S \text{ and } St)$

= 99/99

= 1.000

Design Strength Ratio [fd]:

= 1 for full strength welds

Fillet Weld Strength [Ff]:

= $0.55 \cdot \pi \cdot af (do + 0.67 \cdot af) Sw$ (but not > Ft)

= $0.55 \cdot \pi \cdot 2.11 (19 + 0.67 \cdot 2.11) 99$

= 753.0200 Kgf

Minimum Length of Weld [ar]:

= $2 \cdot \sqrt{((0.75 \cdot do)^2 + 1.07 \cdot t \cdot (do - t) \cdot fw \cdot fd) - 0.75 \cdot do}$

= $2 \cdot \sqrt{((0.75 \cdot 19)^2 + 1.07 \cdot 2.11 \cdot (19 - 2.11) \cdot 1 \cdot 1) - 0.75 \cdot 19}$

= 2.562 mm

Minimum Required Fillet Weld Leg [afr]: 1.2810 mm

Minimum Required Groove Weld Leg [agr]: 1.2810 mm

Tube-Tubesheet Jt allowable, 1133 is >= tube strength 1133 Kgf

Note: This tube-tubesheet joint is a Full Strength joint

Stress/Force summary for loadcase D3 corr. (Psd,max + Ptd,max):

Stress Description	Actual	Allowable	Pass/Fail
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Tubesheet bend. stress	204.5	<=	294.0 N./mm ²	Ok
Tubesheet shear stress	15.5	<=	117.6 N./mm ²	Ok
Stress in Shell at Tubesheet	101.1	<=	205.5 N./mm ²	Ok
Stress in Channel at Tubesheet	382.0	<=	446.0 N./mm ²	Ok

Thickness results for loadcase D3 corr. (Psd,max + Ptd,max):

Thickness (mm)	Required	Actual	P/F
Tubesheet Thickness :	140.930	170.000	Ok
Tube-Tubesheet Fillet Weld Leg :	1.281	2.110	Ok
Tube-Tubesheet Groove Weld Leg :	1.281	2.110	Ok

U-Tube Tubesheet results per ASME Div. 2 Part 4.18.7 2025

Results for 6 Load Cases:

Case#	--Reqd. Thk. + CA Tbsht Extnsn	----	Tubesheet Bend Allwd	Stresses Shear Allwd	Case Type	Pass/ Fail
D1uc	150.099 ...	232	294	18	118	Ps+Pt D1 Ok
D2uc	65.844 ...	43	294	3	118	Ps+Pt D2 Ok
D3uc	133.855 ...	187	294	15	118	Ps+Pt D3 Ok
D1c	157.731 ...	254	294	19	118	Ps+Pt-c D1 Ok
D2c	68.511 ...	47	294	3	118	Ps+Pt-c D2 Ok
D3c	140.930 ...	205	294	16	118	Ps+Pt-c D3 Ok
Max:	157.7312 ... mm	0.865	0.162	(Str. Ratio)		

Load Case Definitions:

[Ps & Pt]:

Shell-side and Tube-side Design or Operating Pressures
derived from Psd,min Ptd,max, Psox,min, Ptox,max etc. per the
Load Case Tables

[c]:

With or Without Corrosion Allowance

[D1, D2, D3]:

Design Load Cases using the Maximum and Minimum Design Pressures

[D4]:

Design Load Case using the Minimum (Vacuum) Pressures (if specified)

Shell and Channel Stress Summary:

Case#	-- Integral Cyl. Req'd. Thk. Shell Channel	Shell Stress	Stress Allwd	Channel Stress	Stress Allwd	Pass Fail
D1uc	...	...	149	205	364	446 Ok
D2uc	...	...	97	205	34	221 Ok
D3uc	...	...	84	205	330	446 Ok
D1c	...	...	179	205	423	446 Ok
D2c	...	...	118	205	41	221 Ok
D3c	...	...	101	205	382	446 Ok
Max	...	...	0.873	0.948		

Note:

Because there was no net pressure differential across the tubesheet due to
the specified pressures psd,min and ptd,min; load case D4 was not processed as
the resulting tubesheet bending stress would be zero.



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Summary of Thickness Comparisons for 6 Load Cases:

Thickness (mm)	Required	Actual	P/F
Tubesheet thickness :	157.731	170.000	Ok
Tube thickness :	0.555	2.110	Ok
Tube-tubesheet fillet weld leg :	1.281	2.110	Ok
Tube-tubesheet groove weld leg :	1.281	2.110	Ok

Given Shell Thickness: 18.0000 mm

Given Channel Thickness: 32.0000 mm

Min Shell length of thk, (18.000) adj. to tubesheet: 253.282 mm

Min Channel length of thk, (32.000) adj. to tubesheet: 337.710 mm

Note: This is a full strength Tube to Tubesheet Joint.

Tubesheet MAWP used to Compute Hydrotest Pressure:

Stress / Force Condition	Tubeside MAWP	Ø shellside Stress Rat.	Shellside MAWP	Ø tubeside Stress Rat.
Tubesheet Bending Stress	7075.48	1.000	6790.57	1.000
Tubesheet Shear Stress	36523.24	1.000	36523.24	1.000
Shell Stress (Axial, Junction)	7075.48	0.512	3859.99	1.000
Tubesheet-Channel Junction	6218.15	0.999	3859.99	0.575
Tube Pressure Stress	24057.10	1.000	12139.80	1.000
Minimum MAWP	6218.15		3859.99	

Tubesheet MAPnc used to Compute Hydrotest Pressure:

Stress / Force Condition	Tubeside MAPnc	Ø shellside Stress Rat.	Shellside MAPnc	Ø tubeside Stress Rat.
Tubesheet Bending Stress	7408.15	0.999	7408.15	0.999
Tubesheet Shear Stress	38115.14	1.000	38115.14	1.000
Shell Stress (Axial, Junction)	7408.15	0.999	5868.25	1.000
Tubesheet-Channel Junction	7408.15	0.999	5868.25	0.804
Tube Pressure Stress	28678.83	1.000	19233.21	1.000
Minimum MAPnc	7408.15		5868.25	

(*) All load cases were analyzed to compute the MAWP for determining the test pressure.

Tubesheet MDMT Calculations:

Shell Side MDMT calculation:

Governing thickness on the shell side per figure UCS-66.3 (e):

$$\begin{aligned}
 &= \max(\text{tubesheet thk}/4, \min(\text{tubesheet thk}, \text{shell thickness})) \\
 &= \max(170/4, \min(170, 18)) \\
 &= 42.500 \text{ mm}
 \end{aligned}$$

MDMT reduction ratio per UCS 66 (b)(1)(b):

$$\begin{aligned}
 &= \min(\text{ps}/\text{Shellside MAPnc}, 1) \\
 &= \min(1080/5868, 1) \\
 &= 0.184
 \end{aligned}$$

Thickness Ratio = 0.18, Temperature Reduction per Fig. UCS 66.1 = 78 °C



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ASME TS Calc: DESIGN 1

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Min Metal Temp. w/o impact per UCS-66, Curve B	13 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Channel Side MDMT calculation:

Governing Channel side thickness per figure UCS-66.3 (e):
= max(tubesheet thk/4, min(tubesheet thk, channel thickness))
= max(170/4, min(170, 32))
= 42.500 mm

MDMT reduction ratio per UCS 66 (b)(1)(b):
= min(pt/Tubeside MAPnc, 1)
= min(5900/7408, 1)
= 0.796

Thickness Ratio = 0.8, Temperature Reduction per Fig. UCS 66.1 = 11 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	33 °C
Min Metal Temp. at Required thickness (UCS 66.1)	21 °C

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Problems/Failures Summary:

Step: 33

9:45am

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Noted issues for the current model:

Listed below are the known problem areas for the current design. If one or more of the design flags are turned on, please re-run the analysis. Some of these issues may be resolved when using updated input values.

**** Warning: UCS-57 Joint Efficiency Violation!**

**** Warning: A standard flange seems to be governing the Shellside MAWP!**

**** Warning: The net force on one of the saddles putting it in tension!**

Please review all reports carefully!

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PV Elite® Pressure Vessel Design Notes

Project Information	
Project Number	
Client	
Date	
Designer	
Design Code	
Additional Standards	

Design Conditions		
Internal Design Pressure:		
External Design Pressure:		
Design Int. Temperature:		
Design Ext. Temperature:		
Corrosion Allowance:		

Vessel Geometry		
Vessel Type		
Diameter		
Tangent-to-Tangent Length		
Head Type		
Number of Segments		

Material Specifications	
Shell:	
Head:	
Nozzle:	

Nozzle/Opening Details	ID/OD	Size	Rating	Matl.	Reinf.

Miscellaneous		
Support Type		
Number of Supports		
Support Material		
Seismic Code		
Wind Code		
Insulation		
Pressure Test		
Radiography		
Fatigue Analysis		

Analysis Notes: